

PZ195E User Manual

E-711/E-712 Modular Digital

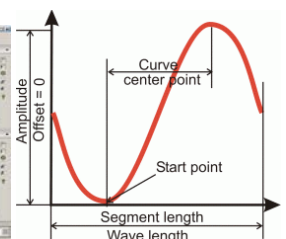
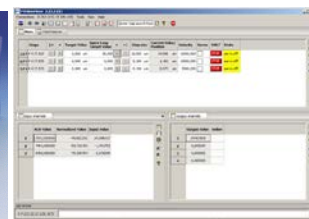
Multi-Channel Controller System

Release: 2.0.1 Date: 06 September 2016



This document describes the following products:

- E-711 and E-712
Digital Multi-Channel Controller System; Modules and Predefined Configurations



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About this Document

This manual is valid for the hardware components and predefined configurations listed in "Model Survey". Furthermore, it is also valid for all customized hardware configurations provided that they are not accompanied by special documentation which supersedes this manual.

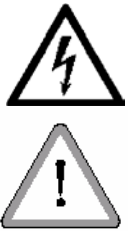
Users of this Manual

This manual is designed to help the reader to operate the E-711/E-712 Modular Digital Multi-Channel Controller System. It assumes that the reader has a fundamental understanding of basic servo systems, as well as motion control concepts and applicable safety procedures.

The manual describes the physical specifications and dimensions of the controller system as well as the software and hardware installation procedures which are required to put the associated motion system into operation.

Symbols and Typographic Conventions

The notes and symbols used in this manual have the following meanings:



WARNING

Calls attention to a procedure, practice or condition which, if not correctly performed or adhered to, could result in injury or death.



CAUTION

Calls attention to a procedure, practice, or condition which, if not correctly performed or adhered to, could result in damage to equipment.

NOTE

Provides additional information or application hints.



Warning signs affixed to the product that refer to detailed information in this manual.

Symbol for the protective earth conductor, affixed to the product.

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1 Introduction

- Modular Design for Highest Flexibility
- PC-based; 50 kHz Sensor Sampling; Low-Noise, 20-bit D/A Converters
- Ethernet (TCP/IP) for Remote Control Capability, USB, RS-232, SPI
- Optional High-Bandwidth Analog Control Inputs / Sensor Inputs
- Optional Analog Outputs for Control of External Amplifiers or Position Monitoring
- Digital I/O Lines for Task Triggering
- Comprehensive Software Package
- Support of Capacitive Sensors, Strain Gauge Sensors (SGS) and Encoders (incremental, absolute measuring)

In this document, the E-711/E-712 Modular Digital Multi-Channel Controller System is also referred to as "E-711/E-712 system" or "controller".

Modular Design

The number of axes and channels and the functionality of the controller can be configured flexibly:

- Digital processor modules with firmware for nanopositioning systems and PiezoWalk® systems; optional firmware upgrades for alternative control algorithm and advanced linearization
- Amplifier modules with output voltage ranges of -10 to +45 V, -30 to 135 V and -250 to 250 V (with or without sensor input); external high-voltage amplifiers with up to 1100 volts output can be integrated in the system
- DC motor driver module, 2 channels, with incremental encoder input
- Sensor modules for single- and dual-electrode capacitive sensors, strain gauge sensors and encoders; digital signal transmission for longer distances is possible
- Interface modules for analog and digital in- and output and SPI
- 9.5" and 19" chassis with power supplies suitable for the individual amplifier output voltage ranges

The individual modules and chassis can be ordered separately. Furthermore, predefined system configurations (E-712.3CD, .3CDA, .6CD, .6CDA, .1AM) are available. See "Model Survey" on p. 17 for more information.

In this document, the predefined system configurations are also referred to as "controller".

Digital Linearization and Control Algorithms for Highest Accuracy

Linearization algorithms based on higher-order polynomials will not only improve the positioning accuracy to 0.001% of the travel range, but make possible simple replacement of the mechanics as well. With its outstanding computing power, the processor provides for optimal linearization procedure.

During fast periodic motion, as typical for scanning applications, the tracking accuracy can be further improved with Dynamic Digital Linearization (DDL). This optionally available control algorithm reduces the tracking error by a factor of up to 1000.

Higher Velocity and Bandwidth for Dynamic Applications

The controller is perfectly suited for high-dynamics operation, thanks to its high-resolution DA-converters and high-performance voltage amplifiers. The high-speed processor of the PC-based system with a sensor sampling rate of 50 kHz assures settling times in the millisecond range and below.

Flexibility for a Variety of Applications

PI nanopositioning systems which are equipped with an ID-chip and calibrated with a digital controller have the mechanics-related calibration and servo-control parameters stored in the chip. The controller automatically adapts to the connected mechanics by the appropriate use of this data, so that recalibration is not necessary when system components are replaced.

The interfaces enable command of the mechanics either from a PC, an SPI master or with an analog control signal connected directly to the device. The digital I/O can be programmed for the most diverse types of triggering tasks.

The integrated wave generators can save and output periodic motion profiles. In addition to sine and triangle waves, arbitrary, user-defined profiles can be created.

The flexibly configurable data recorder enables simultaneous recording and read-out of up to 8 input and output signals, such as for sensor positions or control voltages. This makes the later association of events with saved data points possible.

Simple System Integration

Last but not least, all parameters can be checked and reset via software. System setup and configuration is done with the included PIMikroMove user-interface software. Interfacing to custom software is facilitated with included LabView drivers and DLLs. System programming is the same with all PI controllers, so controlling a system with a variety of different controllers is possible without difficulty.

1.1 Other Applicable Documents

This manual is valid for the hardware components and predefined configurations listed in “Model Survey” on p. 17. Furthermore, it is also valid for all customized hardware configurations provided that they are not accompanied by special documentation which supersedes the information given in this manual.

GCS commands

The commands of the PI General Command Set (GCS) which is used to command the E-711/E-712 system are described in a separate manual, PZ233E.

Some of the hardware modules and firmware features and all software tools mentioned in this manual are described in detail in their own manuals or Technical Notes (see tables below).

Hardware modules and firmware features

Hardware / Firmware	Document
E-712.N1 Digital PiezoWalk® Processor Module	E712T0010_PiezoWalk_DESCRIPTION
SPI interface of E-712.M1 and E-712.N1 digital processor and interface modules	SPIST0001 SPI Slave Description
E-712.U1 Advanced Piezo Control option	E712T0007_Advanced_Piezo_Control
E-712.U4 Firmware Update Advanced Linearization Option	E712T0009
E-711.IP PIO Interface Module	E711T0001_PIO_Interface
E-711.OCT and .OCT0 Digital Sensor Signal Transmission	E711T0002_OCT_OCT0_Digital_Sensor_Transmission
E-711.OET and .OET0 Digital Sensor Signal Transmission	E711T0007_OET_OET0_Digital_Sensor_Transmission
Hexapod Firmware	E712T0011_Hexapod_Description
Autofocus Functionality	E712T0017_Autofocus

PC-Software tools

Tool	Document
PIMikroMove	PIMikroMoveUserManual_SM148E
LabVIEW drivers	E712_GCSTLabVIEW_PZ196E
GCS library	PIGCS_2_0_DLL_SM151E
GCS data format	GCSDData_User_SM146E
PI Update Finder: Search and download updates	A000T0028 Technical Note
PI Update Finder: Updating PC without Internet connection	A000T0032 Technical Note

The latest versions of the relevant manuals and Technical Notes for your system are available for download on our website (www.pi.ws).

For inquiries and orders, contact your PI sales engineer or send us an e-mail (info@pi.ws).

1.2 Intended Use

The E-711/E-712 system is a laboratory device according to DIN EN 61010. It is intended to be used in interior spaces and in an environment which is free of dirt, oil and lubricants.

Corresponding to its design, the E-711/E-712 system is intended for driving capacitive loads (e. g. piezo ceramic actuators) and/or DC motors. The drive type to be used depends on the hardware configuration of the E-711/E-712 system.

The E-711/E-712 system must not be used for purposes other than those named in this user manual.

The E-711/E-712 system can be used for static as well as dynamic applications.

Position sensors must be used for closed-loop operation. The sensor type to be used depends on the hardware configuration of the E-711/E-712 system (single- and dual-electrode capacitive sensors, strain gauge sensors (SGS) and encoders (incremental, absolut measuring)). PI stages intended for closed-loop operation already have the corresponding sensors. Other sensors can only be used with PI approval.

Only authorized and qualified personnel must install, operate, maintain and clean the E-711/E-712 system. Only PI service personnel must repair the E-711/E-712 system.

Observe the safety precautions given in this manual.

1.3 Safety Precautions



WARNING — READ INSTRUCTION

Install and operate the E-711/E-712 system only when you have read the operating instruction. Keep the instruction readily available close to the device in a safe place. When the instruction is lost or has become unusable, ask the manufacturer for a new copy. Add all information given by the manufacturer to the instruction, e.g. supplements or Technical Notes.



WARNING

The chassis of the E-711/E-712 system requires a supply voltage of 100 to 240 VAC. The supply voltage may cause serious or even lethal injury. Strictly observe the following:

Install the E-711/E-712 system near the power source so that the plug of the power cord can be quickly and easily disconnected from the mains supply socket.

Use the supplied power cord (p. 62) to connect the E-711/E-712 system to the power source. If the supplied power cord has to be replaced, use a sufficiently dimensioned power cord.

To disconnect the E-711/E-712 system from the power source completely, remove the plug of the power cord from the mains supply socket, or remove the power cord from the chassis of the E-711/E-712 system.

See "AC Power and Line Fuses" (p. 259) for how to replace the line power fuses.

WARNING



If a sensor box is used that also has connections for piezo actuators (digital sensor signal transmission with E-711.0CT, E-711.0CT0, E-711.0ET, E-711.0ET0):

If a protective earth conductor is not or not properly connected to the sensor box, dangerous touch voltages can occur on the sensor box in the case of malfunction or failure. If touch voltages exist, touching the sensor box can result in serious injury or death from electric shock.

- Connect the sensor box to a protective earth conductor before start-up (p. 71).
- Do not remove the protective earth conductor during operation.
- If the protective earth conductor has to be removed temporarily (e. g. in the case of modifications or repair), reconnect the sensor box to the protective earth conductor before starting it up again.

WARNING



If stages with piezo actuators are used:

Temperature changes and compressive stresses can induce charges in piezo actuators. After being disconnected from the electronics, piezo actuators can also stay charged for several hours. Touching or short-circuiting the contacts in the connector of the piezo actuator can lead to minor injuries. In addition, piezo actuators can be destroyed by an abrupt contraction.

- Do **not** touch the contacts in the connector of the stage.
- Do **not** pull out the connector of the stage from the E-711/E-712 system during operation.
- Secure the connector of the stage with screws against accidental disconnection.

If a sensor box is used that also has connections for piezo actuators (digital sensor signal transmission with E-711.0CT, E-711.0CT0, E-711.0ET, E-711.0ET0):

- Do **not** touch the contacts in the PZT panel plug of the sensor box when piezo actuators are connected to the sensor box.

- Before you connect the sensor box to the corresponding sensor or interface module, make sure that the E-711/E-712 system is switched off.
- During operation, do **not** disconnect the sensor box from the corresponding sensor or interface module.
- Secure the connections on the sensor box and on the corresponding sensor or interface module with screws against accidental disconnection.



WARNING

The E-711/E-712 system does not contain any user-serviceable parts. Any system modification or configuration may be done solely by PI service personnel.

- Never disassemble any part of the E-711/E-712 system. Hazardous voltage can be present on the internal components.



CAUTION

Both line power fuses are active and have to be checked if there is a fault.



CAUTION

Place the system in a location with adequate ventilation to prevent internal heat build-up. Allow at least 15 cm clearance from the front and the rear of the unit and 1 cm from the bottom (ensured by the feet of the chassis).

Never cover any ventilation openings as this will impede ventilation.



CAUTION

Thermally stable systems have the best performance. For a thermally stable system, power on the E-711/E-712 system at least one hour before you start working with it.

CAUTION

If the internal temperature goes out of range, the voltage output of the E-711/E-712 system will be deactivated. In that case communication with the E-711/E-712 system is still possible, but any output-control values will be ignored, and the connected mechanics will not move.

!

CAUTION

Make sure that boxes for sensor-signal transmission or splitting are connected to the corresponding sensor or interface module before powering up the E-711/E-712 system. Connecting a box or removing it while the system is already powered on can cause damage to the electronics.

!

CAUTION

Incorrect parameter values may lead to improper operation or damage to your hardware. Be careful when changing parameters.

It is strongly recommended to save the parameter values of the E-711/E-712 system to a file on the host PC before you make any changes. This way the original settings can be restored if the new parameter settings will not prove satisfactory. To save the parameter values and to load them back to the E-711/E-712 system, use the *Device Parameter Configuration* window of PIMikroMove. See "Creating Backup Files for Controller Parameters" (p. 86) for more information.

!

CAUTION

If the stage starts oscillating (humming noise):

In closed-loop operation, switch off the servo immediately. Adjust the notch filter(s) and the servo-control parameters (P-term, I-term); see "Servo-Controller Dynamic Tuning" (p. 218) for more information).

In open-loop operation, stop the axis motion immediately. Do not operate the stage at its resonant frequency because the notch filters are not active in open-loop operation by default. You can measure the resonant frequency using the Piezo Dynamic Tuner window of PIMikroMove; see "Servo-Controller Dynamic Tuning" (p. 218) for more information.

Otherwise the stage could be irreparable damaged.

!



CAUTION

Before you work with Dynamic Digital Linearization (DDL), eliminate any residual oscillations by adjusting the notch filter(s) and the servo-control parameters (P-term, I-term; see "Servo-Controller Dynamic Tuning" (p. 218)). Using DDL could be critical if there are any residual oscillations in the system; DDL will then cause the oscillations to build up—the more wave generator cycles are used for DDL initialization, the stronger the effect.

1.4 Model Survey

The E-711/E-712 digital multi-channel controller system can be closely adapted to many different applications. All devices are assembled and calibrated at the factory and come ready for use.

1.4.1 Customized Hardware Configurations

Contact your PI Sales Engineer or write info@pi.ws, if you need a customized configuration or want to upgrade your hardware with additional modules. Any system modification or configuration may be done solely by PI service personnel.

NOTES

Customized hardware configurations are not listed here in detail.

The product code for a customized configuration consists of "E-712K", followed by three digits. See the type label on the rear panel of the chassis for the product code of your configuration.

The individual modules, the chassis and the external components (e.g. sensor boxes) of the system can be identified via the configuration list which is delivered separately.

1.4.2 Standard Hardware Components

Chassis		Details on
E-712.R1	Digital Modular Piezo Controller System, 3 to 6 Channels, 9.5" Chassis with P/S	p. 21
E-712.R2	Digital Modular Piezo Controller System, 3 to 6 Channels, 19" Chassis with Power Supply for +/- 250 V Piezo Voltage	
E-712.R3	Digital Modular Piezo Controller System, 3 to 6 Channels, 19" Chassis with Power Supply for Low Voltage	

Chassis		Details on
E-712.R4	Digital Modular Piezo Controller System, 3 to 6 Channels, 9.5" Chassis with Power Supply for +/- 250 V Piezo Voltage	
E-712.R5	Digital Modular Piezo Controller System, Universal 19" 4U Chassis with cPCI Bus	p. 24

Digital Processor Modules		Details on
E-712.M1	Digital Processor and Interface Module with Ethernet Interface, USB, RS-232	p. 26
E-712.N1	Digital PiezoWalk® Processor and Interface Module with Ethernet Interface, USB, RS-232	

Amplifier Modules		Details on
E-711.AL4P	High-Power Amplifier Module, 4 Channels, 8 W, for E-712 Modular, Digital Controller System, -30 to 135 V	p. 29
E-711.AL41	High-Power Amplifier Module, 4 Channels, 8 W, for E-712 Modular, Digital Controller System, -30 to 135 V, Sub-D 25W3 (f) Socket	
E-711.AM4	Amplifier Module for NEXLINE®, for E-712 Modular, Digital Controller System, -250 V to +250 V, with Sub-D Mix connector	p. 32
E-711.AM5	Amplifier Module for NEXLINE®, for E-712 Modular, Digital Controller System, -250 V to +250 V, with Sub-D 25-pin (f) connector	
E-711.AM6	Amplifier Module for NEXLINE®, for E-712 Modular, Digital Controller System, -250 V to +250 V, Without External Connector	
E-711.AM5A	Amplifier Module for NEXLINE®, for E-712 Modular, Digital Controller System, -250 V to +250 V, Incremental Sin/Cos Sensor Input	p. 35
E-711.AM5B	Amplifier Module for NEXLINE®, for E-712 Modular, Digital Controller System, -250 V to +250 V, Absolut Encoder via BiSS Interface	
E-711.AN4	NEXACT® Amplifier Module, for E-712 Modular, Digital Controller System	p. 37
E-711.AN40	NEXACT® Amplifier Module, for E-712 Modular, Digital Controller System, Without External Connector	

Driver Modules		Details on
E-711.C82	DC Motor Driver, 2 Channels, for E-712 Modular, Digital Controller System	p. 39

Sensor Electronics		Details on
E-711.SA3	Module for Incremental Sensors, 3 Channels, for E-712 Modular, Digital Controller System	p. 41
E-711.SA6	Module for Incremental Sensors, 6 Channels, for E-712 Modular, Digital Controller System	
E-711.SAP	Sensor Signal Splitter, 3 Channels, PIOne Incremental Sensors, for E-712 Digital Controller	p. 42
E-711.SAH	Sensor Signal Splitter, 3 Channels, Incremental Sensors, for E-712 Digital Controller, Version H	
E-711.SAN	Sensor Signal Splitter, 3 Channels, Incremental Sensors, for E-712 Digital Controller, Version N	
E-711.SAX1	Connection Cable for Digital Piezo Controllers and Sensor Signal Splitter, HD-Sub-D 26, 3 m	p. 43
E-711.SAX2	Connection Cable for Digital Piezo Controller and Sensor Signal Splitter, HD-Sub-D 26, 5 m	
E-711.SAX5	Connection Cable for Digital Piezo Controller and Sensor Signal Splitter, HD-Sub-D 26, 15 m	
E-711.0ATS	Digital Sensor Signal Transmission, 8 Channels, Encoders with BiSS Interface, for E-712 Digital Controller, Without Cable	p. 44
E-711.SS3	Module for Strain Gauge Sensors, 3 Channels, for E-712 Modular, Digital Controller System	p. 46
E-711.SC3H	Module for Capacitive Sensors, 3 Channels, for E-712 Modular, Digital Controller System	p. 48
E-711.SE3	Module for PISeCa Capacitive Single-Electrode Sensors, 3 Channels, for E-712 Modular, Digital Controller System	p. 49
E-711.0CT	Digital Sensor Signal Transmission, 3 Channels, Capacitive Sensors, for E-712 Digital Controller	p. 50
E-711.0CT0	Digital Sensor Signal Transmission, 3 Channels, Capacitive Sensors, for E-712 Digital Controller, Without Cable	
E-711.0ET	Digital Sensor Signal Transmission, 3 Channels, PISeCa Sensors, for E-712 Digital Controller, Including 10 m Cable	p. 53
E-711.0ET0	Digital Sensor Signal Transmission, 3 Channels, PISeCa Sensors, for E-712 Digital Controller, Without Cable	

Analog Interface Module		Details on
E-711.IA4	Analog Interface Module, 4 I/O	p. 56

SPI Interface Modules		Details on
E-711.iS3	SPI Interface Module for E-712 Modular, Digital Controller System, 3 Channels	p. 57
E-711.iS6	SPI Interface Module for E-712 Modular, Digital Controller System, 6 Channels	

PIO Interface Module		Details on
E-711.IP	PIO Interface Module for E-712 Modular, Digital Controller System	p. 57

Predefined System Configurations		Details on
E-712.3CD	Modular Digital Multi-Channel Piezo Controller, 3 Channels, Capacitive Sensors	p. 59
E-712.3CDA	Modular Digital Multi-Channel Piezo Controller, 3 Channels, Capacitive Sensors, Analog INs and OUTs	
E-712.6CD	Modular Digital Multi-Channel Piezo Controller, 6 Channels, Capacitive Sensors	
E-712.6CDA	Modular Digital Multi-Channel Piezo Controller, 6 Channels, Capacitive Sensors, Analog INs and OUTs	
E-712.1AM	Digital Motion Controller, 1 Channel, for NEXLINE® Nanopositioning Linear Drives with Incremental Encoder	p. 61

1.5 Specifications and Product Descriptions

1.5.1 Ambient Conditions and Classifications

The E-711/E-712 system meets the following ambient conditions and classifications:

- Indoor use only
- Max. altitude for operation 2000 m
- Operating temperature range 5 °C to 40 °C (up to 50 °C possible with loss of performance)
- Storage temperature range 0 °C to +70 °C
- Shipping temperature range -25 °C to +85 °C

- Max. relative humidity 80 % for temperatures up to 31 °C, decreasing linearly to 50 % relative humidity at 40 °C
- Line voltage fluctuations not greater than $\pm 10\%$ of the line voltage
- Transient overvoltages as typical for public power supply
Note: The nominal level of the transient overvoltage is the standing surge voltage according to the overvoltage category II (IEC 60364-4-443).
- Overvoltage category: II
- Protection class: I
- Degree of pollution: 2
- Degree of protection according to IEC 60529: IP20

1.5.2 E-712.R1, .R2, .R3, .R4 Chassis

The chassis can be identified via the configuration list which is delivered separately with the E-711/E-712 system.

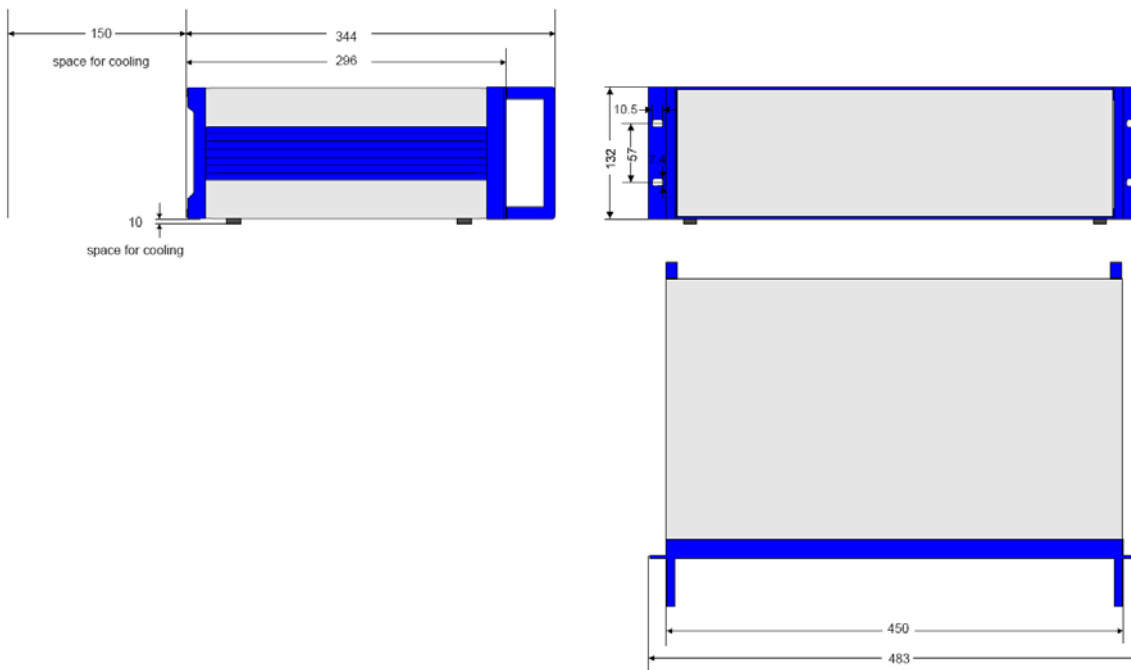


Figure 1: E-712.R2 and E-712.R3 dimensions in millimeters

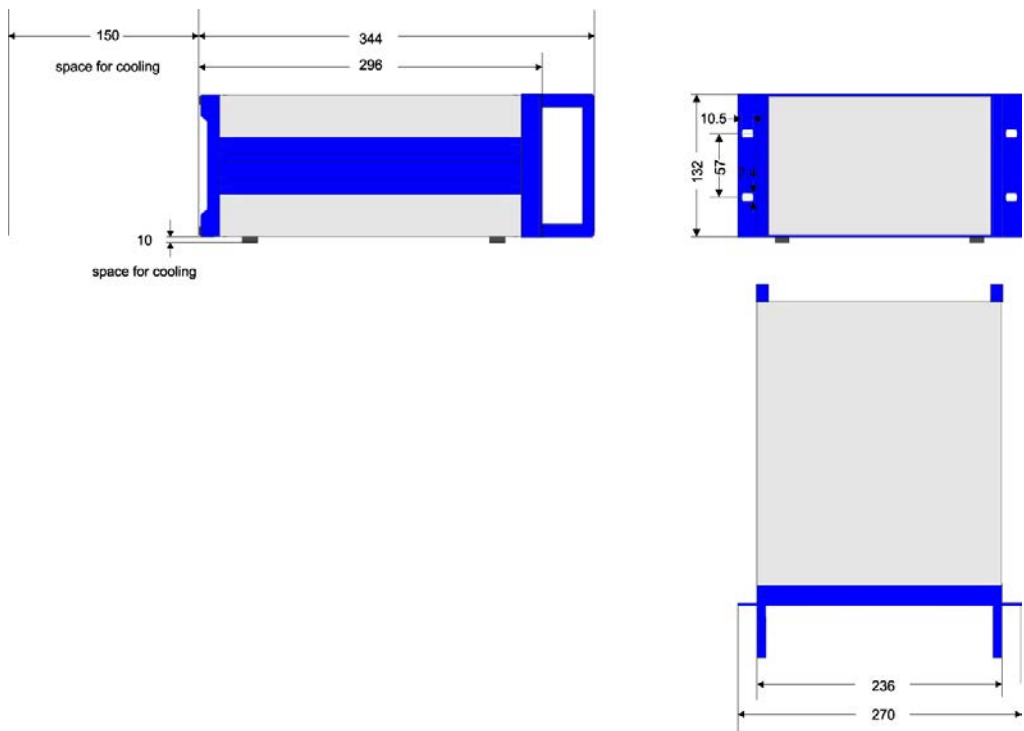


Figure 2: E-712.R1 and E-712.R4 dimensions in millimeters

Operating Elements of E-712.R1, .R2, .R3 and .R4 chassis

All elements described below are located on the rear panel.

Element	Function	
		ON/Off switch (— means ON, O means OFF), integrated in line fuse carrier. See “Power On / Reboot Sequence” on p. 91 and “AC Power and Line Fuses” on p. 259 for more information.
		AC power cord socket See “AC Power Connection” on p. 70 for details.
SERVICE	This connector is for PI service personnel only.	

Specifications

	E-712.R1 / .R2 / .R3 / .R4
Function	Chassis with power supply
Supported piezo voltage range	-30 to 135 V / -250 to 250 V / -30 to 135 V / -250 to 250 V
Channels	3 to 6 channels
Dimensions	9.5" / 19" / 19" / 9.5" for details see figures above
Mass	4.16 kg / 6.7 kg / 6.7 kg / 4.16 kg
Operating voltage	100 to 240 VAC, 50-60 Hz
Line power fuses	2 x T1.6AH, 250 V*
Power consumption	max. 225 VA
Output power	max. 100 W

* Unless otherwise noted on the type label on the rear panel of the chassis.

1.5.3 E-712.R5 Chassis

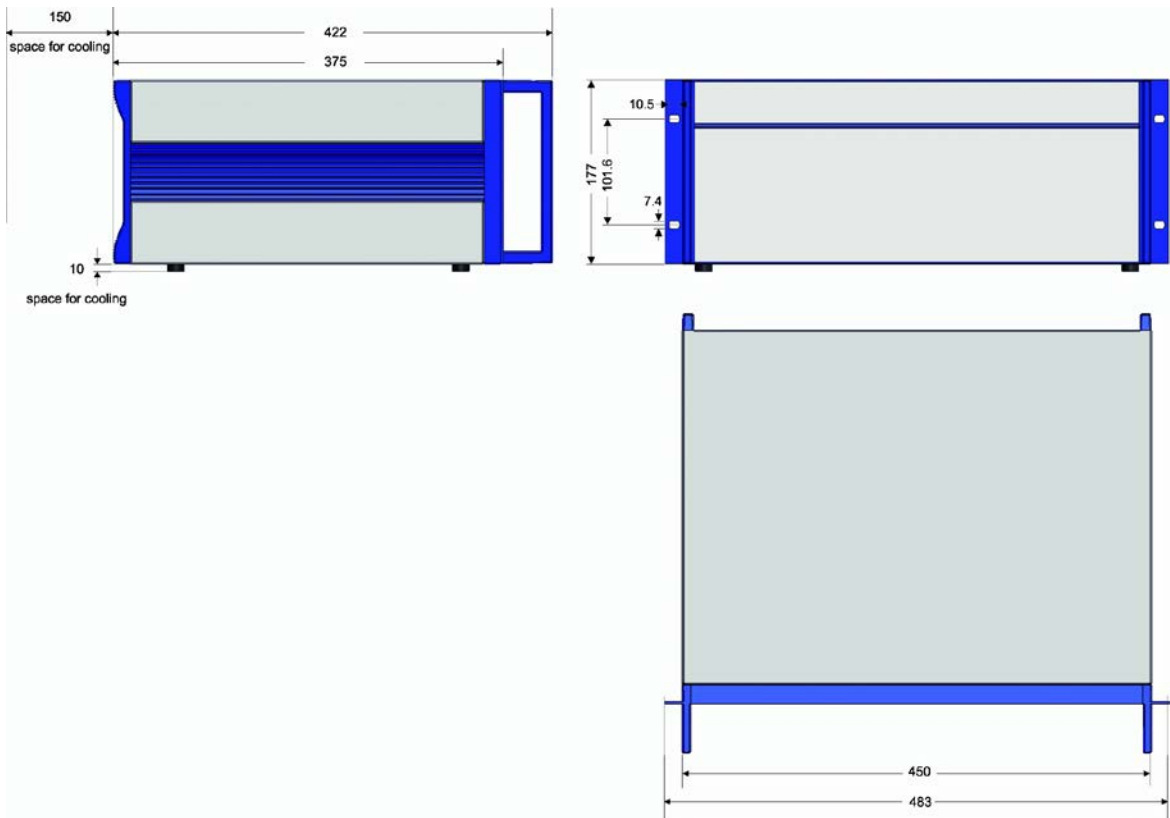


Figure 3: E-712.R5 dimensions in millimeters

Operating Elements of E-712.R5 chassis

The following elements are located on the rear panel:

Element	Function	
		Line fuse carrier. See “Power On / Reboot Sequence” on p. 91 and “AC Power and Line Fuses” on p. 259 for more information.
		AC power cord socket See “AC Power Connection” on p. 70 for details.
SERVICE	This connector is for PI service personnel only.	

The following elements are located on the front panel:

Element	Function
	ON/Off switch (O means OFF; green light means ON) See "Power On / Reboot Sequence" on p. 91 for more information.
Actuators -250V to +250V 	This connector is present only if amplifier modules which do not have their own output sockets are present in the E-711/E-712 system. The output voltage can be: With NEXLINE® (E-711.AM6 modules present): -250 to +250 V With NEXACT® (E-711.AN40 modules present): -10 to 45 V See "E-712.R5: Socket for NEXLINE® Drives" on p. 274 and "E-712.R5: Socket for NEXACT® Drives" p. 275 for pinout.

Specifications

	E-712.R5
Function	Chassis with cPCI bus
Supported piezo voltage range	Up to -250 to 250 V
Channels	Up to 12 channels
Dimensions	19", 4 U; for details see figure above
Mass	4.4 kg
Operating voltage**	100 to 240 VAC, 50-60 Hz
Line power fuses**	2 x T3.15AH, 250 V*
Power consumption**	max. 450 VA
Output power**	max. 200 W

* Unless otherwise noted on the type label on the rear panel of the chassis.

** A suitable power supply is added by PI according to the requirements of the E-711/E-712 digital multi-channel controller system.

1.5.4 E-712.M1 and .N1 Digital Processor Modules

The digital processor module can be identified via the configuration list which is delivered separately with the system.

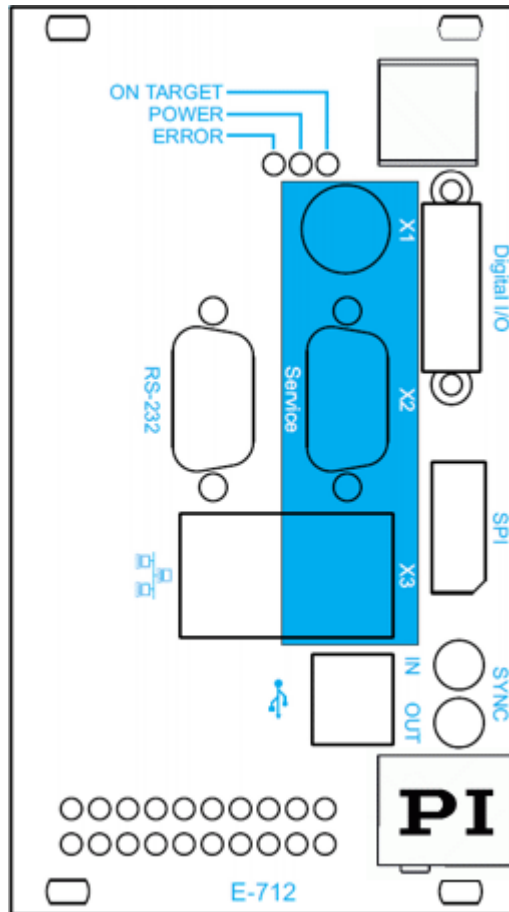


Figure 4: Front panel of E-712.M1 and E-712.N1 digital processor and interface modules

Operating elements of E-712.M1 and.N1 digital processor modules

Name	Function
ON TARGET LED (yellow)	On-target indicator (only relevant in closed-loop operation); lights up if all axes are on target; the on-target state can also be queried with the ONT? command (p. 230).
POWER LED (green)	Power on and ready indicator. The LED blinks during the power-on sequence which takes about 40 seconds (the more modules, the longer the duration). When power-on is finished, the LED glows permanently. Note that depending on the communication interface used, it might take another period of about 30 seconds before communication can be established.
ERROR LED (red)	Error indicator; when LED lights up, error code is non-zero and can be queried and cleared using the ERR? command (p. 230).

Name	Function
RS-232	Serial connection to host PC. See "RS-232 Socket" (p. 286) for pinout.
	Network connection over TCP/IP. See "TCP/IP Connection" (p. 130) for more information.
	Universal Serial Bus (USB-B socket) for connection to host PC. See "USB Connection" (p. 136) for more information.
Service X1 X2 X3	The interfaces in the Service pane are for Physik Instrumente personnel only (system setup and service). The E-711/E-712 system cannot be used like a PC.
DIGITAL I/O	MDR20 socket for digital input and output lines; can be used to trigger external devices and to send start/stop signals to the wave generator. See "Digital I/O Socket" (p. 285) for the availability of the lines and for pinout.
SPI	DisplayPort socket for SPI slave interface. Via the SPI slave interface, the E-711/E-712 system receives the target positions from an SPI master and sends the real positions of the axes to this SPI master. The SPI slave interface is described in the separate SPIST0001 Technical Note.
SYNC IN	Lemo socket for synchronization of multiple E-711/E-712 systems. Connects to previous E-711/E-712 system ("Synchronization of Multiple Controllers" (p. 101)). See "Sync In Socket" (p. 286) for pinout.
SYNC OUT	Lemo socket for synchronization of multiple E-711/E-712 systems. Connects to next E-711/E-712 system ("Synchronization of Multiple Controllers" (p. 101)). See "Sync Out Socket" (p. 287) for pinout.

Specifications

	E-712.M1 / E-712.N1
Function	Digital processor and interface module / Digital PiezoWalk® processor and interface module
Supported positioning systems	Nanopositioning systems / nanopositioning systems and PiezoWalk® systems
Axes	Support of up to 16 axes
Processor	PC-based
Sampling rate, servo-control	max. 50 kHz
Sampling rate, sensor	max. 50 kHz
Ext. sensor synchronization	Yes
Communication interfaces	Ethernet, USB, RS-232, SPI
Digital input	8 x TTL on MDR-connector, 20-pin
Digital output	8 x TTL on MDR-connector, 20-pin

	E-712.M1 / E-712.N1
Command set	PI General Command Set (GCS) 2.0
Supported functionality	Wave generator, trigger I/O, data recorder Several control algorithms can be selected, e.g.: ▪ PI, two notch filters ▪ Advanced Piezo Control, must be ordered separately (order # E-712.U1) ▪ PID position control with subordinated velocity control, two notch filters ▪ PID velocity control, two notch filters
Display	LEDs for OnTarget, Err, Power
Linearization	4th order polynomials DDL (Dynamic Digital Linearization) option, must be ordered separately (order # E-710.SCN) Advanced Linearization option, must be ordered separately (order # E-712.U4)
Mass	500 g

For the support of PiezoWalk® systems (i.e. systems driven by NEXLINE® or NEXACT® drives) by the E-712.N1 Digital PiezoWalk® processor and interface module see the separate E712T0010 Technical Note.

1.5.5 E-711.AL4P, E-711.AL41 Amplifier Modules

The E-711.AL4P and .AL41 amplifier modules are intended for driving up to four piezo actuators.



Operating elements of E-711.AL4P amplifier modules

Name	Function
PWR LED (green)	Power on and ready indicator.
OVER TEMP LED (red)	Provided for future applications and not yet supported.

NOTE

For each E-711.AL4P amplifier module, an E-711.SS3 (p. 46) or E-711.SC3H (p. 48) sensor module is required too because the socket of the sensor module carries the output voltage lines.

Figure 5: Front panel of E-711.AL4P amplifier modules



Figure 6: Front panel of E-711.AL41 amplifier modules

Operating elements of E-711.AL41 amplifier modules

Name	Function
PWR LED (green)	Power on and ready indicator.
OVER TEMP LED (red)	Provided for future applications and not yet supported.
-30 to +135 V 	Sub-D mixed 25W3 socket for piezo stage(s); carries the voltage lines (-30 to +135 V) for up to four piezo actuators. See " E-711.AL41: Socket for Piezo Actuators " (p. 269) for pinout.

Specifications

	E-711.AL4P / .AL41
Min. output voltage	-30 V
Max. output voltage	135 V
Amplifier channels	4
Peak output power per channel	25 W*
Average output power per channel	8 W
Peak output current per channel	250 mA
Average output current per channel	150 mA
Current limitation	Short-circuit proof
Resolution DAC	20 bit (effective)
Temperature sensor	Max. 75 °C, deactivation of the piezo voltage output
Connector	No connector / Sub-D mixed 25W3
Dimensions	8 TE
Mass	515 g

* The maximum output power is limited by the power supply of the chassis and by the number of channels present. See chassis specifications for more information.

1.5.6 E-711.AM4, .AM5, .AM6 Amplifier Modules

The E-711.AM4, .AM5 and .AM6 amplifier modules are intended for driving one NEXLINE® drive.

In combination with an E-712.M1 digital processor module and an E-712.R5 chassis, each of the amplifier modules can also be used to drive up to four conventional piezo actuators.

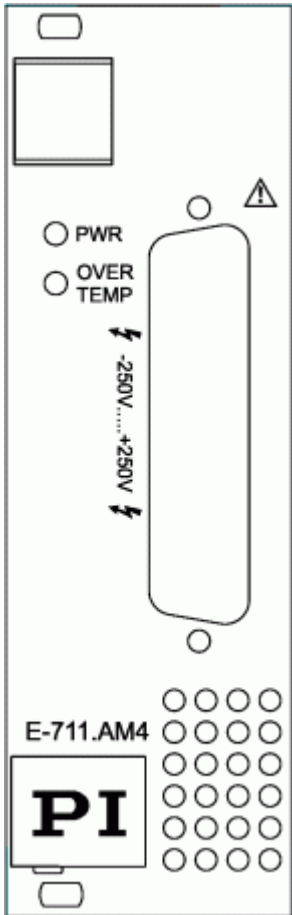


Figure 7: Front panel of E-711.AM4 amplifier modules

Operating elements of E-711.AM4 and .AM5 amplifier modules

Name	Function
PWR LED (green)	Power on and ready indicator.
OVER TEMP LED (red)	Provided for future applications and not yet supported.
 -250 V.....+250 V	Connector for a NEXLINE® drive or for up to four conventional piezo actuators. The output lines carry piezo voltages in the range of -250 to +250 V. See “E-711.AM4: Socket for NEXLINE® Drive”, p. 271 or “E-711.AM5: Socket for NEXLINE® Drive” on p. 272 for pinout.

NOTE

The front panel of the E-711.AM5 module is identical with that of the E-711.AM4 module. Both modules differ only in the connector type:

E-711.AM4: Sub-D mixed 24W7 socket for 7 coax lines and 17 single pins

E-711.AM5: Sub-D 25-pin (f)



Figure 8: Front panel of E-711.AM6 amplifier modules

Operating elements of E-711.AM6 amplifier modules

Name	Function
PWR LED (green)	Power on and ready indicator.
OVER TEMP LED (red)	Provided for future applications and not yet supported.

NOTES

The E-711.AM6 amplifier module is identical in function with the E-711.AM4 module but has no connector on the front panel.

The output voltage of up to 6 E-711.AM6 amplifier modules is available via a Sub-D 50-pin (f) High Density connector on the E-712.R5 chassis (see p. 274 for pinout).

Specifications

	E-711.AM4 / .AM5 / .AM6
Min. output voltage	-250 V
Max. output voltage	+250 V
Amplifier channels	4
Peak output power per channel	45 W*
Average output power per channel	15 W
Peak output current per channel	180 mA
Average output current per channel	60 mA
Current limitation	Short-circuit proof
Resolution DAC	20 bit
Temperature sensor	Max. 75 °C, deactivation of the piezo voltage output
Connector	Sub-D mixed 24W7 / Sub-D 25-pin (f) / no connector
Dimensions	8 TE
Mass	515 g / 515 g / 500 g

* The maximum output power is limited by the power supply of the chassis and by the number of channels present. See chassis specifications for more information.

1.5.7 E-711.AM5A, .AM5B Amplifier Modules

The E-711.AM5A and .AM5B amplifier modules are intended for driving one NEXLINE® drive which is equipped with an encoder.

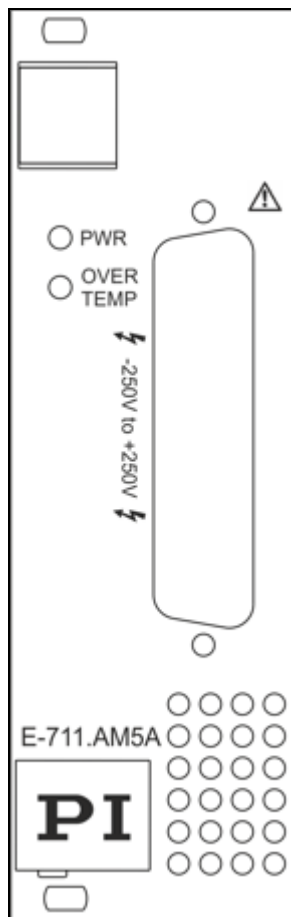


Figure 9: Front panel of E-711.AM5A amplifier modules

Operating elements of E-711.AM5A and .AM5B amplifier modules

Name	Function
PWR LED (green)	Power on and ready indicator.
OVER TEMP LED (red)	Provided for future applications and not yet supported.
-250 V to +250 V 	Sub-D 25-pin socket for a NEXLINE® drive with encoder; the output lines carry the piezo voltages for the shearing and clamping segments of the NEXLINE® stack actuator pairs in the NEXLINE® drive, in the range of -250 to +250 V. See "E-711.AM5A and .AM5B: Socket for NEXLINE® Drive with Encoder" on p. 273 for pinout.

NOTE

The front panel of the E-711.AM5B module is identical with that of the E-711.AM5A module. Both modules differ only in the supported sensor type:

E-711.AM5A: Incremental encoder

E-711.AM5B: Absolute measuring encoder via BiSS interface

Specifications

	E-711.AM5A / .AM5B
Amplifier	
Amplifier channels	4
Min. output voltage	-250 V
Max. output voltage	+250 V
Peak output power per channel	45 W*
Average output power per channel	15 W
Peak output current per channel	180 mA
Average output current per channel	60 mA
Current limitation	Short-circuit proof
Resolution DAC	20 bit, interpolated
Temperature sensor	Max. 75 °C, deactivation of the piezo voltage output
Sensor	
Sensor channels	1
Encoder input	E-711.AM5A: Incremental encoder (analog sin/cos; 1 Vpp, 2.5 V offset) E-711.AM5B: Absolute measuring encoder via 32-bit BiSS C interface
Interpolation factor	2 ¹⁶ -fold interpolation / 16 to 32 bit
Reference point switch and limit switches	TTL
Miscellaneous	
Connector	Sub-D 25-pin (f)
Dimensions	8 TE
Mass	515 g

* The maximum output power is limited by the power supply of the chassis and by the number of channels present. See chassis specifications for more information.

1.5.8 E-711.AN4, .AN40 Amplifier Modules

The E-711.AN4 and .AN40 amplifier modules are intended for driving one NEXACT® drive.

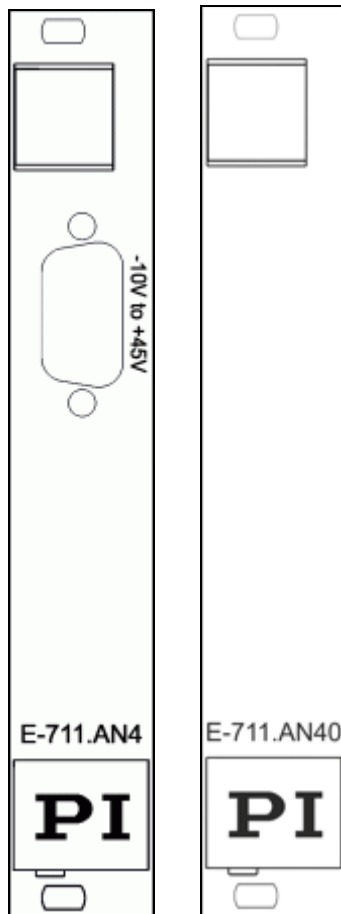


Figure 10: Front panels of E-711.AN4 and .AN40

Operating elements of E-711.AN4 amplifier modules

Name	Function
-10 V to +45 V	Sub-D 15-pin (f) High Density connector for a NEXACT® drive; the output lines carry the piezo voltages for the actuator segments of the NEXACT® drive, in the range of -10 to +45 V. See "E-711.AN4: Socket for NEXACT® Drive", p. 276 for pinout.

NOTES

The E-711.AN40 amplifier module is identical in function with the E-711.AN4 module but has no connector on the front panel.

The output voltage of up to 6 E-711.AN40 amplifier modules is available via an HD Sub-D 78-pin socket on the E-712.R5 chassis (see p. 275 for pinout).

Specifications

	E-711.AN4 / .AN40
Min. output voltage	-10 V
Max. output voltage	+45 V
Amplifier channels	4
Peak output power per channel	30 W*
Average output power per channel	4 W
Peak output current per channel	750 mA
Average output current per channel	250 mA
Current limitation	Short-circuit proof
Resolution DAC	16 bit
Temperature sensor	Max. 160 °C
Connector	Sub-D 15-pin (f) High Density / no connector
Dimensions	8 TE
Mass	347 g

* The maximum output power is limited by the power supply of the chassis and by the number of channels present. See chassis specifications for more information.

1.5.9 E-711.C82 DC Motor Driver Module

The E-711.C82 DC motor driver module is intended for driving two DC motor channels which are each equipped with an incremental encoder.

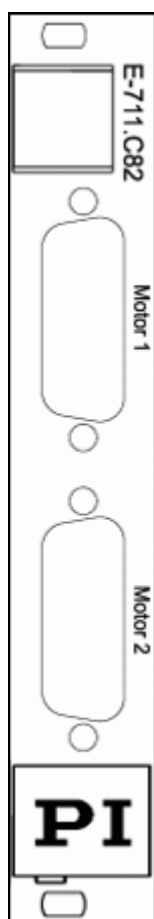


Figure 11: Front panel of E-711.C82 DC motor driver modules

Operating elements of E-711.C82 modules

Name	Function
Motor 1	Sub-D 15(f) socket for stage axis. Only for DC motors! • Output of PWM signals for the motor • Input of the signals of the incremental position sensor • Input of the signals from the limit switches and reference point switch
Motor 2	

CAUTION

Connecting a stage with stepper motor to a DC motor driver can cause irreparable damage.

Only connect to the E-711.C82 a stage with DC motor.

NOTE

The E-711.C82 supports stages with a PWM amplifier as well as stages without a PWM amplifier. Separate lines on the Motor sockets (p. 277) are available for both stage versions. The selection of the proper lines must be ensured via the connector of the stage. With stages from PI the proper connection is ensured.

!

Specifications

	E-711.C82
Function	DC motor driver, 2 channels
Channels	2
Encoder input	Selectable per parameter: ▪ Incremental sensor: AB (quadrature) single-ended or differential TTL signal acc. to RS-422; 60 MHz ▪ Incremental sensor: Analog sin/cos; 1 Vpp ▪ Absolute measuring sensor via 32-bit BISS interface
Interpolation factor	2 ¹⁶
Limit switches	2 × TTL (polarity programmable)
Reference point switch	1 × TTL
Motor brake	1 × TTL, software controlled
Max. output voltage	0 to 24 V for direct control of DC motor
Max. output power	30 W
Current limitation	2 A
Motor connector	Sub-D 15-pin (f)
Dimensions	4 TE
Mass	0.3 kg
Measurement category	II

1.5.10 E-711.SA3 and .SA6 Modules for Incremental Sensors

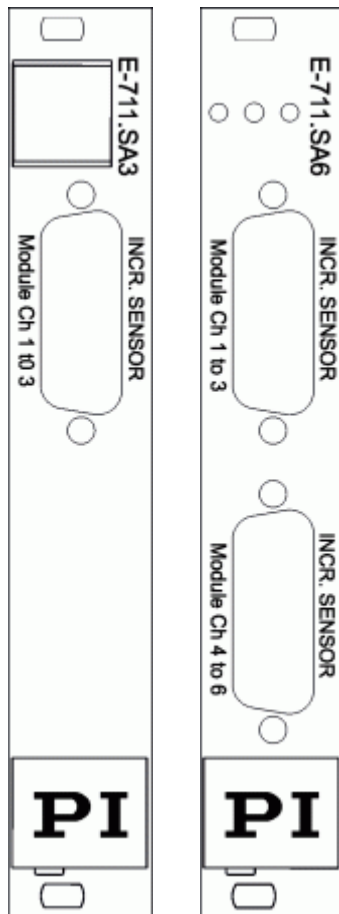


Figure 12: Front panels of
E-711.SA3 (left) and
E-711.SA6 (right)

Operating elements of E-711.SA3 and .SA6 sensor modules

Name	Function
INCR. SENSOR Module Ch 1 to 3	Sub-D 26-pin (m) High Density connector for a sensor signal splitter box. Sensors cannot be connected directly to that connector but must be connected to an E-711.SAx sensor signal splitter box of the suitable type (x = P, H or N depending on the sensor type) and an E-711.SAX connection cable (see p. 76). See “E-711.SA3 and E-711.SA6: Socket for Incremental Sensors”, p. 277 for pinout.
INCR. SENSOR Module Ch 4 to 6	With E-711.SA6 only; Sub-D 26-pin (m) High Density connector for a second sensor signal splitter box. See above for details.

Specifications

	E-711.SA3 / .SA6
Function	Module for incremental sensors
Channels	3 / 6
Interpolation factor	4095 to 262144
Adjustment	Automatic adjustment for highest accuracy, see p. 94
Sensor connector	1 x Sub-D 26-pin (m) High Density / 2 x Sub-D 26-pin (m) High Density Sensors must be connected via an E-711.SAx sensor signal splitter box of the suitable type (x = P, H or N depending on the sensor type) and an E-711.SAX connection cable.
Dimensions	4 TE
Mass	150 g
Measurement category	II

1.5.11 E-711.SAP, .SAH, .SAN Splitter Boxes for Incremental Sensors

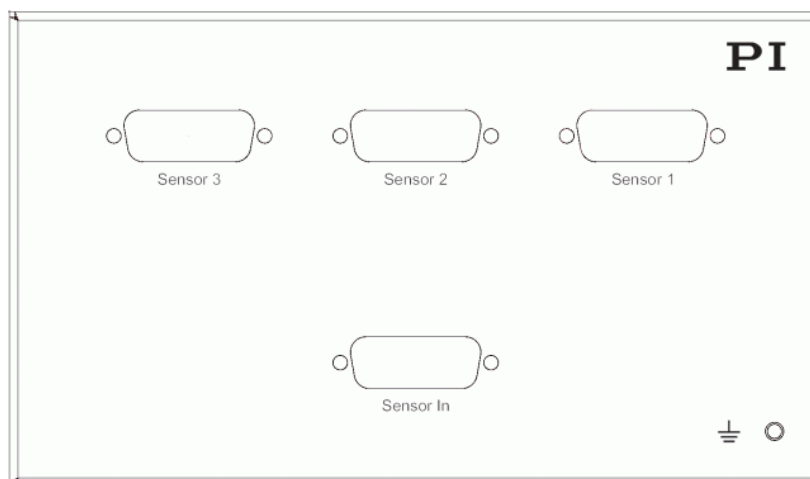


Figure 13: Front panel of E-711.SAP, E-711.SAH and E-711.SAN sensor signal splitter box for incremental sensors

Operating elements of E-711.SAP, .SAH and .SAN sensor splitter boxes

Name	Function
Sensor In	Sub-D 26-pin (f) High Density connector for connection to a sensor module via an E-711.SAX connection cable (see p. 76). See “E-711.SA3 and E-711.SA6: Socket for Incremental Sensors”, p. 277 for pinout.
Sensor 1 Sensor 2 Sensor 3	Sub-D 15-pin (f) connectors for sensor connection; each connector is for one sensor of the suitable type. For pinouts see p. 279 (E-711.SAH), p. 280 (E-711.SAN) or p. 281 (E-711.SAP).

Specifications

	E-711.SAP / .SAH / .SAN
Function	Distribution of incremental sensor signals
Channels	3
Sensor type	PIOne / type H / type N
Sensor connector	3 x Sub-D 15-pin (f) for connection of up to three sensors 1 x Sub-D 26-pin (m) High Density for connection to E-711.SA3 or E-711.SA6 via an E-711.SAX connection cable.
Dimensions	See figure below
Mass	582 g
Measurement category	II

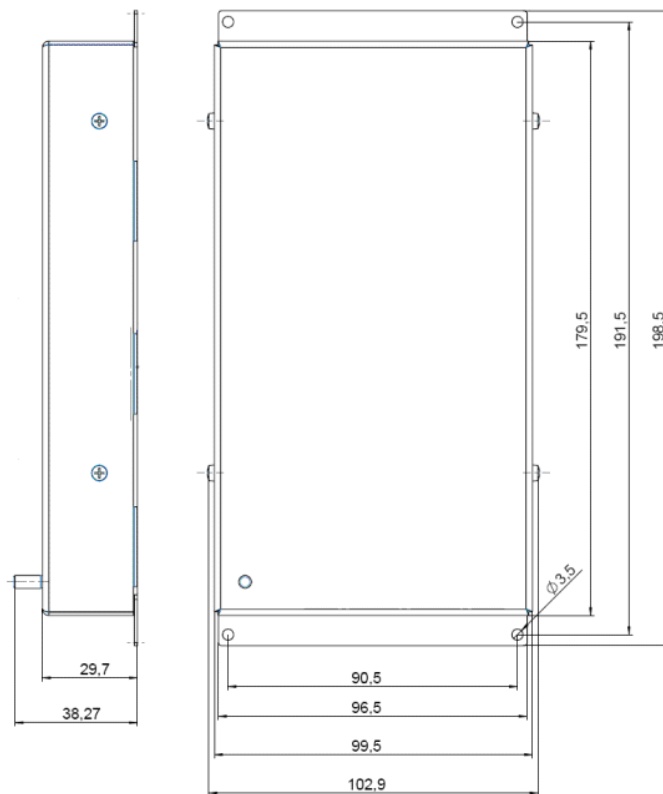


Figure 14: Dimensions of E-711.SAP, E-711.SAH and E-711.SAN sensor signal splitter box for incremental sensors; decimal places separated by commas

1.5.12 E-711.SAX, .SAX1, .SAX2, .SAX5 Connection Cables for Incremental Sensor Modules and Splitter Boxes

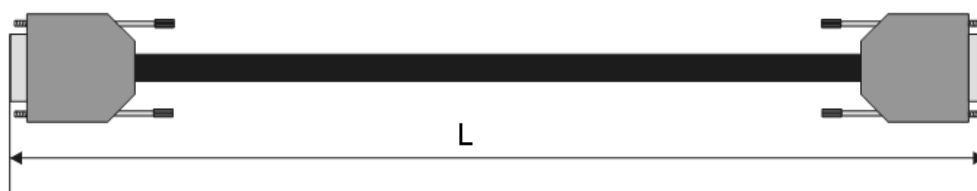


Figure 15: Connection cable for incremental sensor module and sensor signal splitter

E-711.SAX: $L = 1\text{ m}$

E-711.SAX1: $L = 3\text{ m}$

E-711.SAX2: $L = 5\text{ m}$

E-711.SAX5: $L = 15\text{ m}$

1.5.13 E-711.0ATS Sensor Box for Encoders with BiSS Interface

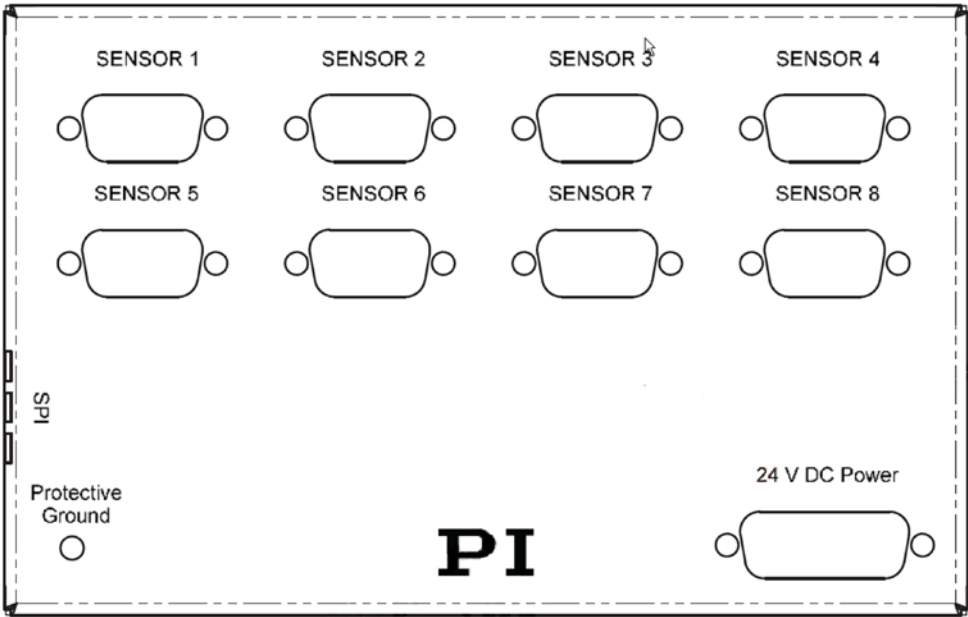


Figure 16: Front panel of E-711.0ATS sensor box for encoders with BiSS interface

Operating elements of E-711.0ATS sensor box

Name	Function
Sensor 1 Sensor 2 Sensor 3 Sensor 4 Sensor 5 Sensor 6 Sensor 7 Sensor 8	Sub-D 9-pin (f) connectors for sensor connection. For pinouts see p. 282.
24 V DC Power	Sub-D 3W3 panel plug for the supply voltage.
SPI	DisplayPort socket on the side panel of the sensor box. For connection to an SPI master unit via a DisplayPort cable.
-	Mini-USB type B on the side panel of the sensor box. For Physik Instrumente personnel only (setup and service).

NOTE

The SPI interface of the E-711.0ATS sensor box runs in slave mode. The master unit connected to the E-711.0ATS box can be an E-711.iS3 or E-711.iS6 SPI interface module (p. 57) from PI.

Specifications

	E-711.0ATS
Function	Digital sensor signal transmission
Channels	8
Sensor type	Encoder via 32-bit BiSS interface (absolute measuring or incremental)
Sensor connector	8 x Sub-D 9-pin (f) for connection of up to eight sensors 1 x DisplayPort socket for connection to E-711.iS3 or E-711.iS6 via a DisplayPort cable (not included).
Dimensions	See figure below
Mass	338 g
Operating voltage	24 V DC, max. 0.6 A from external power supply (not included)
Measurement category	II

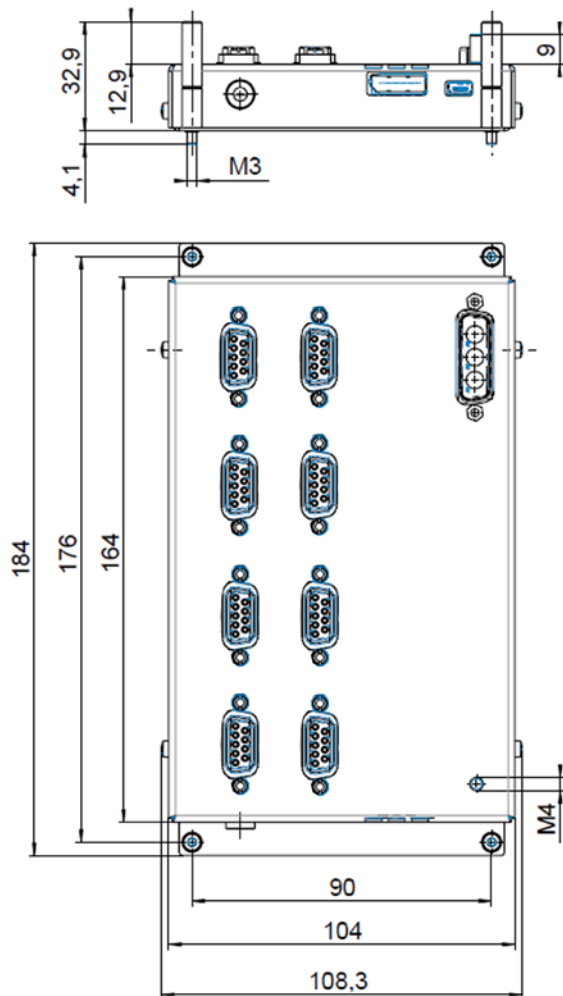
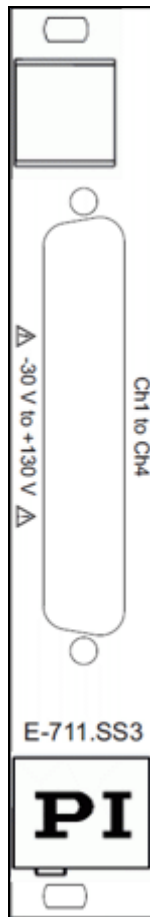


Figure 17: Dimensions of E-711.0ATS sensor box for encoders with BiSS interface;
decimal places separated by commas

1.5.14 E-711.SS3 Module for Strain Gauge Sensors



Operating elements of E-711.SS3 sensor modules

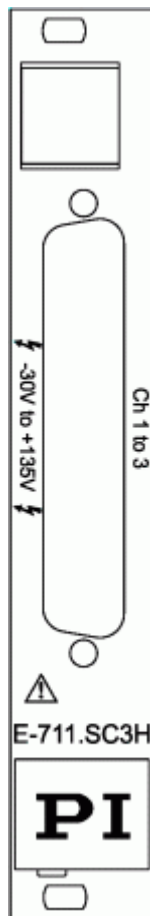
Name	Function
Ch 1 to 4 -30 to +130 V 	Sub-D37 (f) socket for piezo stage(s); carries the signals of up to four strain gauge sensors, one PT1000 temperature sensor and the voltage lines (-30 to +130 V) for up to four piezo actuators (E-711.AL4P must be present in the system). See p. 270 for pinout.

Figure 18: Front panel of E-711.SS3

Specifications

	E-711.SS3
Function	Module for transmission of strain gauge and temperature sensor signals
Sensor channels	4
Connector type	Sub-D37 (f), carries also the 4 piezo voltage lines of E-711.AL4P amplifier modules
SGS / Position	
Sensor resolution, 1 σ , @ 300 Hz	1:20000
Sensor bandwidth (-3dB)	max. 10 kHz
Reference signal, per channel	5 V, max. 15 mA
PT1000 / Temperature	
Sensor resolution	8 bit, -50 °C to 150 °C
Current of reference signal	0.25 mA
Miscellaneous	
Dimensions	4 TE
Mass	164 g
Measurement category	II

1.5.15 E-711.SC3H Module for Capacitive Sensors

**Operating elements of E-711.SC3H sensor modules**

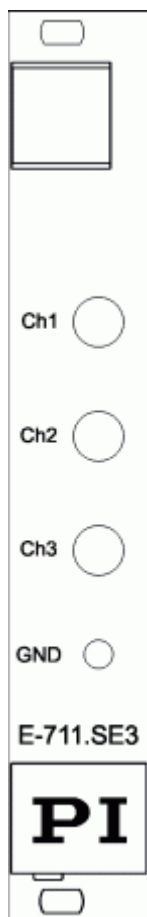
Name	Function
Ch 1 to 3 -30 to +135 V 	Sub-D mixed 25W3 socket for piezo stage(s); carries the signals of up to three capacitive sensors and the voltage lines (-30 to +135 V) for up to four piezo actuators (E-711.AL4P must be present in the system). See "CH1-3 Socket for Piezo Actuators and Capacitive Sensors" (p. 268) for pinout.

Figure 19: Front panel of
E-711.SC3H

Specifications

	E-711.SC3H
Function	Module for capacitive sensors
Sensor type	Capacitive dual-electrode
Sensor channels	3
Sensor bandwidth (-3dB)	10 kHz
Sensor resolution	18 bit (interpolated: 20 bit)
Ext. synchronization	Yes, via E-712.M1 or E-712.N1 digital processor module
Sensor connector	Sub-D mixed 25W3, carries also the 4 piezo voltage lines of E-711.AL4P amplifier modules
Dimensions	4 TE
Mass	177 g
Measurement category	II

1.5.16 E-711.SE3 Module for Capacitive Sensors

**Operating elements of E-711.SE3 sensor modules**

Name	Function
Ch1 Ch2 Ch3	Triaxial LEMO connectors for connection of up to three sensor probes (with active shielding). See "E-711.SE3: Sockets for Capacitive Sensor Probe", p. 282 for pinout.
GND	Common grounding stud for all target surfaces.

Figure 20: Front panel of E-711.SE3

Specifications

	E-711.SE3
Function	Module for PISeca capacitive single-electrode sensors
Sensor type	Capacitive single-electrode
Sensor channels	3
Sensor bandwidth (-3dB)	max. 5 kHz
Static resolution*	0.0005 % of nominal measurement range
Dynamic resolution*	0.002 % of nominal measurement range
Ext. synchronization	Yes, via E-712.M1 or E-712.N1 digital processor module
Sensor connector	3 x Lemo Triax socket, grounding stud for target surfaces
Dimensions	4 TE
Mass	195 g
Measurement category	II

*Static: bandwidth 10 Hz; dynamic: bandwidth 5 kHz; cable length 1 m

1.5.17 E-711.0CT, .0CT0 Digital Sensor Signal Transmission for Capacitive Sensors

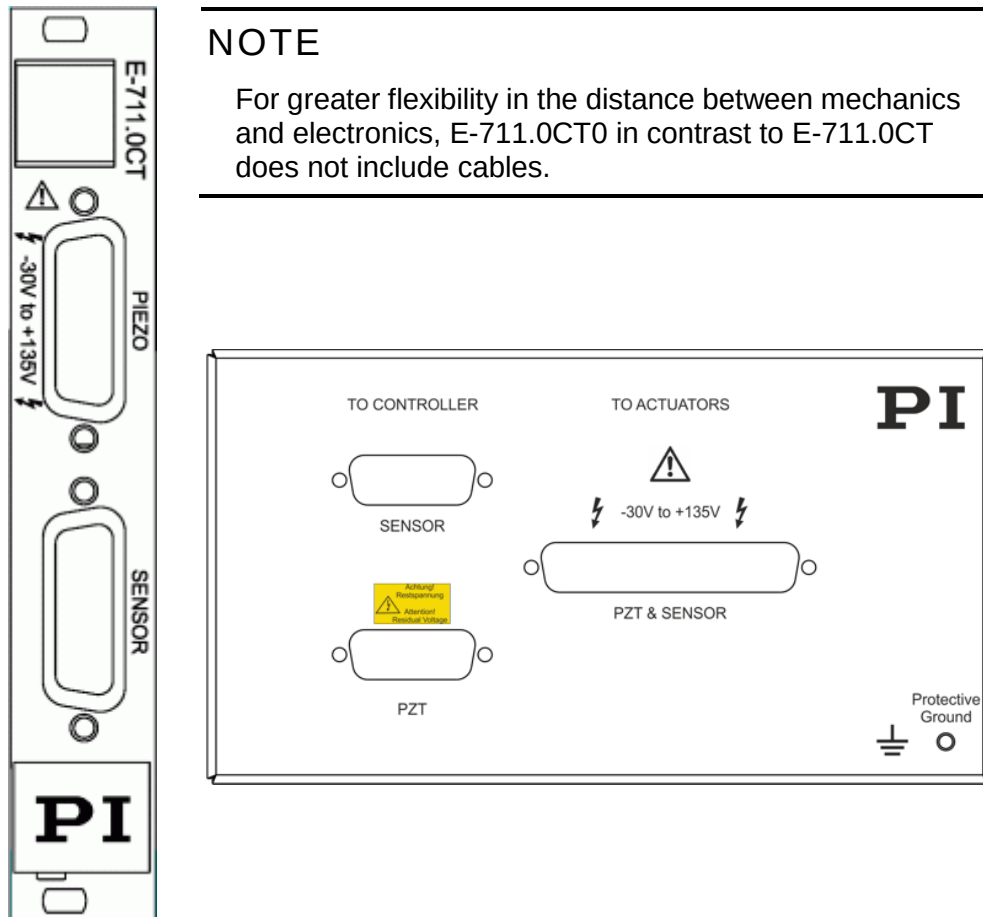


Figure 21: Front panels of E-711.0CT and .0CT0 digital sensor signal transmission for capacitive sensors: interface module (left) and sensor box (right)

Operating elements of the interface module

Name	Function
PIEZO -30V to +135V 	Sub-D 15-pin (f) connector for connection to the PZT socket of the sensor box using a suitable piezo cable (with E-711.0CT, K030B0328 15f/m 1:1, 10 m is included). Four output voltage lines (-30 to +135 V; E-711.AL4P must be present in the system). See separate E711T0002 Technical Note for pinout.
SENSOR	Sub-D 26-pin (f) High Density connector for connection to the SENSOR socket of the sensor box using a suitable sensor cable (with E-711.0CT, K030B0322 26f/m 1:1, 10 m is included). Three sensor input lines. See separate E711T0002 Technical Note for pinout.

Operating elements of the sensor box

Name	Function
SENSOR	Sub-D 26-pin (m) High Density connector for connection to the SENSOR socket of the interface module using a suitable sensor cable (with E-711.0CT, K030B0322 26f/m 1:1, 10 m is included). Three sensor output lines. See separate E711T0002 Technical Note for pinout.
 PZT	Sub-D 15-pin (m) connector for connection to the PIEZO socket of the interface module using a suitable piezo cable (with E-711.0CT, K030B0328 15f/m 1:1, 10 m is included). Four input voltage lines (-30 to +135 V). See separate E711T0002 Technical Note for pinout.
-30V to +135 V PZT & SENSOR 	Sub-D mixed 25W3 socket for piezo stage(s). Three capacitive sensor lines and four output voltage lines (-30 to +135 V; E-711.AL4P must be present in the system). See "CH1-3 Socket for Piezo Actuators and Capacitive Sensors" (p. 268) for pinout.

Specifications

	E-711.0CT / .0CT0
Function	Set for digital sensor signal transmission, consists of: - Interface module - Sensor box with processing electronics - E-711.0CT only: Connection cables
Sensor type	Capacitive dual-electrode
Sensor channels	3
Sensor bandwidth (-3dB)	10 kHz
Sensor resolution	18 bit (interpolated: 20 bit)
Ext. synchronization	Yes, via E-712.M1 or E-712.N1 digital processor module
Connection between interface module and sensor box	Via sensor cable (Sub-D 26-pin (f/m), 1:1) and piezo cable (Sub-D 15-pin (f/m), 1:1) With E-711.0CT, appropriate cables with 10 m length are included.
Sensor connector	Sub-D mixed 25W3, carries also the 4 piezo voltage lines of E-711.AL4P amplifier modules
Dimensions sensor box	198.5 x 102.9 x 38.3 mm (incl. mounting rail and ground stud); see also figure below
Mass sensor box	650 g
Dimensions interface module	4 TE
Mass interface module	150 g
Power consumption sensor box	8 W
Measurement category	II

For more information see the separate E711T0002 Technical Note.

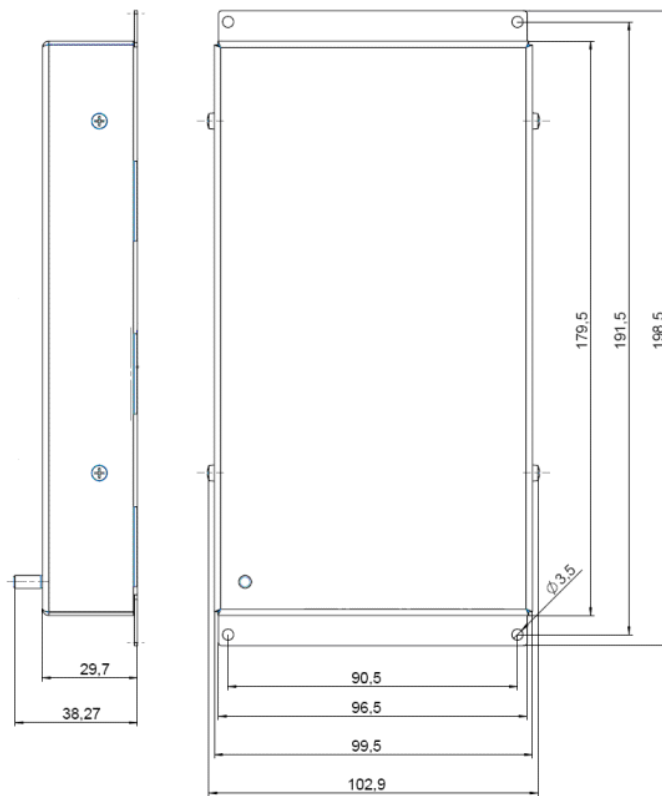


Figure 22: Dimensions of sensor box for E-711.0CT and .0CT0 digital sensor signal transmission; decimal places separated by commas

1.5.18 E-711.0ET, .0ET0 Digital Sensor Signal Transmission for PISeca Sensors

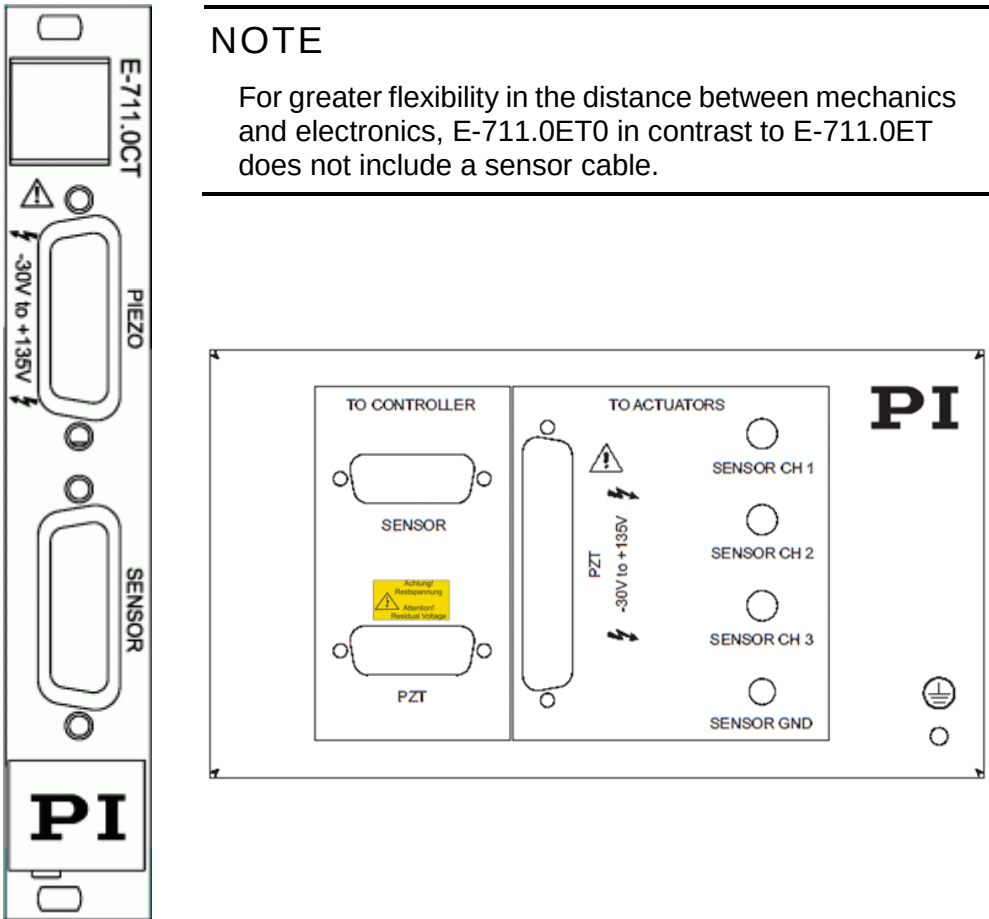


Figure 23: Front panels of E-711.0ET and .0ET0 digital sensor signal transmission for PISeca sensors: interface module (left) and sensor box (right)

Operating elements of the interface module

Name	Function
PIEZO -30V to +135V 	Sub-D 15-pin (f) connector for connection to the PZT socket of the sensor box using a suitable piezo cable (not included). Four output voltage lines (-30 to +135 V; E-711.AL4P must be present in the system). See separate E711T0007 Technical Note for pinout.
SENSOR	Sub-D 26-pin (f) High Density connector for connection to the SENSOR socket of the sensor box using a suitable sensor cable (with E-711.0ET, K030B0322 26f/m 1:1, 10 m is included). Three sensor input lines. See separate E711T0007 Technical Note for pinout.

Operating elements of the sensor box

Name	Function
SENSOR	Sub-D 26-pin (m) High Density connector for connection to the SENSOR socket of the interface module using a suitable sensor cable (with E-711.0ET, K030B0322 26f/m 1:1, 10 m is included). Three sensor output lines. See separate E711T0007 Technical Note for pinout.
 PZT	Sub-D 15-pin (m) connector for connection to the PIEZO socket of the interface module using a suitable piezo cable (not included). Four input voltage lines (-30 to +135 V). See separate E711T0007 Technical Note for pinout.
-30V to +135 V PZT 	Sub-D mixed 25W3 socket for piezo stage(s). Four output voltage lines (-30 to +135 V; E-711.AL4P must be present in the system). See separate E711T0007 Technical Note for pinout.
SENSOR CH1 SENSOR CH2 SENSOR CH3	Lemo triaxial sockets, for input of three sensor probe signals (PISeca sensors)
SENSOR GND	Banana socket for ground connection of the sensor target surface(s)

Specifications

	E-711.0ET / .0ET0
Function	Set for digital sensor signal transmission, consists of: - Interface module - Sensor box with processing electronics - E-711.0ET only: Sensor cable
Sensor type	Capacitive single-electrode
Sensor channels	3
Sensor bandwidth (-3dB)	max. 5 kHz
Static resolution*	0.0005 % of nominal measurement range
Dynamic resolution*	0.002 % of nominal measurement range
Ext. synchronization	Yes, via E-712.M1 or E-712.N1 digital processor module
Connection between interface module and sensor box	Sensor cable (included with E-711.0ET: Sub-D 26-pin (f/m), 1:1, 10 m length; must be ordered separately with E-711.0ET0) If piezo actuators are to be connected to the Sub-D mixed 25W3 socket of the sensor box (E-711.AL4P amplifier module present in the system): Piezo cable for the piezo voltage lines, must be ordered separately.

	E-711.0ET / .0ET0
Sensor connector	3 x Lemo triaxial socket Banana socket for grounding of the target surfaces
Dimensions sensor box	See dimensional drawing
Mass sensor box	776 g
Dimensions interface module	4 TE
Mass interface module	150 g
Power consumption sensor box	10.8 W
Measurement category	II

*Static: bandwidth 10 Hz; dynamic: bandwidth 5 kHz; cable length 1 m

For more information see the separate E711T0007 Technical Note.

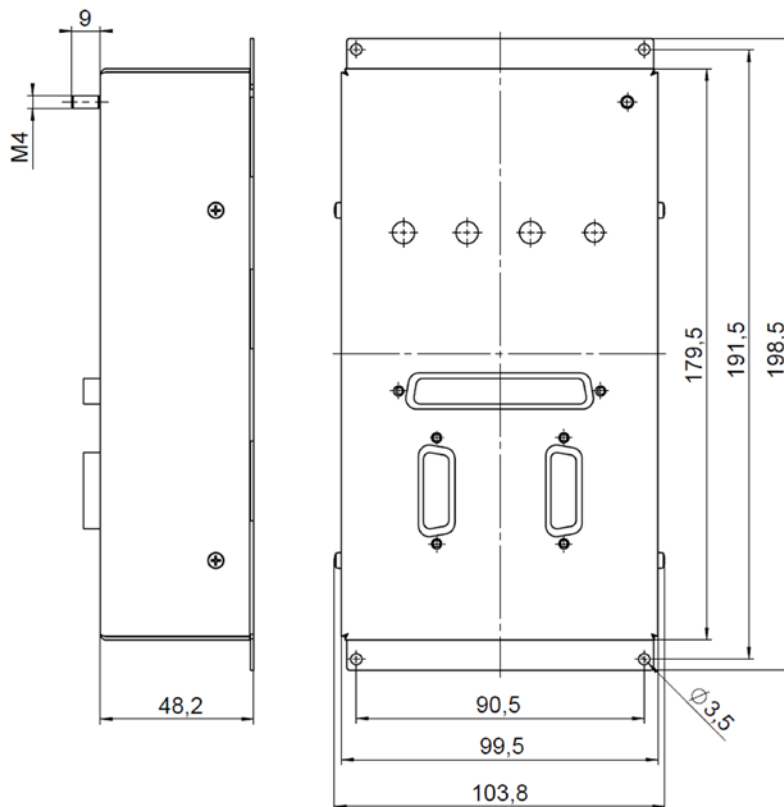
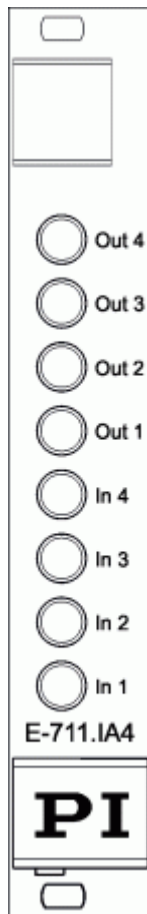


Figure 24: Dimensions of sensor box for E-711.0ET and .0ET0 digital sensor signal transmission; decimal places separated by commas

1.5.19 E-711.IA4 Analog Interface Module

**Operating elements of E-711.IA4 analog interface module**

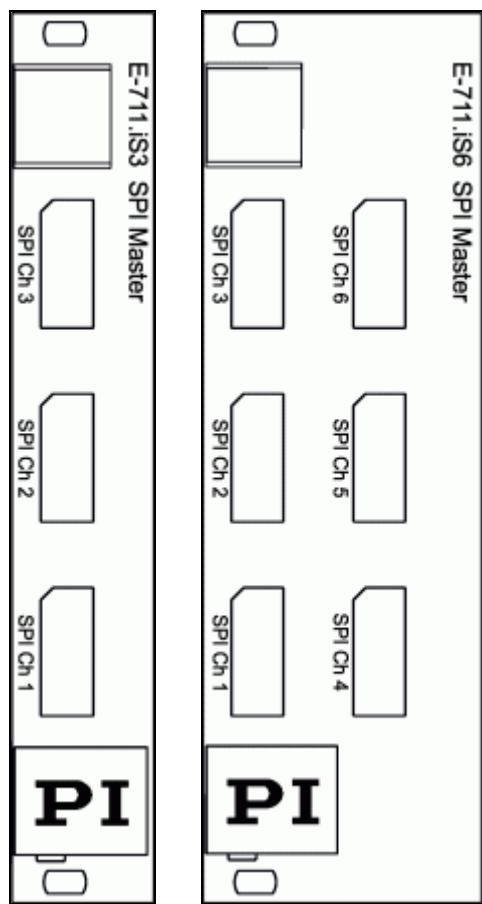
Name	Function
Out 1 to Out 4	4 Lemo sockets for analog output lines; connect external amplifiers and/or measurement devices ("How to work with the Analog Output" (p. 100)). See "Analog Out Sockets" (p. 283) for pinout and specifications.
In 1 to In 4	4 Lemo sockets for analog input lines; connect external sensors and/or control-signal sources ("How to work with the Analog Input" (p. 100)). See "Analog In Sockets" (p. 283) for pinout and specifications.

Figure 25: Front panel of E-711.IA4

Specifications

	E-711.IA4
Function	Analog interface module
Channels	4 inputs, 4 outputs
Resolution	Input: 18 bit Output: 16 bit
Voltage range	Input: ± 10 V differential, max. impedance: 250 ohm Output: ± 12.775 V differential
Bandwidth	Input: max. 25 kHz Output: max. 12 kHz
Connector	8 x LEMO EPG.00.302.NLN
Supported functionality	Input: control signal / sensor signal Output: axis position monitor / amplifier control signal
Dimensions	4 TE
Mass	162 g
Measurement category	II

1.5.20 E-711.iS3 and .iS6 SPI Interface Modules



Operating elements of E-711.iS3 and .iS6 SPI interface modules

Name	Function
SPI Ch 1 to 3	DisplayPort sockets for SPI interface. Each socket connects to one SPI slave unit.
SPI Ch 4 to 6	With E-711.iS6 only; DisplayPort sockets to connect three further SPI slave units.

For pinout, see " E-711.iS3 and E-711.iS6: SPI Socket " (p. 284).

NOTE

E-711.iS3 and E-711.iS6 SPI interface modules run in master mode. The slave unit connected to an E-711.iS3 or .iS6 SPI interface module can be, for example, a sensor box from PI.

Figure 26: Front panels of E-711.iS3 (left) and E-711.iS6 (right)

Specifications

	E-711.iS3 / .iS6
Function	SPI master interface module
Channels	3 / 6
Connector type	DisplayPort
Dimensions	4 TE / 8 TE
Mass	150 g / 170 g
Measurement category	II

1.5.21 E-711.IP PIO Interface Module



Operating elements of E-711.IP PIO interface module

Name	Function
PIO	Sub-D 62-pin (f) High Density connector for connection to a digital interface card in the host PC via the included cable (K040B0121). For pinout see the separate E711T0001 Technical Note.

For more information (data format, reading and writing data) see the separate E711T0001 Technical Note. For the E-712.U3 PI RealTime system see the separate PZ232E Manual.

Figure 27: Front panel of E-711.IP

Specifications

	E-711.IP
Function	PIO interface module for high-speed data exchange
Channels	15
Resolution	32 bit
Operation timing	500 ns read / 1200 ns write
Connector	Sub-D 62-pin (f) High Density for connection to a digital interface card in the host PC via the included cable (K040B0121)
Supported functionality	E-712.U3 PI RealTime system
Dimensions	4 TE
Mass	145 g

1.5.22 E-712.3CD, .3CDA, .6CD, .6CDA Predefined System Configurations

Overview

The predefined system configurations for multi-axis piezo nanopositioning systems with dual-electrode capacitive sensors consist of the following components:

	E-712.M1	E-711.SC3H	E-711.AL4P	E-711.IA4	E-712.R1
E-712.3CD	1	1	1	0	1
E-712.3CDA	1	1	1	1	1
E-712.6CD	1	2	2	0	1
E-712.6CDA	1	2	2	1	1

Specifications

	E-712.3CD / E-712.3CDA / E-712.6CD / E-712.6CDA	Tolerance
Function	Modular digital controller for multi-axis piezo nanopositioning systems with dual-electrode capacitive sensors	
Axes	3 / 3 / 6 / 6	
Processor	PC-based	
Sampling rate, servo-control	max. 50 kHz / 50 kHz / 20 kHz / 20 kHz	
Sampling rate, sensor	max. 50 kHz / 50 kHz / 20 kHz / 20 kHz	
Servo characteristics	PI, two notch filters	
Sensor		
Sensor type	Capacitive	
Sensor channels	3 / 3 / 6 / 6	
Sensor bandwidth (-3dB)	10 kHz	max.
Sensor resolution	18 bit (interpolated: 20 bit)	
Ext. synchronization	Yes	
Amplifier		
Min. output voltage	-30 V	±3 V
Max. output voltage	135 V	±3 V
Amplifier channels	4 / 4 / 8 / 8	
Peak output power per channel	25 W*	max.
Average output power per channel	8 W	max.
Current limitation	Short-circuit proof	
Resolution DAC	20 bit (effective)	
Temperature sensor	Max. 75 °C, deactivation of the piezo voltage output	

	E-712.3CD / E-712.3CDA / E-712.6CD / E-712.6CDA	Tolerance
Interfaces and operation		
Communication interfaces	Ethernet, USB, RS-232, SPI	
Piezo connector	Sub-D mixed 25W3 connector	
Sensor connector	Sub-D mixed 25W3 connector	
Analog input	E-712.3CD & E-712.6CD: none E-712.3CDA & E-712.6CDA: 4 x ± 10 V differential; bandwidth: max. 25 kHz resolution: 18 bit max. impedance: 250 ohm for control signal / sensor signal; Lemo connectors	
Analog output	E-712.3CD & E-712.6CD: none E-712.3CDA & E-712.6CDA: 4 x ± 10 V differential; bandwidth: max. 25 kHz resolution: 16 bit for axis position monitor / amplifier control signal; Lemo connectors	
Digital input	8 x TTL on MDR-connector, 20-pin	
Digital output	8 x TTL on MDR-connector, 20-pin	
Command set	PI General Command Set (GCS) 2.0	
User software	PIMikroMove	
Software drivers	LabVIEW Driver, DLLs	
Supported functionality	Wave-Gen, Trigger I/O, macros	
Display	LEDs for OnTarget, Err, Power	
Linearization	4th order polynomials, DDL (Dynamic Digital Linearization)	
Miscellaneous		
Mass	5.35 kg / 5.53 kg / 5.78 kg / 5.96 kg	
Dimensions	9.5" chassis, 236 x 132 x 296 mm + handles (47 mm length)	
Output power	100 W	max.
Line power fuses	2 x T1.6AH, 250 V**	
Power consumption	225 VA	max.
Operating voltage	100 to 240 VAC, 50-60 Hz	

* The maximum output power is limited by the power supply of the chassis and the number of modules present.

** Unless otherwise noted on the type label on the rear panel of the chassis.

1.5.23 E-712.1AM Predefined System Configuration

Overview

The E-712.1AM predefined system configuration for single-axis NEXLINE® systems with incremental encoder consists of the following components:

	E-712.N1	E-711.AM5A	E-712.R4
E-712.1AM	1	1	1

Specifications

	E-712.1AM
Function	Modular digital controller for NEXLINE® piezo stepping drives
Axes	1
Processor	PC-based, real-time operating system
Servo frequency	50 kHz
Servo characteristics	P-I, two notch filters
Sensor	
Sensor type	Incremental, analog signals (sin/cos)
Sensor resolution	16 bit
Ext. synchronization	Yes
Amplifier	
Amplifier channels	4
Output voltage	-250 to 250 V
Peak output power per channel	45 W
Average output power per channel	15 W
Peak output current per channel	180 mA
Average output current per channel	60 mA
Current limitation	Short-circuit proof
Resolution DAC	20 bit, interpolated
Overheat protection	Output voltage switch-off at 75 °C
Interfaces and operation	
Interface / communication	Ethernet, USB, RS-232, SPI
Piezo / sensor connection	Sub-D 25-pin (f)
Digital input / output	MDR20; 8 × IN, 8 × OUT; TTL
Command set	PI General Command Set (GCS)
User software	PIMikroMove
Software drivers	LabVIEW Driver, shared libraries for Windows and Linux
Supported functionality	NEXLINE® servo algorithms, data recorder, wave generator, trigger I/O, macros
Display and indicators	LEDs for OnTarget, Error, Power
Linearization	4th order polynomials, DDL option (Dynamic Digital Linearization)

	E-712.1AM
Miscellaneous	
Operating temperature range	5 to 50 °C
Mass	5.35 kg
Dimensions	9.5" chassis, 236 mm x 132 mm x 296 mm + handles (47 mm length)
Max. output power	100 W
Line power fuses	2 x T2.5AH, 250 V*
Max. power consumption	225 VA
Operating voltage	100 to 240 VAC, 50-60 Hz

* Unless otherwise noted on the type label on the rear panel of the chassis.

1.6 Scope of Delivery

Unpack the E-711/E-712 modular digital multi-channel controller system with care. Compare the contents against the items covered by the contract and against the packing list.

With the E-712.3CD, .3CDA, .6CD, .6CDA and .1AM predefined system configurations, the following components are included:

- E-711/E-712 system with power cord
- C-815.34 RS-232 cable for connecting controller and host PC (null-modem cable)
- C-815.563 cross-over network cable, can be used to connect the E-712 to a network access point or to a PC with Ethernet socket
- 000011448 USB cable (USB-A/USB-B)
- E-712.CD Distribution CD, containing host software (see "Software Description" (p. 66)) and manuals as PDF files
- E-712 User Manual in printed form (PZ195E; this document)
- GCS Commands Manual in printed form (PZ233E)
- Only with E-712.1AM: E712T0010 Technical Note with PiezoWalk® description

With customized hardware configurations, the following components are included:

- Configuration list (in printed form) with the codes of the individual modules, the chassis and the external components (e.g. sensor boxes) of the system
- 3763 power cord
- C-815.34 RS-232 cable for connecting controller and host PC (null-modem cable)
- C-815.563 cross-over network cable, can be used to connect the E-712 to a network access point or to a PC with Ethernet socket
- 000011448 USB cable (USB-A/USB-B)
- E-712.CD Distribution CD, containing host software (see "Software Description" (p. 66)) and manuals as PDF files
- E-712 User Manual in printed form (PZ195E; this document)
- GCS Commands Manual in printed form (PZ233E)
- Only if E-712.N1 is present: E712T0010 Technical Note with PiezoWalk® description
- Only if E-711.0CT or E-711.0CT0 is present: E711T0002 Technical Note for digital sensor signal transmission
- Only if E-711.0ET or E-711.0ET0 is present:
 - o E711T0007 Technical Note for digital sensor signal transmission
 - o 3214 banana plug mating to the "SENSOR GND" socket, for ground connection of the sensor target surface(s)
- Only if E-711.IP is present:
 - o E711T0001 Technical Note for PIO interface
 - o K040B0121 cable (Sub-D 62-pin (m) High Density / solderable end) for connection to digital interface card in the host PC
- Optional: special documentation for customized configurations and additional components

Inspect the contents for signs of damage. If parts are missing or you notice signs of damage, contact PI immediately.
Save all packing materials in case the product needs to be shipped again.

If controller and mechanics were ordered together, make sure a label with the serial number(s) of the mechanics is affixed to the controller case.

1.7 Additional Components

Contact your PI Sales Engineer or write info@pi.ws, if you need one of the following options / accessories / modules:

E-710.SCN Dynamic Digital Linearization (DDL) option. The DDL makes it possible to achieve significantly better position accuracy for dynamic applications with periodic motion. It is used in conjunction with the wave generator output and in addition to the "normal" servo algorithm in closed-loop operation. You can activate the DDL after purchase and without opening the device. See "Dynamic Digital Linearization (DDL)" (p. 189) for more information.

E-712.U1 Advanced Piezo Control option. Advanced Piezo Control (APC) is an alternative control algorithm for closed-loop operation of piezo actuator systems. You can activate the license after purchase and without opening the device. See the separate E712T0007 Technical Note for more information.

E-712.U4 Firmware Update Advanced Linearization Option. Using the Advanced Linearization option, the crosstalk nonlinearity of axes can be minimized. To install the option, a firmware update is required. See the separate E712T0009 Technical Note for more information.

Analog cables E-711.i1B Lemo 2-pin/BNC and E-711.i10 Lemo 2-pin/open end, 1.5 m. If the E-712 is equipped with an E-711.IA4 analog interface module, four analog input lines and four analog output lines are available. Use the cables to connect signals / external devices to these lines. See "Using the Analog Input" (p. 153) and "Using the Analog Output" (p. 165) for more information.

E-712.01 adapter cable for E-711.SC3H and E-711.AM4; 0.3 m. For connection of NEXLINE® PiezoWalk® systems which are equipped with capacitive sensors.

E-711.AS1 adapter cable from multi-axis piezo stage to digital controller (Sub-D Mix) with external amplifier (LEMO); 1 m.

P-895.2DDC adapter cable for two channels,
Sub-D mixed 25W3 (m) to 2 x Sub-D mixed 7W2 (f); 0.3 m.

P-895.3DDC adapter cable for three channels,
Sub-D mixed 25W3 (m) to 3 x Sub-D mixed 7W2 (f); 0.3 m.

P-895.2D1DDC adapter cable for three channels,
Sub-D mixed 25W3 (m) to 1 x Sub-D mixed 25W3 (f) and 1 x Sub-D
mixed 7W2 (f); 0.5 m.

P-895.3LDC adapter cable for three channels,
Sub-D mixed 25W3 (m) to 10 x Lemo (4 x piezo, 3 x sensor probe, 3 x
sensor target); 0.3 m.

P-895.4LDS adapter cable for four channels,
Sub-D 37 (m) to 8 x Lemo (4 x piezo, 4 x sensor); 0.3 m.

E-500.HCD Access to HyperBit Functionality for Enhanced System
Resolution (supports certain D/A boards). PI's patented Hyperbit
technology for providing position resolution higher than that of the D/A
board is in the LabVIEW analog driver set. Activating Hyperbit requires
purchase of the password, which can be obtained from PI under Order
No. E-500.HCD.

K040B0107 Sync cable. If multiple E-712s are used, their sensor
signals can be synchronized. To synchronize $n+1$ E-712s you need n
synchronization cables. See "Synchronization of Multiple Controllers"
(p. 101) for more information.

1.8 Motion System Requirements

To start working with the E-711/E-712 digital piezo controller system, your
motion system must also include the following components:

- A PC with Windows operating system (Vista Service Pack 1 (32 bit, 64 bit), 7 (32 bit, 64 bit), 8 (32 bit, 64 bit) or 10 (32 bit, 64 bit)) or Linux operating system (kernel 2.6, GTK 2.0, glibc 2.4). Note that not all software components are available for Linux PCs. See "Software Description" (p. 66) for more information.
- Communications interface to the PC:
A free COM port on the PC or
A free USB port or
An Ethernet card in the PC or
A free access point on a network to which the PC is connected
- A suitable Physik Instrumente mechanics. Mechanics equipped with capacitive sensors must have been calibrated with the E-711/E-712 system, or must contain an ID-chip.

1.9 Software Description

Operating system details:

- "Windows" stands for Vista Service Pack 1 (32 bit, 64 bit), Windows 7 (32 bit, 64 bit), Windows 8 (32 bit, 64 bit) and Windows 10 (32 bit, 64 bit)
- "Linux" stands for kernel 2.6, GTK 2.0, glibc 2.4

The table below lists the software tools which are on the E-712 product CD with application recommendations.

For more information see the corresponding software manuals.

Software Tool	Supported Operating System	Short Description	Recommended for
GCS Library	Windows, Linux	Allows program access to the E-711/E-712 system from languages like C++. The functions in the library are based on the PI General Command Set (GCS). Windows operating systems: PI GCS2 DLL; Linux operating systems: libpi_pi_gcs2.so.x.x.x and libpi_pi_gcs2-x.x.x.a where x.x.x gives the version of the library	Recommended for customers who want to use a library for their applications. The dynamic version of the library is needed by the LabVIEW driver set, by PIMikroMove.
LabVIEW drivers	Windows, Linux	LabVIEW is a software tool (available separately from National Instruments) for data acquisition and process control. The E-712 LabVIEW software consists of a collection of virtual instrument (VI) drivers for the E-711/E-712 system. This driver set supports the PI General Command Set (GCS). Included are Vis for GCS commands and high-level Vis for various tasks.	Users who want to use LabVIEW for programming their applications based on the GCS. See the GCS LabVIEW manual of your controller for more information.
LabVIEW Merge Tool	Windows	The LabVIEW Merge Tool allows you to combine product-specific LabVIEW drivers from PI with each other.	For users who want to operate several products from PI at the same time while using LabVIEW.

Software Tool	Supported Operating System	Short Description	Recommended for
LabVIEW analog driver set	Windows, Linux	Each axis of the mechanics connected to the E-711/E-712 system can be controlled via an analog input line (see "Using the Analog Input" (p. 153) for more information). This analog signal can, for example, be generated using a D/A board in a PC. PI offers a LabVIEW driver set which can be used with certain D/A boards. It is compatible with the PI General Command Set (GCS) LabVIEW driver set available for all newer controllers from PI.	Users who want to use LabVIEW for programming their applications based on the GCS. See the GCS LabVIEW manual for the analog driver set for more information. The PI LabVIEW drivers support all D/A converter boards from National Instruments that are compatible with DAQmx8.3. LabVIEW compatibility is given from version 7.1 upwards.
PIMikroMove	Windows	Graphic user interface with which the E-711/E-712 system and other controllers from PI can be used: The system can be started without programming effort Graph of motions in open-loop and closed-loop operation Macro functionality for storing command sequences on the PC (host macros) Support of HID devices Complete environment for command entry, for trying out different commands No command knowledge is necessary to operate PIMikroMove. PIMikroMove uses the dynamic program library to supply commands to the controller. To provide the <i>Device Parameter Configuration</i> window, PIMikroMove requires the NI LabVIEW Run-Time Engine which is installed with the setup from the E-712 product CD.	For users who want to perform simple automation tasks or test their equipment before or instead of programming an application. A log window showing the commands sent makes it possible to learn how to use the commands.
PITerminal	Windows	PITerminal is a Windows GUI which can be used as a simple terminal with almost all PI controllers.	Users who want to send the commands of the PI General Command Set (GCS) directly.
PI Update Finder	Windows	Checks the PI software installed on the PC. If more current versions of the PC software are available on the PI server, downloading is offered.	For users who want to update the PC software.
Firmware Update Wizard	Windows	The Firmware Update Wizard guides you through the update of the firmware for the hardware modules of your E-711/E-712 system.	Users who want to update the firmware.
USB driver	Windows	Driver for the USB interface	For all users.

2 Installation

2.1 Installing the Software on the Host PC

For an overview of the host software provided see "Software Description" (p. 66).

Windows operating systems:

- 1 Start the installation wizard by double-clicking the PI_E-712.CD_Setup.exe file in the installation directory (main directory of the CD).

The InstallShield Wizard window for the installation of programs and manuals for the E-711/E-712 system opens.

- 2 Follow the instructions on the screen and select the "complete" installation.

Linux operating systems:

- 1 Unpack the tar archive from the /linux directory of the product CD to a directory on your PC.
- 2 Open a terminal and go to the directory to which you have unpacked the tar archive.
- 3 Log on as a superuser (root rights).
- 4 Enter ./INSTALL to start the installation.
Pay attention to lower and upper case when entering commands.
- 5 Follow the instructions on the screen.

2.2 Installing Software Updates

PI is constantly improving the PC software. Always install the latest version of the PC software.

Updating the PC software on Windows

Use the PI Update Finder:

- When the PC to be updated is directly connected to the Internet:
Follow the instructions in the A000T0028 Technical Note.
- When the PC to be updated is not directly connected to the Internet:
Follow the instructions in the A000T0032 Technical Note.

Updating the PC software on Linux

NOTE

If software is missing or problems occur with downloading: Contact our customer service department (p. 266).

- 1 Open the website <http://www.pi.ws>.
- 2 Log in with the user name and the password from the "xxx_Releasenews.pdf" file on the product CD.
- 3 Start a search for "E-712".
- 4 In the resulting list, navigate to the "CD-Mirror E-712" archive file.
- 5 Copy the "CD Mirror" archive file to your PC.
- 6 Unpack the archive file to a separate installation directory.
- 7 In the directory with the unpacked files, go to the linux subdirectory.
- 8 Unpack the archive file in the linux directory by entering the command `tar -xvpf <name of the archive file>` on the console.
- 9 Read the accompanying information on the software update (readme file and/or "xxx_Releasenews.pdf" file) and decide whether the update makes sense for your application.
 - If no: Stop the update procedure.
 - If yes: Perform the following steps.
- 10 Log onto the PC as a superuser (root rights).
- 11 Install the update.

2.3 Set Up the E-711/E-712 System

CAUTION

Place the system in a location with adequate ventilation to prevent internal heat build-up. Allow at least 15 cm clearance from the front and the rear of the unit and 1 cm from the bottom (ensured by the feet of the chassis).

Never cover any ventilation openings as this will impede ventilation.

If the required space for ventilation is adhered to, the E-711/E-712 system

- can be set up as free-standing device, e.g. on a desktop or
- can be installed in a cabinet

in any orientation.

For further requirements concerning the place of installation see “Ambient Conditions and Classifications” on p. 20.

When lifting or carrying the E-711/E-712 system use only the intended handles of the chassis.

2.4 AC Power Connection

WARNING

The chassis of the E-711/E-712 system requires a supply voltage of 100 to 240 VAC. The supply voltage may cause serious or even lethal injury. Strictly observe the following:

Install the E-711/E-712 system near the power source so that the plug of the power cord can be quickly and easily disconnected from the mains supply socket.

Use the supplied power cord (p. 62) to connect the E-711/E-712 system to the power source. If the supplied power cord has to be replaced, use a sufficiently dimensioned power cord.

To disconnect the E-711/E-712 system from the power source completely, remove the plug of the power cord from the mains supply socket, or remove the power cord from the chassis of the E-711/E-712 system.

See “AC Power and Line Fuses” (p. 259) for how to replace the line power fuses.



The power connection and line fuses are located on the rear panel of the chassis.

No settings need be changed when connecting an E-711/E-712 system to a different supply voltage.

If it ever should be necessary to change fuses, follow the instructions in "AC Power and Line Fuses" (p. 259).

2.5 How to Interconnect the System



WARNING

If stages with piezo actuators are used:

Temperature changes and compressive stresses can induce charges in piezo actuators. After being disconnected from the electronics, piezo actuators can also stay charged for several hours. Touching or short-circuiting the contacts in the connector of the piezo actuator can lead to minor injuries. In addition, piezo actuators can be destroyed by an abrupt contraction.

- Do **not** touch the contacts in the connector of the stage.
- Do **not** pull out the connector of the stage from the E-711/E-712 system during operation.
- Secure the connector of the stage with screws against accidental disconnection.

If a sensor box is used that also has connections for piezo actuators (digital sensor signal transmission with E-711.0CT, E-711.0CT0, E-711.0ET, E-711.0ET0):

- Do **not** touch the contacts in the PZT panel plug of the sensor box when piezo actuators are connected to the sensor box.
- Before you connect the sensor box to the corresponding sensor or interface module, make sure that the E-711/E-712 system is switched off.
- During operation, do **not** disconnect the sensor box from the corresponding sensor or interface module.

- Secure the connections on the sensor box and on the corresponding sensor or interface module with screws against accidental disconnection.

CAUTION

Make sure that boxes for sensor-signal transmission or splitting are connected to the corresponding sensor or interface module before powering up the E-711/E-712 system. Connecting a box or removing it while the system is already powered on can cause damage to the electronics.

!

CAUTION

When interconnecting the system, respect any assignment of mechanics and electronics given by separate documentation accompanying the system or by labelings on the devices. Labelings may be attached to connectors, cables, front or rear panels of the devices.

!

- 1 Make sure that the ON/OFF switch of the E-711/E-712 system is in the OFF position (0).
- 2 **Connect the E-711/E-712 system either to the same TCP/IP network as the host PC, or to the host PC directly:**
 - To connect directly to the host PC, use a cross-over network cable or a USB-A/USB-B cable or a null-modem RS-232 cable (all cables come with the E-711/E-712 system).
 - Use the included cross-over network cable or a straight-through network cable if connecting to a network access point (newer switches accept both cross-over and straight-through network cables). Note that many network administrators have set their networks to forbid unknown devices like the E-711/E-712 system to log on.

See "Communication" (p. 128) for more information.
- 3 **If you want to use the digital I/O lines of the E-711/E-712 system** to trigger external devices and/or to send start/stop signals to the integrated wave generator, connect these devices/signals to the "DIGITAL I/O" socket (p. 285) on the E-712.M1 or .N1 module, whichever is present. See "External Triggering / Signaling" (p. 142) for more information.

4 If an E-711.IA4 analog interface module is present:

- If you want to use the analog input lines for external sensors and/or for control value generation, connect the appropriate input to the "In 1" to "In 4" sockets (p. 283) on the front panel of the E-711.IA4 module. See "How to work with the Analog Input" (p. 100) for more information.
- If you want to use the analog output lines to monitor axis positions and/or to control external amplifiers, connect the appropriate devices to the "Out 1" to "Out 4" sockets (p. 283) on the front panel of the E-711.IA4 module. See "How to work with the Analog Output" (p. 100) for more information.

5 If an E-711.IP PIO interface module is present:

For high-speed data exchange, connect to E-711.IP to a digital interface card in the host PC via the included cable (K040B0121). For more information see the separate E711T0001 Technical Note.

6 Connect the stage to the E-711/E-712 system:

- With **piezo nanopositioning systems** (i.e. mechanics with conventional piezo actuators and capacitive or strain gauge sensors):

Mechanics with capacitive sensors: Connect the piezo stage(s) to the "CH 1-3" socket(s) (p. 268) of the E-711.SC3H module(s), possibly using an adapter.

Mechanics with strain gauge sensors: Connect the piezo stage(s) to the "CH 1-4" socket(s) (p. 270) of the E-711.SS3 module(s), possibly using an adapter.

A label on the case of the E-711/E-712 system indicates which piezo stage (axis) was calibrated with which controller (axis). Be sure to respect this assignment when connecting the stage to the controller. When you are using a piezo stage with ID-chip together with the E-711/E-712 system, piezo stages can be easily exchanged because the calibration data is in the ID-chip; for details see "ID-Chip Support / Stage Replacement" (p. 216).

Note: When you connect a stage when the controller is powered on, the ID-chip of the stage is not read by the controller. To read the ID-chip data, the controller must be power-cycled or rebooted using the RBT command (p. 230) or the corresponding host software functions.

- If **digital signal transmission** via E-711.OCT, .OCT0, .OET or .OET0 is required, i.e. when the **distance between the piezo nanopositioning system with capacitive sensors and the E-711/E-712 system is greater than 3 m**:

Install the sensor box near the mechanics. If multiple sensor boxes are used for the same mechanics, mount them side by side.

Connect the ground stud on the sensor box to a protective earth conductor (cross-sectional area of the cable $\geq 0.75 \text{ mm}^2$). The contact resistance must be $< 0.1 \text{ ohm}$ at 25 A at all connection points relevant to the protective earth conductor function. If multiple sensor boxes are used for the same mechanics, they must share a common protective earth conductor (min. 2 mm^2 conductor cross section).

Only when the E-711/E-712 system is powered down: Connect the sensor box(s) to the interface module(s) using the included piezo and sensor cables.

Connect the piezo stage(s) to the "PZT & SENSOR" socket(s) on the sensor box(es). See also the above-mentioned notes on calibration and ID-chip. With PISeca capacitive single-electrode sensors, see also "How to Install PISeca Capacitive Single-Electrode Sensors" on p. 77.

See the corresponding Technical Note (E711T0002 or E711T0007) for more information.

- **With NEXLINE® PiezoWalk® drives:**

Connect the drive to an E-711.AM4, ..AM5, .AM5A or .AM5B amplifier module, or to the Sub-D 50-pin socket on the front panel of the chassis, whichever is present.

- **With NEXACT® PiezoWalk® drives:**

Connect the drive to an E-711.AN4 amplifier module, or to the Sub-D 78-pin socket on the front panel of the chassis, whichever is present.

- **With DC motor drives:**

Connect the drive to an E-711.C82 DC motor driver module.

- **With sensor signal splitter boxes for incremental sensors:**

Connect the sensor to a Sub-D 15-pin (f) connector on the E-711.SAP, .SAH or .SAN sensor signal splitter box, whichever is present. Make sure that the box type matches the sensor type.

Connect the splitter box to the E-711.SA3 or .SA6 sensor module using an E-711.SAX, .SAX1, .SAX2 or .SAX5 cable.

See Figure 28 on p. 76 for details.

- **With PISeca capacitive single-electrode sensors:**

See "How to Install PISeca Capacitive Single-Electrode Sensors" on p. 77.

- 7 **Connect the E-711/E-712 system to a wall socket** using the included power cord.

The E-711/E-712 system is equipped with a wide-range power supply and with fuses that are admissible for both 115 V and 230 V operation. See "AC Power Connection" (p. 70) and "AC Power and Line Fuses" (p. 259) for further information.

This section refers to a stand-alone E-711/E-712 system. Synchronization of multiple E-711/E-712 systems is described in "Synchronization of Multiple Controllers" (p. 101).

Incremental sensors via sensor signal splitter box:

How to interconnect sensor modules, splitter boxes and sensors:

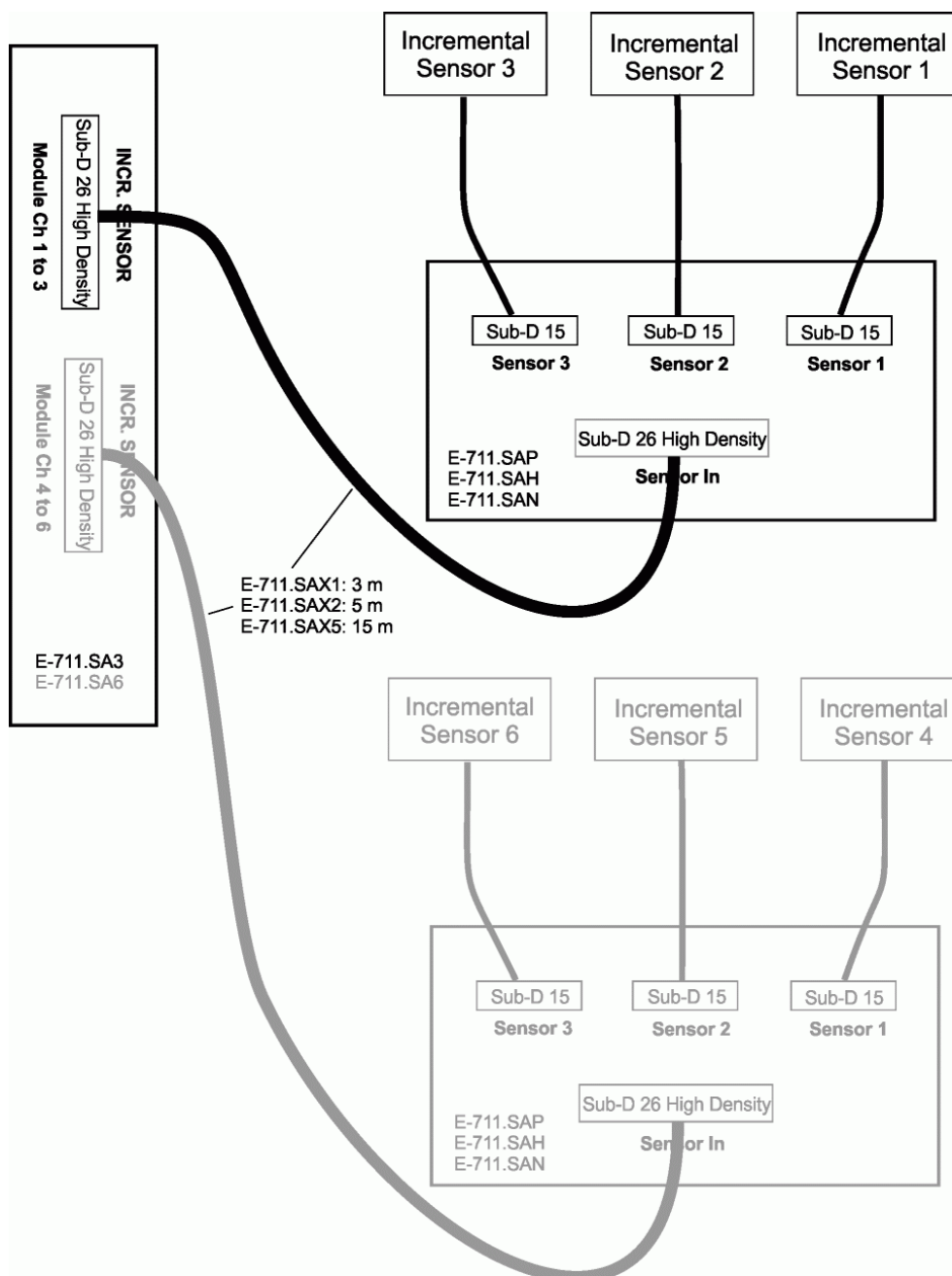


Figure 28: Interconnection of sensor module and sensor signal splitter box(es) for incremental sensors

2.6 How to Install PISeca Capacitive Single-Electrode Sensors



CAUTION

When interconnecting the system, respect any assignment of mechanics and electronics given by separate documentation accompanying the system or by labelings on the devices. Labelings may be attached to connectors, cables, front or rear panels of the devices.



CAUTION

Take care not to shift the sensor probe when connecting the cable!

NOTE

Motion of the connecting cable should be avoided because of capacitive influences. If possible, the application should be designed with sensor probe always at rest and the target the moving part of the system.

NOTE

In the firmware of the E-711/E-712 system, the Sensor Range parameter (ID 0x02000100) scales the measurement range (by changing the capacitance defined as the mid-position value). Example: The nominal measurement range of a sensor is 100 μm . With a factor of 2, the measurement range is enlarged to 200 μm , and with a factor of 5 enlarged to 500 μm . Possible parameter values for PISeca capacitive single-electrode sensors (parameter value = factor):

1 = 10x

2 = 5x

3 = 2x

4 = 1x (default)

This parameter also selects automatically the corresponding polynomial for electronics linearization.

With the E-711/E-712 system, all measurement ranges are calibrated by PI and therefore ready for use.

- 1 Mount the sensor probe in your intended application as described in the separate D510T0001 Technical Note. Note that the minimum allowable probe-to-target gap is 10 μm . A minimum gap of 15 μm is recommended.

- 2 Connect the sensor probe to a LEMO socket of the E-711.SE3 sensor module using the D-891.01E cable (comes with the sensor). If digital signal transmission via E-711.0ET or .0ET0 is required, connect the sensor probe to a LEMO socket of the sensor box.

Make sure that each sensor probe is connected to the channel with which it was calibrated (see label affixed to back of chassis or to sensor box).

- 3 Connect the target surface to the GND ground stud of the E-711.SE3 module. If digital signal transmission via E-711.0ET or .0ET0 is required, connect the target surface to the SENSOR GND banana socket of the sensor box.
Up to three sensor probes can be connected. If they measure different target surfaces: Make sure that each target surface is connected to GND.

- 4 Perform steps 1 to 8 of the “Starting the System in PIMikroMove” Section on p. 81.

- 5 Close the *Start up controller* window by clicking *Close*.

The PIMikroMove main window is open.

- 6 If you want to change the measurement range (the *Sensor Range* factor by default is 1):
 - a. In the PIMikroMove main window, open the *Device Parameter Configuration* window using the *E-712 ⇒ Parameter Configuration* menu sequence.
 - b. Select the *Sensor Electronics* parameter group for the channel to which the PISeca sensor is connected.
 - c. Select a new value for the Sensor Range factor in the *Edit Mask* field.
 - d. Copy the new parameter value from the *Edit Mask* field to the RAM (volatile memory) or non-volatile memory of the E-711/E-712 system using one of the three *Write selected edit values...* buttons in the top right-hand corner of the window.

Note that changing the Sensor Range factor requires command level 1. In the dialog which pops up automatically when you try to copy the value from the *Edit Mask* column to the E-711/E-712 system, enter the password "advanced" to switch to command level 1.

- 7 In the PIMikroMove main window, make sure that the axis to be measured by the PISeca sensor is in open-loop operation (*Servo* must be unchecked).
- 8 Adjust the sensor probe until the *Current Value / Position* column for the axis to be measured by the PISeca sensor shows the value 0. The *Current Value / Position* column is displayed on the Axes tab card of the PIMikroMove main window.

The sensor is now adjusted. The gap between sensor probe and target is equal to the measurement range value. This means that the sensor probe is located in the middle of the measurement range.

$$\text{Measurement Range} = \text{Nom. Range} * \text{Sensor Range Factor}$$

$$\text{Upper Meas. Range Limit} = \text{Meas. Range} + \frac{\text{Meas. Range}}{2}$$

$$\text{Lower Meas. Range Limit} = \text{Meas. Range} - \frac{\text{Meas. Range}}{2}$$

Example: The nominal measurement range of the sensor is 100 µm:

- With the default Sensor Range factor of 1, the gap between sensor probe and target is 100 µm after the adjustment, and the measurement range is 50 to 150 µm.
 - With a Sensor Range factor of 5, the gap between sensor probe and target is 500 µm after the adjustment, and the measurement range is 250 to 750 µm.
- 9 Secure the sensor probe. See the D510T0001 Technical Note for mounting examples.

2.7 Synchronization of Multiple Controllers

If multiple E-711/E-712 systems are used, they can be synchronized.

To synchronize $n+1$ E-711/E-712 systems, you need n special synchronization cables which can be obtained from PI (order# K040B0107, see "Additional Components" (p. 64)). Connect the Sync Out socket of the first device to the Sync In socket of the second device, the Sync Out socket of the second device to the Sync In socket of the third device and so on (see "Sync Out Socket" (p. 287) and "Sync In Socket" (p. 286) for pinout and specifications).

Each connector of a synchronization cable matches to only one of the sockets on the E-711/E-712 system, so do not apply force when connecting them.

NOTE

If you apply external synchronization signals to the Sync In socket, make sure that these signals match the following specifications:

The tolerance of the 100 kHz input signal is ± 2 kHz.

The 4.8 MHz signal must be synchronized with the 100 kHz signal, and both signals must be in a fixed ratio to each other.

3 Quick Start



WARNING

If a sensor box is used that also has connections for piezo actuators (digital sensor signal transmission with E-711.0CT, E-711.0CT0, E-711.0ET, E-711.0ET0):

If a protective earth conductor is not or not properly connected to the sensor box, dangerous touch voltages can occur on the sensor box in the case of malfunction or failure. If touch voltages exist, touching the sensor box can result in serious injury or death from electric shock.

- Connect the sensor box to a protective earth conductor before start-up (p. 71).
- Do not remove the protective earth conductor during operation.
- If the protective earth conductor has to be removed temporarily (e. g. in the case of modifications or repair), reconnect the sensor box to the protective earth conductor before starting it up again.

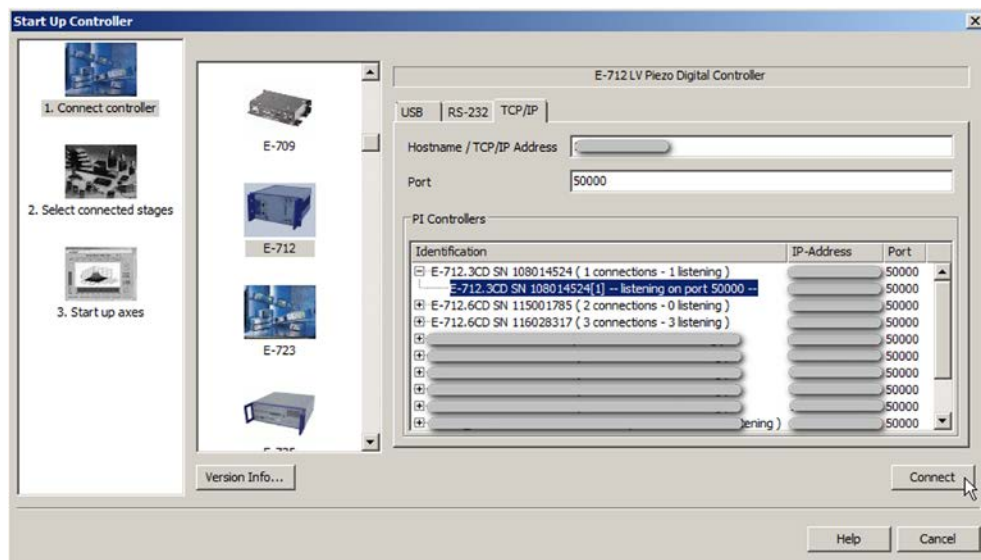
3.1 Starting the System in PIMikroMove

Proceed as follows to start the E-711/E-712 system with the stage(s) in PIMikroMove:

- 1 Read "Safety Precautions" (p. 13) and the documentation of the stage(s).
- 2 Install the following on the PC:
 - The PC software and the USB drivers from the product CD. For further details, see "Installing the Software on the Host PC" (p. 68).
 - Updates for PC software. For further details, see "Software Updates" (p. 69).
- 3 Make sure that the ON/OFF switch of the E-711/E-712 system is in the OFF position (0).

- 4 Install the E-711/E-712 system:
 - For further details, see “Set Up the E-711/E-712 Sytem” (p. 70).
- 5 Connect the following to the E-711/E-712 system:
 - The stage(s), the PC (direct or via network) and the power source. For further details, see “How to Interconnect the System” (p. 71).
 - If you want to synchronize multiple E-711/E-712 systems, see also “Synchronization of Multiple Controllers” (p. 101).
- 6 Power on the E-711/E-712 system. For further details, see “Power On / Reboot Sequence” (p. 91).
- 7 Start PIMikroMove on the PC.
- 8 Establish communication between the E-711/E-712 system and the PC in PIMikroMove via RS-232 or USB or TCP/IP:
 - 8.1 Select E-712 in the field for controller selection.
 - 8.2 Select the tab card that matches the interface used (TCP/IP in the example shown below).
 - 8.3 Depending on the interface used, select your controller on the tab card (TCP/IP, USB) or select the interface settings (RS-232).
 - 8.4 Click *Connect*.

For further details, see “Communication” (p. 128).

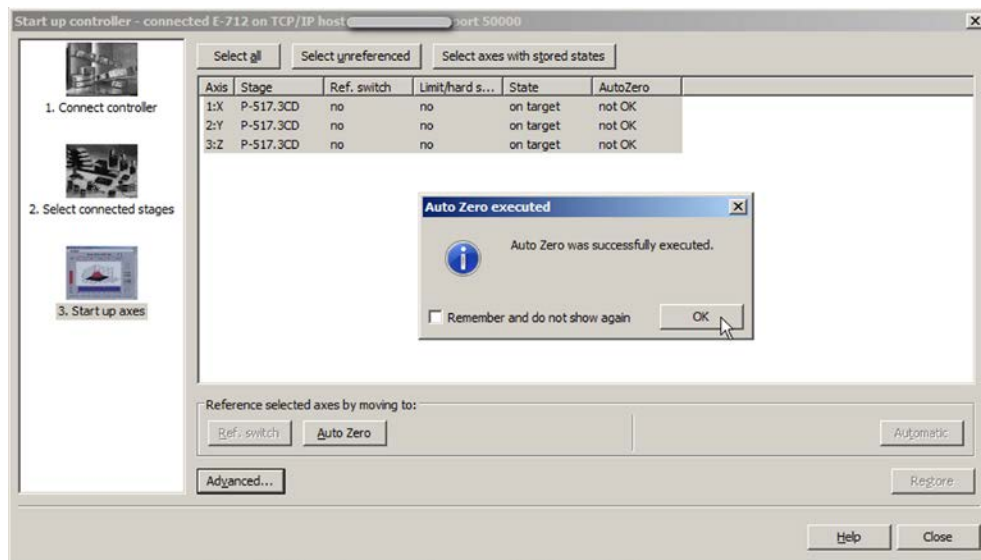


- 9 For systems with conventional piezo actuators and capacitive or strain gauge sensors:
In the *Start up axes* step of the *Start Up Controller* window, execute the AutoZero procedure for all linear axes of the stage(s) that are connected.

- 9.1 Mark the linear axes in the list.
- 9.2 Click *Auto Zero*. The *Auto Zero* dialog opens.
- 9.3 In the *Auto Zero* dialog, start the AutoZero procedure by clicking *Start*.
- 9.4 Click *OK* in the *Auto Zero executed* dialog.

For further details, see "AutoZero Procedure" (p. 92)).

Note that starting AutoZero for rotation axes will fail and cause the error code 74 („No sensor channel or no piezo channel connected to selected axis (sensor and piezo matrix)").



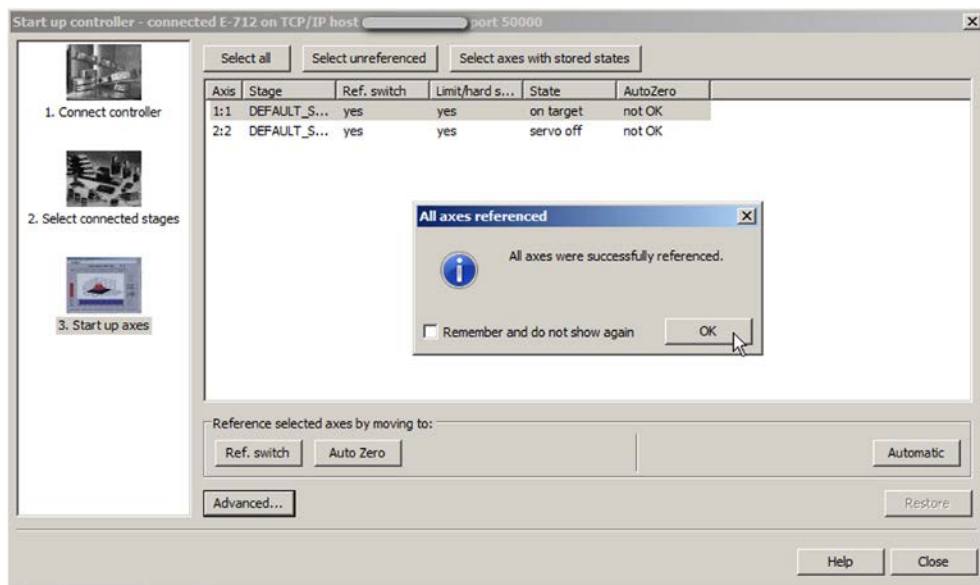
- 10 If necessary, reference the axes measured by incremental sensors:
 - 10.1 Mark the axes to be referenced in the list.
 - 10.2 Use the *Ref. switch* button or the *Automatic* button to start the reference move.
 - 10.3 In the *Reference Axes* dialog, start the reference move by clicking *Start*.
 - 10.4 When the reference move is successfully finished, click *OK* in the *All axes referenced* dialog.

For further details, see “How to Reference an Axis” (p. 94).

Notes:

The reference move is only possible in open-loop operation (servo off).

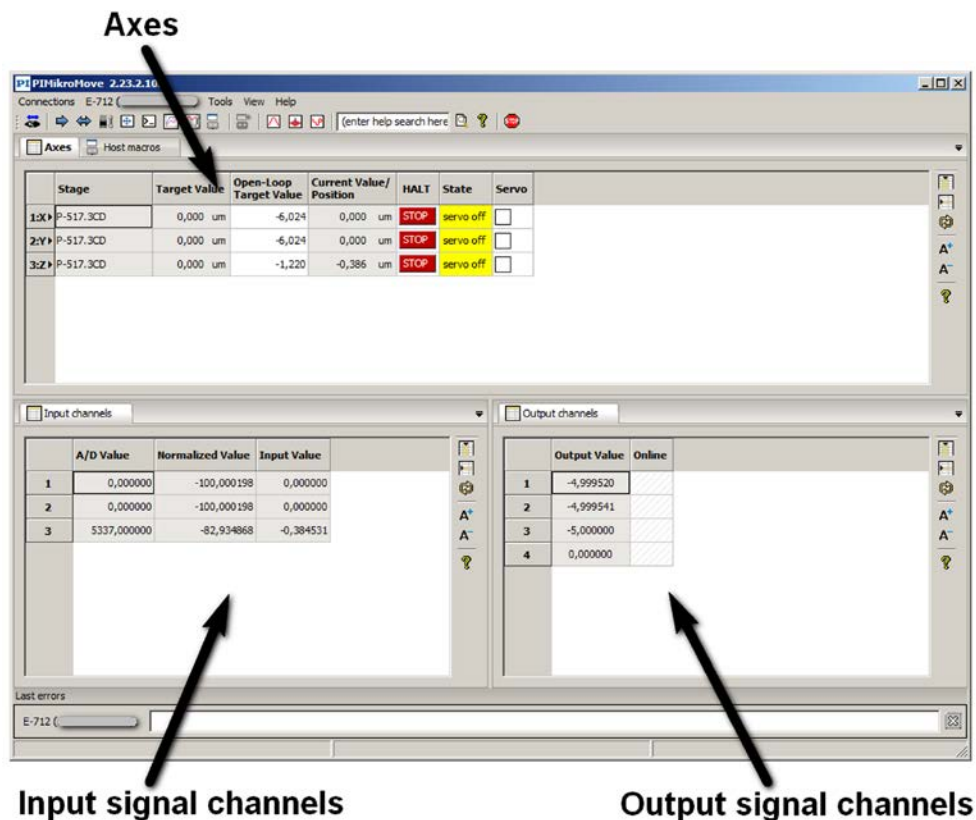
For axes measured by incremental sensors, switching to closed-loop operation is only possible after they have been referenced.



- 11 In the *Start up controller* window, click *Close*.
The main window of PIMikroMove opens.
- 12 Optionally: Configure the PIMikroMove main window.

It is recommended to see the tab cards for axes, input signal channels and output signal channels (see figure below). You can arrange them by dragging them with the left mouse button pressed so that they become docked e.g. to the bottom border of the window.

On the Axes tab card, amongst others you can start axis motion. The channel tab cards show the current values of the input signal channels (sensors) and output signal channels (output voltages for piezo actuators).



Note that the input and output signal channels of the E-711/E-712 system are allocated to the logical axes via matrices, for details, see “Processing Steps” (p. 108). Depending on the connected stage type, an axis may be driven by more than one actuator/drive and measured by more than one sensor.

3.2 Creating Backup Files for Controller Parameters

NOTES

The properties of the E-711/E-712 system and the connected stage(s) are stored in the E-711/E-712 system as parameter values.

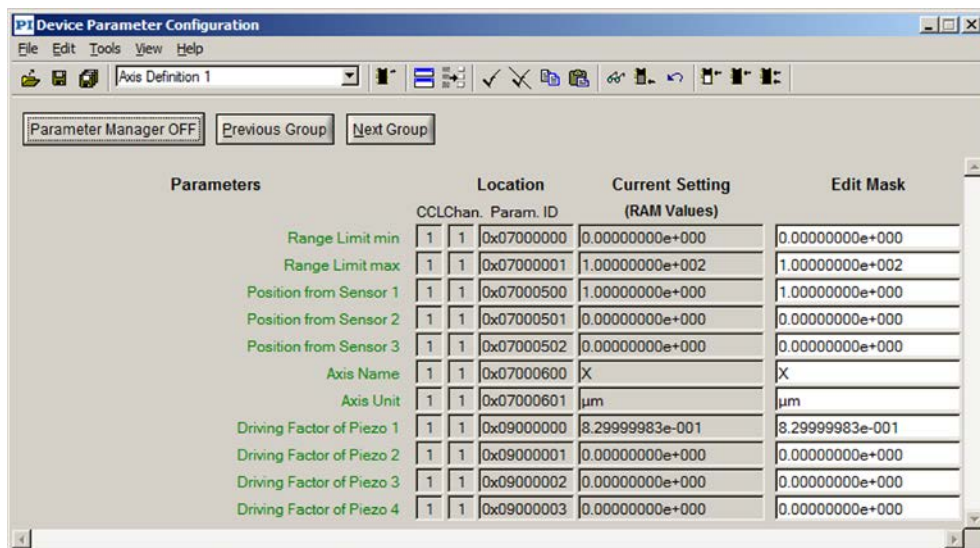
- ➔ Create a backup copy on the PC before changing the parameter values of the E-711/E-712 system. You can then restore the original settings at any time.
- ➔ Create an additional backup copy with a new file name each time you have optimized the parameter values.

To save the parameter values and to load them back to the E-711/E-712 system, use the *Device Parameter Configuration* window provided by PIMikroMove.

Proceed as follows to create a parameter file:

- 1 In the main window of PIMikroMove, open the *Device Parameter Configuration* window via the *E-712... ⇒ Parameter Configuration ...* menu item.

In the figure below, the *Device Parameter Configuration* window shows the *Axis Definition 1* parameter group (includes the parameters for the input and output matrices for channel allocation).



- 2 Save the parameter values from the *Edit Mask* column of the *Device Parameter Configuration* window in a parameter file (file extension .pam) on your PC. Use one of the following options:
 - *File ⇒ Save Edit Values* or *File ⇒ Save Edit Values As* menu item
 -  (Save) or  (Save As) button in the icon bar

3.3 Executing Test Motions in Open-Loop Operation

3.3.1 Systems with Conventional Piezo Actuators and Capacitive or Strain Gauge Sensors

The first moves should be made in open-loop operation. With the factory default settings of the E-711/E-712 system, open-loop commanding means to give open-loop values which correspond approximately to axis positions.

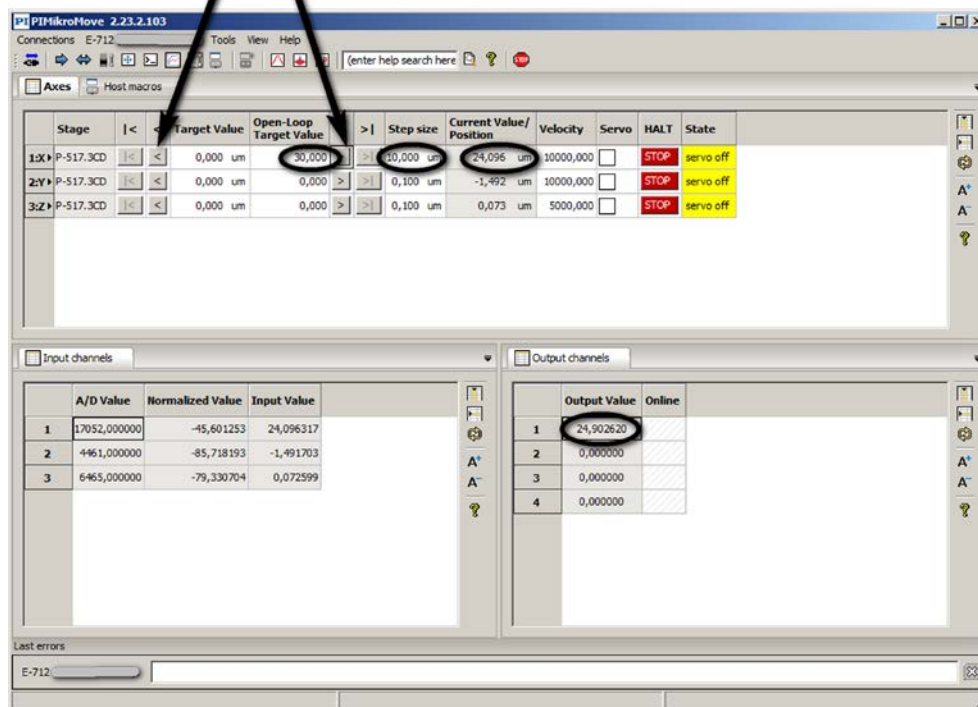
- 1 In the main window of PIMikroMove, make some test moves with the individual axes using the controls on the *Axes* tab card. During the test moves, observe the position display for the axes (in the *Current Value / Position* fields) and the current output voltage(s) for the piezo actuator(s) in the stage(s) (in the *Output Value* fields of the *Output Channels* tab card).

Proceed as follows for each linear axis (for rotation axes, the given open-loop values correspond approximately to positions in μrad , and the step size can be set to a larger value, for example $100 \mu\text{rad}$):

- 1.1 Make sure that the *Servo* box is unchecked.
- 1.2 Command an open-loop value of 0 (μm) by entering 0 in the *Open-Loop Target Value* field of the axis and pressing Enter on your keyboard.
- 1.3 Enter the value 10 (μm) in the *Step size* field of the axis and press Enter.
- 1.4 Use the > button next to the *Open-Loop Target Value* field to increment the commanded value by the value given in the *Step size* field (10). Increment the open-loop value this way step by step up to the upper travel range limit of the axis.
- 1.5 Use the < button next to the *Open-Loop Target Value* field to decrement the commanded value by the value given in the *Step size* field (10). Decrement the open-loop value this way step by step back to zero.

The values for position and output voltage should follow the commanded open-loop values: The axis position should always correspond approximately to the commanded value, and the output voltage(s) should become noticeably different from 0 V and then go back to zero again during the procedure.

Arrow buttons causing motion



In the example shown in the figure above, the open-loop value for axis 1 was increased to 30 by clicking the > button three times (step size value is 10). The current position approximately corresponds to the commanded open-loop value (24.096 μm). Since axis 1 of the stage is driven by piezo actuator 1 in the example, the output voltage of the corresponding output signal channel 1 has changed to an appropriate value.

- 2 Make open-loop frequency response measurements in the *Piezo Dynamic Tuner* window of PIMikroMove to determine the resonant frequencies of the axes. If there are resonances which are intolerable in your application, adjust the notch filter settings for the axis before you switch to closed-loop operation for the first time (servo on). Furthermore, it might be necessary to readjust the preset servo parameters for the axis. See "Servo Controller Dynamic Tuning" (p. 218) for more information.

3.3.2 System with NEXLINE® or NEXACT® Drives

After starting a system with PiezoWalk® drives (NEXLINE® or NEXACT®), you can proceed with test motions in open-loop or closed-loop operation.

- In the main window of PIMikroMove, make some test moves with the individual axes using the controls on the Axes tab card (if necessary, select the columns to be displayed):
 - For motions in open-loop operation, make sure that the *Servo* box is unchecked, and use the controls in the *OL Steps: <*, *OL Steps:>* and *Open-Loop Steps* columns.
 - For motions in closed-loop operation, make sure that the *Servo* box is checked, and use the controls in the *|<, <, >, >|*, *Target Value* and *Step size* columns.
 - During the test moves, observe the position display for the axes (in the *Current Value / Position* fields).

For optimum performance, it may be necessary to adapt parameters of the E-711/E-712 system. See the separate E712T0010 Technical Note for more information.

3.3.3 Systems with DC Motors

After starting a system with DC motors, you can proceed with test motions in closed-loop operation.

- In the main window of PIMikroMove, make some test moves with the individual axes using the controls on the Axes tab card (if necessary, select the columns to be displayed):
 - For motions in closed-loop operation, make sure that the *Servo* box is checked, and use the controls in the *|<, <, >, >|*, *Target Value* and *Step size* columns.
 - During the test moves, observe the position display for the axes (in the *Current Value / Position* fields).

4 Operation



WARNING

If a sensor box is used that also has connections for piezo actuators (digital sensor signal transmission with E-711.0CT, E-711.0CT0, E-711.0ET, E-711.0ET0):

If a protective earth conductor is not or not properly connected to the sensor box, dangerous touch voltages can occur on the sensor box in the case of malfunction or failure. If touch voltages exist, touching the sensor box can result in serious injury or death from electric shock.

- Connect the sensor box to a protective earth conductor before start-up (p. 71).
- Do not remove the protective earth conductor during operation.
- If the protective earth conductor has to be removed temporarily (e. g. in the case of modifications or repair), reconnect the sensor box to the protective earth conductor before starting it up again.

4.1 Power On / Reboot Sequence



CAUTION

Thermally stable systems have the best performance. For a thermally stable system, power on the E-711/E-712 system at least one hour before you start working with it.

Power on the E-711/E-712 system using the ON/OFF switch of the chassis.

On power-on or reboot (with the RBT command), the E-711/E-712 system performs firmware verification and copies information from non-volatile memory to volatile memory. The power-on/reboot sequence takes about 40 seconds (the more hardware modules, the longer the duration). During this time, the POWER LED blinks with different intervals. Permanent glow of the POWER LED indicates that the controller has finished the power-on/reboot sequence and is now in the normal operating state.

When a different stage with ID-chip is connected to the controller, the stage parameters from the ID-chip will be written both to non-volatile and volatile memory upon the next controller power-on or reboot. On subsequent boots

with the same stage, the complete parameters will be rewritten only if the "Power Up Read ID-Chip" option is enabled. By default, this option is disabled to maintain optimized parameter settings on the controller. See "ID-Chip Support / Stage Replacement" (p. 216) for more information.

NOTE

Before communication between E-711/E-712 system and host PC can be established, the power-on sequence of the controller must have finished successfully (takes approx. 40 s; the more modules, the longer the duration; when finished the Power LED glows permanently). TCP/IP communication: If no DHCP server is available on the network or if a point-to-point connection between host PC and controller is being used, it might take another period of about 30 seconds before communication is possible.

4.2 AutoZero Procedure

For systems with conventional piezo actuators and capacitive or strain gauge sensors, the AutoZero procedure performs automatic zero point adjustment of the sensors.

NOTES

During the AutoZero procedure, the axis will move, and the motion can cover the whole travel range.

AutoZero is to be performed with linear axes only. Starting AutoZero for rotation axes will fail and cause the error code 74 („No sensor channel or no piezo channel connected to selected axis (sensor and piezo matrix)").

AutoZero changes the mechanical zero position of the piezo stage.

The AutoZero procedure has the highest priority, i.e. it will overwrite the control values given by all other sources. When the analog control input is enabled, it will be disabled automatically at the start of the AutoZero procedure and reenabled again when AutoZero is finished. See "Control Value Generation" (p. 114) for more information.

AutoZero works in open-loop operation only. If servo-control is on, it will be switched off automatically at the start of the AutoZero procedure and switched on again when it is finished.

Objective of AutoZero:

- Make the entire travel range available:
Changes in temperature or changes in the mechanical load can cause small deviations of the sensor zero point. When the sensor zero-point is set correctly, the complete output voltage range of the amplifier can be used in closed-loop operation.
- Prevent the piezo actuators from damage:
In open-loop operation, the stage displacement with 0 V piezo voltage should already be about 10 % of the travel range. Then the average applied voltage is reduced which lengthens the lifetime of the piezo actuator in the stage without reducing the nominal travel range.

Prerequisites for AutoZero:

- LowVoltage < HighVoltage
LowVoltage is given by the value of the AutoZero Low Voltage parameter (ID 0x07000a00);
HighVoltage is given by the value of the AutoZero High Voltage parameter (ID 0x07000a01)
- The value of the AutoZero High Voltage parameter (ID 0x07000a01) should be identical with the piezo voltage that is required for maximum displacement of the axis.

Settings Changed by AutoZero:

The AutoZero procedure changes the values of the parameters Sensor Mech. Correction 1 (ID 0x02000200). With strain gauge sensors (E-711.SS3 sensor module present in the system), the AutoZero procedure also changes the Sensor Offset factor (ID 0x02000102) parameters.

Starting AutoZero via Command Entry:

Via command entry, you have the following options to start AutoZero:

- Use the ATZ command to perform the AutoZero procedure once (see the GCS commands manual (PZ233E) for ATZ details). Afterwards save the values of the parameters Sensor Mech. Correction 1 (ID 0x02000200) and Sensor Offset factor (ID 0x02000102) to non-volatile memory.
- Send the ATZ command after every start or reboot of the E-711/E-712 system.
- Set the value of the Power Up AutoZero Enable parameter (ID 0x07000802) to 1 for all axes so that the AutoZero procedure is

performed automatically with every start or reboot of the E-711/E-712 system.

Starting AutoZero in PIMikroMove:

See “Starting the System in PIMikroMove” (p. 81) for how to perform the AutoZero procedure in PIMikroMove.

4.3 How to Reference an Axis

With incremental sensors, the encoder signals used for position feedback provide only relative motion information. Hence the controller cannot know the absolute position of an axis upon startup. This is why a referencing procedure is required before absolute target positions can be commanded and reached.

NOTES

For axes measured by incremental sensors, switching to closed-loop operation is only possible after they have been referenced.

NOTES

With E-711.SA3 and .SA6 modules for incremental sensors (p. 41), an automatic adjustment of the offset and gain for the sin/cos sensor signals is performed by the interpolation electronics. Using the adjustment, the absolute accuracy within one signal period is optimized for best performance. The adjustment requires a motion over at least 16 signal periods of the sensor.

If the reference move comprises at least 16 signal periods, the adjustment is performed during the reference move. If the reference move is shorter than 16 signal periods, an additional motion over the whole travel range is recommended to make sure that the adjustment is performed. Like the referencing procedure, the adjustment has to be repeated after each power-on/reboot.

4.3.1 Set Referencing Mode

The current referencing mode setting of the axis (ask with RON?) determines how referencing can be performed. By default, a reference move must be performed, but it is also possible to set absolute positions manually. To switch between the two referencing modes, use the RON command.

4.3.2 Perform a Reference Move

When the reference mode is set to "1" (factory default), referencing is done by performing a reference move with FRF.

NOTES

The referencing velocity is set to the value of the Referencing Velocity parameter, ID 0x7030300.

Proceed as follows to perform a reference move for an axis:

1. Make sure that the axis is in open-loop operation (SVO <AxisID> 0).
2. Only necessary if the current sensor settings do not match the connected hardware for the input signal channels which are assigned to the axis (via the input matrix; see "Input Signal Processing" in the E-712 User Manual):
 - a. Select the type of referencing motion. This is done via the Sensor Reference Mode parameter, ID 0x02000a00, using the SPA command. See the parameter list beginning on p. 233 for possible parameter values.
 - b. Set the Sensor Mech. Correction 1 parameter, ID 0x02000200, to the value which is to be set as new sensor position after a reference move.
3. Start the reference move for the axis using the FRF command.
4. Check with FRF? if referencing was successful.

If referencing was not successful:

- Check the response to the SRG? command and value of the Lim, Ref Signals Detectable parameter (ID 0x02001900) (p. 126).
- See "Signals of Reference and Limit Switches" (p. 126) for further details.

See "Starting the System in PIMikroMove" (p. 81) for how to perform a reference move in PIMikroMove.

4.3.3 Set Absolute Position

When the referencing mode is set to "0", referencing is done by entering an absolute position value using the POS command.

NOTE

The referencing mode can only be set to 0 for axes which are measured by incremental sensors only. This means that for all input signal channels involved in axis measurement, the value of the Sensor Reference Mode parameter (ID 0x02000a00) must be one of 2, 3, 4 or 5 (but not 1).

4.4 How to Command Axis Motion

The axes of the E-711/E-712 system can be operated in open-loop or closed-loop mode. In closed-loop operation, sensor feedback participates in the control value generation, while it is not used in open-loop operation. The operating mode for the axes can be selected with the SVO command, and the axis motion can result from multiple control sources (see "Control Value Generation" (p. 114) for details):

- Move commands: SVA and SVR in open-loop operation; MOV and MVR in closed-loop operation; IMP and STE for both operating modes (see below for examples)
- Wave generator output for periodic motion in either operating mode (see "Wave Generator" (p. 172) for more information and examples)
- Analog control input for motion in either operating mode (see "Using the Analog Input" (p. 153) for more information and examples)

The sources are listed above according to their write priority, starting with the lowest priority. This means that, for example, the wave generator output will overwrite the control value given by a move command.

Furthermore, consider the following when working with PiezoWalk® drives:

- In closed-loop operation, target positions are commanded. Depending on the driving mode selection made with the PiezoWalk Driving Mode parameter and on the commanded target, motion is realized in full-step and/or analog mode or in nanostepping mode. The internal control value is a velocity and hence influenced by the settings for step size per cycle and minimum cycle time.

- In open-loop operation, the control sources described above will always cause motion in analog mode. Stepping motion must be started with the PiezoWalk-channel-related OSM command. See the separate E712T0010 Technical Note for more information on driving modes.

To increase the lifetime of the PiezoWalk® drives, perform a relax procedure whenever there is no need for closed-loop positioning. See separate E712T0010 Technical Note for details.

Axis motion can also be commanded via SPI. Depending on the SPI data segment used, target values (with same write priority as analog control input) or move commands can be sent from an SPI master. See separate SPIST0001 Technical Note for details.

The following examples apply to a conventional nanopositioning system with piezo actuators and capacitive sensors. They can be used in a terminal, e.g. in the *Command Entry* window of PIMikroMove or in the PI Terminal.

A first test that can be made after unpacking your new system: Install, interconnect and power on the system as described in this User Manual. Then perform first open-loop moves and check the position values for the individual axes:

Command String to Send	Action Performed
SVO?	Check servo-control state. All axes should be in open-loop operation (response = 0 for all axes), i.e. there is no correction of drift or other effects.
SVA 1 0	Send this command to move axis 1 to an absolute open-loop value of 0. Note that with the factory default settings of the system, this value should correspond to 0 µm (approximately).
POS?	The current position value of axis 1 should be approximately 0 µm, but due to the calibration settings of the system, the axis position can differ from 0 by about 20% of the axis travel range.
SVR 1 10	With this command, axis 1 moves relative by an open-loop value of 10 (corresponds approximately to 10 µm with the factory default system settings).
POS?	The current position value of axis 1 should be approximately 10 µm.

Perform the motion also for the other axes of the system. Note that for rotation axes, the position unit is µrad.

If no load is applied to the piezo stage or if the system was calibrated at the factory with a load equal to the current one, you can perform first closed-loop moves, in the example shown for axis 1:

Command String to Send	Action Performed
SVO 1 1	Set servo-control on (closed-loop operation) for axis 1; this also sets the current axis position as the target position, to avoid jumps of the mechanics.
POS?	Get current position of axis 1. The current axis position value should be approximately 10 μm .
MOV 1 10	Axis 1 moves to an absolute position of 10 μm .
POS?	The current position of axis 1 should be exactly 10 μm .
MVR 1 4	Axis 1 moves relative by 4 μm .
POS?	The new position should be exactly 14 μm .

4.5 How to Customize the System

NOTES

Values stored in non-volatile memory are power-on defaults, so that the system can be used in the desired way immediately. Note that PI records the data files of every E-711/E-712 system calibrated at the factory for easy restoration of original settings should that ever be necessary.

For stages with ID-chip, to make the optimized settings available in the future, the option "Power Up Read ID-Chip" must have "disabled" as its power-on default (value of parameter 0X0F000000 = 0 in non-volatile memory). See "ID-Chip Support / Stage Replacement" (p. 216) for more information.

4.5.1 Set System Parameters



CAUTION

Incorrect parameter values may lead to improper operation or damage to your hardware. Be careful when changing parameters.

It is strongly recommended to save the parameter values of the E-711/E-712 system to a file on the host PC before you make any changes. This way the original settings can be restored if the new parameter settings will not prove satisfactory. To save the parameter values and to load them back to the E-711/E-712 system, use the *Device Parameter Configuration* window of PIMikroMove. See "Create Backup File for Controller Parameters" (p. 86) for more information.

To adapt the E-711/E-712 system to your application, you can modify parameter values. The parameters available depend on the controller firmware. With HPA? you can obtain a list of all available parameters with information about each (e.g. short descriptions). The volatile and non-volatile memory parameter values can be read with the SPA? or SEP? commands, respectively.

Note that many parameters are "protected" by higher command levels, as indicated in the "Command Level" column in the "Parameter Overview" table (p. 233). By going to command level 1 using the CCL command, it is possible to change level-1 parameters. Parameters with level 2 or higher are reserved for service personnel.

Using the "general" modification commands SPA, RPA, SEP and WPA, all parameters for which the currently active command level has write permission can be changed in volatile memory (SPA, RPA) or in non-volatile memory (SEP, WPA). It is recommended that any modifications be first made with SPA, and when the controller runs well, saved using WPA. Do not change the current interface settings with SPA—except of the baud rate—because it will not be possible to maintain communication afterwards.

In addition to the "general" modification commands, there are commands which change certain specific parameters. All the commands listed below, except of IFS, change the parameter value only in volatile memory, and WPA must be used to save the value to non-volatile memory. IFS changes and saves the interface parameters directly in non-volatile memory only.

AOS (analog input offset)
ATZ (Autozero Matched Offset as the autozero result)
CTO (trigger output configuration)
DRC (data source and record option)
DPO (internal DDL processing parameters)
IFC (baud rate for RS-232 serial connection)
RTR (record table rate)
VEL (Servo Loop Slew-Rate)
WOS (wave generator output offset)
WTR (wave table rate)

IFS (interface parameters in non-volatile memory)

The *Device Parameter Configuration* window of PIMikroMove gives access to parameter values in a more convenient way. Use this window to check/edit the individual parameters. See the PIMikroMove manual for more information.

See "Controller Parameters" (p. 231) for detailed information.

NOTE

When a TCP/IP connection is established at which the IP address is obtained from DHCP server, this address will be automatically written to the IP Address parameter (ID 0x11000600) in the E-711/E-712 systems volatile memory. Provided that the current command level is 1 (see CCL command) and you send WPA 100, the new address setting will be saved to non-volatile memory as the new default value (together with all other current valid parameter values with level-1 write access).

4.5.2 Adjustment for Load Changes

If the controller and the attached stages are ordered together and if PI has sufficient knowledge of your application, then the parameters of notch filters and standard control algorithm (servo parameters) will be set to suitable values at the factory, and, if present, saved in the stage's ID-chip (p. 216). For axes driven by conventional piezo actuators, modification of those parameters will, however, be necessary if the load applied to the stage is changed. See "Control Details" (p. 120) and "Servo-Controller Dynamic Tuning" (p. 218) for detailed information.

5 System Description

5.1 Basic Elements

For successful operation of the E-711/E-712 system, you should familiarize yourself with the following features of the device.

Logical Axes:

Digital controllers from PI control logical axes. Multiple piezo amplifiers, DC motor drivers ("output signal channels") or PiezoWalk channels and multiple sensors ("input signal channels") can be involved in the motion of one logical axis, just as one amplifier/driver, one PiezoWalk channel or one sensor can participate in the motion of more than one logical axis.

See "Accessible Items and Their Identifiers" (p. 104) and "Processing Steps" (p. 108) for more information.

E-712.3CD and .3CDA models can control up to 3 logical axes of a mechanics, E-712.6CD and .6CDA models up to 6 axes, and the E-712.1AM model controls one axis.

Input and Output Signals:

Most of the input and output signals provided by the E-711/E-712 system are used as interface to the connected mechanics (sensor signals and amplifier output). In addition, some signals can be used as control input / output, for external measurement devices or for triggering purposes. See "Accessible Items and Their Identifiers" (p. 104) for details.

Communication Interfaces:

The E-711/E-712 system can be controlled from a host computer (not included) with ASCII commands sent via:

- TCP/IP
- RS-232 serial connection
- USB connection

In addition, the E-711/E-712 system can be controlled by an SPI master, see separate SPIST0001 Technical Note for details.

All interfaces can be active simultaneously. The commands from the interfaces are queued in the order the completed command lines are received.

See "Communication" (p. 128) for more information.

Controller Firmware:

The firmware comprises the ASCII command set and the controller parameters and also includes some special features. For version information and updates see "Firmware Update" (p. 260).

■ ASCII Commands:

The E-711/E-712 system understands the PI General Command Set (GCS; version 2.0).

The PI General Command Set (GCS) is supported by a wide range of PI systems. This command set is well-suited for positioning tasks with one or more axes. The command set itself is independent of the specific hardware (controller or attached stages).

Commands are used, for example, to set operating modes, to initiate motion of the mechanics and to query system and motion values. See the GCS commands manual PZ233E for more information.

■ Controller Parameters:

The key features of the controller are mirrored in parameters. They represent the hardware basics and the calibration setup of the system. Some of the parameters are protected so that their factory settings can not be changed, other parameters can be modified by the user to adapt the system to the individual application. See "Controller Parameters" (p. 231) for more information.

The stage which is connected to the E-711/E-712 system may contain an ID-chip (located in the stage connector). This ID-chip holds selected parameters which will be written to non-volatile and volatile memory when the stage is connected for the first time. See "ID-Chip Support / Stage Replacement" (p. 216) for more information.

Note that PI records data files of every E-711/E-712 system calibrated at the factory for easy restoration of original parameter settings after shipping.

■ Command Levels:

"Command levels" determine the availability of commands and the write access to the controller parameters. Changing the current active command level may require a password and can be done with the CCL command.

■ Special Features:

Wave generator: Each axis can be controlled by a "wave generator" which outputs user-specified patterns, so-called "waveforms". This feature is especially important in dynamic applications which require

periodic, synchronous motion of the axes. See "Wave Generator" (p. 172) for more information.

Data recorder: The E-711/E-712 system comprises a real-time data recorder. It is able to record several input and output signals (e.g. current position, sensor input, output voltage) from different data sources (e.g. logical axes or input and output signal channels). See "Data Recording" (p. 138) for more information.

Macros: The E-711/E-712 system can save macros—command sequences can be defined and permanently stored in the nonvolatile memory of the device via the macro function. A start-up macro can be defined that is executed each time when the E-711/E-712 system is switched on or rebooted. This simplifies stand-alone operation (operation without a connection to the PC). Further information can be found in the "Controller Macros" section (p. 200).

■ Control algorithm for Closed-Loop Operation:

For better position accuracy and performance, the E-711/E-712 system can be operated in closed-loop mode. A control algorithm (with sensor feedback) will then apply corrections to the control value. See "Control Details" (p. 120), "Control Value Generation" (p. 114) and "Servo-Controller Dynamic Tuning" (p. 218) for more information.

In addition to the control algorithm, the Dynamic Digital Linearization (DDL) feature can be used for dynamic applications calling for periodic motion (wave generator operation). The Dynamic Digital Linearization (DDL) feature must be expressly ordered. See "Dynamic Digital Linearization (DDL)" (p. 189) and "Wave Generator" (p. 172) for more information.

The Advanced Piezo Control option (APC) is an alternative control algorithm for closed-loop operation of piezo actuator systems. You can activate the license after purchase (order number E-712.U1) and without opening the device. See the separate E712T0007 Technical Note for more information.

Software on Host PC

Usually, a host computer is used to operate or at least configure the E-711/E-712 system. Therefore an ample array of software tools for installation on the host computer comes with the E-711/E-712 system. For a complete list of all software on the E-712 CD, see "Software Description" (p. 66).

5.2 Accessible Items and Their Identifiers

The identifiers listed below are used to address the appropriate items with the commands of the PI General Command Set (GCS; supported by the firmware of the E-711/E-712 system). The identifiers are factory defaults and cannot be changed by the user.

- **Logical axes:**

A logical axis is an axis of a linear, orthogonal coordinate system and represents a basic direction of motion in the firmware of the E-711/E-712 system (max. 16). All closed-loop motion of the mechanics is commanded for logical axes (MOV, MVR commands). With PiezoWalk® systems, in open-loop operation an axis-related motion command is only available for analog mode motion (SVA command) while stepping motion must be started with the PiezoWalk-channel-related OSM command.

The number of axes is given by the Number Of System Axes parameter, ID 0x0E000B02.

See "Processing Steps" (p. 108) for more information regarding the interrelation of logical axes and input / output signal channels.

E-712.3CD, .3CDA: three axes, the identifiers are 1 to 3

E-712.6CD, .6CDA: six axes, the identifiers are 1 to 6

E-712.1AM: one axis, the identifier is 1

- **PiezoWalk channels:**

A PiezoWalk channel represents a NEXLINE® or NEXACT® drive connected to the controller.

Open-loop stepping motion must be started with a PiezoWalk-channel-related motion command (OSM).

The maximum possible number of PiezoWalk channels is given by the Number Of PiezoWalk Channels parameter, ID 0x0E000B06. The number of PiezoWalk channels which are actually configured for use is given by the Number Of Configured PiezoWalk Channels parameter, ID 0x0E000B07.

The piezo amplifier channels required for control of a NEXLINE® or NEXACT® drive are assigned to the corresponding PiezoWalk channel via parameters (IDs 0x09000100, 0x09000101, 0x09000200, 0x09000201). Standard NEXLINE® and NEXACT® drives require four piezo amplifier channels. See the separate E712T0010 Technical Note for details and for the assignment of the PiezoWalk channels to the logical axes.

The identifiers of the PiezoWalk channels start with 1.

E-712.1AM: one channel, the identifier is 1

- **Input signal channels:**

In the firmware of the E-711/E-712 system, the following input signals are represented by input signal channels (max. 32):

“Internal” sensors of the stage(s)

Analog input lines

Input lines of the SPI slave interfaces

The counting of the channels always starts with the internal sensors, followed by the analog input lines, and ends with the SPI input lines.

Internal sensors:

In the firmware of the E-711/E-712 system, the first input signal channels always represent the lines provided for the internal sensors which are integrated in the connected stage(s). These signals are connected to sensor modules or boxes.

Analog input lines:

If an E-711.IA4 analog interface module is present in the system, its four analog input lines (In 1 to In 4 sockets (p. 283)) are always represented by the input signal channels that follow the internal sensor channels. These lines can be used for external sensors and/or as sources for control value generation (see "Using the Analog Input" (p. 153) for more information).

Input lines of the SPI slave interface (on E-712.M1/.N1 digital processor module):

In the E-711/E-712 system, the target positions received for the axes via the SPI slave interface are handled as additional input signal channels (i.e. as if it would be analog input). You can configure the number of additional input signal channels to be provided by the E-711/E-712 system for the SPI slave interface: Set the SPI Slave Number Of Inputs parameter (ID 0x0e001300) to the desired value (max. 15).

The total number of input signal channels is given by the Number Of Input Channels parameter, ID 0x0E000B00. The Number Of Sensor Channels parameter, ID 0x0E000B03, gives the number of sensor channels and is always less than or equal to the number of input signal channels.

E-712.3CD: three channels, the identifiers are 1 to 3, all for capacitive sensors

E-712.3CDA: seven channels, the identifiers 1 to 3 are for capacitive sensors, 4 to 7 for analog input lines

E-712.6CD: six channels, the identifiers are 1 to 6, all for capacitive sensors

E-712.6CDA: ten channels, the identifiers 1 to 6 are for capacitive sensors, 7 to 10 for analog input lines

E-712.1AM: one channel, the identifier is 1 (incremental encoder)

■ Output signal channels:

In the firmware of the E-711/E-712 system, the following output signals are represented by output signal channels (max. 64):

“Internal” amplifiers and DC motor drivers for the stage(s)

Analog output lines

Output lines of the SPI slave interface

The counting of the channels always starts with the internal amplifiers/drivers, followed by the analog output lines, and ends with the SPI output lines.

Internal amplifiers/drivers:

In the firmware of the E-711/E-712 system, the first output signal channels always represent the lines which carry the voltages for the actuators/motors in connected stage(s). These voltages are provided by the amplifier or DC motor driver module(s).

Analog output lines:

If an E-711.IA4 analog interface module is present in the system, its four analog output lines (Out 1 to Out 4 sockets (p. 283)) are always represented by the output signal channels that follow the internal amplifier/driver channels. These lines can be used to control external amplifiers and/or to monitor axis positions (see "Using the Analog Output" (p. 100) for more information).

Output lines of the SPI slave interface (on E-712.M1/.N1 digital processor module):

In the E-711/E-712 system, the current axis positions to be sent via the SPI slave interface are handled by the controller as additional output signal channels (i.e. as if it would be analog output). You can configure the number of additional output signal channels to be provided by the E-711/E-712 system for the SPI slave interface: Set the SPI Slave Number Of Outputs parameter (ID 0x0e001400) to the desired value (max. 15).

The total number of output signal channels is given by the Number Of Output Channels parameter, ID 0x0E000B01. The Number Of Piezo Channels parameter, ID 0x0E000B04, gives the number of piezo amplifier channels and is always less than or equal to the number of output signal channels.

E-712.3CD: four channels, the identifiers are 1 to 4, all for piezo amplifiers

E-712.3CDA: eight channels, the identifiers 1 to 4 are for piezo amplifiers, 5 to 8 for analog output lines

E-712.6CD: eight channels, the identifiers are 1 to 8, all for piezo amplifiers

E-712.6CDA: twelve channels, the identifiers 1 to 8 are for piezo amplifiers, 9 to 12 for analog output lines

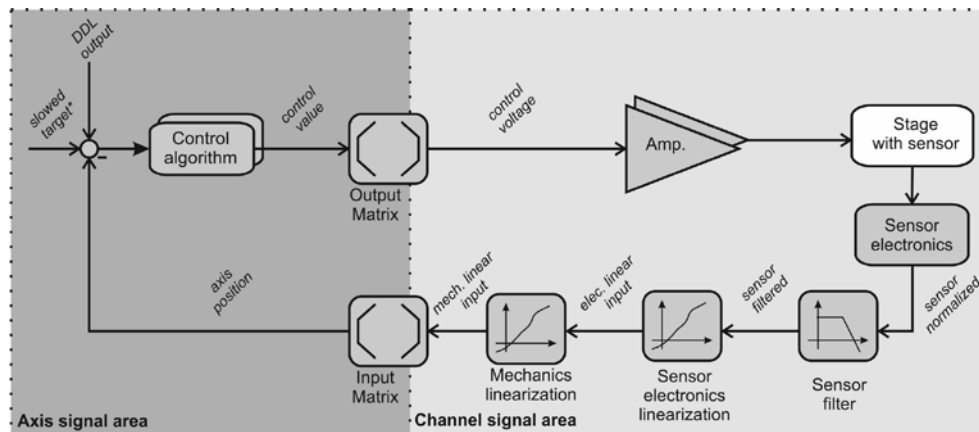
E-712.1AM: four channels, the identifiers are 1 to 4, all for piezo amplifiers

- **Digital output lines:** eight lines (ServoTrigger and OUT1 to OUT7, for pinout see "Digital I/O Socket" (p. 285)), the identifiers for OUT1 to OUT7 are 1 to 7; the ServoTrigger line is not accessible by commands.
The number of digital output lines that are accessible by commands is given by the Number Of Trigger Outputs parameter, ID 0x0E000B05. See "Configuring Trigger Output" (p. 142) for more information.
- **Digital input lines:** eight lines (IN1 to IN8, for pinout see "Digital I/O Socket" (p. 285)), the identifiers are 1 to 8.
See "Using Digital Input" (p. 142) for more information.
- **Wave generators:**
The number of wave generators is the same as the number of logical axes, and each wave generator is dedicated to one axis. See "Wave Generator" (p. 172) for more information.
E-712.3CD, .3CDA: three wave generators, the identifiers are 1 to 3
E-712.6CD, .6CDA: six wave generators, the identifiers are 1 to 6
E-712.1AM: one wave generator, the identifier is 1
- **Wave tables (memory tables for waveform data):** 90 tables with a total of 2^{23} points, the identifiers are 1 to 90
The number of wave tables is given by the Number of Waves parameter, ID 0x1300010A. See "Wave Generator" (p. 172) for more information.
- **DDL tables (memory tables for the data of the Dynamic Digital Linearization (DDL) feature):**
The number of DDL tables is the same as the number of logical axes, and each DDL table is dedicated to one axis. The total number of points provided for the DDL tables is 2^{23} , given by the Max DDL Points parameter, ID 0x1400000B. See "Dynamic Digital Linearization (DDL)" (p. 189) for more information.
E-712.3CD, .3CDA: three DDL tables, the identifiers are 1 to 3
E-712.6CD, .6CDA: six DDL tables, the identifiers are 1 to 6
E-712.1AM: one DDL table, the identifier is 1
- **Data recorder tables (memory tables for recorded data):** up to 12 tables with a total of 2^{23} points, the identifiers start with 1 and continue sequentially up to the number of tables.
The number of data recorder tables can be set via the Data Recorder Chan Number parameter, ID 0x16000300. The maximum number of tables is limited by the Max Number Of Data Recorder Channels parameter, ID 0x16000100.
See "Data Recording" (p. 138) for more information.
- **Whole system:** the E-711/E-712 system as a whole, the identifier is 1

- **Hardware modules:** the slots available in the chassis, the identifiers start with 1. Counting (front view): 1 = leftmost slot. The actual number of hardware modules present in the chassis depends on the system configuration.
- **Firmware units:** the number of different firmware units present in the E-711/E-712 system, the identifiers start with 1 and continue sequentially up to the number of firmware units.
E-712.M1 and .N1 digital computer and interface module have 2 different firmware units installed (CPU and FPGA firmware), and each of the other modules in the E-711/E-712 system has one firmware unit installed. All firmware units can be updated separately, see "Firmware Updates" (p. 260) for more information.

5.3 Processing Steps

The E-711/E-712 system controls the motion of the logical axes of the connected stage(s) in open-loop or closed-loop operation. The block diagram below shows the signal path for an axis in closed-loop operation.



Signal names are written in *italics*

* slowed target = target value manipulated by slew rate limitation and profile generator

Figure 29: Block diagram of an axis in closed-loop operation

The individual processing steps and the corresponding controller parameters are described in the subsections below.

NOTES

It is strongly recommended to save the parameter values of the E-711/E-712 system to a file on the host PC before you make any changes. This way the original settings can be restored if the new parameter settings will not prove satisfactory. To save the parameter values and to load them back to the E-711/E-712 system, use the *Device Parameter Configuration* window of PIMikroMove. See "Create Backup File for Controller Parameters" (p. 86) for more information.

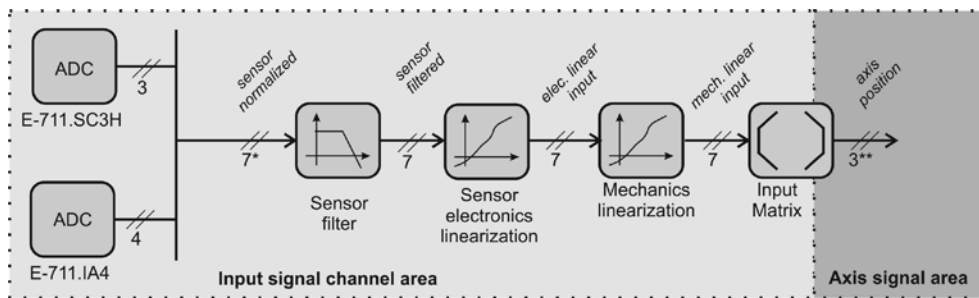
Wherever changing parameter values is mentioned, you can do this using SPA ((volatile memory) or SEP (non-volatile memory). Furthermore, you can use WPA to copy the current values from volatile memory to non-volatile memory, where they become the power-on defaults. To have write access to certain parameter(s), it might be necessary to switch to a higher command level using CCL. To read parameter values, query with the SPA? or SEP? commands.

The *Device Parameter Configuration* window of PIMikroMove gives access to parameter values in a more convenient way. Use this window to check/edit the individual parameters. See the PIMikroMove manual for more information.

5.3.1 Input Signal Processing

The following processing is applied to all input signal channels of the E-711/E-712 system (channels for internal sensors of the mechanics and for analog input lines):

- Analog to digital conversion
- Digital processing (filtering and linearization / scaling)
- Allocation of input signal channels to axes via the input matrix to calculate the axis positions from the input signals



Signal names are written in italics

* E-712.3CDA is shown as an example. It has a total of 7 input signal channels:

3 sensor channels (IDs 1 to 3) and 4 analog inputs (IDs 4 to 7)

** The positions of 3 axes are calculated from the 3 sensor channels via the input matrix.

Figure 30: Input signal processing for E-712.3CDA

Analog to digital conversion:

The results of the analog to digital conversion can be queried with the TAD? command for all channels.

For capacitive sensors: The following settings influence the analog sensor signal and can be selected by parameters:

■ Sensor Range

parameter ID 0x02000100

Scales the measurement range (by changing the capacitance defined as the mid-position value). Example: The nominal measurement range of a sensor is 100 μm . With a factor of 1.27, the measurement range is enlarged to 127 μm , and with a factor of 3.26 enlarged to 326 μm .

Possible parameter values for capacitive dual-electrode sensors (parameter value = factor):

1 = 3.26x

2 = 2.15x

3 = 1.27x

4 = 1.00x

This parameter also selects automatically the corresponding polynomial for electronics linearization (see "Digital processing").

■ Sensor Cable Compensation,

parameter ID 0x02000103

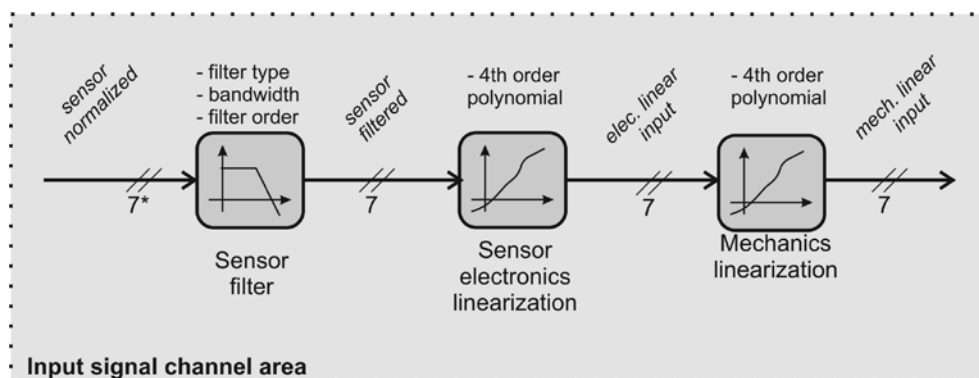
When set to zero, cable compensation is disabled. Otherwise the cable compensation is enabled and the electronics polynomial for cable compensation is selected. The value is determined at the factory and should only be changed when an additional extension cable or adapter is used, or when the length of the stage cable is changed subsequent to the calibration at the factory. In this case, contact your Physik Instrumente Sales Engineer or write info@pi.ws to obtain a suitable value.

Note that in PIMikroMove, the parameters are available in the *Sensor Electronics* parameter groups in the *Device Parameter Configuration* window.

Digital processing:

The digital processing of the input signals comprises the following steps:

- Digital filtering
- Electronics linearization
- Mechanics linearization



Signal names are written in italics

* E-712.3CDA is shown as an example. It has a total of 7 input signal channels:
3 sensor channels (IDs 1 to 3) and 4 analog inputs (IDs 4 to 7)

Figure 31: Digital processing of the input signals, shown for E-712.3CDA

The following parameters determine the digital filter settings:

- Digital Filter Type
parameter ID 0x05000000
0 = no filter
1 = IIR low-pass filter, 2nd order
2 = moving-average filter
- Digital Filter Bandwidth
parameter ID 0x05000001
Gives the cut-off-frequency f_g of the IIR low-pass filter. Only used if "Digital Filter Type" is set to "IIR low-pass filter, 2nd order".
Note that the duration of the signal processing for a capacitive sensor results from two portions:
 - 1) Duration of the analog sensor processing, which takes about 80 μ s
 - 2) Duration of the digital filtering which depends on the f_g setting:

for signal frequencies $f < f_g/2$, the duration of the filtering can be estimated as follows:

$$t \approx 0.216 / f_g$$

- **Digital Filter Order**
parameter ID 0x05000002
Filter order of moving-average filter, gives the number of previous values used in determining the present output. Only used if "Digital Filter Type" is set to "moving-average filter" (for the IIR filter, the order is always 2).

In PIMikroMove, the digital filter parameters are available in the *Sensor Mechanics* parameter groups in the *Device Parameter Configuration* window.

Polynomial linearization is used to correct system performance. The basic form of the polynomials is as follows:

$$y = a_0 + a_1 \cdot x + a_2 \cdot x^2 + a_3 \cdot x^3 + a_4 \cdot x^4 + a_5 \cdot x^5$$

x	–	<i>filtered sensor ADC value</i>
y	–	<i>linearized sensor value</i>

To make the system components easily replaceable, sensor (i.e. mechanics) and electronics use separate polynomials. The coefficients of the polynomials are determined at the factory. Some terms of the polynomials are provided for future application and presently set to zero. The following terms are currently in use:

- **Electronics linearization:** offset, gain, 2nd, 3rd and 4th order correction. The corresponding coefficients of the polynomial are given by the parameters 0x03000100 to 0x03000500. They are independent of the connected mechanics and may not be changed by the user. In PIMikroMove, these parameters are available in the *Sensor Electronics* parameter groups in the *Device Parameter Configuration* window.
- **Mechanics linearization:** offset, gain, 2nd, 3rd and 4th order correction. The corresponding coefficients of the polynomial are given by the parameters 0x02000200 to 0x02000600. They depend on the connected mechanics. In PIMikroMove, these parameters are available in the *Sensor Mechanics* parameter groups in the *Device Parameter Configuration* window.
For the internal sensors in the mechanics, the parameters should not be changed by the user. For analog input lines (last four input signal channels if an E-711.IA4 analog interface module is present), changing the offset and gain values is required to scale the analog

input to suitable position values (see "Using the Analog Input" (p. 153) for more information and examples).

If the connected mechanics has an ID-chip, the coefficients will be read in from the ID-chip (see "ID-Chip Support / Stage Replacement" (p. 216) for more information).

The TNS? command reports the result after the linearization for the electronics (normalized value, dimensionless), while the TSP? command reports the result after the linearization for the mechanics (scaled value, the unit is μm).

Allocation of input signal channels to axes:

Multiple sensors can be used to monitor the position of an axis, especially with rotation axes. The internal sensors in the mechanics are active by default, while additional, external sensors can optionally be connected to the analog input lines if an E-711.IA4 analog interface module is present in the E-711/E-712 system. The axis positions are calculated from the position values of the input signal channels using the input matrix. The number of rows and columns of the input matrix depends on the configuration of the E-711/E-712 system. The example below shows the matrix for an E-712.3CDA model which controls 3 axes and has 3 channels for internal sensors and 4 additional analog input lines (the last four input signal channels):

$$\begin{pmatrix} \text{Axis1} \\ \text{Axis2} \\ \text{Axis3} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} & a_{16} & a_{17} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} & a_{26} & a_{27} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} & a_{36} & a_{37} \end{pmatrix} * \begin{pmatrix} \text{InputCh1} \\ \text{InputCh2} \\ \text{InputCh3} \\ \text{InputCh4} \\ \text{InputCh5} \\ \text{InputCh6} \\ \text{InputCh7} \end{pmatrix}$$

In equation form:

$$\begin{aligned} \text{Axis}_1 &= a_{11}\text{InputCh}_1 + a_{12}\text{InputCh}_2 + a_{13}\text{InputCh}_3 + a_{14}\text{InputCh}_4 + a_{15}\text{InputCh}_5 + \\ &\quad a_{16}\text{InputCh}_6 + a_{17}\text{InputCh}_7 \\ \text{Axis}_2 &= a_{21}\text{InputCh}_1 + a_{22}\text{InputCh}_2 + a_{23}\text{InputCh}_3 + a_{24}\text{InputCh}_4 + a_{25}\text{InputCh}_5 + \\ &\quad a_{26}\text{InputCh}_6 + a_{27}\text{InputCh}_7 \\ \text{Axis}_3 &= a_{31}\text{InputCh}_1 + a_{32}\text{InputCh}_2 + a_{33}\text{InputCh}_3 + a_{34}\text{InputCh}_4 + a_{35}\text{InputCh}_5 + \\ &\quad a_{36}\text{InputCh}_6 + a_{37}\text{InputCh}_7 \end{aligned}$$

The matrix coefficients are given by the Position From Sensor n parameters (n depends on the number of input signal channels present in the E-711/E-712 system). With an E-712.3CDA, for example, the following parameters are used (with $i = 1$ to 3 for the three axes of the system, i.e. each

parameter has a different value for each of the logical axes):

- a_{i1} = Position From Sensor 1, parameter ID 0x07000500,
 a_{i2} = Position From Sensor 2, parameter ID 0x07000501,
 a_{i3} = Position From Sensor 3, parameter ID 0x07000502
These coefficients are for the internal sensors in the mechanics
(sensor channels on the E-711.SC3H sensor module).
- a_{i4} = Position From Sensor 4, parameter ID 0x07000503
 a_{i5} = Position From Sensor 5, parameter ID 0x07000504
 a_{i6} = Position From Sensor 6, parameter ID 0x07000505
 a_{i7} = Position From Sensor 7, parameter ID 0x07000506
These coefficients are for the analog input lines on the E-711.IA4
analog interface module (In 1 to In 4 sockets).

In PIMikroMove, these parameters are available in the *Axis Definition* parameter groups in the *Device Parameter Configuration* window.

NOTES

The coefficients of the input matrix are determined during calibration at the factory. The preset values of the coefficients for the internal sensors should not be changed unless the internal sensors are to be excluded from the position feedback of the axes, e.g. if external sensors are connected to the analog input lines. The coefficients for the analog input lines on an E-711.IA4 analog interface module should be set to zero as long as no external sensors are connected to the analog input lines or when the analog input is used for control value generation (see "Using the Analog Input" (p. 153) for more information).

If the connected mechanics has an ID-chip, the coefficients will be read in from the ID-chip (see "ID-Chip Support / Stage Replacement" (p. 216) for more information).

While TSP? reports the position values of the input signal channels, the POS? command reports the axis positions calculated via the input matrix (the unit is μm or μrad).

5.3.2 Control Value Generation

The control value for the motion of an axis can result from multiple sources (see below). Furthermore, the feedback from multiple sensors can be used to maintain the axis position, depending on the current operating mode. The interpretation of the control values depends on the settings of the output matrix (see "Output Generation" (p. 118) for more information). By default, the output matrix is set up so that control values correspond numerically to axis position values (for exceptions regarding PiezoWalk® systems see below).

The E-711/E-712 system provides the following operating modes:

- Open-loop control (also referred to as "servo-off state" in this document): **No** control algorithm is used, and the sensor feedback does **not** participate in the control value generation.
- Closed-loop control (also referred to as "servo-on state" in this document): Sensor feedback participates in the control value generation.
For each logical axis, a control algorithm is used to generate corrections to the control value (default: PID algorithm). In addition, two notch filters are used for each axis (default: only active in closed-loop operation). The settings for control algorithm and notch filters are accessible as parameters. See "Control Details" (p. 120), "Servo Controller Dynamic Tuning" (p. 218) and "Controller Parameters" (p. 231) for more information.

The operating mode can be selected with the SVO command for each axis. By default, open-loop control is active after power-on. Using the Power Up Servo On Enable parameter (ID 0x07000800), you can set up the individual axes to start with closed-loop control. When switching from open-loop to closed-loop control, the behaviour depends on the setting made with parameter 0x0e002000. Default: The current axis position is set as the target position. For further details, see "Parameter Overview" (p. 233). Switching from closed-loop to open-loop control sets the current closed-loop control value as the open-loop control value.

The E-711/E-712 system supports the following control sources:

- Move commands, sent from the command line or from a macro:
MOV and MVR in closed-loop operation
SVA and SVR in open-loop operation
IMP and STE commands generate an impulse or step response
- Wave Generator: The wave generator is enabled with WGO.
An offset value can be added to the wave generator output using the WOS command.
- Analog Input (only possible when an E-711.IA4 analog interface module is present in the E-711/E-712 system): The analog control input is enabled via parameter settings, see "How to work with the Analog Input" (p. 153) for more information.
An offset value can be added to the analog input scaled value using the AOS command.
- AutoZero procedure: this procedure is started by the ATZ command and performed in open-loop operation only (if servo is on, it will be switched off automatically during the AutoZero procedure and on

again afterwards). The AutoZero procedure has the highest priority, i.e. it will overwrite the control values given by all other sources. When the analog control input is enabled, it will be disabled automatically at the start of the AutoZero procedure and reenabled again when AutoZero is finished. See "AutoZero Procedure" (p. 92) for details.

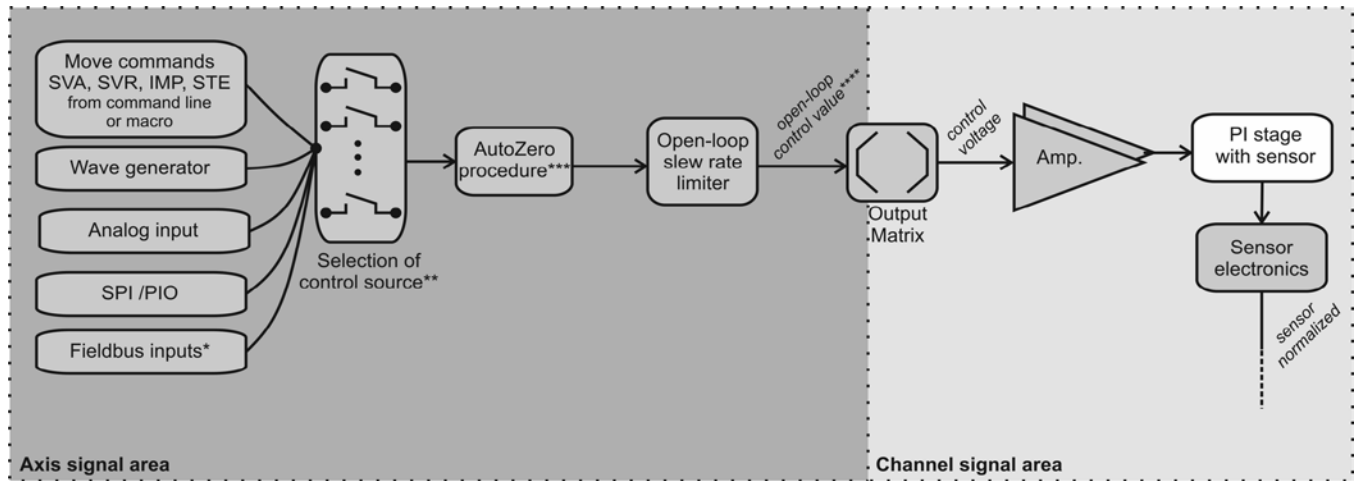
The E-711/E-712 system can also be commanded via SPI. Depending on the SPI data segment used, target values (with same write priority as analog control input) or GCS commands can be sent from an SPI master. See separate SPIST0001 Technical Note for details.

Furthermore, consider the following when working with PiezoWalk® systems (i.e. systems driven by NEXLINE® or NEXACT® drives):

- In closed-loop operation, target positions are commanded. Depending on the driving mode selection made with the PiezoWalk Driving Mode parameter and on the commanded target, motion is realized in full-step and/or analog mode or in nanostepping mode. The internal control value is a velocity and hence influenced by the settings for step size per cycle and minimum cycle time.
- In open-loop operation, the control sources described above command voltage values and will always cause motion in analog mode. Stepping motion must be started with the PiezoWalk-channel-related OSM command.

See the separate E712T0010 Technical Note for more information on PiezoWalk® driving modes and operation.

Open-Loop Control:



Signal names are in italics

* For future applications.

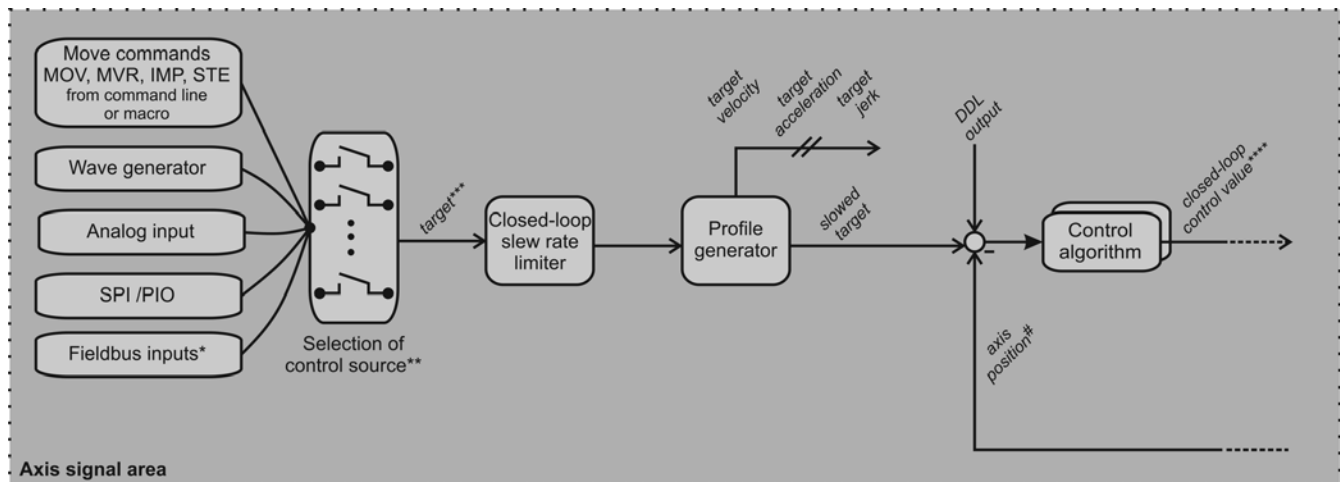
** Only one control source can be used at a time.

*** The AutoZero procedure is triggered with the ATZ command and has the highest priority for signal manipulation. If the AutoZero procedure is not running, it is bypassed. For further details, see „AutoZero Procedure“.

**** The open-loop control value is reported by the SVA? command.

Figure 32: Control sources for an axis of a conventional nanopositioning system in open-loop operation

Closed-Loop Control:



Signal names are in *italics*

* For future applications.

** Only one control source can be used at a time.

*** The target position is reported by the MOV? command.

**** The closed-loop control value is the input signal for the output matrix, see block diagram of closed-loop operation for details.

The axis position is the output signal of the input matrix, see block diagram of closed-loop operation for details.

Figure 33: Control sources for an axis of a conventional nanopositioning system in closed-loop operation

5.3.3 Output Generation

For PiezoWalk® systems (i.e. systems driven by NEXLINE® or NEXACT® drives), see the separate E712T0010 Technical Note for the assignment of axes to output channels.

With conventional nanopositioning systems, multiple piezo actuators can be used to execute the motion of an axis, i.e. multiple output signal channels (piezo amplifiers) can be involved. The control value for an axis is transformed to control voltage values for the output channels via the output matrix. After the digital-to-analog conversion, the resulting control voltage values are sent to the piezo amplifiers whose output drives the actuators in the mechanics.

If an E-711.IA4 analog interface module is present in the E-711/E-712 system, the control voltage values can also be output by the analog output lines to drive external amplifiers (see "Using the Analog Output" (p. 100) for more information).

The number of rows and columns of the output matrix depends on the configuration of the E-711/E-712 system. The example below shows the matrix for an E-712.3CDA model which controls 3 axes and has 4 piezo

amplifier channels and 4 additional analog output lines (the last four output signal channels):

$$\begin{pmatrix} \text{OutputCh1} \\ \text{OutputCh2} \\ \text{OutputCh3} \\ \text{OutputCh4} \\ \text{OutputCh5} \\ \text{OutputCh6} \\ \text{OutputCh7} \\ \text{OutputCh8} \end{pmatrix} = \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \\ p_{41} & p_{42} & p_{43} \\ p_{51} & p_{52} & p_{53} \\ p_{61} & p_{62} & p_{63} \\ p_{71} & p_{72} & p_{73} \\ p_{81} & p_{82} & p_{83} \end{pmatrix} * \begin{pmatrix} \text{Axis1} \\ \text{Axis2} \\ \text{Axis3} \end{pmatrix}$$

In equation form:

$$\begin{aligned} \text{OutputCh}_1 &= p_{11}\text{Axis}_1 + p_{12}\text{Axis}_2 + p_{13}\text{Axis}_3 \\ \text{OutputCh}_2 &= p_{21}\text{Axis}_1 + p_{22}\text{Axis}_2 + p_{23}\text{Axis}_3 \\ \text{OutputCh}_3 &= p_{31}\text{Axis}_1 + p_{32}\text{Axis}_2 + p_{33}\text{Axis}_3 \\ \text{OutputCh}_4 &= p_{41}\text{Axis}_1 + p_{42}\text{Axis}_2 + p_{43}\text{Axis}_3 \\ \text{OutputCh}_5 &= p_{51}\text{Axis}_1 + p_{52}\text{Axis}_2 + p_{53}\text{Axis}_3 \\ \text{OutputCh}_6 &= p_{61}\text{Axis}_1 + p_{62}\text{Axis}_2 + p_{63}\text{Axis}_3 \\ \text{OutputCh}_7 &= p_{71}\text{Axis}_1 + p_{72}\text{Axis}_2 + p_{73}\text{Axis}_3 \\ \text{OutputCh}_8 &= p_{81}\text{Axis}_1 + p_{82}\text{Axis}_2 + p_{83}\text{Axis}_3 \end{aligned}$$

The matrix coefficients are given by the Driving Factor of Piezo m parameters (m depends on the number of output signal channels present in the E-711/E-712 system). With an E-712.3CDA, for example, the following parameters are used (with $i = 1$ to 3 for the three axes of the system, i.e. each parameter has a different value for each of the logical axes):

- p_{1i} = Driving Factor of Piezo 1, parameter ID 0x09000000,
 p_{2i} = Driving Factor of Piezo 2, parameter ID 0x09000001,
 p_{3i} = Driving Factor of Piezo 3, parameter ID 0x09000002
 p_{4i} = Driving Factor of Piezo 4, parameter ID 0x09000003
 These coefficients are for the piezo amplifiers in the E-711.AL4 amplifier module which drive the piezo actuators in the mechanics.
- p_{5i} = Driving Factor of Piezo 5, parameter ID 0x09000004
 p_{6i} = Driving Factor of Piezo 6, parameter ID 0x09000005
 p_{7i} = Driving Factor of Piezo 7, parameter ID 0x09000006
 p_{8i} = Driving Factor of Piezo 8, parameter ID 0x09000007
 These coefficients are for the analog output lines on the E-711.IA4 analog interface module (Out 1 to Out 4 sockets).

In PIMikroMove, these parameters are available in the *Axis Definition* parameter groups in the *Device Parameter Configuration* window.

NOTES

During calibration of conventional nanopositioning systems at the factory, the coefficients of the output matrix are set numerically to the number of volts which are required per axis unit by the attached piezo actuators (i.e. the unit of the coefficients is V/ μm). Thus both the closed-loop control value and the open-loop control value correspond numerically to axis position values. This means that all control sources always command with axis position values, irrespective of the current operating mode. You should not change the coefficients for the piezo amplifier channels.

If the connected mechanics has an ID-chip, the coefficients will be read in from the ID-chip (see "ID-Chip Support / Stage Replacement" (p. 216) for more information).

The VOL? command reports the current voltage output of the output signal channel (in volts).

With systems that support DC motors (E-711.C82 motor driver module is present), the output current can be limited for the output signal channels via the Max Output Current (A) parameter (ID 0x0B000900).

5.4 Control Details

5.4.1 Profile Generator

In closed-loop operation, a profile generator can be used to specify the target position, jerk and acceleration of the axis for any point in time (dynamics profile). Each axis has its own profile generator which can be enabled/disabled via the Target Generator Enable parameter (ID 0x06010300; 0 = off, 1 = on). The profile generator should only be enabled for axes driven by DC motors and disabled for all other axes.

NOTE

For axes with disabled profile generator, the target signal can be manipulated by a slew rate limitation (parameter 0x07000200). For fastest possible settling, the slew rate limitation can be switched off with the VCO command.

The dynamics profile generated by the profile generator depends on the target position which is to be reached at the end of the motion and on the following motion parameters:

- Target Generator Maximum Acceleration (ID 0x06010000)

- Target Generator Maximum Jerk (ID 0x06010100)
- Target Generator Maximum Velocity (ID 0x06010400; read only and identical with the value of the Servo Loop Slew Rate (ID 0x07000200))

5.4.2 Control Algorithms for Closed-Loop Operation

The control algorithm to be used in closed-loop operation can be selected for each axis via the Closed-Loop Control Mode parameter (ID 0x07030100). Possible values of the parameter:

- 0 = Hardware, e.g. control via an external analog control module (position control)
- 1 = PID algorithm (position control)
- 2 = Advanced Piezo Control (APC) algorithm, licence must be ordered separately, see “Additional Components” (p. 64)
- 6 = PID velocity control (a position sensor signal is used, the target is given as velocity)
- 7 = PID position control with subordinate velocity control (a position sensor signal is used, the target is given as position)

With the PID algorithms and all derived algorithms (see below), up to two notch filters are available.

A PID algorithm in principle has the following structure:

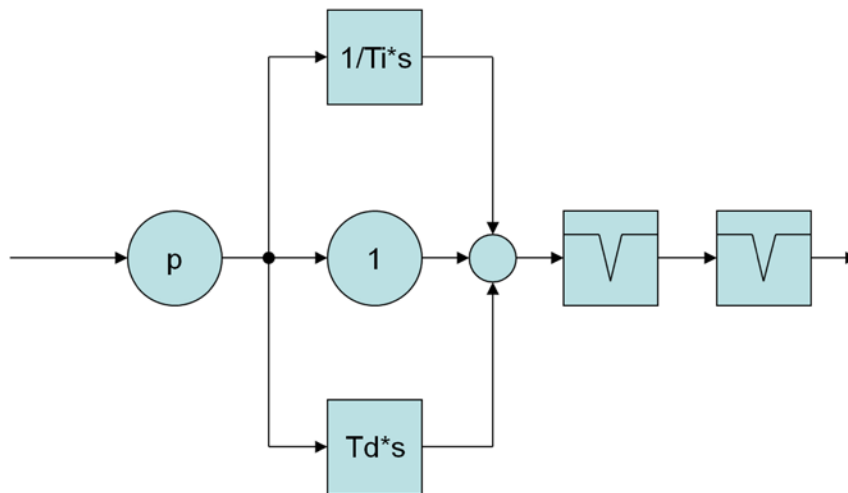


Figure 34: Structure of a PID algorithm; two notch filters are available

Other algorithms can be derived as follows from a PID algorithm:

- P algorithm: I-term and D-term are set to zero
- I algorithm: P-term and D-term are set to zero
- PI algorithm: The D-term is set to zero

Axes driven by conventional piezo actuators, and PiezoWalk®-driven axes normally use the PID algorithm for position control (mode 1) for closed-loop operation, see figure below.

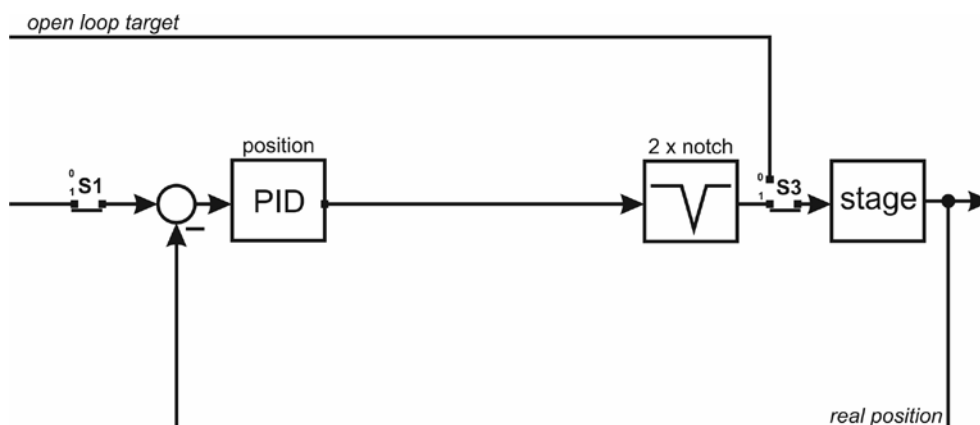


Figure 35: Control structure for position control (mode 1)

Control mode	Servo-control state	S1	S3
Position control (mode 1)	Open-loop operation	0	0
	Closed-loop operation	1	1

Axes driven by DC motors normally use the position control with subordinate velocity control (mode 7), see figure below.

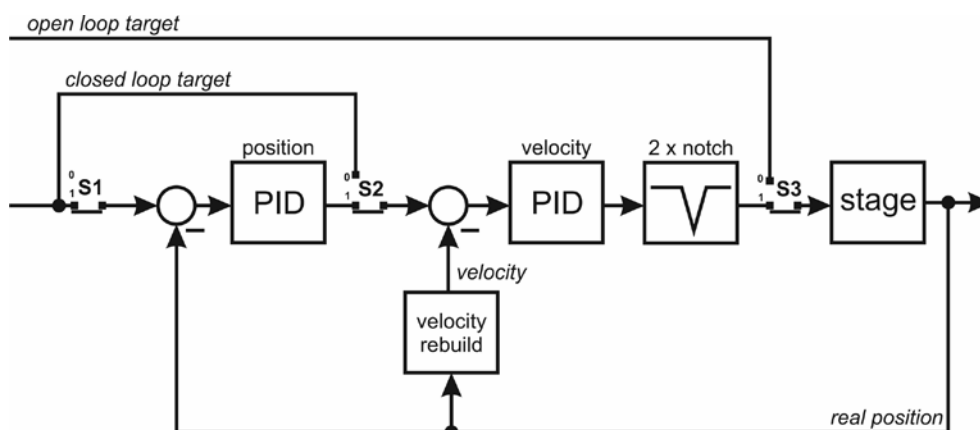


Figure 36: Control structure for position control with subordinate velocity control (mode 7) and for velocity control (mode 6)

Control mode	Servo-control state	S1	S2	S3
Position control with subordinate velocity control (mode 7)	Open-loop operation	0	0	0
	Closed-loop operation	1	1	1
Velocity control (mode 6)	Open-loop operation	0	0	0
	Closed-loop operation	0	0	1

The position control and velocity control can be configured with the following parameters:

Parameter	Notes
Servo-Loop P-Term, ID 0x07000300	P constant for position control Must be > 0. For further details, see “Servo-Controller Dynamic Tuning” (p. 218).
Servo-Loop I-Term, ID 0x07000301	Integrator time constant T_i for position control output = $T_s / T_i \cdot \sum \text{input}$ where T_s is the servo update time (parameter 0x0E000200). When the time constant T_i is zero, then the integrator is turned off. For further details, see “Servo-Controller Dynamic Tuning” (p. 218).
Servo-Loop D-Term, ID 0x07000302	Differentiator time constant T_d for position control output = $T_d / T_s \cdot \Delta \text{input}$ where T_s is the servo update time (parameter 0x0E000200). Must be > 0. For further details, see “Servo-Controller Dynamic Tuning” (p. 218).
Servo-Loop P-Term for Velocity, ID 0x07000307	P constant for velocity control Must be > 0.
Servo-Loop I-Term for Velocity, ID 0x07000308	Integrator time constant T_i for velocity control output = $T_s / T_i \cdot \sum \text{input}$ where T_s is the servo update time (parameter 0x0E000200). When the time constant T_i is zero, then the integrator is turned off
Servo-Loop D-Term for Velocity, ID 0x07000309	Differentiator time constant T_d for velocity control output = $T_d / T_s \cdot \Delta \text{input}$ where T_s is the servo update time (parameter 0x0E000200). Must > 0.
On-Target Tolerance To Fix I-Term, ID 0x07030a00	[μm] Gives a position window which is centered around the target position. If the current position is in this window, the I-term of the control algorithm is fixed to prevent a servo jitter of the axis. For axes driven by DC motors.

Parameter	Notes
Zeroing Control Value If I-Term Is Fixed, ID 0x07030d00	Determines the control value when the I-term is fixed (current position is in the window given by parameter 0x07030a00; see above): 0 = The control value is fixed to the last valid value. 1 = The control value is set to zero. The purpose is to avoid heat generation in static operation. For axes driven by DC motors.

5.4.3 Feedforward

In closed-loop operation, for positioning systems with piezo actuators the main component regarding the tracking error (tracking error = commanded position – real position) is the phase shift between the commanded position and the real position. By adding a feedforward signal to the control algorithm this phase shift can be reduced.

For every axis, an input signal channel of the controller can be selected as feedforward signal using the Feed Forward Input Channel Index parameter (ID 0x07030900; S1 in the figure below). This way, an external preshaping sequence can be used to compensate for a known error, for example. If no input signal channel is selected, the target signal is used as feedforward signal (value of parameter 0x07030900 is zero).

The feedforward signal is fed in before the notch filters to avoid exciting the resonances of the system.

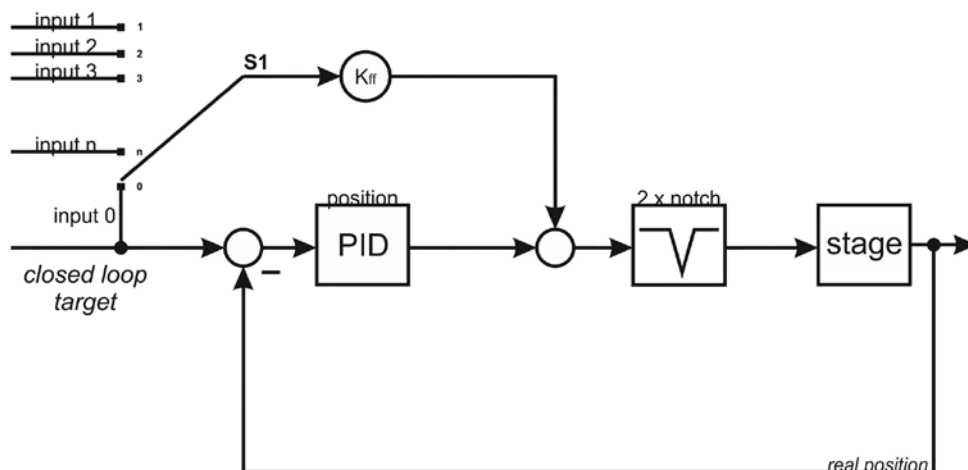


Figure 37: Control structure for position control with feedforward signal

For every axis, the usage of the feedforward signal can be configured using the Feedforward Gain parameter (ID 0x07030600).

5.4.4 Notch Filters

The E-711/E-712 system provides two notch filters per axis. The corrections by a notch filter only take place in closed-loop operation by default, but can also be enabled for open-loop operation. The appropriate frequency component is reduced in the control value to compensate for undesired resonances in the mechanics.

The transfer function of a notch filter is as follows:

$$G(s) = k^2 \times \frac{s^2 + 2 \times \omega \times r \times s + \omega^2}{s^2 + 2 \times \omega \times k \times s + \omega^2 \times k^2}$$

Where

$G(s)$ is the transfer function of the notch filter

k is the bandwidth of the notch filter

s is the input signal

ω is the angular frequency, with $\omega = 2\pi \times f_0$, where f_0 is the notch filter frequency in Hz

r is the notch rejection

The notch filters can be configured per axis using the following parameters:

Parameter	Notes
Notch frequency 1, ID 0x08000100	Frequency f_0 of notch filter 1 and notch filter 2, in Hz. The maximum value is: $f_{0\max} = 0.45 \times f_{\text{sample}}$ where f_{sample} is the servo rate in Hz (1/Servo Update Time (ID 0x0e000200)) Adjusting the notch filter frequency can be useful, particularly in the case of very high loads. For further details, see “Adjusting the Notch Filter(s) in Open-Loop Operation” (p. 221).
Notch frequency 2, ID 0x08000101	
Notch Rejection 1, ID 0x08000200	Notch rejection value r for notch filter 1 and notch filter 2. 0 to 0.98 Recommended value is 0.05. A notch rejection value of 1 deactivates the notch filter. The notch rejection value determines the filter width of the notch filter, i.e. it scales the damping done by the notch filter: The greater the rejection value, the wider the frequency spectrum of the damping, but the smaller the damping effect.
Notch Rejection 2, ID 0x08000201	

Parameter	Notes
Notch Bandwidth 1, ID 0x08000300	Bandwidth k of notch filter 1 and notch filter 2 ≥ 0.1 The notch filter bandwidth determines the effect of the low-pass filtering: The smaller the bandwidth, the smaller the low-pass filter frequency.
Notch Bandwidth 2, ID 0x08000301	
Creep factor T1/sec, ID 0x08000400	Currently not used; provided for future applications.
Creep factor T2/sec, ID 0x08000401	
Enable Notch in Open Loop, ID 0x08000500	Enables usage of notch filter in open-loop operation. In closed-loop operation, the notch filters are always enabled. 0 = disable notch filter in open-loop operation (default) 1 = enable notch filter in open-loop operation
Notch Filter Calculation Method, ID 0x08000600	0 = bilinear 1 = zero-order hold 2 = frb (rejection rate is independent of bandwidth)

5.5 Signals of Reference and Limit Switches

The signals of reference and limit switches can be fed into the E-711/E-712 system and evaluated only when one of the modules E-711.C82, .AM5A, .AM5B, .SA3 and/or .SA6 is present. The activation state of the evaluation can be checked with the LIM? command or via the value of the Lim, Ref Signals Detectable parameter (ID 0x02001900).

While the reference switch signals are only used for reference moves of the axes (see “How to Reference an Axis”, p. 94), the limit switch signals can also be used for the safety shutdown functionality (see “Safety Shutdown”, p. 211).

The current state of reference and limit switch signals can be queried using the SRG? command. The signal polarity can be inverted using the Sensor Lim, Ref Signal Inversion parameter (ID 0x02001000).

There are no separate channels for reference and limit switch signals in the firmware of the E-711/E-712 system, but each of the signals belongs to a certain input signal channel (i.e. to a sensor). For that reason, all parameters and commands for evaluation of reference and limit switch signals refer to input signal channels.

Overview of the parameters which are relevant for the reference and limit switch signals:

Parameter	Notes								
Sensor Reference Mode, ID 0x02000a00	Determines how the sensor is to be referenced: 1 = no reference (absolute sensors; no referencing required, RON cannot be set to 0) 2 = reference on negative hardstop 3 = reference on positive hardstop 4 = reference on signed reference mark 5 = reference on impulse								
Sensor Lim, Ref Signal Inversion, ID 0x02001000	Determines if the E-711/E-712 system inverts the reference switch and limit switch signals. The value is given in hexadecimal format and is based on a combination of 4 bits: <table border="1"><tr><td>bit 3</td><td>bit 2</td><td>bit 1</td><td>bit 0</td></tr><tr><td>[neg_lim]</td><td>[pos_lim]</td><td>[ref_sign]</td><td>[ref_slope]</td></tr></table> bit is set (1) = the signal is inverted bit is not set (0) = the signal is not inverted Examples: 0xC (default) = 0b1100 = negative and positive limit switch signals are inverted (active low) to ensure normal operation for stages that have no limit switches. 0x8 = 0b1000 = the negative limit signal is inverted (active low), all other signal are not inverted 0x3 = 0b0011 = hardware is referencing on falling edge, the reference signal will be inverted	bit 3	bit 2	bit 1	bit 0	[neg_lim]	[pos_lim]	[ref_sign]	[ref_slope]
bit 3	bit 2	bit 1	bit 0						
[neg_lim]	[pos_lim]	[ref_sign]	[ref_slope]						
Lim, Ref Signals Detectable, ID 0x02001900	Flag that indicates if signals of limit and reference point switches are evaluated by the firmware (does not indicate if switches are actually present) 0 = no evaluation of switch signals 1 = switch signals are evaluated The parameter is read only.								

6 Communication

6.1 Interfaces Available

The E-711/E-712 system can be controlled from a host computer (not included) with ASCII commands sent via:

- TCP/IP
- RS-232 serial connection
- USB connection

All interfaces can be active simultaneously. The commands from the interfaces are queued in the order the completed command lines are received.

NOTES

Before communication between E-711/E-712 system and host PC can be established, the power-on sequence of the controller must have finished successfully (takes approx. 40 s; the more modules, the longer the duration; when finished the Power LED glows permanently). TCP/IP communication: If no DHCP server is available on the network or if a point-to-point connection between host PC and controller is being used, it might take another period of about 30 seconds before communication is possible.

Up to 8 command streams can be connected to the E-711/E-712 system via TCP/IP (default: 1). The maximum number of connections can be configured via the Max. TCP/IP Connections parameter (ID 0x11000c00).

With TCP/IP and USB connections, communication cannot be maintained after the E-711/E-712 system is power-cycled or rebooted. The connection must then be closed and reopened.

Using multiple interfaces simultaneously may cause problems with the host software.

It is also possible to set up the E-711/E-712 system so that it can be commanded directly by an analog input signal. See "Using the Analog Input" (p. 153) for more information.

The E-711/E-712 system can also be commanded via SPI. Depending on the SPI data segment used, target values (with same write priority as analog control input) or ASCII commands (GCS) can be sent from an SPI master. See separate SPIST0001 Technical Note for details.

With PIMikroMove, it is possible to connect to multiple controllers in one program instance. With PITerminal, you need a new program instance for each E-711/E-712 system you want to connect to.

6.2 Default and Current Settings

The default communication parameters are stored on the E-711/E-712 system. You can read the default settings using the IFS? command and change them with IFS. Changes become active with the next power-on or reboot, when the default values are loaded into the controllers volatile memory (RAM).

The currently active baud rate for RS-232 serial connections can be temporarily changed with IFC. The new setting becomes active immediately and the host PC interface configuration may need to be changed to maintain communication. When the controller is powered down, the baud rate setting made with IFC is lost if it was not saved with WPA.

To read all current active communication parameters use the IFC? command.

NOTE

The IFS, IFS?, IFC and IFC? commands affect the controller side only.

It is also possible to change the default settings with SEP and to read them with the SEP? command, but do not activate them with RPA and do **not** change the current settings with SPA—except of the baud rate—because it will not be possible to maintain communication afterwards. The appropriate parameter IDs are given below.

The factory defaults of the communication settings stored in the controller are as follows (response to IFS?):

- RSBAUD: gives the baud rate to be used for RS-232 communication, default is 115200
also accessible as Uart Baudrate parameter, ID 0x11000400
- IPADR: the first four portions specify the default IP address for TCP/IP communication, the last portion specifies the default port to be used, default is 192.168.168.10:50000
also accessible as IP Address parameter, ID 0x11000600
Note: While the IP address can be changed, the port must always be 50000!

- IPSTART: defines the startup behavior for configuration of the IP address for TCP/IP communication, default is 1 = "use DHCP to obtain IP address, if this fails, use IPADR" also accessible as IP Configuration parameter, ID 0x11000800
- IPMASK: gives the IP mask to be used for TCP/IP communication, default is 255.255.255.0 also accessible as IP Mask parameter, ID 0x11000700
- MACADR: is the unique address of the network hardware in the E-711/E-712 system, read-only, example: 0-d0-c9-a7-1f-86 also accessible as MAC Address parameter, ID 0x11000B00
- IPMAXCONN: gives the maximum number of allowed IP connections for TCP/IP communication, default is 1 (maximum is 8) also accessible as Max. TCP/IP Connections parameter, ID 0x11000C00

NOTE

When a TCP/IP connection is established at which the IP address is obtained from DHCP server, this address will automatically be written to the IP Address parameter (ID 0x11000600) in the E-711/E-712 systems volatile memory. Provided that the current command level is 1 (see CCL command) and you send WPA 100, the new address setting will be saved to non-volatile memory as the new default value (together with all other current valid parameter values with level-1 write access).

6.3 TCP/IP Connection

The TCP/IP connection is available on the front panel of the controller (E-712.M1 or E-712.N1 digital processor module), via the RJ 45 socket with the network icon. The access differs depending on the network availability, which can be as follows:

- Network with DHCP server
- PC equipped with an Ethernet connection or network without DHCP server

For the default IP address, IP mask and startup behaviour settings of the TCP/IP interface see "Default and Current Settings" (p. 129).



CAUTION

If the communication between host PC and E-711/E-712 system is done via TCP/IP, do not use RPA after you have changed the parameters of the TCP/IP communication with IFS or SEP in non-volatile memory, because it will not be possible to maintain communication afterwards.

With TCP/IP and USB connections, communication cannot be maintained after the E-711/E-712 system is power-cycled or rebooted. The connection must then be closed and reopened.

NOTES

Make sure that your network administrator has not set the network to forbid unknown devices like the E-711/E-712 system from logging on.

The maximum number of TCP/IP connections can be configured via the Max. TCP/IP Connections parameter (ID 0x11000c00; default: 1; maximum: 8).

The default port setting (50000) cannot be changed.

6.3.1 Network with DHCP Server

NOTE

For successful connection, the correct startup behaviour for the IP address configuration must be selected. The default selection is "Use DHCP to obtain IP address", so that nothing has to be changed.

If you want to check the startup behaviour setting, establish a connection via USB or RS-232 interface, and ask with the IFS? command. In the answer, IPSTART must be 1. If the IPSTART value differs from 1, send
IFS 100 IPSTART 1

If a network with DHCP server is available, connect the controller to a network access point and power cycle it (newer switches accept both cross-over and straight-through network cables). Factory default: The controller will automatically obtain an IP address over DHCP. If this should not be successful within 30 seconds, the default IP address of the controller

will automatically be used (see "Default and Current Settings" (p. 129)).

In the host software (e.g. PIMikroMove, PITerminal or LabView drivers), all available E-711/E-712 systems with their IP address and available connections are listed. In PIMikroMove you have, for example, to click on the controller type (1). Then select the *TCP/IP* tab card (2). On the interface tab card, click on the E-711/E-712 system to which you want to connect (3). To establish the connection, click *Connect* (4).

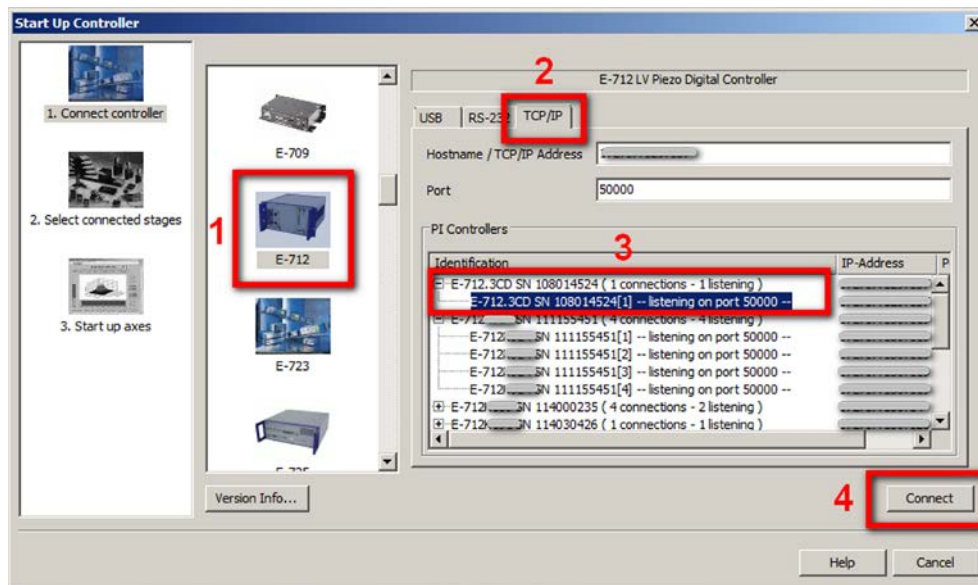


Figure 38: The connection dialog for TCP/IP in PIMikroMove

NOTE

In the controller selection list, you can identify the controllers which are currently connected via TCP/IP.

6.3.2 PC with Ethernet Connection or Network without DHCP Server

The IP address and IP mask settings of PC and controller must be compatible with each other in the following cases:

- The E-711/E-712 system is directly connected to an Ethernet connection of the PC.
- E-711/E-712 system and the host PC both are connected to the same network where no DHCP server is available (in this case, the settings must also be compatible with those of any other devices in the same network).

Otherwise no connection can be established. You can configure either the PC or the controller settings to be compatible. If you have a network with multiple E-711/E-712 systems, the settings of the individual controllers must be changed to have unique IP addresses for all devices in the network. See below for how to proceed.

NOTE

After the power-on / reboot sequence which takes about 40 seconds (the more modules, the longer the duration), you might have to wait for another period of about 30 seconds before communication is possible.

If you want to change the PC settings:

Configure the connection on the PC according to the IP address and IP mask settings of the controller (see "Default and Current Settings" (p. 129) for the default controller settings). Note that the following steps may vary in some details depending on the version of your Windows operating system:

- 1 Open the window in which the properties of the TCP/IP Internet protocol are displayed and set, in a suitable way on your PC. The necessary steps depend on the operating system used.

If your operating system distinguishes between Internet protocol version 4 (TCP/IPv4) and version 6 (TCP/IPv6), open the window for version 4.

- 2 In the *Internet Protocol (TCP/IP) Properties* window, activate *Use the following IP address*. Make a note of the current *IP address* and *Subnet mask* settings, if any, in case they need to be restored later. Then adapt the *IP address* and *Subnet mask* settings to make them compatible with the settings of your E-711/E-712 system:

Set the first three portions of *IP address* identical to those of the IP address of the E-711/E-712 system, while the last portion must be different. One possible *IP address* setting would be, for example, 192.168.168.2 (the default IP address of the E-711/E-712 system is 192.168.168.10). Do not use "255" for the last portion.

Set *Subnet mask* to 255.255.255.0 (if the IP mask of the E-711/E-712 system is 255.255.255.0).

Confirm with *OK*. An example is shown in the figure below.

- 3 Connect the E-711/E-712 system to the Ethernet socket of the PC using the included, special, cross-over cable ("point-to-point" connection). If you connect the E-711/E-712 system to a free access point (e.g. to a hub) on a network to which the PC is connected, it might be necessary to use a straight-through network cable.

- 4 Power on the E-711/E-712 system.
- 5 Establish the connection between PC and the E-711/E-712 system as described in "Network with DHCP Server" (p. 131).

NOTE

If the connection fails, change the last portion of the *IP address* setting on the PC and try again to connect.

Adapt *IP address* and *Subnet mask*; do not change the *Default gateway* setting

Press *OK*

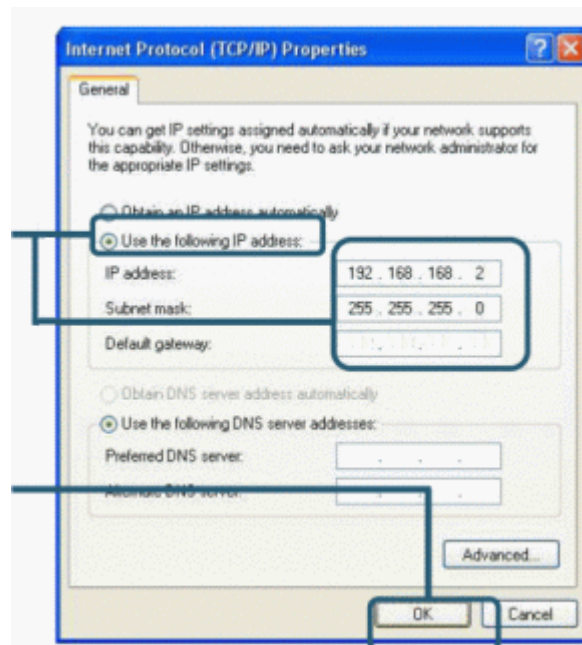


Figure 39: Internet Protocol (TCP/IP) Properties window, the settings shown are only examples, maybe they do not match that of your controller

If you want to change the controller settings:

- 1 Establish a serial connection between PC and E-711/E-712 system as described in "RS-232 Serial Connection" (p. 135).
- 2 Use the IFS command in the command entry facility of the program to adapt the IP address and IP mask settings of the E-711/E-712 system to those of the PC (to check the PC settings, you can open

the *Internet Protocol (TCP/IP) Properties* window as described above):

To change the IP mask (subnet mask), send
IFS 100 IPMASK *mask*
mask must be identical to the Subnet mask setting of the PC.

To change the IP address, send
IFS 100 IPADR *address*
At least the last portion of the IP address must be different from that of the PC and any other device in the same network (the applicable address settings depend on the IP mask setting). If, for example, the PC has the IP address 172.21.0.1, send
IFS 100 IPADR 172.21.0.2:50000
Do not use "255", and do not change the port setting (must always be 50000).

- 3 Close the connection.
- 4 Connect the E-711/E-712 system to the Ethernet socket of the PC using the included, special, cross-over cable ("point-to-point" connection). If you connect the E-711/E-712 system to a free access point (e.g. to a hub) on a network to which the PC is connected, it might be necessary to use a straight-through network cable.
- 5 Power-cycle the E-711/E-712 system.
- 6 Establish the connection between PC and the E-711/E-712 system as described in "Network with DHCP Server" (p. 131).

6.4 RS-232 Serial Connection

The serial communications port is accessed via the sub-D 9 (m) "RS-232" connector (p. 286) on the front panel of the controller (E-712.M1 or E-712.N1 digital processor and interface module). Use the included null-modem cable to connect the controller to the host PC; if the PC has only one COM port, it is probably COM 1.

The serial port on the E-711/E-712 system is preset to the following parameters:
115,200 baud, 8 bits, no parity, RTS/CTS.

In the connection dialog of the host software (e.g. PIMikroMove, PITerminal or LabView drivers), you make the settings on the host PC side. In PIMikroMove you have, for example, to click on the controller type (1). Then select the RS-232 tab card (2). On the interface tab card, select the correct

COM port and baud rate of the PC (3). To establish the connection, click *Connect* (4).

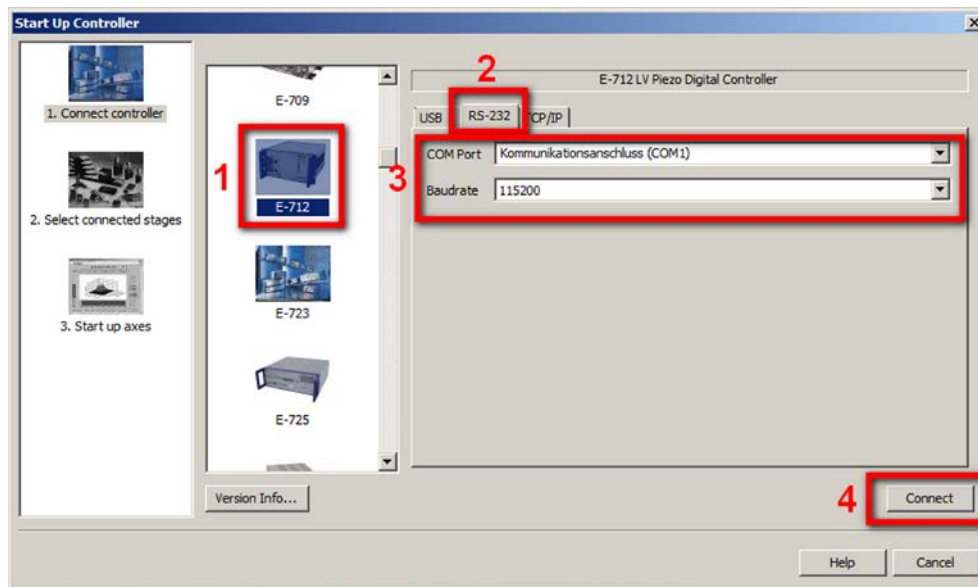


Figure 40: RS-232 configuration of the host PC side in PIMikroMove

If you have established a connection and want to change the currently active baud rate, proceed as follows:

- 1 Use the IFC command in the command entry facility of the program, e.g. by sending IFC RSBAUD 57600.
- 2 Close the connection.
- 3 Open the connection again with the baud rate you just set with IFC (in the example 57600).

6.5 USB Connection

The USB interface is available on the front panel of the controller (E-712.M1 or E-712.N1 digital processor and interface module) via the type B USB socket. Use the included USB cable (USB-A/USB-B) to connect the controller to the host PC.

In the host software (e.g. PIMikroMove, PITerminal or LabView drivers) all E-711/E-712 system which are connected to the USB sockets of the host PC are listed. In PIMikroMove you have, for example, to click on the controller type (1). Then select the *USB* tab card (2). On the interface tab card, click on the E-711/E-712 system to which you want to connect (3). To establish the connection, click *Connect* (4).

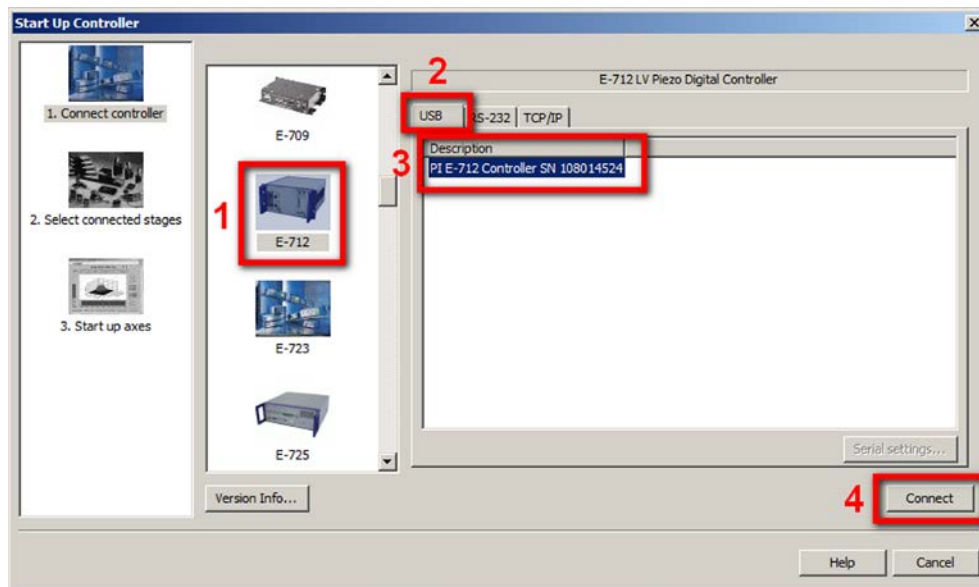


Figure 41: The USB connection dialog in PIMikroMove

NOTE

With TCP/IP and USB connections, communication cannot be maintained after the E-711/E-712 system is power-cycled or rebooted. The connection must then be closed and reopened.

7 Data Recording

7.1 How to Use the Data Recorder

The E-711/E-712 system includes a real-time data recorder. It is able to record several input and output signals (e.g. current position, sensor input, output voltage) from different data sources (e.g. controller axes or input and output channels). The gathered data is stored (temporarily) in "data recorder tables"—each table contains the signal from one data source. You can configure the data recorder flexibly, e.g. select the type of data and the data source. Furthermore, you can choose the number of record tables and hence influence their size.

For general information regarding the data recording you can send HDR?, which lists available options, and gives information about additional parameters and commands concerned with data recording.

How to Define What to Record—Set Record Options

The data recorder configuration, i.e. the assignment of data sources and record options to the recorder tables, can be read with the DRC? command. The answer gives the values of the parameters DRC Data Source (ID 0x16000700) and DRC Record Option (ID 0x16000701). Use the DRC command or change the parameter values directly in volatile memory. The default setting is that the current positions of the axes are recorded.

How to Start Recording—Set Trigger Options

Recording can be triggered in several ways. Ask with DRT? for the current trigger option and use DRT to change it. Irrespective of the DRT settings, data recording is always triggered by the following four commands:

- STE (step response measurement),
- IMP (impulse response measurement),
- WGO (wave generator start)
- WGR (restarts recording when the wave generator is running).

Recording always takes place for all data recorder tables and ends when the data recorder tables are completely filled.

If digital input lines are used to trigger data recording (configuration with DRT): For reliable triggering, the pulse width of the input signal has to be at

least 2 x the servo update time of the E-711/E-712 system. The servo update time is given in seconds by parameter 0x0E000200.

How to Read Recorded Data

The last recorded data can be read with the DRR? command. The data is reported in GCS array format. For details regarding GCS array see the separate manual (SM146E), which is provided on the E-712 CD. Reading out recorded data can take some time, depending on the number of points to be read! It is possible to read the data while recording is still in progress.

The number of points comprised by the last recording can be read with the DRL? command. This can be useful, for example, if you restart recording with WGR and want to read data while recording is still in progress.

How to Configure Number of Tables and Sampling Period

The number of available data recorder tables can be read with the TNR? command. The answer gives the value of the Data Recorder Chan Number parameter, ID 0x16000300. You can change the parameter value to increase or decrease the number of data recorder tables. For the E-711/E-712 system, the number of tables must be in the range of 1 to 12.

The total number of points available for data recording is given by the Data Recorder Max Points parameter, ID 0x16000200. The controller allocates these points in equal shares to the available tables (i.e. to the number of tables given in the TNR? answer). For the E-711/E-712 system, the total number of points is 2^{23} . If, for example, TNR? replies 8, each table is comprised of 1,048,576 points.

The data recorder sampling period can be read with the RTR? command. The answer gives the value of the Data Recorder Table Rate parameter (ID 0x16000000) whose default value is one servo cycle. You can cover longer periods by increasing this value. Use the RTR command or change the parameter value directly in volatile memory.

How to Save Settings to Non-Volatile Memory

Wherever changing parameter values is mentioned, you can do this using SPA (volatile memory) or SEP (non-volatile memory). Furthermore, you can use WPA to copy the current values from volatile memory to non-volatile memory, where they become the power-on defaults. To have write access to certain parameter(s), it might be necessary to switch to a higher command level using CCL. To read parameter values, query with the SPA? or SEP? commands.

When the controller is powered down, the contents of the data recorder tables and all settings which were only made in volatile memory are lost. On power on, all settings are reset to their power-on defaults.

7.2 Data-Recorder Related Commands and Parameters

Command	Description	Notes
DRC	Set Data Recorder Configuration	Assigns data sources and record options to data recorder tables in volatile memory (DRC Data Source parameter, ID 0x16000700, and DRC Record Option parameter, ID 0x16000701)
DRC?	Get Data Recorder Configuration	Reads current data recorder settings
DRR?	Get Recorded Data Values	Reading can take some time, depending on the number of points.
DRT	Set Data Recorder Trigger Source	Determines how recording is to be triggered. Settings will be lost on controller power down or reboot.
DRT?	Get Data Recorder Trigger Source	Reads current trigger option
HDR?	Get All Data Recorder Options	Lists available record options, gives information about additional parameters and commands concerned with data recording
IMP	Start Impulse and Response Measurement	Triggers recording
RTR	Set Record Table Rate	Changes the data recorder table rate in volatile memory (Data Recorder Table Rate parameter, ID 0x16000000)
RTR?	Get Record Table Rate	Reads the current setting of the data recorder table rate (Data Recorder Table Rate parameter, ID 0x16000000)
STE	Start Step and Response Measurement	Triggers recording
TNR?	Get Number of Record Tables	Reads the number of available data recorder tables (Data Recorder Chan Number parameter, ID 0x16000300)
WGO	Set Wave Generator Start/Stop Mode	Triggers recording
WGR	Start Recording Synchronous to Wave Generator	Triggers recording

See "How to use the Data Recorder" (p. 138) for more information. For detailed command descriptions see the GCS commands manual PZ233E. For the identifiers of the items which can be addressed with the commands see "Accessible Items and Their Identifiers" (p. 104).

Parameter ID	Command Level	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description
0x16000000	1	System	1	INT	Data Recorder Table Rate
0x16000100	3	System	1	INT	Max Number of Data Recorder Channels
0x16000200	3	System	1	INT	Data Recorder Max Points
0x16000300	1	System	1	INT	Data Recorder Chan Number; the available data recorder points are allocated in equal shares to the number of tables given by this parameter
0x16000700	1	Data recorder table	12	INT	DRC Data Source
0x16000701	1	Data recorder table	12	INT	DRC Record Option

See "Controller Parameters" (p. 231) for more information regarding the controller parameters and their handling.

8 External Triggering / Signaling

The digital I/O lines of the E-711/E-712 system are available on the "Digital I/O Socket" (see p. 285 for the lines and pinout).

8.1 Using Digital Input

The IN1 to IN8 input lines can be used to start data recording, see the DRT command and "Data Recording" (p. 138) for details.

The IN1 and IN2 input lines can be used in conjunction with the WGO command to trigger the wave generator output (IN1 and IN2) and to stop it (IN2). See "Wave Generator Started by Trigger Input" (p. 184) for an example.

The IN1 to IN8 lines can be used for safety shutdown of the E-711/E-712 system. Configure the lines via parameters. It is not recommended to define lines which are used to start the data recorder or to start/stop the wave generators. See "Safety Shutdown" (p. 211) for details.

8.2 Configuring Trigger Output

You can program the digital output lines OUT1 to OUT7 of the E-711/E-712 system to trigger other devices using the CTO command. The ServoTrigger output line carries the servo cycle pulses and is not accessible by command.

The general format of the CTO command is as follows, i.e. all trigger-related settings for a digital output line can normally be made with one command line (the number of arguments following a command mnemonic is limited to 32):

CTO {<TrigOutID> <CTOPam> <Value>}

The following trigger modes are supported by the E-711/E-712 system:

- 0 = Position Distance; a trigger pulse is written whenever the axis has covered a given distance. Optionally, values for StartThreshold and StopThreshold can be defined to enable the trigger output for a limited position range and a certain direction of motion only (negative or positive). When StartThreshold and StopThreshold are set to the same value, they will not be used. See "Example—"Position Distance" Trigger Mode" (p. 145).
- 2 = OnTarget; the on-target status of the selected axis is written to the selected trigger output line (this status can also be read with the ONT? command). See "Example—"On Target" Trigger Mode" (p. 148).

- 3 = MinMaxThreshold; values for MinThreshold and MaxThreshold must be defined. When the axis position of the selected axis is inside the band specified by the MinThreshold and MaxThreshold values, the selected trigger output line is set high, otherwise it is set low. See "Example—"MinMax Threshold" Trigger Mode" (p. 150).
- 4 = Generator Level Trigger; the trigger line action must be defined with TWS. The trigger output will be synchronized with the wave generator output. The length of a single trigger pulse is the same as the duration of one servo cycle. If the signal level is set to HIGH with TWS for consecutive points of a wave table, the signal level therefore does not change back to LOW between the points. See "Example—"Generator Level Trigger" Mode" (p. 151) and "Trigger Output Synchronized with Wave Generator" (p. 183).
- 9 = Generator Pulse Trigger; the trigger line action must be defined with TWS. The trigger output will be synchronized with the wave generator output. The length of a single trigger pulse is shorter than the servo cycle duration. If the signal level is set to HIGH with TWS for consecutive points of a wave table, the signal level therefore changes back to LOW after each point. See "Example—"Generator Pulse Trigger" Mode" (p. 151).

To select the mode, set <CTOPam> = 3 and <Value> to the code of the mode; default selection is On Target (2).

Furthermore, it is possible to select the signal polarity for the digital output line (active high / active low). See "Example—Polarity Setting" (p. 152).

CTO changes the values of the parameters listed below in volatile memory. The current values in volatile memory can be read with the CTO? command. You can also change these parameters using SPA (volatile memory) or SEP (non-volatile memory). Furthermore, you can use WPA to copy the current values from volatile memory to non-volatile memory, where they become the power-on defaults. When using SPA, SEP or WPA, it is necessary to switch to command level 1 with CCL to have write access to the parameters. To read the parameter values, you can also query with the SPA? or SEP? commands.

Parameter ID	Command Level	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description
0x18000201	1	Digital Output Line	7	FLOAT	CTO Trigger Step (<CTOPam> = 1)
0x18000202	1	Digital Output Line	7	INT	CTO Axis (<CTOPam> = 2)
0x18000203	1	Digital Output Line	7	INT	CTO Trigger Mode (<CTOPam> = 3)

Parameter ID	Command Level	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description
0x18000205	1	Digital Output Line	7	FLOAT	CTO Min.Threshold (<CTOPam> = 5)
0x18000206	1	Digital Output Line	7	FLOAT	CTO Max.Threshold (<CTOPam> = 6)
0x18000207	1	Digital Output Line	7	INT	CTO Polarity (<CTOPam> = 7)
0x18000208	1	Digital Output Line	7	FLOAT	CTO Start Threshold (<CTOPam> = 8)
0x18000209	1	Digital Output Line	7	FLOAT	CTO Stop Threshold (<CTOPam> = 9)

The following examples can be reproduced using the command entry facilities of PIMikroMove or PI Terminal.

8.2.1 Example—"Position Distance" Trigger Mode

The "Position Distance" trigger mode is designed for scanning applications. A trigger pulse is written whenever the axis has covered the distance set with CTO (<TriggerStep>). The time interval of the trigger pulses depends on the position distance set with <TriggerStep> and on the current speed of the axis.

The unit of <TriggerStep> is μm or μrad .

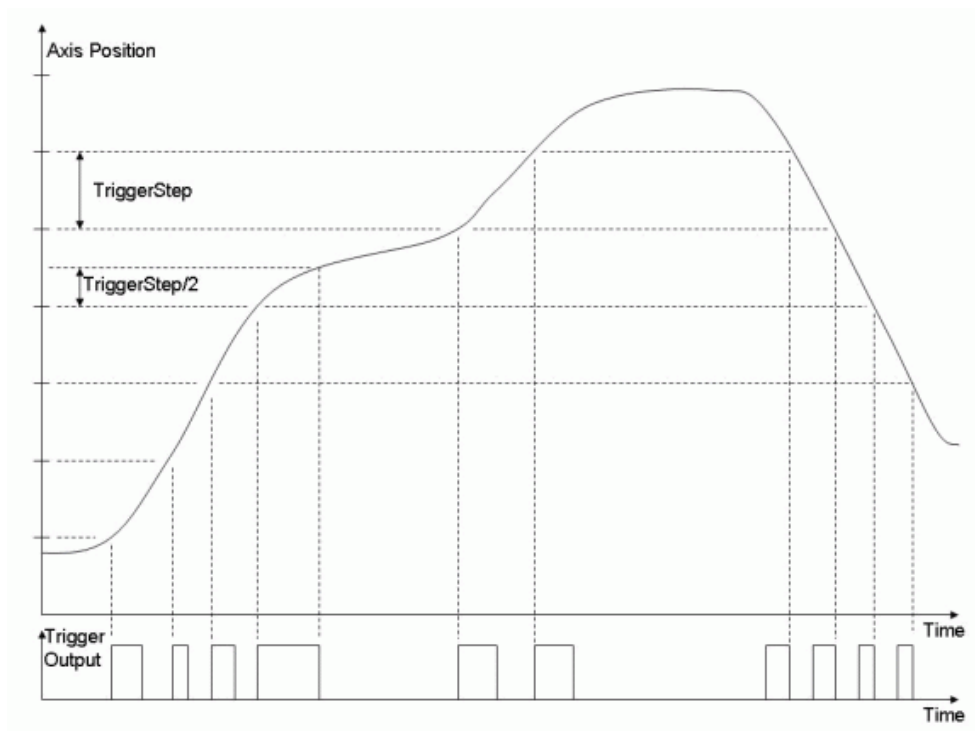


Figure 42: "Position Distance" Trigger Mode

The following parameters must be set for the digital output line which is to be used for trigger output (<TrigOutID>):

- Axis (<CTOPam> = 2)
- TriggerMode (<CTOPam> = 3)
- TriggerStep (<CTOPam> = 1)

General notation of the CTO command for this mode (in fact, the command arguments can be divided in three "portions", each starting with the <TrigOutID> declaration):

Command mnemonic	Axis selection	Trigger mode selection	Step size setting
CTO	<TrigOutID> 2 <i>Axis</i>	<TrigOutID> 3 0	<TrigOutID> 1 <i>Stepsize</i>

Instead of using the CTO command, you can also set the values of the corresponding parameters with SPA or SEP, see “Configuring Trigger Output” (p. 142) for a parameter list.

NOTE

For reliable triggering, the TriggerStep setting has to be at least 4.5 times larger than the noise of the axis position (peak-to-peak value). To detect the noise, you can record, for example, the position error of the axis using the data recorder (p. 138).

Example: A pulse on the digital output line 1 is to be generated whenever axis 1 of the stage has covered a distance of 0.1 μm . Send:

CTO 1 2 1 1 3 0 1 1 0.1

Optionally, start and stop values can be set with CTO (<StartThreshold> and <StopThreshold>) to enable the trigger output for a limited position range and a certain direction of motion only (positive or negative). When <StartThreshold> and <StopThreshold> are set to the same value, they will not be used. Should the motion direction be reversed before the axis position has reached the stop threshold, trigger pulses will continue to be generated.

The following parameters must then be set for the digital output line which is to be used for trigger output (<TrigOutID>):

- Axis (<CTOPam> = 2)
- TriggerMode (<CTOPam> = 3)
- TriggerStep (<CTOPam> = 1)
- StartThreshold (<CTOPam> = 8)
- StopThreshold (<CTOPam> = 9)

General notation of the CTO command for this option (in fact, the command arguments can be divided in five "portions", each starting with the <TrigOutID> declaration):

Command mnemonic	Axis selection	Trigger mode selection	Step size setting	Start threshold setting	Stop threshold setting
CTO	<TrigOutID> 2 <i>Axis</i>	<TrigOutID> 3 0	<TrigOutID> 1 <i>Stepsize</i>	<TrigOutID> 8 <i>Startpos.</i>	<TrigOutID> 9 <i>Stoppos.</i>

NOTES

For reliable activation of the trigger output, the axis has to move through the “Trigger ready, not active” position range, or the motion has to start in this range (see Figure 43 and Figure 44).

For reliable deactivation of the trigger output, the axis has to move through the “Trigger stopped” position range, or the motion has to end in this range (see Figure 43 and Figure 44).

The distance between the start position of the axis and the StartThreshold setting has to be at least 3 times larger than the noise of the axis position (peak-to-peak value), as well as the distance between the end position of the axis and the StopThreshold setting.

See the examples 2 and 3 below for details.

Example 2: A pulse on the digital output line 1 is to be generated whenever axis 1 of the stage has covered a distance of $0.1\text{ }\mu\text{m}$, as long as axis 1 moves in positive direction in the range of $0.2\text{ }\mu\text{m}$ to $0.55\text{ }\mu\text{m}$ (start threshold < stop threshold). For reliable activation and deactivation of the trigger output, the axis motion should start at $0.15\text{ }\mu\text{m}$ and end at $0.6\text{ }\mu\text{m}$. Send:

CTO 1 2 1 1 3 0 1 1 0.1 1 8 0.2 1 9 0.55

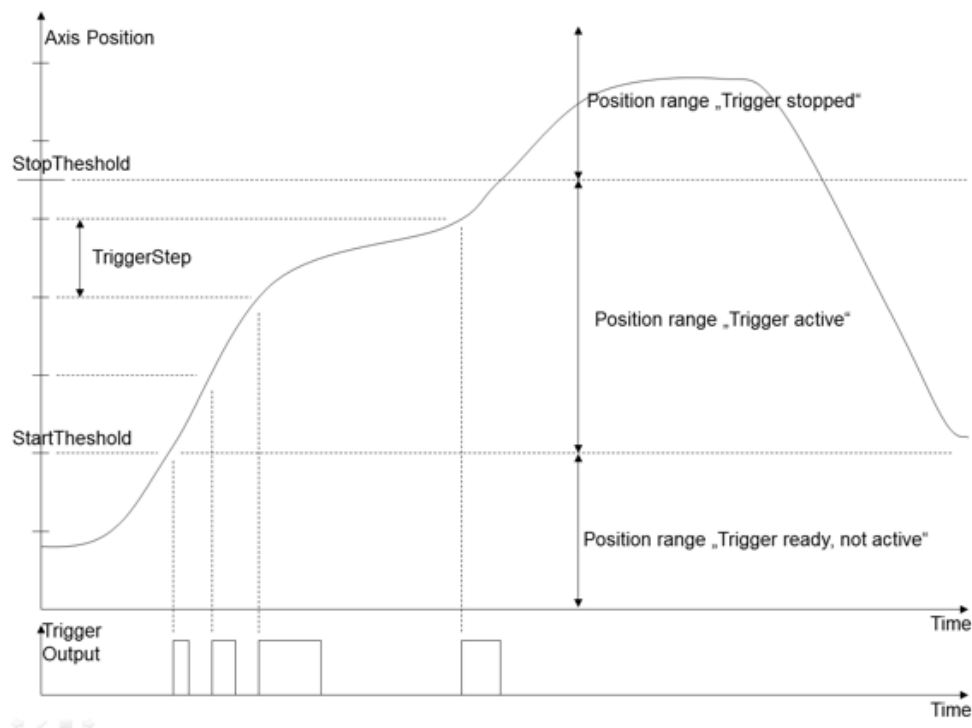


Figure 43: "Position Distance" Trigger Mode with threshold settings for positive direction of motion

Example 3: A pulse on the digital output line 1 is to be generated whenever axis 1 of the stage has covered a distance of $0.1\text{ }\mu\text{m}$, as long as axis 1 moves in negative direction in the range of $0.55\text{ }\mu\text{m}$ to $0.2\text{ }\mu\text{m}$ (start threshold > stop threshold). For reliable activation and deactivation of the trigger output, the axis motion should start at $0.6\text{ }\mu\text{m}$ and end at $0.15\text{ }\mu\text{m}$. Send:

CTO 1 2 1 1 3 0 1 1 0.1 1 8 0.55 1 9 0.2

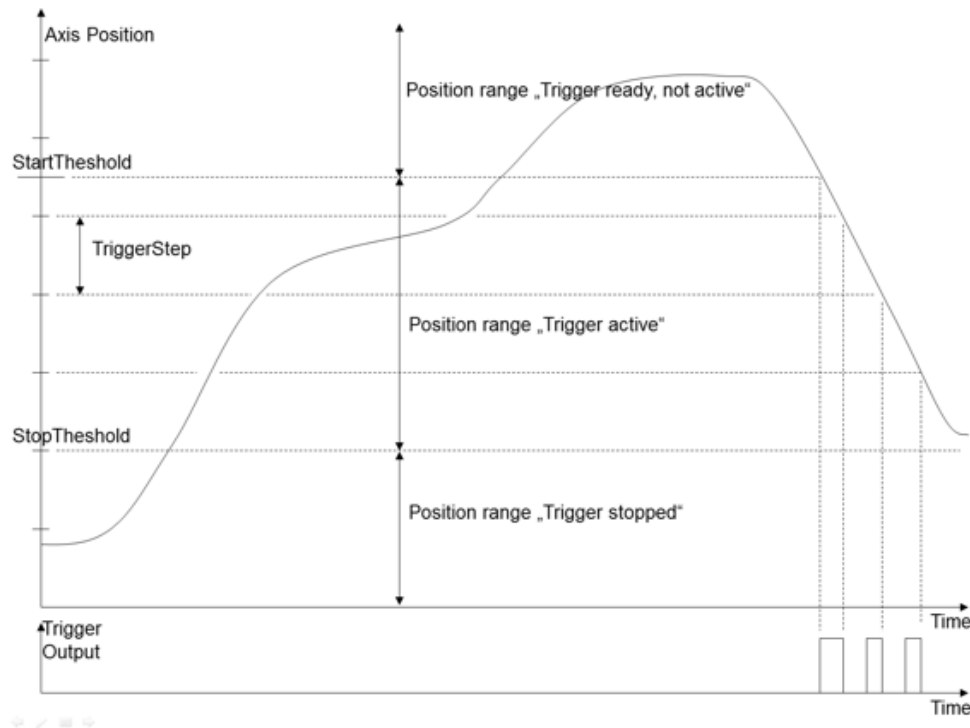


Figure 44: "Position Distance" Trigger Mode with threshold settings for negative direction of motion; here, the trigger output is not yet disabled reliably at the end of the curve)

8.2.2 Example—"On Target" Trigger Mode

With the "On Target" trigger mode, the on-target status of the selected axis is written to the selected trigger line. It is the same on-target status flag which can also be read by the ONT? command. The on-target status is influenced by two parameters: settling window (On Target Tolerance, ID 0x07000900) and settling time (Settling Time, ID 0x07000901). The on-target status is true when the current position is inside the settling window and stays there for at least the settling time. The settling window is centered around the target position.

The following parameters must be set for the digital output line which is to be used for trigger output (<TrigOutID>):

- Axis (<CTOPam> = 2)
- TriggerMode (<CTOPam> = 3)

General notation of the CTO command for this mode (in fact, the command arguments can be divided in two "portions", each starting with the <TrigOutID> declaration):

Command mnemonic	Axis selection	Trigger mode selection
CTO	<TrigOutID> 2 Axis	<TrigOutID> 3 2

Instead of using the CTO command, you can also set the values of the corresponding parameters with SPA or SEP, see "Configuring Trigger Output" (p. 142) for a parameter list.

Example: The On-Target status flag of axis 1 is to be written to the digital output line 1. Send:

CTO 1 2 1 1 3 2

8.2.3 Example—"MinMax Threshold" Trigger Mode

With the "MinMax Threshold" trigger mode, a band is specified with MinThreshold and MaxThreshold (<CTOPam> IDs 5 and 6). When the axis position is inside the specified band then the trigger output line is set high, otherwise it is set low.

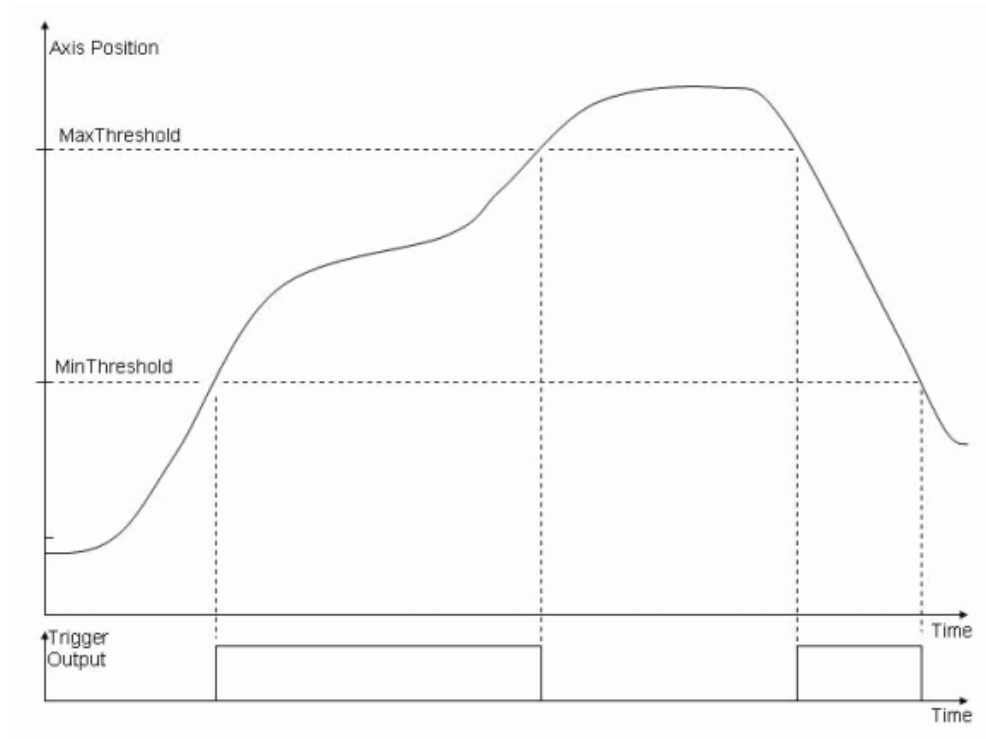


Figure 45: "MinMax Threshold" Trigger Mode

The following parameters must be set for the digital output line which is to be used for trigger output (<TrigOutID>):

- Axis (<CTOPam> = 2)
- TriggerMode (<CTOPam> = 3)
- MinThreshold (<CTOPam> = 5)
- MaxThreshold (<CTOPam> = 6)

General notation of the CTO command for this mode (in fact, the command arguments can be divided in four "portions", each starting with the <TrigOutID> declaration):

Command mnemonic	Axis selection	Trigger mode selection	Min threshold setting	Max threshold setting
CTO	<TrigOutID> 2 Axis	<TrigOutID> 3 3	<TrigOutID> 5 <i>min.pos.</i>	<TrigOutID> 6 <i>max.pos.</i>

Instead of using the CTO command, you can also set the values of the corresponding parameters with SPA or SEP, see "Configuring Trigger Output" (p. 142) for a parameter list.

Example: The digital output line 1 is to be set high whenever the axis position of axis 1 is higher than 0.3 μm and lower than 0.6 μm . Send:

CTO 1 2 1 1 3 3 1 5 0.3 1 6 0.6

8.2.4 Example—"Generator Level Trigger" Mode

With the "Generator Level Trigger" mode, the trigger output will be synchronized with the wave generator output, and CTO must be used in combination with TWS.

The length of a single trigger pulse is the same as the duration of one servo cycle. If the signal level is set to HIGH with TWS for consecutive points of a wave table, the signal level therefore does **not** change back to LOW between the points.

The following parameter must be set for the digital output line which is to be used for trigger output (<TrigOutID>):

- TriggerMode (<CTOPam> = 3)

General notation of the CTO command for this mode:

Command mnemonic	Trigger mode selection
CTO	<TrigOutID> 3 4

Instead of using the CTO command, you can also set the value of the corresponding parameter with SPA or SEP, see "Configuring Trigger Output" (p. 142) for a parameter list.

See "Trigger Output Synchronized with Wave Generator" (p. 183) for a detailed example.

8.2.5 Example—"Generator Pulse Trigger" Mode

With the "Generator Pulse Trigger" mode, the trigger output will be synchronized with the wave generator output, and CTO must be used in combination with TWS.

The length of a single trigger pulse is shorter than the servo cycle duration. If the signal level is set to HIGH with TWS for consecutive points of a wave table, the signal level therefore changes back to LOW after each point. This way, the trigger output can be used, for example, to count the waveform points that are output by the wave generator.

The following parameter must be set for the digital output line which is to be used for trigger output (<TrigOutID>):

- TriggerMode (<CTOPam> = 3)

General notation of the CTO command for this mode:

Command mnemonic	Trigger mode selection
CTO	<TrigOutID> 3 9

Instead of using the CTO command, you can also set the value of the corresponding parameter with SPA or SEP, see “Configuring Trigger Output” (p. 142) for a parameter list.

8.2.6 Example—Polarity Setting

It is possible to select the signal polarity (active high = 1, default / active low = 0) for the digital output line which is to be used for trigger output.

The following parameter must be set for the digital output line (<TrigOutID>):

- Polarity (<CTOPam> = 7)

General notation of the CTO command for polarity selection:

Command mnemonic	Trigger mode selection
CTO	<TrigOutID> 7 <i>pol.code</i>

Instead of using the CTO command, you can also set the value of the corresponding parameter with SPA or SEP, see “Configuring Trigger Output” (p. 142) for a parameter list.

Example: The signal polarity for the digital output line 1 is to be set to "active low". Send:

CTO 1 7 0

9 Using the Analog Input

9.1 How to Work with the Analog Input - Overview

The E-711.IA4 analog interface module provides four analog input lines on its front panel, labeled as "In 1" to "In 4" (e.g. present in E-712.3CDA and E-712.6CDA models). See "Analog In Sockets" (p. 283) for pinout and specifications. For highest resolution, it is recommended to use the full range of ± 10 V.

You can use an analog input line as follows:

- Connect an external sensor
- Connect a source for control value generation

Irrespective of the intended usage, the analog input values must first be scaled to suitable position values (see "Scaling the Analog Input" (p. 155)). Then, to set the usage of the analog input, it is furthermore necessary to change certain controller parameters. See "Use as Control Value Generation Source" (p. 158) or "Use as External Sensor Input" (p. 160) for details. Analog input lines which are not used should be deactivated to avoid interferences, see "Deactivation of Unused Analog Input Lines" (p. 161) for details.

The analog input lines are a subset of the E-711/E-712 systems input signal channels (Number Of Input Channels parameter, ID 0x0E000B00) which comprise also the sensor channels for the internal sensors in the mechanics (Number Of Sensor Channels parameter, ID 0x0E000B03). In the firmware of the E-711/E-712 system, the internal sensor channels are always represented by the first input signal channels, while the analog input lines are always represented by the last input signal channels. Examples:

- If the E-711/E-712 system is equipped with three internal sensor channels, they are accessible as input signal channels 1 to 3, while the analog input lines In 1 to In 4 have the channel identifiers 4 to 7.
- If the E-711/E-712 system is equipped with six internal sensor channels, they are accessible as input signal channels 1 to 6, while the analog input lines In 1 to In 4 have the identifiers 7 to 10.

See "Accessible Items and Their Identifiers" (p. 104) for more information.

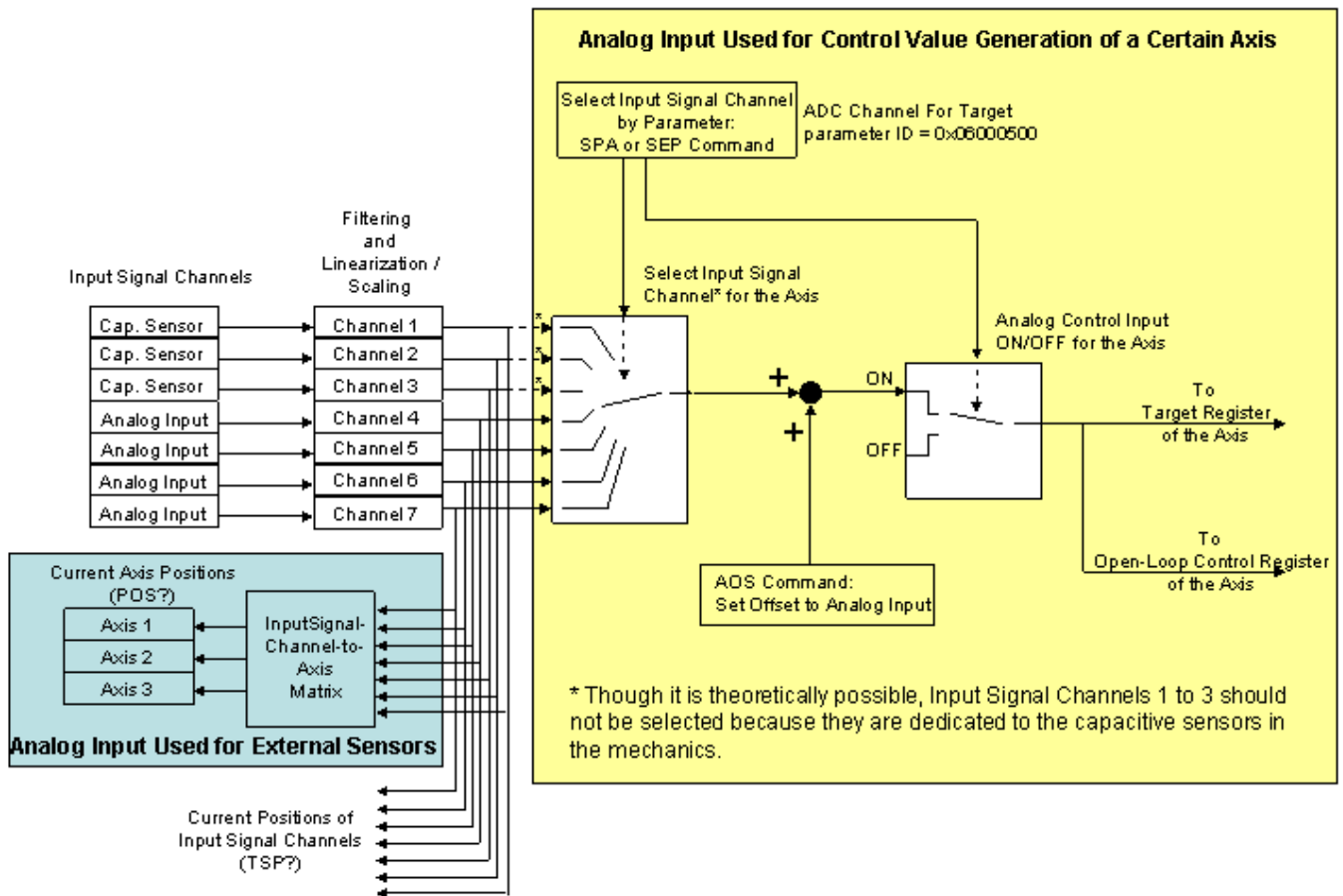


Figure 46: Overview over the usage of the analog input lines, exemplified by E-712.3CDA

NOTES

It is strongly recommended to save the parameter values of the E-711/E-712 system to a file on the host PC before you make any changes. This way the original settings can be restored if the new parameter settings will not prove satisfactory. To save the parameter values and to load them back to the E-711/E-712 system, use the *Device Parameter Configuration* window or PIMikroMove. See "Create Backup File for Controller Parameters" (p. 86) for more information.

Wherever changing parameter values is mentioned, you can do this using SPA (volatile memory) or SEP (non-volatile memory). Furthermore, you can use WPA to copy the current values from volatile memory to non-volatile memory, where they become the power-on defaults. To have write access to certain parameter(s), it might be necessary to switch to a higher command level using CCL. To read parameter values, query with the SPA? or SEP? commands.

The *Device Parameter Configuration* window of PIMikroMove gives access to parameter values in a more convenient way. Use this window to check/edit the individual parameters. See the PIMikroMove manual for more information.

9.2 Scaling the Analog Input

Before the analog input line can be used with an external sensor or with a control-signal source, the input levels must be associated with suitable position values. To do this, adjust the OFFSET (parameter ID 0x02000200) and the GAIN (parameter ID 0x02000300) of the Mechanics linearization polynomial according to the travel range of the axis and the input signal range. See below for details. The TSP? command reports the analog input values after the scaling as position values in μm .

In addition, the digital filter parameters can be adjusted. See "Digital processing" (p. 111) for details.

How to adjust OFFSET and GAIN to map the analog input voltage to a suitably scaled position value for a certain axis:

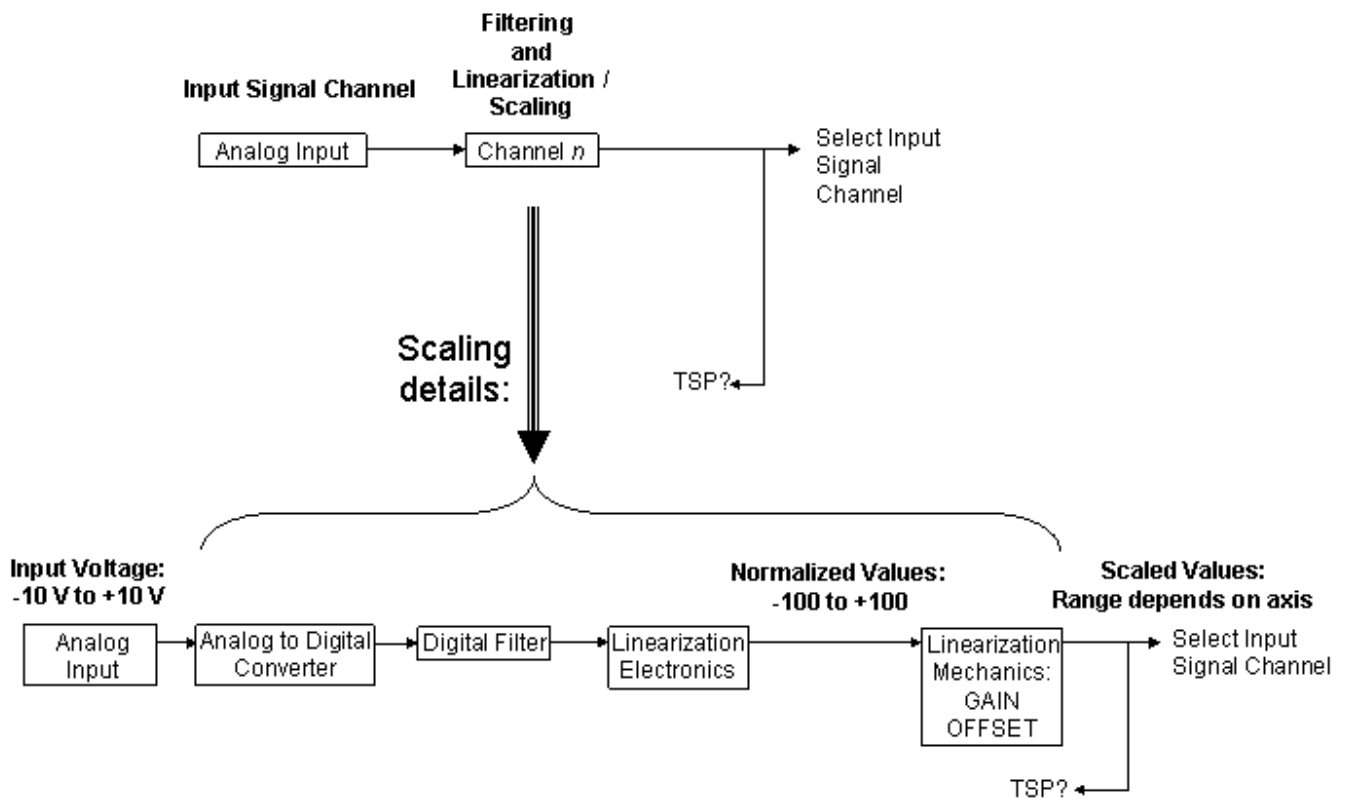


Figure 47: Processing of an analog input signal, detail from the overview figure above

Input Voltage:
the maximum range is -10 to +10 V

Normalized Value:
The polynomial used for electronics linearization (see "Digital processing" for details) converts the analog input voltage to a number in the range of -100 to +100. An input voltage value of -10 V always corresponds to -100, and +10 V corresponds to +100 respectively.

Scaled Value:
The range depends on the axis and can be set by the coefficients of the polynomial used for Mechanics linearization (see "Digital processing" for details):

$$\text{ScaledValue} = \text{OFFSET} + \text{GAIN} * \text{NormalizedValue}$$

where
OFFSET corresponds to the Sensor Mech. Correction 1 parameter, ID 0x02000200

GAIN corresponds to the Sensor Mech. Correction 2 parameter, ID 0x02000300

If no linearization is necessary, the other coefficients of the Mechanics linearization polynomial can be set to zero (parameter IDs 0x02000400, 0x02000500, 0x02000600).

Note that in PIMikroMove, these parameters are available in the *Sensor Mechanics* parameter groups in the *Device Parameter Configuration* window (E-712.3CDA: *Sensor Mechanics 4* to *Sensor Mechanics 7*; E-712.6CDA: *Sensor Mechanics 7* to *Sensor Mechanics 10*).

How to calculate the values to set for OFFSET and GAIN:

$$\text{GAIN} = (\text{MaxScaledValue} - \text{MinScaledValue}) / (\text{MaxNormalizedValue} - \text{MinNormalizedValue})$$

$$\text{OFFSET} = \text{MaxScaledValue} - \text{GAIN} * \text{MaxNormalizedValue}$$

The values of "MinScaledValue" and "MaxScaledValue" depend on the travel range of the axis with which the analog input line is to be used: "MinScaledValue" is given by the TMN? answer (is defined by the Range Limit min parameter, ID = 0x07000000), and "MaxScaledValue" is given by the TMX? answer (is defined by the Range Limit max parameter, ID = 0x07000001).

The values of "MinNormalizedValue" and "MaxNormalizedValue" depend on the range of the external signal applied to the analog input line. See the examples below. For all examples, it is assumed that you have an E-712.3CDA, and the analog input line In 1 (input signal channel 4) is to be used with axis 1 which has the following travel range:

MinScaledValue = -20 μm
MaxScaledValue = +120 μm

Example 1:

The full range of -10 V to +10 V is to be used (this is recommended for highest resolution).

MinNormalizedValue = -100
MaxNormalizedValue = +100
 $\text{GAIN} = (120 - (-20)) / (100 - (-100)) = 0.7$
 $\text{OFFSET} = 120 - 0.7 * 100 = 50$
ScaledValue = 50 + 0.7 * NormalizedValue

So you have to send

SPA 4 0x02000200 50

SPA 4 0x02000300 0.7

to adjust the GAIN and OFFSET parameters for input signal channel 4 (= analog input line In 1) in the E-712.3CDA.

Example 2:

Only positive input voltages are to be used, i.e. the range is 0 V to +10 V.

MinNormalizedValue = 0
MaxNormalizedValue = +100
 $\text{GAIN} = (120 - (-20)) / (100 - 0) = 1.4$
 $\text{OFFSET} = 120 - 1.4 * 100 = -20$

$\text{ScaledValue} = -20 + 1.4 * \text{NormalizedValue}$

Send:

SPA 4 0x02000200 -20

SPA 4 0x02000300 1.4

Example 3:

Positions with positive sign shall correspond to positive input voltages, and positions with negative sign shall correspond to negative input voltages.

The positive input voltage ranges to +10 V.

Then, the following is valid provided that the absolute value of the negative positions will never be greater than the positive positions.

$\text{MinNormalizedValue} = 0$

$\text{MaxNormalizedValue} = +100$

$\text{GAIN} = (120 - 0) / (100 - 0) = 1.2$

$\text{OFFSET} = 120 - 1.2 * 100 = 0$

$\text{ScaledValue} = 1.2 * \text{NormalizedValue}$

Send:

SPA 4 0x02000200 0

SPA 4 0x02000300 1.2

Note that these OFFSET and GAIN values would also be valid if axis 1 had a travel range of 0 to 120 μm and if there were only positive input voltages to +10 V.

Example 4:

The same conditions as in example 3 are valid, but the positive input voltages range to +5 V.

$\text{MinNormalizedValue} = 0$

$\text{MaxNormalizedValue} = +50$

$\text{GAIN} = (120 - 0) / (50 - 0) = 2.4$

$\text{OFFSET} = 120 - 2.4 * 50 = 0$

$\text{ScaledValue} = 2.4 * \text{NormalizedValue}$

Send:

SPA 4 0x02000200 0

SPA 4 0x02000300 2.4

9.3 Use as Control Value Generation Source

To enable the analog control input for an axis, an input signal channel must be connected to that axis. This is done with the ADC Channel For Target parameter (ID 0x06000500). If the connection of axis and input signal channel is saved as the power-on default, the axis can be commanded via analog input immediately after controller start-up, and no host PC is required.

Example: With an E-712.3CDA, the analog input line In 1 (= input signal channel 4) is to be used to command axis 1. Send:

SPA 1 0x06000500 4

to enable the connection in volatile memory. Note that in PIMikroMove, this parameter is available in the *Target Manipulation 1* parameter group in the *Device Parameter Configuration* window.

When the analog control input is enabled for an axis, then it overwrites the values of all other control sources for that axis except those from the AutoZero procedure. The AutoZero procedure has the highest priority, i.e. it will overwrite the control values given by all other sources. When the analog control input is enabled, it will be disabled automatically at the start of the AutoZero procedure and reenabled again when AutoZero is finished. See "Control Value Generation" (p. 114) for more information.

An offset value can be added to the analog input scaled value for an axis using the AOS command.

When no input signal channel is connected to an axis (i.e. the value of the ADC Channel For Target parameter is 0), the analog control input is disabled for that axis (including the offset set with AOS).

When the analog input is used as control source and the axis motion is stopped with STP or #24, the behaviour depends on the value of the Disconnect Analog Target Input When Stopping parameter (ID 0x0E001E00): 1 = the analog input channel is disconnected from the axis; 0 = the analog input channel remains connected to the axis. If the analog input channel is disconnected from the axis: To recommence commanding the axis via the analog input, the corresponding input signal channel must be reconnected to the axis. See the description above.

When the analog input is being used as control source and servo is switched off, the axis motion will continue in open-loop mode.

NOTES

The analog input values must be scaled to suitable position values. See "Scaling the Analog Input" (p. 155) for more information.

Make sure that the analog input line which is used to control an axis is not used as external sensor for the same axis. This means that in the input matrix, the coefficient of the appropriate analog input line must be set to zero for that axis. With an E-712.3CDA, the coefficients of the analog input lines are represented by the values of the Position From Sensor 4 to Position From Sensor 7 parameters (ID 0x07000503 to 0x07000506); E-712.6CDA: Position From Sensor 7 to Position From Sensor 10 parameters (ID 0x07000506 to 0x07000509). In PIMikroMove, these parameters are available in the *Axis Definition* parameter groups in the *Device Parameter Configuration* window.

9.4 Use as External Sensor Input

To let the sensor on the analog input line participate in the position signal of an axis, set the corresponding coefficient in the input matrix to 1 for that axis. Example: With an E-712.3CDA, the analog input line In 1 (input signal channel 4) is to be used to measure the position of axis 1, i.e. the Position From Sensor 4 parameter (ID 0x07000503) must be set to 1 for axis 1. Send: SPA 1 0x07000503 1

to change the coefficient in volatile memory. In PIMikroMove, this parameter is available in the *Axis Definition 1* parameter group in the *Device Parameter Configuration* window.

If only the external sensor on the analog input line is to be used for position control of an axis, the signals of all other sensors must be excluded from the position monitoring of that axis, especially the signals of the internal sensors integrated in the mechanics. To do this, set the corresponding coefficients in the input matrix to zero for that axis (with the E-712.3CDA, the internal sensors are represented by the Position from Sensor 1 to Position from Sensor 3 parameters; E-712.6CDA: Position from Sensor 1 to Position from Sensor 6). Example: To deactivate the first internal sensor for axis 1, send: SPA 1 0x07000500 0

The position of the axis (i.e. the POS? response) will then be based on the external sensor only, but it is still possible to read the signals of all sensors using the TSP? command.

NOTES

The analog input values must be scaled to suitable position values. See "Scaling the Analog Input" (p. 155) for more information.

Make sure that internally, the analog input line used to monitor the position of an axis is not connected to the same axis for control value generation. This means that the value of the ADC Channel For Target parameter (ID 0x06000500) for an axis must be different from the identifier of the analog input line which is used as external sensor for that axis. In PIMikroMove, you can check this in the *Target Manipulation* parameter groups in the *Device Parameter Configuration* window.

9.5 Deactivation of Unused Analog Input Lines

Analog input lines which are not used should be deactivated to avoid interferences. To deactivate an unused analog input line, the following settings must be done:

- 1 Exclude the analog input line from the calculation of axis positions by setting its coefficients for all axes to zero in the input matrix (Position from Sensor *n* parameters).
- 2 Make sure that the analog input line is not connected to an axis for control value generation. This means that the value of the ADC Channel For Target parameter, ID 0x06000500, for all axes must be different from the ID of the analog input line.

Example: The analog input line In 4 of an E-712.3CDA model is to be deactivated temporarily (i.e. in volatile memory). Because the E-712.3CDA has three internal sensor channels, the analog input line In 4 is the 7th input signal channel and its input matrix coefficients corresponds to the Position from Sensor 7 parameters (ID 0x07000506) for axis 1 to axis 3.

Send:

```
SPA 1 0x07000506 0 2 0x07000506 0 3 0x07000506 0
```

to exclude the analog input line from the axis position calculation. Then send:

```
SPA? 1 0x06000500 2 0x06000500 3 0x06000500
```

to check the RAM settings for the axis control value generation. The response must be different from 7 for all axes, i.e. if the E-711/E-712 system replies

```
1 0x06000500=7
2 0x06000500=3
3 0x06000500=0
```

the analog input line In 4 is still connected to axis 1, and you have to send

```
SPA 1 0x06000500 0
```

to disconnect it.

9.6 Analog-Input-Related Commands and Parameters

Command	Description	Notes
AOS	Set Analog Input Offset	Adds an offset value to the analog input scaled value (Analog Target Offset, ID 0x06000501). This offset is active as long as the analog input is enabled as control source for this axis.
AOS?	Get Analog Input Offset	Reads the current value of Analog Target Offset, parameter ID 0x06000501, from volatile memory
SEP	Set Nonvolatile Memory Parameters	Can be used to set the power-on default configuration for analog input usage.
SEP?	Get Nonvolatile Memory Parameters	Reads the current parameter values from non-volatile memory
SPA	Set Temporary Memory Parameters	Can be used to set a temporary configuration for analog input usage.
SPA?	Get Temporary Memory Parameters	Reads the current parameter values from volatile memory (RAM)
TAD?	Get ADC Value Of Input Signal	Reports the current ADC value of the analog input, dimensionless
TNS?	Get Normalized Input Signal Value	Reports the resulting value for the analog input after the electronics linearization, dimensionless
TSP?	Get Input Signal Position Value	Reports the resulting value for the analog input after the mechanics linearization (scaling), the unit is μm
WPA	Save Parameters To Nonvolatile Memory	Can be used to save the currently active configuration (including analog input usage) to non-volatile memory, where it becomes the power-on default.

See "How to work with the Analog Input - Overview" (p. 153) for more information. For detailed command descriptions see the GCS commands

manual PZ233E. For the identifiers of the items which can be addressed with the commands see "Accessible Items and Their Identifiers" (p. 104).

Parameter ID	Command Level	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description
0x02000200	1	Input Signal Channel	E-712.3CDA: 7 E-712.6CDA: 10	FLOAT	Sensor Mech. Correction 1 (Offset)
0x02000300	1	Input Signal Channel	E-712.3CDA: 7 E-712.6CDA: 10	FLOAT	Sensor Mech. Correction 2 (Gain)
0x05000000	1	Input Signal Channel	E-712.3CDA: 7 E-712.6CDA: 10	INT	Digital Filter Type
0x05000001	1	Input Signal Channel	E-712.3CDA: 7 E-712.6CDA: 10	FLOAT	Digital Filter Bandwidth
0x05000002	1	Input Signal Channel	E-712.3CDA: 7 E-712.6CDA: 10	INT	Digital Filter Order
0x06000500	1	Logical Axis	E-712.3CDA: 3 E-712.6CDA: 6	INT	ADC Channel for Target; if 0, then the analog control input is disabled for the axis
0x06000501	1	Logical Axis	E-712.3CDA: 3 E-712.6CDA: 6	FLOAT	Analog Target Offset
0x07000500	1	Logical Axis	E-712.3CDA: 3 E-712.6CDA: 6	FLOAT	Position from Sensor 1
0x07000501	1	Logical Axis	E-712.3CDA: 3 E-712.6CDA: 6	FLOAT	Position from Sensor 2
0x07000502	1	Logical Axis	E-712.3CDA: 3 E-712.6CDA: 6	FLOAT	Position from Sensor 3
0x07000503	1	Logical Axis	E-712.3CDA: 3 E-712.6CDA: 6	FLOAT	Position from Sensor 4
0x07000504	1	Logical Axis	E-712.3CDA: 3 E-712.6CDA: 6	FLOAT	Position from Sensor 5
0x07000505	1	Logical Axis	E-712.3CDA: 3 E-712.6CDA: 6	FLOAT	Position from Sensor 6
0x07000506	1	Logical Axis	E-712.3CDA: 3 E-712.6CDA: 6	FLOAT	Position from Sensor 7
0x07000507	1	Logical Axis	E-712.3CDA: 3 E-712.6CDA: 6	FLOAT	Position from Sensor 8
0x07000508	1	Logical Axis	E-712.3CDA: 3 E-712.6CDA: 6	FLOAT	Position from Sensor 9
0x07000509	1	Logical Axis	E-712.3CDA: 3 E-712.6CDA: 6	FLOAT	Position from Sensor 10

Parameter ID	Command Level	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description
0x0e000b00	3	System	1	INT	Number of input channels
0x0e000b03	3	System	1	INT	Number of sensor channels

See "Controller Parameters" (p. 231) for more information regarding the controller parameters and their handling.

10 Using the Analog Output

10.1 How to Work with the Analog Output - Overview

The E-711.IA4 analog interface module provides four analog output lines on its front panel, labeled as "Out 1" to "Out 4" (e.g. present with E-712.3CDA and E-712.6CDA models). The output range is ± 12.775 V, see "Analog Out Sockets" (p. 283) for pinout and further specifications.

You can use an analog output line as follows:

- Control an external amplifier
- Monitor an axis position

The usage of the analog output is set via the parameters Select Output Type (ID 0x0a000003) and Select Output Index (ID 0x0a000004). See "Use Analog Output to Control External Amplifier" (p. 167) or "Use Analog Output to Monitor Axis Position" (p. 168) for details. If the analog output is set up to control an external amplifier, the VOL? command reports the output voltage.

The analog output lines are a subset of the E-712.3CDAs and .6CDAs output signal channels (Number Of Output Channels parameter, ID 0x0E000B01) which comprise also the piezo amplifier channels for the piezo actuators in the mechanics (Number Of Piezo Channels parameter, ID 0x0E000B04). In the firmware of the E-711/E-712 system, the piezo amplifier channels are always represented by the first output signal channels, while the analog output lines are always represented by the last output signal channels. Examples for counting of the output signal channels:

- The E-712.3CDA is equipped with four piezo amplifier channels, they are accessible as output signal channels 1 to 4, while the analog output lines Out 1 to Out 4 have the channel identifiers 5 to 8.
- The E-712.6CDA is equipped with eight piezo amplifier channels, they are accessible as output signal channels 1 to 8, while the analog output lines Out 1 to Out 4 have the identifiers 9 to 12.

See "Accessible Items and Their Identifiers" (p. 104) for more information.

The parameters Select Output Type and Select Output Index can only be changed for analog output lines. For piezo amplifier channels, they are read-only: the output type is fixed to "control voltage of output signal channel", and the output index is fixed to a certain output signal channel.

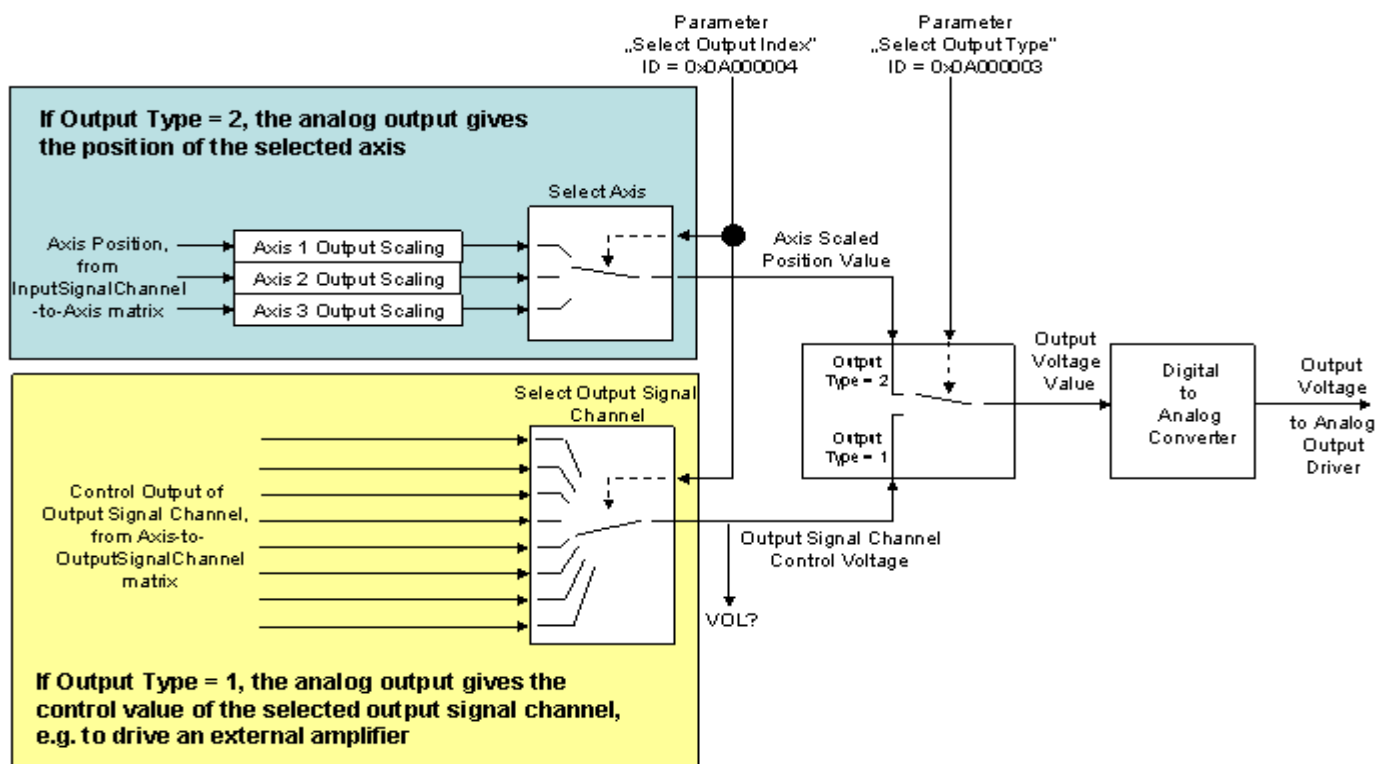


Figure 48: Overview over the usage of an analog output line, exemplified by E-712.3CDA

NOTES

It is strongly recommended to save the parameter values of the E-711/E-712 system to a file on the host PC before you make any changes. This way the original settings can be restored if the new parameter settings will not prove satisfactory. To save the parameter values and to load them back to the E-711/E-712 system, use the *Device Parameter Configuration* window of PIMikroMove. See "Create Backup File for Controller Parameters" (p. 86) for more information.

Wherever changing parameter values is mentioned, you can do this using SPA (volatile memory) or SEP (non-volatile memory). Furthermore, you can use WPA to copy the current values from volatile memory to non-volatile memory, where they become the power-on defaults. To have write access to certain parameter(s), it might be necessary to switch to a higher command level using CCL. To read parameter values, query with the SPA? or SEP? commands.

The *Device Parameter Configuration* window of PIMikroMove gives access to parameter values in a more convenient way. Use this window to check/edit the individual parameters. See the PIMikroMove manual for more information.

10.2 Use Analog Output to Control External Amplifier

Proceed as follows if you want to control an external amplifier via an analog output line:

- 1 Select output type 1 = "control voltage of output signal channel" for the analog output line using the Select Output Type parameter, ID 0x0A000003.

Example: With an E-712.6CDA, the analog output line Out 2 (output signal channel 10) is to be used to control an external amplifier. To select the corresponding output type in volatile memory, send:
SPA 10 0x0A000003 1

Note that in PIMikroMove, this parameter is available in the *DAC 10* parameter group in the *Device Parameter Configuration* window.

- 2 Connect the output signal channel whose control value is to be output to the analog output line using the Select Output Index parameter, ID 0x0A000004.

Note:

The control value of an output signal channel results from the output matrix, see "Output Generation" (p. 118) for more information.

In the example, the control value of output signal channel 1 (which is the first internal piezo amplifier of the E-712.6CDA) is to be connected to the Out 2 line (output signal channel 10). Send SPA 10 0x0A000004 1

In PIMikroMove, this parameter is also available in the *DAC 10* parameter group in the *Device Parameter Configuration* window. Note that for piezo amplifier channels, the output index is fixed to a certain output signal channel.

- 3 Check the output matrix (Driving Factor of Piezo *n* parameters, ID 0x09000000 to 0x09000000*n-1*; in PIMikroMove available in the *Axis Definition* parameter groups in the *Device Parameter Configuration* window) for feasible settings.

10.3 Use Analog Output to Monitor Axis Position

Proceed as follows if you want to output axis position values on an analog output line:

- 1 Select output type 2 = "current position of axis" for the analog output line using the Select Output Type parameter, ID 0x0A000003.

Example: With an E-712.6CDA, the analog output line Out 1 (output signal channel 9) is to be used to monitor the position of axis 3. To select the corresponding output type in volatile memory, send:
SPA 9 0x0A000003 2

In PIMikroMove, this parameter is available in the *DAC 9* parameter group in the *Device Parameter Configuration* window. Note that for piezo amplifier channels, the output type is fixed to "control voltage of output signal channel" (1).

- 2 Connect the desired axis to the analog output line using the Select Output Index parameter, ID 0x0A000004.

Note:

An axis position results from the input matrix, see "Input Signal Processing" (p. 109) for more information.

In the example, send
SPA 9 0x0A000004 3
to connect the axis 3 position to the analog output line Out 1 (output signal channel 9).

In PIMikroMove, this parameter is also available in the *DAC 9* parameter group in the *Device Parameter Configuration* window.

- 3 Scale the output value, i.e. associate the axis position values with suitable output levels (= scaled position values). To do this, set the Position Report Scaling parameter, ID 0x07001005, and the Position Report Offset parameter, ID 0x07001006 to suitable values for the appropriate axis. See also the figure below.

$$\text{ScaledPositionValue} = \text{PositionReportScaling} * (\text{PositionReportOffset} + \text{PositionValue})$$

Example:

The position range of axis 3 with which the analog output line Out 1 is to be used is given by the TMN? answer (is defined by the Range Limit min parameter, ID = 0x07000000) and by the TMX? answer (is defined by the Range Limit max parameter, ID = 0x07000001), it is -20 µm to +120 µm in the example. Furthermore, the output range to be used is -10 V to +10 V. The resulting parameter values for the axis position scaling are as follows:

Position Report Scaling = 0.143

Position Report Offset = -50

i.e. you have to send:

SPA 3 0x07001005 0.143

SPA 3 0x07001006 -50

In PIMikroMove, these parameters for axis 3 are available in the *Servo 3* parameter group in the *Device Parameter Configuration* window.

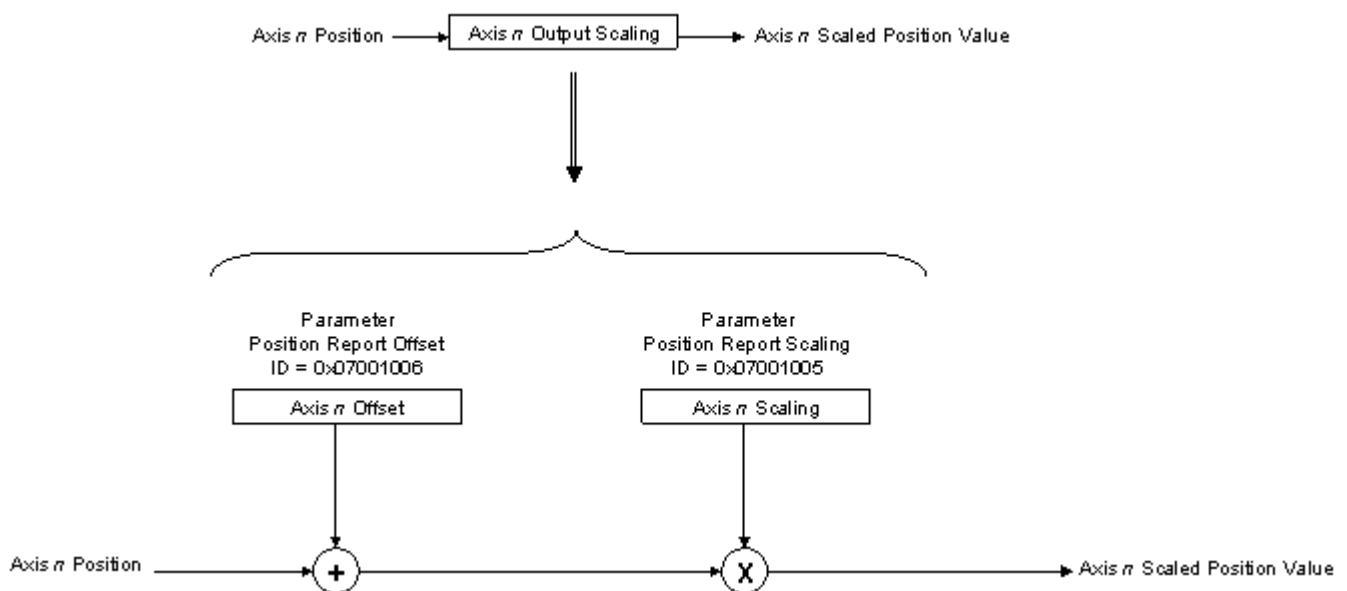


Figure 49: Processing of axis position output, detail from the overview figure

10.4 Analog-Output-Related Commands and Parameters

Command	Description	Notes
SEP	Set Nonvolatile Memory Parameters	Can be used to set the power-on default configuration for analog output usage.
SEP?	Get Nonvolatile Memory Parameters	Reads the current parameter values from non-volatile memory
SPA	Set Temporary Memory Parameters	Can be used to set a temporary configuration for analog output usage.
SPA?	Get Temporary Memory Parameters	Reads the current parameter values from volatile memory (RAM)
VOL?	Get Voltage Of Output Signal Channel	Reads output voltage value of the given output signal channel, if the output type is set to "control voltage of output signal channel"
WPA	Save Parameters To Nonvolatile Memory	Can be used to save the currently active configuration (including analog output usage) to non-volatile memory, where it becomes the power-on default.

See "How to work with the Analog Output - Overview" (p. 165) for more information. For detailed command descriptions see the GCS commands manual PZ233E. For the identifiers of the items which can be addressed with the commands see "Accessible Items and Their Identifiers" (p. 104).

Parameter ID	Command Level	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description
0x07001005	1	Logical Axis	E-712.3CDA: 3, E-712.6CDA: 6	FLOAT	Position Report Scaling, required if the axis position is to be output (output type = 2)
0x07001006	1	Logical Axis	E-712.3CDA: 3, E-712.6CDA: 6	FLOAT	Position Report Offset, required if the axis position is to be output (output type = 2)

Parameter ID	Command Level	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description
0x09000000 to 0x0900000d	1	Logical Axis	E-712.3CDA: 3, E-712.6CDA: 6	FLOAT	Driving Factor of Piezo 1 to Driving Factor of Piezo 14, give the output matrix
0x0A000003	1	Output Signal Channel	E-712.3CDA: 8 (ch. 1 to 4 read-only), E-712.6CDA: 12 (ch. 1 to 8 read-only)	INT	Select Output Type; 1 = control voltage of output signal channel 2 = current position of axis
0x0A000004	1	Output Signal Channel	E-712.3CDA: 8 (ch. 1 to 4 read-only), E-712.6CDA: 12 (ch. 1 to 8 read-only)	INT	Select Output Index; the selected object can be an output signal channel or an axis (depends on the selected output type)
0x0e000b01	3	System	1	INT	Number of output channels
0x0e000b04	3	System	1	INT	Number of piezo channels

See "Controller Parameters" (p. 231) for more information regarding the controller parameters and their handling.

11 Wave Generator

Each axis can be controlled by a "wave generator" which outputs user-specified patterns, so-called "waveforms". This feature is especially important in dynamic applications which require periodic, synchronous motion of the axes. The waveforms to be output are stored in "wave tables" in the controllers volatile memory—one waveform per wave table. Waveforms can be created based on predefined "curve" shapes. This can be sine, ramp or single scan line curves. Additionally you can freely define curve shapes. Programmable trigger inputs and outputs facilitate synchronization of external events.

In "How to Work with the Wave Generator" (p. 172) and "Wave Generator Examples" (p. 178) you will learn how to use the wave generator, and "Wave-Generator-Related Commands and Parameters" gives an overview.

The Dynamic Digital Linearization (DDL) option can be used in conjunction with the wave generator output in addition to the "normal" servo algorithm. See "Dynamic Digital Linearization (DDL)" (p. 189) for more information.

During the wave generator output, data is recorded in "record tables" on the controller. See "Data Recording" p. 138) for more information.

11.1 How to Work with the Wave Generator

The following subsections describe the wave generator handling in detail. See also "Wave Generator Examples" (p. 178).

11.1.1 Basic Data

The number of wave tables can be queried using the SPA? command, parameter ID 0x1300010A. The E-711/E-712 system has 90 wave tables for creating and (temporarily) storing arbitrary waveforms (identifiers are 1 to 90).

To ask for the number of wave generators, use the TWG? command. The assignment of wave generators and axes to each other is fixed: wave generator 1 is connected to axis 1, wave generator 2 to axis 2, ..., wave generator n to axis n .

The available wave tables can be flexibly assigned to the wave generators and hence to the axes using the WSL command. A wave table can be used by multiple wave generators at the same time.

A certain amount of the controllers memory space is reserved for the waveform data (ask with the SPA? command, parameter ID 0x13000004). The E-711/E-712 system provides 2^{23} data points for waveform definition. This memory space is (temporarily) allocated to the individual wave tables during the waveform definition.

11.1.2 Basic Operation

- 1 Define the waveform segment-by-segment using the WAV command. The waveform will be written to the selected wave table.
- 2 Connect the wave generator to the wave table using the WSL command.
- 3 Start the wave generator output and hence the motion of the axis using the WGO command. You can choose several start options (e.g. start/stop by external trigger, initialization/usage of the Dynamic Digital Linearization (DDL) feature; see the description of the WGO command and "Dynamic Digital Linearization (DDL)" (p. 189) for more information).
When starting the wave generator, data recording is started automatically.
- 4 Stop the wave generator output with WGO or #24 or STP.

A simple example for your first steps (using the command entry facilities of PIMikroMove or PITerminal):

Command String to Send	Action Performed
WAV 4 X SIN_P 2000 20 10 2000 0 1000	Define a sine waveform for Wave Table 4; see WAV description for details
WSL 1 4	Connect the Wave Generator 1 (axis 1) to Wave Table 4
WGO 1 1	Start output of Wave Generator 1 immediately (synchronized by servo cycles)
WGO 1 0	Stop output of Wave Generator 1

11.1.3 Additional Steps and Settings

You can calculate the memory space remaining if you ask with WAV? for the current wave table length. To release memory space, delete the content of selected wave tables with the WCL command.

After you send the waveform definition to the wave table (with WAV), it is always a good idea to check it by reading back the waveform sequence from the controller before actually outputting it. This can be done by the GWD? command. Note that the response to GWD? does not contain any offset set with WOS to the wave generator output.

You can add an offset to the output of a wave generator using the WOS command. Thereafter, the output of the specified wave generator is the sum of the offset value and the wave value:

Generator Output = Offset + Current Wave Value

If the wave generator is started with the option "start at the endpoint of the last cycle", the E-711/E-712 system at the end of each output cycle equates the WOS offset value with the current generator output.

WOS sets the value of the Wave Offset parameter, ID 0x1300010b, in volatile memory. You can also change this parameter with SPA or SEP and save the value to non-volatile memory with WPA (switch to command level 1 before with the CCL command).

Deleting wave table content with WCL has no effect on the WOS settings.

For triggering purposes, the wave generator output can be coupled with the digital output lines OUT1 to OUT7 of the controller (see "Digital I/O Socket" (p. 285)). You should first use TWC to set the signal state of the output lines to "low" for all waveform points ("low" is also the power-on default). Then use the TWS command to define the trigger line actions by setting the desired signal states (high or low) of selected output lines for selected waveform points. At last, use the CTO command to activate the Generator Trigger mode for the selected output lines.

The #9 single-character command can be used to query the current activation state of the wave generators. The reply shows if a wave generator is running or not, but does not contain any information about the wave generator start mode (e.g. with DDL). With WGO? you can ask for the last-commanded wave generator start options (WGO settings).

You can limit the duration of the wave generator output by setting the number of output cycles with WGC. The waveform itself remains unchanged.

Using the WTR command, you can lengthen the individual output cycles of the waveform. The duration of one output cycle for the waveform can be calculated as follows:

Output Duration = Servo Update Time * WTR value * Number of Points

where

Servo Update Time is given in seconds by parameter 0x0E000200

WTR value gives the number of servo cycles the output of a waveform point lasts, default is 1

Number of Points is the length of the waveform (i.e. the length of the wave table)

WTR sets the value of the Wave Generator Table Rate parameter, ID 0x13000109, in volatile memory. You can change this parameter also with

SPA or SEP and save the value to non-volatile memory with WPA (switch to command level 1 before with the CCL command). The value is always valid for the whole system and cannot be set separately for individual wave generators. The value of the parameter in volatile memory can be read with the WTR? command.

WTR also sets the type of interpolation to use for the wave generator output. If the Wave Generator Table Rate is larger than 1, interpolation helps to avoid sudden position jumps of an axis controlled by the wave generator.

With WGR you can restart data recording while the wave generator is running. The recorded data can be read with the DRR? command. See "Data Recording" (p. 138) for more information.

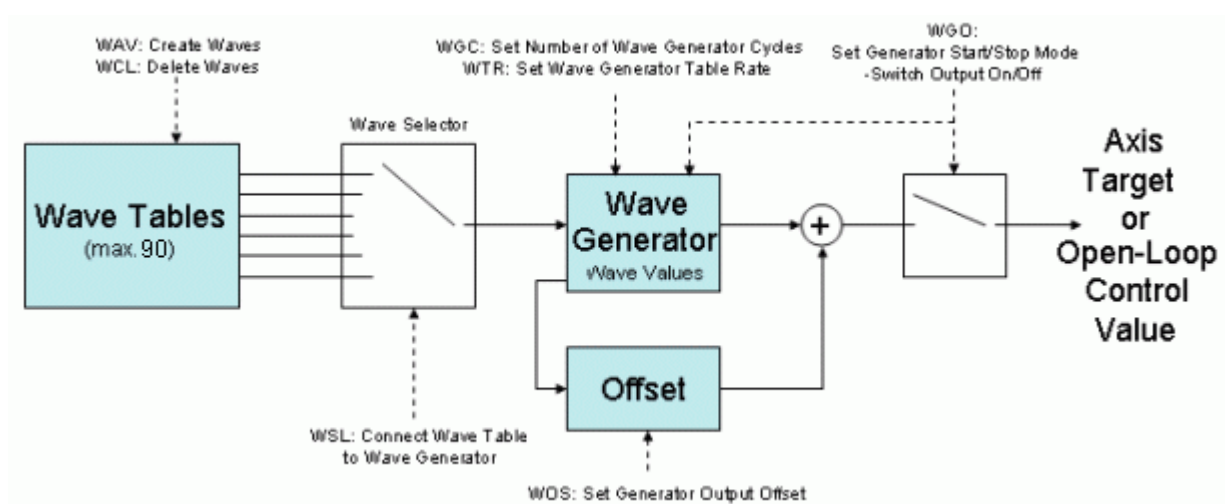


Figure 50: Block diagram of one wave generator

11.1.4 Application Notes

All wave generators can run simultaneously. All waveform output is synchronized because there is a common pulse generator used by all wave generators. For that reasons, wave tables which are supposed to run at the same time (each with one wave generator) should have the same length. If the wave tables have different lengths, an output cycle will comprise only the number of points given by the shortest table. This means that all waveform output is cut to the length of the shortest waveform currently running.

Waveforms cannot be changed while they are being output by a wave generator. If you want to modify a waveform with WAV, first stop any wave generator output from the associated wave table.

The frequency of the wave generator output depends, among other factors, on the wave table length. When you create waveforms, keep in mind that the usable frequency is limited by the available amplifier power. Example: with a

capacitive load of 6.6 μF , the frequency should not exceed 15 Hz if three amplifiers are involved in the motion or 50 Hz if only one amplifier is involved. If the frequency is too high, overheating of the amplifier(s) can occur, and the piezo voltage output will be deactivated automatically. If that occurs, the piezo stage will not move although communication with the controller is still possible.

When you use the wave generator to create white noise for a measurement (WAV command with <WaveType> = "NOISE"), make sure that the wave table segment is at least as long as the duration of the measurement. Otherwise, the noise sequence will be repeated during the measurement and will no longer be white noise.

Wave generator output and analog control input:

It is possible to configure an axis for control by an analog input line while the wave generator output is active for that axis. In that case, the wave generator will continue running, but its output will no longer be used for control value generation. As long as the corresponding axis is set up to be commanded by analog control input, you can stop the wave generator output, but not restart it.

Wave generator output and move commands:

When the wave generator output is active, move commands like MOV or SVA are not allowed for the associated axis.

See "Control Value Generation" (p. 114) for details.

When the wave generator is to be started by an external trigger signal (WGO bit 1 is set): For reliable triggering, the pulse width of the input signal has to be at least 2 x the servo update time of the E-711/E-712 system (the servo update time is given in seconds by parameter 0x0E000200). The value of the Wave Multi Start By Trigger parameter (ID 0x13000202) determines if the trigger is enabled for only one generator start or for multiple starts. See "Wave Generator Started by Trigger Input" (p. 184) for details and an example.

A wave generator outputs absolute values. In closed-loop operation (servo ON), the output is interpreted as target positions in either case. In open-loop operation (servo OFF), the interpretation of the wave generator output depends on the settings of the output matrix (see "Output Generation" (p. 118) for more information). By default, the matrix is set up so that commanded open-loop control values numerically correspond to axis position values.

Servo cannot be switched off (SVO) while a wave generator is running for the axis.

As long as a wave generator is running, it is not possible to change (WSL) or to delete (WCL) the connected wave table (i.e. the waveform). The wave generator table rate (WTR), the number of output cycles (WGC), the wave offset (WOS) and the output trigger settings (TWS) can be modified while a wave generator is running.

When a wave generator finishes by running through a specified number of cycles completely, the final position will be the first point of the waveform, unless the option "start at the endpoint of the last cycle" was selected. In that case, the final position is the sum of the endpoint of the last output cycle and any offset defined with WAV for the waveform.

When the wave generator is stopped within an output cycle by command, the axis will remain at the last output position until a new position is commanded. If the wave generator is then restarted, it will normally continue with the first point of the waveform, unless started with the option "start wave generator output triggered by external signal", and the digital input line IN2 (see "Digital I/O Socket" (p. 285)) is used for triggering. In that case, the generator starts with the first rising edge which is detected on this input line, and it will be stopped when a falling edge is detected on this line. With the next rising edge, the generator output will continue at the waveform point where it was stopped.

Wave generator output will continue even if the terminal or the program from which it was started is quit or if the high voltage output is deactivated. See the WGO command for more information.

The following data is always lost when the controller is powered down or rebooted:

- Wave table content (WAV)
- Assignment of wave tables to wave generators (WSL)
- Output trigger settings (TWS)
- Number of cycles for wave generator output (WGC)

The following settings can be saved with WPA to non-volatile memory, where they become the power-on defaults (switch to command level 1 before with the CCL command):

- Wave offset (WOS)
- Wave generator table rate (WTR)

The different software interfaces provided for the controller also support use of the wave generator. Waveforms can be defined, stored and displayed in and by the software in a more user-friendly way than in a terminal using WAV and WGO. If using the wave generator with the GCS DLL, PIMikroMove or LabView, read the descriptions in the associated software manual first.

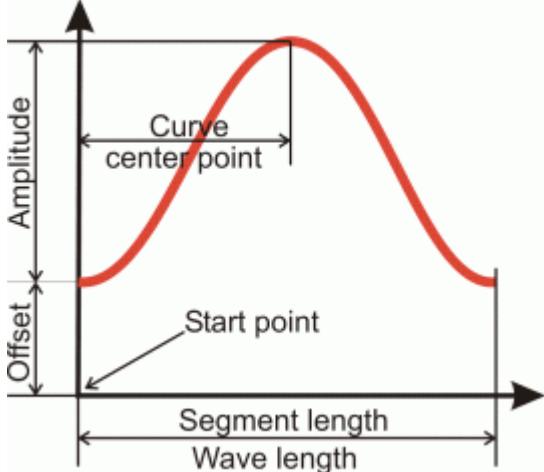
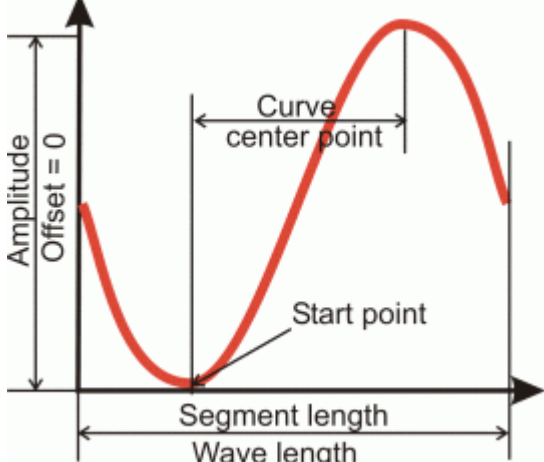
11.2 Wave Generator Examples

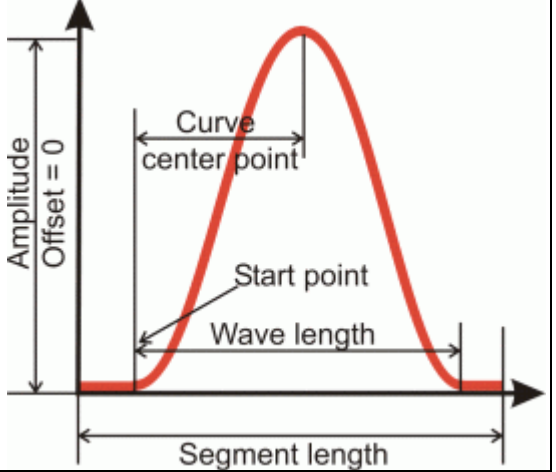
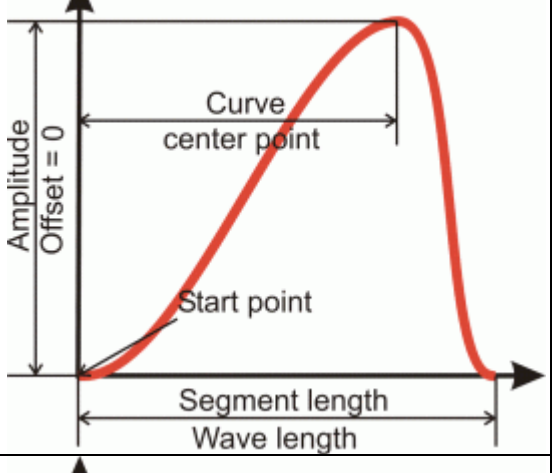
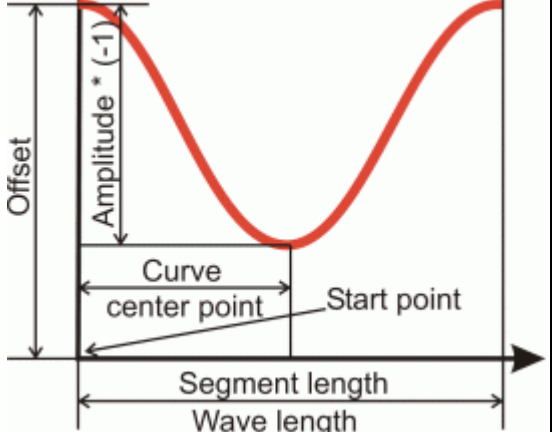
The following examples can be reproduced using the command entry facilities of PIMikroMove or PI Terminal. Note that it might be necessary to adapt them to your hardware configuration.

11.2.1 Defining Waveforms

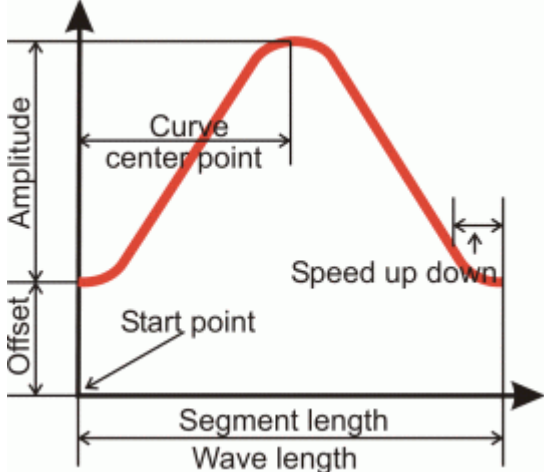
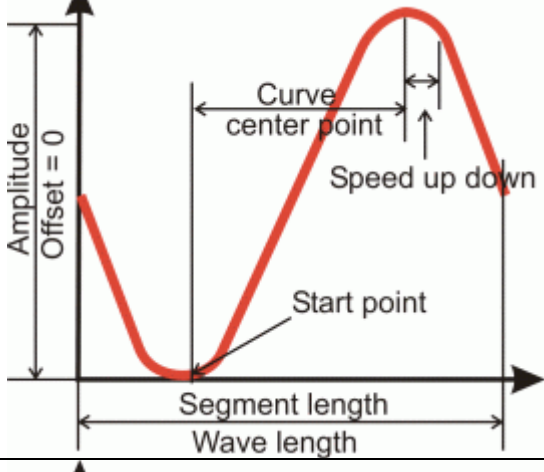
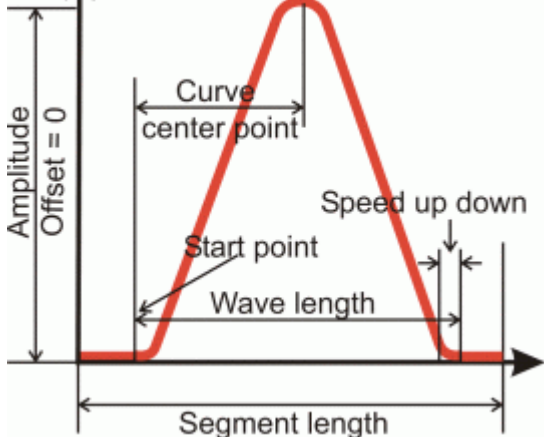
Examples for how to define waveform segments for the wave tables, based on predefined curve shapes (each WAV command defines a waveform segment which either replaces or is appended to the waveform in the specified wave table):

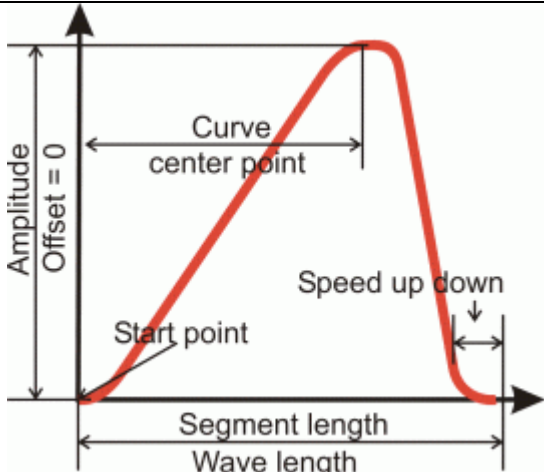
Sine Curves

WAV command	Comments	Waveform Segment
WAV 2 X SIN_P 2000 20 10 2000 0 1000 <WaveTableID> = 2 <AppendWave> = X <WaveType> = SIN_P <SegLength> = 2000 <Amp> = 20 <Offset> = 10 <WaveLength> = 2000 <StartPoint> = 0 <CurveCenterPoint> = 1000	The previous contents of the wave table are overwritten by the new segment, waveform offset = 10 (Do not confuse with the wave generator output offset set with WOS!), symmetric curve	
WAV 2 X SIN_P 2000 30 0 2000 499 1000 <WaveTableID> = 2 <AppendWave> = X <WaveType> = SIN_P <SegLength> = 2000 <Amp> = 30 <Offset> = 0 <WaveLength> = 2000 <StartPoint> = 499 <CurveCenterPoint> = 1000	The previous contents of the wave table are overwritten by the new segment, symmetric curve	

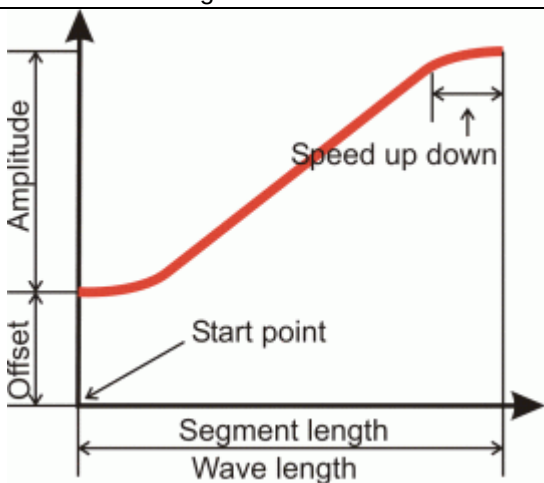
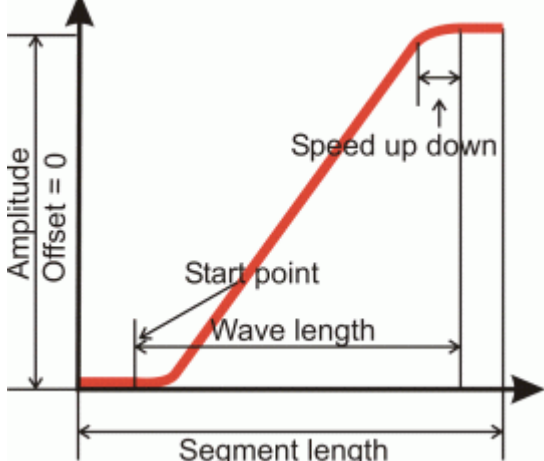
WAV command	Comments	Waveform Segment
WAV 2 & SIN_P 2000 25 0 1800 100 900 <WaveTableID> = 2 <AppendWave> = & <WaveType> = SIN_P <SegLength> = 2000 <Amp> = 25 <Offset> = 0 <WaveLength> = 1800 <StartPoint> = 100 <CurveCenterPoint> = 900	The defined segment will be appended to the existing wave table contents, symmetric curve	
WAV 3 X SIN_P 4000 20 0 4000 0 3100 <WaveTableID> = 3 <AppendWave> = X <WaveType> = SIN_P <SegLength> = 4000 <Amp> = 20 <Offset> = 0 <WaveLength> = 4000 <StartPoint> = 0 <CurveCenterPoint> = 3100	The previous contents of the wave table are overwritten by the new segment, asymmetric curve	
WAV 2 X SIN_P 1000 -30 45 1000 0 500 <WaveTableID> = 2 <AppendWave> = X <WaveType> = SIN_P <SegLength> = 1000 <Amp> = -30 <Offset> = 45 <WaveLength> = 1000 <StartPoint> = 0 <CurveCenterPoint> = 500	The previous contents of the wave table are overwritten by the new segment, negative-amplitude curve, symmetric curve	

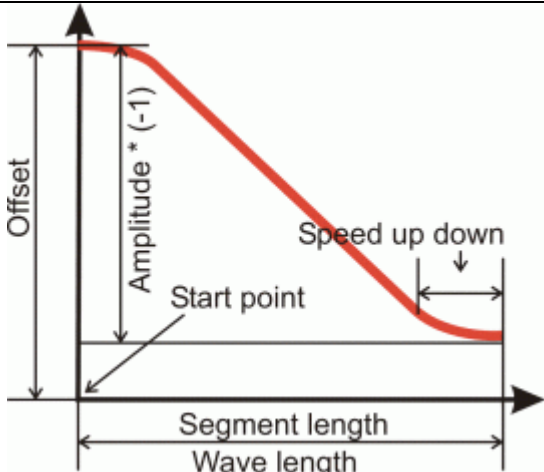
Ramp Curves

WAV command	Comments	Waveform Segment
WAV 4 X RAMP 2000 20 10 2000 0 300 1000 <WaveTableID> = 4 <AppendWave> = X <WaveType> = RAMP <SegLength> = 2000 <Amp> = 20 <Offset> = 10 <WaveLength> = 2000 <StartPoint> = 0 <SpeedUpDown> = 300 <CurveCenterPoint> = 1000	The previous contents of the wave table are overwritten by the new segment, waveform offset = 10 (Do not confuse with the wave generator output offset set with WOS!) symmetric curve	
WAV 4 X RAMP 2000 35 0 2000 499 300 1000 <WaveTableID> = 4 <AppendWave> = X <WaveType> = RAMP <SegLength> = 2000 <Amp> = 35 <Offset> = 0 <WaveLength> = 2000 <StartPoint> = 499 <SpeedUpDown> = 300 <CurveCenterPoint> = 1000	The previous contents of the wave table are overwritten by the new segment, symmetric curve	
WAV 5 X RAMP 2000 15 0 1800 120 150 900 <WaveTableID> = 5 <AppendWave> = X <WaveType> = RAMP <SegLength> = 2000 <Amp> = 15 <Offset> = 0 <WaveLength> = 1800 <StartPoint> = 120 <SpeedUpDown> = 150 <CurveCenterPoint> = 900	The previous contents of the wave table are overwritten by the new segment, symmetric curve	

WAV command	Comments	Waveform Segment
WAV 5 & RAMP 3000 35 0 3000 0 200 2250 <WaveTableID> = 5 <AppendWave> = & <WaveType> = RAMP <SegLength> = 3000 <Amp> = 35 <Offset> = 0 <WaveLength> = 3000 <StartPoint> = 0 <SpeedUpDown> = 200 <CurveCenterPoint> = 2250	The defined segment will be appended to the existing wave table contents, asymmetric curve	

Single Scan Line Curves

WAV command	Comments	Waveform Segment
WAV 1 X LIN 1500 30 15 1500 0 370 <WaveTableID> = 1 <AppendWave> = X <WaveType> = LIN <SegLength> = 1500 <Amp> = 30 <Offset> = 15 <WaveLength> = 1500 <StartPoint> = 0 <SpeedUpDown> = 370	The previous contents of the wave table are overwritten by the new segment, waveform offset = 15 (Do not confuse with the wave generator output offset set with WOS!)	
WAV 2 X LIN 1500 40 0 1100 210 180 <WaveTableID> = 2 <AppendWave> = X <WaveType> = LIN <SegLength> = 1500 <Amp> = 40 <Offset> = 0 <WaveLength> = 1100 <StartPoint> = 210 <SpeedUpDown> = 180	The previous contents of the wave table are overwritten by the new segment	

WAV command	Comments	Waveform Segment
WAV 2 & LIN 3000 -40 50 3000 0 650 <WaveTableID> = 2 <AppendWave> = & <WaveType> = LIN <SegLength> = 3000 <Amp> = -40 <Offset> = 50 <WaveLength> = 3000 <StartPoint> = 0 <SpeedUpDown> = 650	The defined segment will be appended to the existing wave table contents, negative-amplitude curve	

11.2.2 Modifying the Wave Generator Table Rate

An example for how to modify the duration of the wave generator output using the wave table rate:

Command String to Send	Action Performed
WAV 2 X SIN_P 2000 20 10 2000 0 1000	Define a sine waveform for Wave Table 2, the segment length and hence the number of points in the wave table is 2000
SPA? 1 0x0E000200	Ask for the servo update time of the controller (reading the wave table for wave generator output is clocked by servo cycles). An E-712.3CD, for example, has a servo update time of 20 μ s.
WTR?	Ask for the current wave table rate and interpolation settings, default is wave table rate = 1 (i.e. each wave table point will be output for a duration of one servo cycle). The duration of one wave generator output cycle will be: $\text{Servo Update Time (in s)} * \text{WTR value} * \text{Number of Points} = \text{Output Duration (in s)}$ $0.000020 \text{ s} * 1 * 2000 = 0.04 \text{ s}$

Command String to Send	Action Performed
WTR 0 3 1	Set the wave table rate to 3, tripling the duration of one wave generator output cycle, with linear interpolation (each wave table point will now "occupy" 3 servo cycles, but with linear interpolation applied to smooth the output). Duration of one output cycle will now be: $0.000020 \text{ s} * 3 * 2000 = 0.12 \text{ s}$ Note that the WTR command must always specify all wave generators in the system (<WaveGenID> must be 0).

11.2.3 Trigger Output Synchronized with Wave Generator

Using the digital output lines OUT1 to OUT7 of the E-711/E-712 system, it is possible to trigger external devices. See "Digital I/O Socket" (p. 285) for the availability of the lines (pinout) and "Configuring Trigger Output" (p. 142) for further information.

An example for how to generate trigger pulses synchronized with the wave generator in Generator Level Trigger mode:

Command String to Send	Action Performed
WAV 2 X SIN_P 2000 20 10 2000 0 1000	Define a sine waveform for Wave Table 2, the segment length and hence the number of points in the wave table is 2000
TWC	Clears all output trigger settings related to the wave generator by switching the signal state for all points to "low" (the power-on default state is also "low"). It is recommended to use TWC before new trigger actions are defined.
TWS 1 500 1 1 1500 1 1 1900 1 1 2000 1	Set trigger actions for the digital output line OUT1 (identifier is 1): at the waveform points 500, 1500, 1900 and 2000 it is set high; at all other points the state of the line is low (due to the TWC usage).
CTO 1 3 4	The digital output line OUT1 is set to "Generator Level Trigger" mode.
WSL 1 2	Connect Wave Generator 1 (Axis 1) to Wave Table 2

Command String to Send	Action Performed
WGO 1 1	Start output of Wave Generator 1 immediately (synchronized by servo cycle). Now the trigger output action will take place as specified.
WGO 1 0	Stop output of Wave Generator 1 and hence also the trigger output.

NOTE

If a phase shift between the trigger output and the actual position is observed when the wave generator is running with the "Use DDL" or "Use and reinitialize DDL" options (actual position measured with external device is ahead of the trigger signal), check the cut-off-frequency f_g of the IIR low-pass filter used for digital filtering and increase it to shorten the duration of the digital filtering. See below for details.

The duration of the signal processing for a internal sensor results from two portions:

- Duration of the analog sensor processing, which takes about 80 μ s for the E-711/E-712 system
- Duration of the digital filtering. If the Digital Filter Type parameter, ID 0x05000000, is set to "IIR low-pass filter, 2nd order", the required time depends on the cut-off-frequency f_g of the IIR low-pass filter which is given by the Digital Filter Bandwidth parameter, ID 0x05000001. For signal frequencies $f < f_g/2$, the duration of the filtering can be estimated as follows:
 $t \approx 0.216 / f_g$
 If required, increase f_g . Examples:
 $f_g = 300 \text{ Hz: } t = 0.72 \text{ ms}$
 $f_g = 1000 \text{ Hz: } t = 0.216 \text{ ms}$

11.2.4 Wave Generator Started by Trigger Input

Using the digital input lines of the E-711/E-712 system, it is possible to apply start/stop signals for the wave generator output. See the pinout description of the digital I/O socket (p. 285) for the availability of the lines and "Using Digital Input" (p. 142) for an overview.

NOTE

For reliable triggering, the pulse width of the input signal has to be at least 2 x the servo update time of the E-711/E-712 system. The servo update time is given in seconds by parameter 0x0E000200.

An example for how to start / stop the wave generator by external trigger signals:

Command String to Send	Action Performed
WAV 2 X SIN_P 2000 20 10 2000 0 1000	Define a sine waveform for Wave Table 2, the segment length and hence the number of points in the wave table is 2000
WSL 1 2	Connect Wave Generator 1 (Axis 1) to Wave Table 2
WGO 1 2	Start output of Wave Generator 1 triggered by external signal. To provide the external signal, the digital input lines IN1 or IN2 can be used. If IN1 is used: The wave generator output starts with the first rising edge which is detected on this input line. If IN2 is used: The generator output starts with the first rising edge which is detected on this input line, and it will be stopped when a falling edge is detected on this line. With the next rising edge, the generator output will continue at the waveform point where it was stopped. Starting and stopping the wave generator this way can be repeated indefinitely. If output cycle limitations were made with WGC: with each generator restart the counting of the output cycles continues, and the generator will be stopped when the given number of cycles are completed, irrespective of any further trigger pulses. It is possible to mix the usage of both digital input lines.
WGO 1 0	Stop output of Wave Generator 1 (any further trigger pulses will be ignored). You can also use #24 or STP.

NOTE

When the wave generator is to be started by an external trigger signal (WGO bit 1 is set), the value of the Wave Multi Start By Trigger parameter (ID 0x13000202) determines if the trigger is enabled for only one generator start or for multiple starts:

0 = Trigger is enabled for only one generator start. Trigger becomes disabled after the generator has been started. To enable the trigger again, WGO must be sent again with start mode bit 1 set. Default setting.

1 = As long as WGO bit 1 is set, the trigger stays enabled for an unlimited number of generator starts. To disable the trigger, the wave generator output must be stopped with WGO, STP or #24.

11.3 Wave-Generator-Related Commands and Parameters

Command	Description	Notes
CTO	Set Configuration Of Trigger Output	Activates the Generator Trigger output mode which is required for the triggerline actions set with TWS.
DDL?	Get DDL Table Values	Reads the current content of the DDL table(s). DDL initialization and usage are started by the wave generator start command (WGO).
DRR?	Get Recorded Data Values	Reads the last recorded data. Data recording is triggered by the WGO and WGR commands (among others).
GWD?	Get Wave Table Data	Should be used to check the waveform before the wave generator output is started.
TWC	Clear All Wave Related Triggers	Clears only the TWS settings, but not the CTO settings.
TWG?	Get Number Of Wave Generators	Number of wave generators = number of axes
TWS	Set TriggerLine Action To Waveform Point	In addition, the CTO command must be used to activate the Generator Trigger mode for the desired digital output line.
WAV	Set Waveform Definition	A waveform must be defined before the wave generator output can be started.

Command	Description	Notes
WAV?	Get Waveform Definition	Reads the current wave table length.
WCL	Clear Wave Table Data	Clears the wave table content, but not the WSL and WOS settings.
WGC	Set Number Of Wave Generator Cycles	If WGC is not used, the wave generator must be stopped with WGO, #24 or STP.
WGC?	Get Number Of Wave Generator Cycles	
WGO	Set Wave Generator Start/Stop Mode	The WGO command starts the wave generator output. It provides several start options, e.g. "Start wave generator output triggered by external signal", "Use and reinitialize DDL" or "Use DDL".
WGO?	Get Wave Generator Start/Stop Mode	Gets the last commanded start options, but not the activation status (use #9 instead)
WGR	Starts Recording in Sync with Wave Generator	Restarts data recording as long as a wave generator is running.
WOS	Set Wave Generator Output Offset	Sets the value of the Wave Offset parameter, ID 0x1300010b, in volatile memory.
WOS?	Get Wave Generator Output Offset	Gets the value of the Wave Offset parameter, ID 0x1300010b, from volatile memory.
WSL	Set Connection Of Wave Table To Wave Generator	Must be set before the wave generator can be started.
WSL?	Get Connection Of Wave Table To Wave Generator	
WTR	Set Wave Generator Table Rate	Sets the value of the Wave Generator Table Rate parameter, ID 0x13000109, in volatile memory; determines the interpolation type for wave table rate values > 1.
WTR?	Get Wave Generator Table Rate	Gets the value of the Wave Generator Table Rate parameter (ID 0x13000109) from volatile memory and the current interpolation type.

Command	Description	Notes
#9	Get Wave Generator Status	Gets the current activation status of the wave generator, but not the start options (use WGO? instead)

See "How to Work with the Wave Generator" (p. 172) for more information. For detailed command descriptions see the GCS commands manual PZ233E. For the identifiers of the items which can be addressed with the commands see "Accessible Items and Their Identifiers" (p. 104).

Parameter ID	Command Level	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description
0x13000004	3	System	1	INT	Max Wave Points
0x13000109	1	System	1	INT	Wave Generator Table Rate
0x1300010a	3	System	1	INT	Number of Waves
0x1300010b	1	Logical Axis	E-712.3CD, .3CDA: 3 E-712.6CD, .6CDA: 6	FLOAT	Wave Offset
0x13000202	1	System	1	INT	Wave Multi Start By Trigger

See "Controller Parameters" (p. 231) for more information regarding the controller parameters and their handling.

12 Dynamic Digital Linearization (DDL)

Dynamic Digital Linearization (DDL) makes it possible to achieve significantly better position accuracy in dynamic applications with periodic motion. It is used in conjunction with the wave generator output and in addition to the "normal" servo algorithm in closed-loop operation of an axis. DDL "observes" axis motion over one or more wave generator output cycles (DDL initialization). The information gathered is written to "DDL tables" and can then be used to refine the control output signals.

"Working Principle" (p. 189) describes the DDL basics, "How to Activate the DDL Licence" (p. 193) gives information on how to get the DDL started the first time, in "How to work with the DDL" (p. 195) you will learn how to use the DDL feature, and "DDL-Related Commands and Parameters" (p. 198) gives a summary.

See also "Wave Generator" (p. 172).

12.1 Working Principle

The tracking error of a standard piezo servo-controller is practically zero at very low speeds but increases with the operating frequency of the axis.

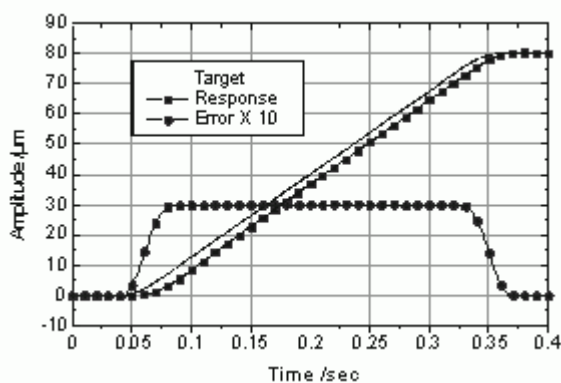


Figure 51: Tracking error for typical linear PID servo system

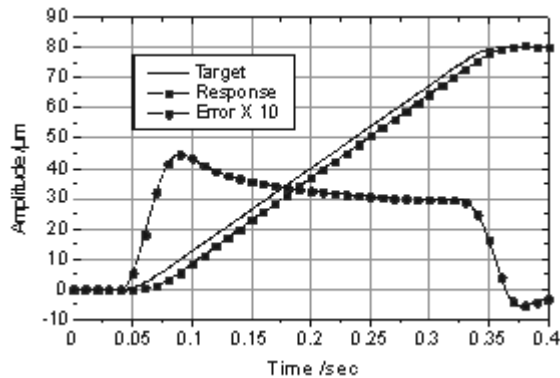


Figure 52: Tracking error for linear bi-directional scanning application

In addition, for linear bi-directional scanning applications, the tracking error is not a constant, as with other types of linear PID servo systems, but changes with the scan position. This is caused by the non-linearity of piezo actuators and the servo-controller's limited dynamic performance. As a result of these factors, the scan position is not exactly proportional to time but exhibits dynamic non-linearity.

For periodic motion it is possible to record the errors for an axis during one or more DDL initialization periods and then compensate for them in all subsequent periods. The principle of DDL is thus the inclusion of a regulating error-feedback loop in addition to the standard proportional-integral (P-I) servo-control loop.

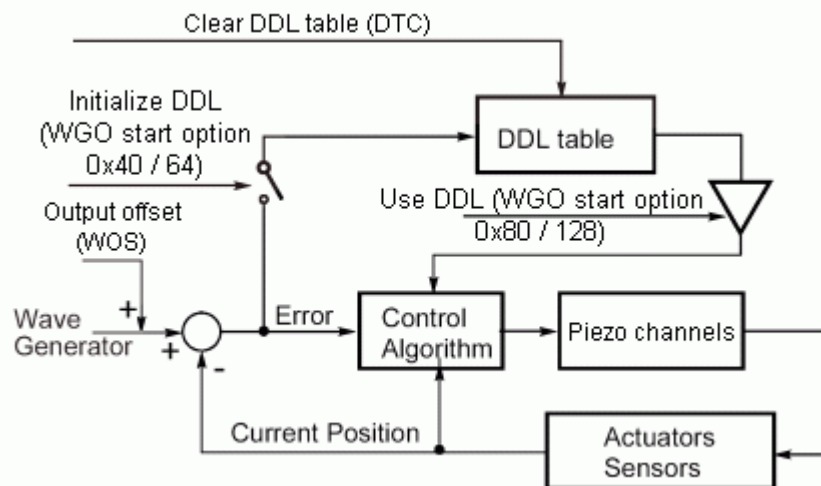


Figure 53: DDL block diagram, for one axis

Example:

A simple 400 µm forward and backward scan at 2.5 Hz. Even though the frequency is not high, there is still a static error of about ± 10 µm and a

dynamic error of about $\pm 5 \mu\text{m}$. The dynamic error indicates that the scan speed is not constant. This error will deteriorate the scanning accuracy.

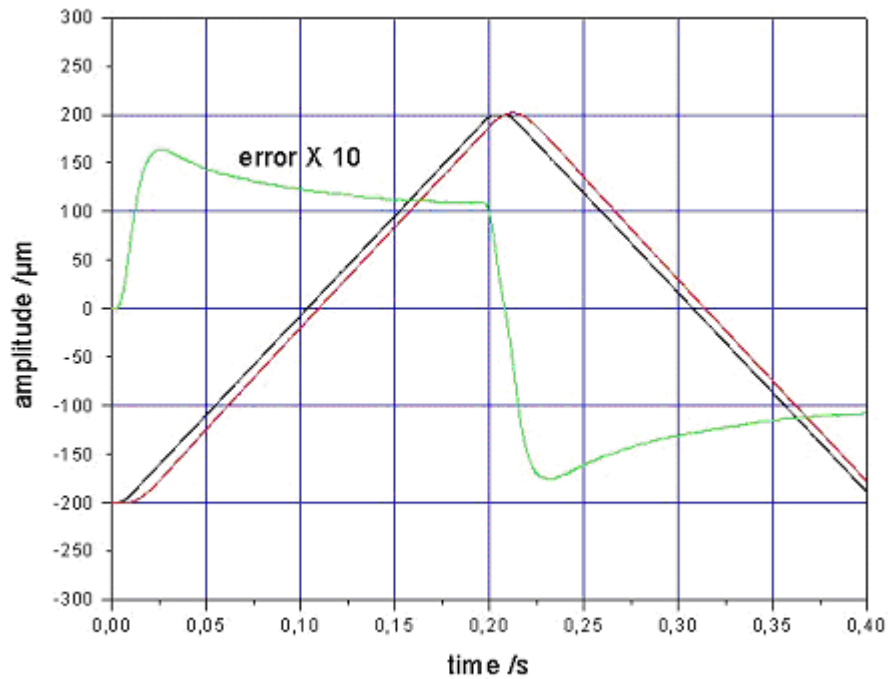


Figure 54: Without DDL

With DDL, after one initialization period the error for constant-speed motion will be reduced by a factor of about 100 and the dynamic error by a factor between 20 and 90.

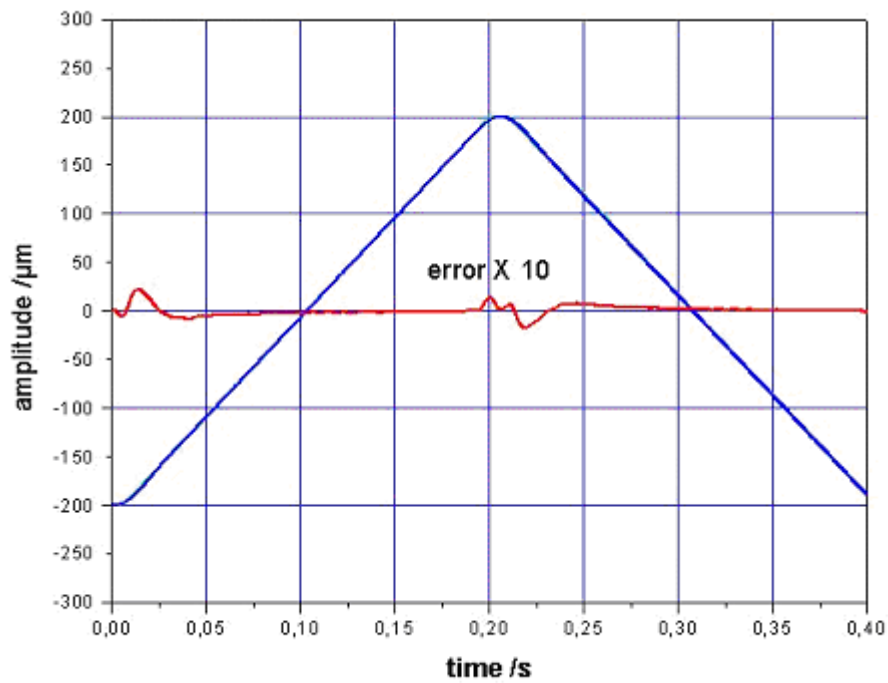


Figure 55: With DDL, one initialization cycle

With more DDL initialization periods, the dynamic error can be reduced by a factor of many hundred.

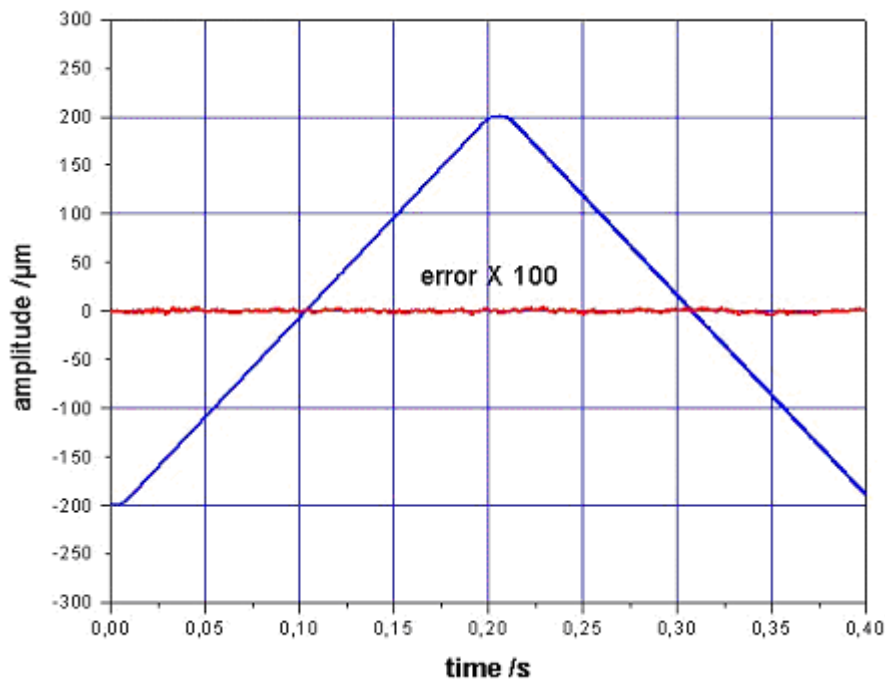


Figure 56: With DDL, after several DDL initialization cycles

12.2 How to Activate the DDL License

The Dynamic Digital Linearization (DDL) feature must be expressly ordered (order# E-710.SCN). You can activate DDL after purchase and without opening the device.

To activate the DDL license, proceed as follows:

- 1 Order DDL for your controller and inform PI of the controller serial number—PI generates your DDL license number based on this information.
The serial number is saved in the controller as the Device S/N parameter, ID 0x0D000000.
You can read it in PIMikroMove or using the SPA? or SEP? command:
Send: SPA? 1 0x0D000000 (reads the value from volatile memory)
or
Send: SEP? 1 0x0D000000 (reads the value from non-volatile memory)
- 2 PI will provide you with the DDL license number.

- 3 Switch to command level 1 using the CCL command:
Send: CCL 1 advanced
- 4 Write the DDL license number to the non-volatile memory of your controller as the DDL License parameter, ID 0x0E000400.
You can do this in PIMikroMove or using the SEP command:
Send: SEP 100 1 0x0E000400 *licensenumber*
- 5 Power cycle the controller.
- 6 Check if the DDL activation was successful. To do this, read the value of the DDL License Valid parameter, ID 0x0E000401 in PIMikroMove or using SPA? or SEP?.
Send: SPA? 1 0x0E000401
or
Send: SEP? 1 0x0E000401
DDL activation was successful if the parameter value is 1.
Otherwise the DDL feature is still deactivated.

If the DDL activation fails:

- Check if the correct serial number was used for license number generation.
- Check whether the correct license number was entered into controller memory:
Send: SEP? 1 0x0E000400
- Make sure that command level 1 is active during license activation.

12.3 How to work with the DDL

NOTE

The Dynamic Digital Linearization (DDL) feature must be expressly ordered (order# E-710.SCN). You can activate it after purchase and without opening the device. See "How to activate the DDL License" (p. 193) for more information.

If you are not sure if the DDL feature is activated on your controller, use the SPA? command to check the value of parameter 0x0E000401 (DDL License Valid). If the parameter value is 1, the DDL feature is activated, otherwise it is deactivated.

DDL related commands will not provoke an error when the DDL feature is deactivated.

CAUTION

Before you work with Dynamic Digital Linearization (DDL), eliminate any residual oscillations by adjusting the notch filter(s) and the servo-control parameters (P-term, I-term; see "Servo-Controller Dynamic Tuning" (p. 218)). Using DDL could be critical if there are any residual oscillations in the system; DDL will then cause the oscillations to build up—the more wave generator cycles are used for DDL initialization, the stronger the effect.

The number of DDL tables can be queried using the TLT? command. The number of DDL tables present in the E-711/E-712 system is the same as the number of logical axes, and each DDL table is dedicated to one axis. Depending on its configuration, the E-711/E-712 system supports 3 or 6 axes.

A certain amount of memory space is reserved for the DDL data (ask with the SPA? command, parameter ID 0x1400000B). 2^{23} memory points are available. The individual DDL tables may use different portions of the complete memory space—the number of points in a DDL table corresponds to the length of the waveform which was output during the DDL initialization. You can ask for the current DDL table length using the DTL? command.

DDL tables are automatically filled with data when a wave generator is started with the "Use and reinitialize DDL" option activated. It is also possible to write data "manually" with the DDL command—in this case, make sure that the DDL table length is correct (see DDL command description for more information).

NOTES

The DDL initialization must be repeated when a new stage is connected, the servo parameters are changed (e.g. due to load changes) or the waveform is changed.

It is recommended to start the DDL initialization for all axes at the same time. Each new initialization will stop all running initialization processes. The initialization process is also stopped by the DDL command.

How to initialize DDL for an axis:

- 1 Define a waveform using the WAV command.
- 2 Optionally: Set the number of wave generator cycles to use for DDL initialization. This can be done using the SPA or SEP command with the DDL Repeat Number parameter (ID 0x14000001). The factory default is 35. (To have write access to the parameter, it might be necessary to switch to command level 1 using CCL).
- 3 Not essential but recommended: Delete DDL table content which is no longer used. This can be done using the DTC command.
- 4 When the servo parameters (notch filter frequency, servo-loop P-term, servo-loop I-term and servo-loop slew rate) have changed: Recalculate the internal DDL processing parameters using the DPO command.
- 5 Assign the waveform to the axis using the WSL command.
- 6 Switch the servo on for the axis using the SVO command (closed-loop operation).
- 7 Start the wave generator with the WGO command, with the "Use and reinitialize DDL" option activated.
- 8 Optionally: Check the content of the DDL table using the DDL? command.

How to use DDL for an axis:

As long as your application does not change, you can use the current DDL table content without new initialization. In this case, start the wave generator with WGO and the "Use DDL" option activated.

NOTE

If the DDL performance does not prove satisfactory or if errors occur during DDL use:

- Recalculate the internal DDL processing parameters using DPO.
- Reduce the frequency of the wave generator output to make sure that the axis can follow the target values. To do this, define a new, optimized waveform with the WAV command. You can also use the WTR command, which sets the wave generator table rate, to optimize the output frequency.

After such changes DDL initialization must be repeated.

NOTE

If a phase shift between the trigger output and the actual position is observed when the wave generator is running with the "Use DDL" or "Use and reinitialize DDL" options (actual position measured with external device is ahead of the trigger signal), check the cut-off-frequency f_g of the IIR low-pass filter used for digital filtering and increase it to shorten the duration of the digital filtering. See below for details.

The duration of the signal processing for a capacitive sensor results from two portions:

- Duration of the analog sensor processing, which takes about 80 μ s for the E-711/E-712 system
 - Duration of the digital filtering. If the Digital Filter Type parameter, ID 0x05000000, is set to "IIR low-pass filter, 2nd order", the required time depends on the cut-off-frequency f_g of the IIR low-pass filter which is given by the Digital Filter Bandwidth parameter, ID 0x05000001. For signal frequencies $f < f_g/2$, the duration of the filtering can be estimated as follows:
 $t \approx 0.216 / f_g$
 If required, increase f_g . Examples:
 $f_g = 300 \text{ Hz: } t = 0.72 \text{ ms}$
 $f_g = 1000 \text{ Hz: } t = 0.216 \text{ ms}$
-

NOTE

The DDL table content and the calculated processing parameters will be lost when the controller is powered down or rebooted.

12.4 DDL-Related Commands and Parameters

Command	Description	Notes
DDL	Set DDL Table Values	Can be used to fill the DDL table(s) "manually" with data
DDL?	Get DDL Table Values	Reads the current content of the DDL tables(s)
DPO	DDL Parameter Optimization	Recalculates the DDL processing parameters (Time Delay Max, ID 0x14000006, and Time Delay Min, ID 0x14000007), required if the servo parameters have changed
DTC	Clear DDL Table Data	It is recommended that the content of DDL tables which are not used be deleted. This avoids errors due to insufficient memory space during the DDL initialization. On controller power down or reboot, the content of the DDL tables will be erased automatically.
DTL?	Get DDL Table Length	Returns the value of the Max DDL Points parameter, ID 0x1400000B
TLT?	Get Number of DDL Tables	The reported value corresponds to the number of axes available on the controller
WAV	Set Waveform Definition	Since DDL works only in conjunction with the wave generator output, the waveform must first be defined
WGO	Set Wave Generator Start/Stop Mode	The WGO command starts the wave generator output and provides two modes for the DDL: "Use and reinitialize DDL" and "Use DDL"

See "How to Work with the DDL" (p. 195) for more information. For detailed command descriptions see the GCS commands manual PZ233E. For the identifiers of the items which can be addressed with the commands see "Accessible Items and Their Identifiers" (p. 104).

Parameter ID	Command Level	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description
0x0e000400	1	System	1	INT	DDL license
0x0e000401	3	System	1	INT	DDL license valid
0x14000001	1	Logical Axis	E-712.3CD, .3CDA: 3 E-712.6CD, .6CDA: 6	INT	DDL repeat number
0x14000006	1	Logical Axis	E-712.3CD, .3CDA: 3 E-712.6CD, .6CDA: 6	FLOAT	Time Delay Max
0x14000007	1	Logical Axis	E-712.3CD, .3CDA: 3 E-712.6CD, .6CDA: 6	FLOAT	Time Delay Min
0x14000008	1	Logical Axis	E-712.3CD, .3CDA: 3 E-712.6CD, .6CDA: 6	INT	Time Delay Change Rule
0x1400000a	1	Logical Axis	E-712.3CD, .3CDA: 3 E-712.6CD, .6CDA: 6	INT	DDL Zero Gain Number
0x1400000b	3	System	1	INT	Max DDL Points

See "Controller Parameters" (p. 231) for more information regarding the controller parameters and their handling.

13 Controller Macros

The E-711/E-712 system can save and process command sequences as macros.

The following functionalities make macros an important tool in many application areas:

- Several macros can be saved at the same time.
- Any macro can be defined as the start-up macro. The start-up macro is executed each time that the E-711/E-712 system is switched on or rebooted.
- Processing a macro and stopping macro execution can be linked to conditions. In this way, loops can be realized as well.
- Macros can call up themselves or other macros in several nesting levels.
- Variables (p. 209) can be set for the macro and in the macro itself and used in different operations.
- Input signals can be evaluated for conditions and variables.

13.1 Commands for Macros

The following commands are specially available for handling macros or for use in macros:

Command	Syntax	Function
ADD	ADD <Variable> <FLOAT1> <FLOAT2>	Adds two values and saves the result to a variable (p. 209). Can only be used for local variables in macros.
CPY	CPY <Variable> <CMD?>	Copies a command response to a variable (p. 209). Can only be used for local variables in macros.
DEL	DEL <uint>	Can only be used in macros. Delays <uint> milliseconds.
JRC	JRC <Jump> <CMD?> <OP> <Value>	Can only be used in macros. Triggers a relative jump of the macro execution pointer depending on a condition.

Command	Syntax	Function
MAC	MAC BEG <macroname>	Starts the recording of a macro with the name macroname on the controller. macroname can consist of up to 15 characters.
	MAC DEF <macroname>	Defines the given macro as the start-up macro.
	MAC DEF?	Gets the start-up macro.
	MAC DEL <macroname>	Deletes the given macro.
	MAC END	Ends the macro recording.
	MAC ERR?	Reports the last error which occurred during macro execution.
	MAC FREE?	Gets the free memory space for macro recording (unit: number of characters).
	MAC NSTART <macroname> <uint> [<String1> [<String2>]]	Starts the given macro n times in succession (n = number of executions). The values of local variables can be set for the macro with <String1> and <String2>.
	MAC START <macroname> [<String1> [<String2>]]	Starts one execution of the specified macro. The values of local variables can be set for the macro with <String1> and <String2>.
MAC?	MAC? [<macroname>]	Lists all macros or the content of a given macro.
MEX	MEX <CMD?> <OP> <Value>	Can only be used in macros. Stops the macro execution depending on a condition.
RMC?	RMC?	Lists macros which are currently running.
VAR	VAR <Variable> <String>	Sets a variable (p. 209) to a certain value or deletes it. Can only be used for local variables in macros.
VAR?	VAR? [{<Variable>}]	Gets variable values.
WAC	WAC <CMD?> <OP> <Value>	Can only be used in macros. Waits until a condition is met.
#8	-	Tests if a macro is running on the controller.

13.2 Working with Macros

Work with macros comprises the following:

- Recording of macros (p. 202)
- Starting macro execution (p. 204)
- Stopping macro execution (p. 207)
- Setting up a start-up macro (p. 207)
- Deleting of macros (p. 208)

NOTE

For working with controller macros, it is recommended to use the *Controller macros* tab in PIMikroMove. There you can conveniently record, start and manage controller macros. Details are found in the PIMikroMove manual.

13.2.1 Recording a Macro

NOTES

A maximum of 5 nesting levels are possible in macros.

Basically all GCS commands can be included in a macro. Exceptions:

RBT for rebooting the E-711/E-712 system
MAC BEG and MAC END for macro recording
MAC DEL for deleting a macro

Query commands can be used in macros in combination with the CPY, JRC, MEX and WAC commands. Otherwise they have no effect, since macros do not send any responses to interfaces.

To make the use of macros more flexible, you can use local and global variables in macros. See “Variables” (p. 209) for more information.

The number of write cycles in the nonvolatile memory is restricted by the limited lifetime of the memory chip.

→ Only record macros when it is necessary.

→ Use variables (p. 209) in macros to make macros more flexible, and give the corresponding variable values when starting macro execution.

→ Contact our customer service department (p. 266) if the E-711/E-712 system shows unexpected behavior.

A macro must be deleted before a macro with the same name can be re-recorded.

1. Start the macro recording.

- If you are working with PITerminal or in the *Command entry* window of PIMikroMove: Send the MAC BEG *macroname* command, where *macroname* indicates the name of the macro.
- If you are working in PIMikroMove on the *Controller macros* tab: Click the *Create new empty macro* icon to create a tab for entering a new macro. Do **not** enter the MAC BEG *macroname* command.

2. Enter the commands to be included in the *macroname* macro line by line, using the normal command syntax.

Macros can call up themselves or other macros in several nesting levels.

3. End the macro recording.

- If you are working with PITerminal or in the *Command entry* window of PIMikroMove: Send the MAC END command.
- If you are working in PIMikroMove on the *Controller macros* tab: Do **not** enter the MAC END command. Click the *Send macro to controller* icon and enter the name of the macro in a separate dialog window.

The macro has been stored in the nonvolatile memory of the E-711/E712 system.

4. If you want to check whether the macro has been correctly recorded:

If you are working with PITerminal or in the *Command entry* window of PIMikroMove:

- Get which macros are saved in the E-711/E-712 system by sending the MAC? command.
- Get the contents of the *macroname* macro by sending the MAC? *macroname* command.

If you are working in PIMikroMove on the *Controller macros* tab:

- Click the *Read list of macros from controller* icon.
- Mark the macro to be checked in the list on the left side and click the *Load selected macro from controller* icon.

13.2.2 Starting a Macro Execution

NOTES

Any commands can be sent from the command line when a macro is running on the controller. The macro content and move commands received from the command line can overwrite each other.

Simultaneous execution of multiple macros is not possible. Only one macro can be executed at a time.

You can link the macro execution to conditions with the JRC and WAC commands. The commands must be included in the macro.

In the following, PITerminal or the *Command entry* window of PIMikroMove is used to enter commands. Details on working with the *Controller macros* tab in PIMikroMove are found in the PIMikroMove manual.

1. Start the macro execution:

- If the macro is to be executed once, send the MAC START *macroname string* command, whereby *macroname* indicates the name of the macro.
- If the macro is to be executed *n* times, send the MAC NSTART *macroname n string* command, whereby *macroname* indicates the name of the macro and *n* indicates the number of executions.

string stands for the values of local variables. The values only have to be given when the macro contains corresponding local variables. The sequence of the values in the input must correspond to the numbering of the appropriate local variables, starting with the value of the local variable 1. The individual values must be separated from each other with spaces.

2. If you want to check the macro execution:

- Get whether a macro is being executed on the controller by sending the #8 command.
- Get the name of the macro that is currently being executed on the controller by sending the RMC? command.

Example: Moving an axis with a variable travel distance back and forth

NOTE

When macros are recorded on the *Controller macros* tab in PIMikroMove, the MAC BEG and MAC END commands must be omitted.

The axis 1 is to move back and forth. The travel to the left and to the right is to be variably adjustable without having to change the used macros. Local and global variables are therefore used.

1. Create the global variables LEFT and RIGHT by sending:

```
VAR LEFT 5  
VAR RIGHT 15
```

LEFT thus has the value 5, and RIGHT has the value 15. These values can be changed at any time, e.g., by sending the VAR command again.

- Create the global variables again each time that the E-711/E-712 system is switched on or rebooted, since they are only written to the volatile memory of the E-711/E-712 system.

2. Record the MOVLR macro by sending:

```
MAC BEG movlr  
MAC START movwai ${LEFT}  
MAC START movwai ${RIGHT}  
MAC END
```

MOVLR successively starts the MOVWAI macro (which is still to be recorded) for both directions of motion. The values of the global variables LEFT and RIGHT are used when MOVWAI is started, to set the value of the local variable 1 contained in MOVWAI (dollar signs and braces are necessary for the local variable 1 in the macro to actually be replaced by the value of the global variable and not by its name).

3. Record the MOVWAI macro by sending:

```
MAC BEG movwai  
MOV 1 $1  
WAC ONT? 1 = 1  
MAC END
```

MOVWAI moves axis 1 to the target position which is given by the value of the local variable 1 and waits until the axis has reached the

target position.

4. Start the execution of the MOVLR macro by sending:

```
MAC NSTART movlr 5
```

The MOVLR macro is executed five times in succession, i.e., axis 1 alternately moves to the positions 5 and 15 five times. You can also select any other value for the number of executions.

Example: Implementing multiple calls of a macro via a loop

NOTE

When macros are recorded on the *Controller macros* tab in PIMikroMove, the MAC BEG and MAC END commands must be omitted.

The TESTDION macro checks the status of the digital input lines on the Digital I/O socket. It uses a local variable to identify the digital input line (1 to 4). So that the TESTDION macro does not have to be called separately for each input line, another macro with a loop is recorded.

- Record the LOOPDION macro by sending:

```
MAC BEG loopdion  
VAR COUNTER 1  
MAC START TESTDION ${COUNTER}  
ADD COUNTER ${COUNTER} 1  
JRC -2 VAR? COUNTER < 5  
MAC END
```

The COUNTER variable is created with the value 1. After this, the TESTDION macro is started for the input line whose identifier is specified via the COUNTER variable. Then the value of the COUNTER is increased by 1. As long as the value of the COUNTER is less than 5, the macro execution pointer subsequently jumps two lines back, so that the TESTDION is now started for the next digital input line.

13.2.3 Stopping a Macro Execution

NOTE

You can link the stopping of the macro execution to a condition with the MEX command. The command must be included in the macro.

In the following, PITerminal or the *Command entry* window of PIMikroMove is used to enter commands. Details on working with the *Controller macros* tab in PIMikroMove are found in the PIMikroMove manual.

- Stop the macro execution with the #24 or STP commands.
- If you want to check whether an error has occurred during the macro execution, send the MAC ERR? command. The response shows the last error that has occurred.

13.2.4 Setting up a Start-up Macro

Any macro can be defined as the start-up macro. The start-up macro is executed each time that the E-711/E-712 system is switched on or rebooted.

NOTE

Deleting a macro does not delete its selection as the start-up macro.

In the following, PITerminal or the *Command entry* window of PIMikroMove is used to enter commands. Details on working with the *Controller macros* tab in PIMikroMove are found in the PIMikroMove manual.

- Define a macro as the start-up macro with the MAC DEF *macroname* command, whereby *macroname* indicates the name of the macro.
- If you want to cancel the selection of the start-up macro and do not want to define another macro as the start-up macro, only send MAC DEF.
- Get the name of the currently defined start-up macro by sending the MAC DEF? command.

Example: Moving an axis via a start-up macro to a certain position in closed-loop operation

NOTE

When macros are recorded on the *Controller macros* tab in PIMikroMove, the MAC BEG and MAC END commands must be omitted.

Setting the servo mode via a start-up macro is not necessary because this can be done via the value of the Power Up Servo ON Enable parameter (ID 0x07000800; 0 = Servo mode is not automatically switched on; 1 = Servo mode is automatically switched on). Furthermore, the AutoZero procedure can be executed automatically according to the value of the Power Up AutoZero Enable parameter (ID 0x07000802; 0 = AutoZero procedure is not automatically executed; 1 = AutoZero procedure is automatically executed). For that reason, the STARTCL macro only starts the motion to the desired position which is 10 in this example.

■ Send:

```
CCL 1 advanced
SEP 100 1 0x07000800 1
SEP 100 1 0x07000802 1
MAC BEG startcl
MOV 1 10
MAC END
MAC DEF startcl
```

13.2.5 Deleting a Macro

NOTE

A running macro may not be deleted.

In the following, PITerminal or the *Command entry* window of PIMikroMove is used to enter commands. Details on working with the *Controller macros* tab in PIMikroMove are found in the PIMikroMove manual.

- Delete a macro with the MAC DEL *macroname* command, whereby *macroname* indicates the name of the macro.

14 Variables

For more flexible programming, the E-711/E-712 system supports variables. While global variables are always available, local variables are only valid for a given macro. Typically, variables are used when working with macros.

Variables are present in volatile memory (RAM) only. The variable values are of the STRING data type.

The following conventions apply to variable names:

- Variable names must not contain special characters (especially no "\$").
- The maximum number of characters is 8.
- Names of global variables can consist of characters A to Z and 0 to 9. They must start with an alphabetic character.
- Names of local variables must not contain alphabetic characters. Possible characters are 0 to 9.
- The variable name can also be given via the value of another variable.
- If the value of a variable is to be used, the notation must be as follows:
- The variable name must be preceded by the dollar sign (\$).
- Variable names consisting of multiple characters must be put in curly brackets.

If the variable name consists of a single character, the curly brackets can be omitted.

Note that if the curly brackets are omitted with variable names consisting of multiple characters, the first character after the "\$" is interpreted as the variable name.

Local variables:

- Local variables can only be used in macros.
- At present, the controller firmware supports three local variables: 0, 1 and 2.

- The values of the local variables 1 and 2 are given as arguments of the MAC START or MAC NSTART command when starting the macro. The command formats are:

MAC START <macroname> [<String1> [<String2>]]

MAC NSTART <macroname> <uint> [<String1> [<String2>]]

<STRING1> and <STRING2> indicate the values for the local variables 1 and 2 used in the macro. <STRING1> and <STRING2> can be given directly or via the values of variables. <uint> defines the number of times the macro is to be run. See the MAC command description for more information.

- The local variable 0 is read-only. Its value gives the number of arguments (i. e. values of local variables) set when starting the macro.
- Inside a macro, the values of local variables can be modified using ADD, CPY or VAR, and can be deleted with VAR (except for the local variable 0).
- As long as the macro is running, the values of the local variables can be queried with
VAR? 0
VAR? 1
VAR? 2
The queries can be sent inside or outside of the macro.

Global variables:

- Global variables can be used inside and outside of macros.
- The maximum number of global variables is 10.
- Global variables are created and modified using ADD, CPY or VAR. They can be deleted with VAR.
- The variable values can be queried with VAR?.

15 Protective Functions

15.1 Safety Shutdown

The firmware of the E-711/E-712 system provides a safety shutdown functionality: The E-711/E-712 system observes several items and compares their current states with abort criteria defined by controller parameters (see table below for details).

The safety shutdown functionality is enabled/disabled for the individual axes using the value of the Safety Shutdown Activation Register parameter (ID 0x1a000600). The parameter value is the hexadecimal sum of bits for up to 32 axes (bit 0: axis 1, bit 1: axis 2, ... bit 31: axis 32). The safety shutdown functionality is enabled for an axis if the corresponding bit is set. Default value: 0xFFFFFFFF (safety shutdown functionality is enabled for axes 1 to 32).

If any of the abort criteria is true, for all axes with enabled safety functionality the E-711/E-712 system will

- Stop all motion
- Set all piezo/drive voltages to zero
- Switch to open-loop operation (servo OFF)
- Activate all brakes (if present)
- Set the error code to a non-zero value

NOTE

The error code can be read and reset with the ERR? command. But as long as any of the abort criteria is true, every ERR? command will report a non-zero value.

To get back to normal operation, proceed as follows:

1. Send ERR? to check the error code.
2. Via the corresponding parameter, disable the criterion which caused the system abort.
3. Eliminate the condition which caused the system abort.
4. Send ERR? to clear the error code.
5. Enable the criterion again.
6. Switch to closed-loop operation (servo ON) for the axes in question. When switching to closed-loop operation, the target position is set to the current axis position and the brakes are deactivated (if present).

Item	Abort Criterion	Controller Parameters
Position error of axis	Position error for the axis is out of given range of tolerance.	0x1A000000 to 0x1A00000n Maximum Position Error: give the range of tolerance for axes 1 to $n+1$. The parameter values are floating-point numbers ("float", the range of tolerance is $-\text{float}$ to $+\text{float}$), float must be > 0; the unit is the axis position unit. 0x1A000400 to 0x1A00040n Use Position Error to Stop: enable/disable the usage of the criterion for axes 1 to $n+1$; parameter values can be: 0 = position error criterion not used 1 = position error criterion used Example: To use the position error criterion with the axis 3, you have to set parameters 0x1A000002 and 0x1A000402 to appropriate values.
Digital input lines	Digital input signal is high.	0x1A000100 Trigger to Stop: gives the digital input line which is to be observed (one of the lines on the "DIGITAL I/O" socket). Active HIGH. The digital input signal has to be high for as long as the safety shutdown is to be effective. Notes: If the parameter value is 0, the criterion is disabled. The digital inputs are pulled to HIGH by internal pullup resistors when nothing is connected to the DIGITAL I/O socket. It is not recommended to define lines which are used to start the data recorder or to start/stop the wave generators.

Item	Abort Criterion	Controller Parameters
Analog input lines	Analog input signal is out of given range of tolerance.	0x1A000200 to 0x1A00020n Maximum Value of Input: give the range of tolerance for input signal channels 1 to $n+1$. The parameter values are floating-point numbers ("float", the range of tolerance is $-\text{float}$ to $+\text{float}$), must not be negative. If the parameter value is 0, the criterion is disabled for the input signal channel. Notes: This criterion should only be enabled with input signal channels provided by an E-711.IA4 analog interface module. These input lines can be used, for example, with external force sensors. This criterion is used irrespective of the settings which determine the usage of the analog input as sensor or for control value generation. Nevertheless, make sure that the analog input is scaled to suitable values (see p. 155 for details). Example: To use the analog input criterion with the input signal channel 4, you have to set parameter 0x1A000203 to an appropriate value.

Item	Abort Criterion	Controller Parameters
Limit switches of input signal channel	Limit switch tripped.	0x1A000300 to 0x1A00030n Use Limit Switch of Input: enable/disable the usage of the criterion for input signal channel 1 to $n+1$; parameter values can be: 0 = limit switch criterion not used 1 = limit switch criterion used Notes: This criterion should only be enabled for sensors which support limit switches. This criterion can be used to detect if the system is connected properly: If a connector is not connected, both negative and positive limit switch will be active. Example: During axis motion the limit switch lines are observed (except for reference moves). The axis trips a limit switch, and the system is aborted. Now you have to identify the limit switch which caused the system abort. Disable the abort criterion for the corresponding input signal channel (using the corresponding parameter) and move the axis away from the limit switch (e.g. using move commands; probably you have to switch servo ON before). Enable the abort criterion again for the input signal channel.
Sensor signal (input signal channel)	Sensor signal invalid.	0x1a000500 Observe Sensor Signal: enable/disable the usage of the criterion for input signal channel 1 to $n+1$; parameter values can be: 0 = sensor signal criterion not used 1 = sensor signal criterion used Notes: The E-711/E-712 system validates the sensor signal based on internal criteria which are not accessible to the user.

15.2 Plausibility Check of Sensor Position

A plausibility check of the sensor position is necessary if the position of multiple axes is measured by the same sensors. If the plausibility check is enabled, the E-711/E-712 system checks if the sensor components resulting from given axis target positions will be inside the permitted sensor ranges. If the sensor ranges will be exceeded by the axis targets, the motion will not be executed, and error code 7 (“Position out of limits”) will be set.

Example: A tip/tilt stage provides motion in axes Z, rotX and rotY. The motion is realized by three actuators, here, referred to as Z1, Z2, and Z3. In addition, the stage is equipped with three position sensors, and each sensor belongs to one of the three actuators. Due to this design, the motion of an axis always results from the motion of multiple actuators, and the axis position value always results from the values of multiple sensors (via matrix calculations; for further details, see “Input Signal Processing”, p. 109, and “Output Generation”, p. 118). The range limits of the sensors could be exceeded when motion is commanded for multiple axes, i.e., as a combination of Z-motion, rotX-motion and/or rotY-motion. Exceeding the sensor range could cause sensor overflow or even sensor damage by internal collisions.

To configure the plausibility check, proceed as follows:

- 1 Enable the plausibility check for the E-711/E-712 system:

Set the Check Sensor Position Plausibility parameter (ID 0x0e001f00) to value 1 (= plausibility check is enabled) with the SEP command.
(Value 0 means that the plausibility check is disabled for the E-711/E-712 system.)
- 2 Reboot the E-711/E-712 system. The parameters 0x02001B00, 0x02001C00 and 0x02001D00 are now available for the individual input signal channels.
- 3 Set the range limits for every input signal channel which represents a sensor that is to be checked:

Set the Maximum Position Of Sensor parameter (ID 0x02001B00) to the maximum permissible sensor position.

Set the Minimum Position Of Sensor parameter (ID 0x02001C00) to the minimum permissible sensor position.
- 4 Enable the plausibility check for every input signal channel which represents a sensor that is to be checked:

Set the Validate Sensor Position Range parameter (ID 0x02001D00) to value 1 (= plausibility check is enabled) for the sensor.
(Value 0 means that the plausibility check is disabled for the sensor.)

16 Adjustment Procedures

16.1 ID-Chip Support / Stage Replacement

The stage which is connected to the E-711/E-712 system may contain an ID-chip (located in the stage connector). The following data is stored in the ID-chip (and cannot be modified there by the customer):

- Stage type
- Serial number of the stage
- Calibration data
- Servo-control data (dynamic tuning, load dependent)

The parameters which are usually stored in ID-chips are marked in the table in "Parameter Overview" (p. 233), but the list can differ slightly among the different mechanics provided by PI.

When a stage with ID-chip is connected to the E-711/E-712 system for the first time, the stage parameters from the ID-chip will be written to non-volatile and volatile memory upon power-on or reboot of the E-711/E-712 system. Afterwards, the complete set of ID-chip parameters will be overwritten on power-on or reboot only if the "Power Up Read ID-Chip" option is enabled. By default, this option is disabled to facilitate maintaining optimized parameter settings in the E-711/E-712 system.

NOTE

When the complete set of ID-chip parameters of the stage is read by the E-711/E-712 system, the axes of the stage are assigned to the logical axes in the firmware of the E-711/E-712 system. The assignment depends on the value of the ID-Chip Chan. to Axis Map Ctrl parameter (ID 0x07030c00). Possible values:

0 = The first axis of the stage is assigned to the first logical axis of the E-711/E-712 system which is free, i.e. not assigned to another stage with ID-chip that is currently connected.

2 (default) = The first axis of the stage is assigned to the logical axis of the E-711/E-712 system that has the same ID as the first input signal channel that is occupied by the stage. Example: A stage is connected to input signal channels 4 to 6. The first axis of the stage therefore is assigned to axis 4 in the E-711/E-712 system.

NOTE

When you connect a stage when the E-711/E-712 system is powered on, the ID-chip of the stage is not read by the E-711/E-712 system. To read the ID-chip data, the E-711/E-712 system must be power-cycled or rebooted using the RBT command or the corresponding host software functions.

A stage can be easily exchanged due to the functionality of the ID-chip. Consider the following when replacing stages with ID-chips.

"Simple" Replacement

Normally, when you replace a stage with a new unit and you are using standard factory settings for all parameters, you do not have to adjust anything. The ID-chip holds all information needed. At power-on of the system, the firmware reads the stage type and serial number stored in the ID-chip and compares this data to the data from the last connected stage, stored in the E-711/E-712 system:

- If there is a new stage type connected to the E-711/E-712 system, all the data in the ID-chip will be read and the corresponding parameters in the E-711/E-712 system overwritten.
- If there is a stage of the same type but with a different serial number connected to the E-711/E-712 system, the calibration data from the ID-chip will be read and only the corresponding parameters overwritten. The servo-control data will not be read, so those parameters will remain unchanged in the E-711/E-712 system.

If you have optimized some parameters for your application, PI recommends that you repeat your optimization routine with any new stage, because there are variations, e.g. in the stiffness and natural frequency, of stages.

Upgrade or Repair of Stages

If you send your stage to PI, e.g. for upgrade or repair, the calibration data stored in the ID-chip might be changed in the process. However, when you reconnect this stage to the E-711/E-712 system to which it was connected before, the firmware will detect that the type and serial number are unchanged and will not read the new ID-chip data.

To force the E-711/E-712 system to read the complete data of the ID-chip of the connected stage when the E-711/E-712 system is switched on, you can enable the "Power Up Read ID-Chip" option (parameter ID 0X0F000000).

This has to be done for each input signal channel separately. Note that it might be necessary to switch to a higher command level to have write access to that parameter (use CCL or the appropriate facilities of PIMikroMove). Proceed as follows:

- 1 In the PIMikroMove main window, open the *Device Parameter Configuration* window (E-712... ⇒ Parameter Configuration) and select the *System Mechanics* groups where you can enable the option. When this is done for all input signal channels associated with the stage, press the "Write selected edit values to default settings" button in the icon bar of the *Device Parameter Configuration* window.

Alternatively you can use the following commands in a terminal to enable the option:

```
SEP 100 1 0X0F000000 1 for input signal channel 1
SEP 100 2 0X0F000000 1 for input signal channel 2 , etc.
```

- 2 Now reboot the E-711/E-712 system by typing the RBT command in the terminal (alternatively you can power-cycle the E-711/E-712 system). This time all data is read from the ID-chip and stored in the E-711/E-712 system.
- 3 To ensure that at next power-on or reboot the E-711/E-712 system will not read all data again and overwrite parameters you may have optimized, you will have to disable the "Power Up Read ID-Chip" option, again for each input signal channel separately.

In PIMikroMove, proceed as described above in step 1 but make sure that the parameter now has the value "disabled".

Alternatively you can use the following commands in a terminal to disable the option:

```
SEP 100 1 0X0F000000 0 for input signal channel 1,
SEP 100 2 0X0F000000 0 for input signal channel 2, etc.
```

If you had optimized parameters before the repair/upgrade, PI recommends you to repeat your optimization routine when the stage is returned.

16.2 Servo-Controller Dynamic Tuning

If the E-711/E-712 system and the attached stages are ordered together and if PI has sufficient knowledge of your application, then the parameters of notch filters and standard control algorithm (servo parameters) will be set to suitable values at the factory, and, if present, saved in the stage's ID-chip (p. 216). For axes driven by conventional piezo actuators, modifications of

those parameters may, however, be necessary if the load applied to the stage is changed.

16.2.1 Parameters to be Modified

The following parameters may need to be modified in the E-711/E-712 system:

- Settings for notch filters 1 and 2:
Notch frequency 1 (parameter ID 0x08000100), notch frequency 2 (parameter ID 0x08000101)

To determine the resonant frequencies and set the notch filters properly, observe the system response to an impulse in open-loop operation (p. 221).

Note that by default, the notch filters are enabled only in closed-loop operation.

- Servo-control parameters:
P-term (parameter ID 0x07000300), I-term (parameter ID 0x07000301):

Normally the proper P-term and I-term settings are found by observing the response of the axis to an abrupt change of the control value (step response) in closed-loop operation (p. 225).

16.2.2 General Notes on Servo-Controller Dynamic Tuning

CAUTION

If the stage starts oscillating (humming noise):

In closed-loop operation, switch off the servo immediately. Switch the servo on only after you have adjusted the notch filter(s) and the servo-control parameters (P-term, I-term).

In open-loop operation, stop the axis motion immediately.

Otherwise the stage could be irreparable damaged.

- Before you change parameter values of the E-711/E-712 system, create a backup file. See "Create Backup File for Controller Parameters" (p. 86) for more information.
- Enter the password "advanced" when prompted to change to command level 1.
- For stages with ID-chip, to make the optimized settings available in the future, the option "Power Up Read ID-Chip" must have "disabled" as

its power-on default (value of parameter 0x0F000000 = 0 in non-volatile memory). See "ID-Chip Support / Stage Replacement" (p. 216) for more information.

- The settling behavior of the axis in closed-loop operation is influenced by the notch filter settings. Set the notch filter(s) **before** you optimize the servo-control parameters (p. 225).
- If you work with the *Piezo Dynamic Tuner* window of PIMikroMove:
 - If you change a parameter value of the E-711/E-712 system by entering a corresponding value: The value is displayed in a blue font until you press Enter on your keyboard. Pressing Enter sends the value to the E-711/E-712 system and changes the font color from blue to black. For fields highlighted by a red background, the parameter values in volatile and non-volatile memory of the E-711/E-712 system differ.
 - When the *Notch Frequency 1* value is set in the *Parameter Settings* panel of the *Piezo Dynamic Tuner* window, the *Servo-Loop I-Term* value can be adjusted automatically in accordance. The adjustment depends on the selection in the *Automatic I-Term calculation* drop down menu.

Default: The I-term is set to a "conservative" value which is calculated with the following formula:

$$I\ term_{conservative} = \frac{P\ term}{0.05 \times 4 \times \pi \times \text{Notch Frequency 1}}$$

Further options:

"Dynamic" I-term value, calculated with the following formula:

$$I\ term_{dynamic} = \frac{0.8 \times P\ term}{0.05 \times 4 \times \pi \times \text{Notch Frequency 1}}$$

"Off", i.e. no automatic I-term calculation.

- The settings for slew rate (*Slew Rate / Velocity* field) and record table rate (*Record Rate* field) can be changed in the *Piezo Dynamic Tuner* window. Entering new values in these fields changes the values of the corresponding parameters in volatile memory: Servo Loop Slew-Rate parameter (ID 0x07000200) or Open Loop Slew-Rate (ID 0x07000201), depending on the current operating mode (open-loop or closed-loop operation); Data Recorder Table Rate (ID 0x16000000). The values are **not** saved or reset when you use the *Save ...* and *Reset ...* buttons in the *Parameter Settings* panel of the *Piezo Dynamic Tuner* window.

16.2.3 Adjusting the Notch Filter(s) in Open-Loop Operation

The corrections by a notch filter only take place in closed-loop operation by default, but can also be enabled for open-loop operation. The appropriate frequency component is reduced in the control value to compensate for undesired resonances in the mechanical system. Adjusting the notch filter frequency can be useful, particularly in the case of very high loads.

For further details, see “Notch Filters”, p. 125.

NOTES

The notch rejection value, which scales the damping done by the notch filter, should always be 0.05. A notch rejection value of 1 deactivates the notch filter.

Before you measure the resonant frequencies as described below, make sure that the notch filters are not enabled in open-loop operation. To do this, check the value of the Enable Notch In Open Loop parameter, ID 0x08000500, for all axes (0 = disable notch filter in open-loop operation; 1 = enable notch filter in open-loop operation). You can do this in the *Servo* parameter groups of the *Device Parameter Configuration* window in PIMikroMove.

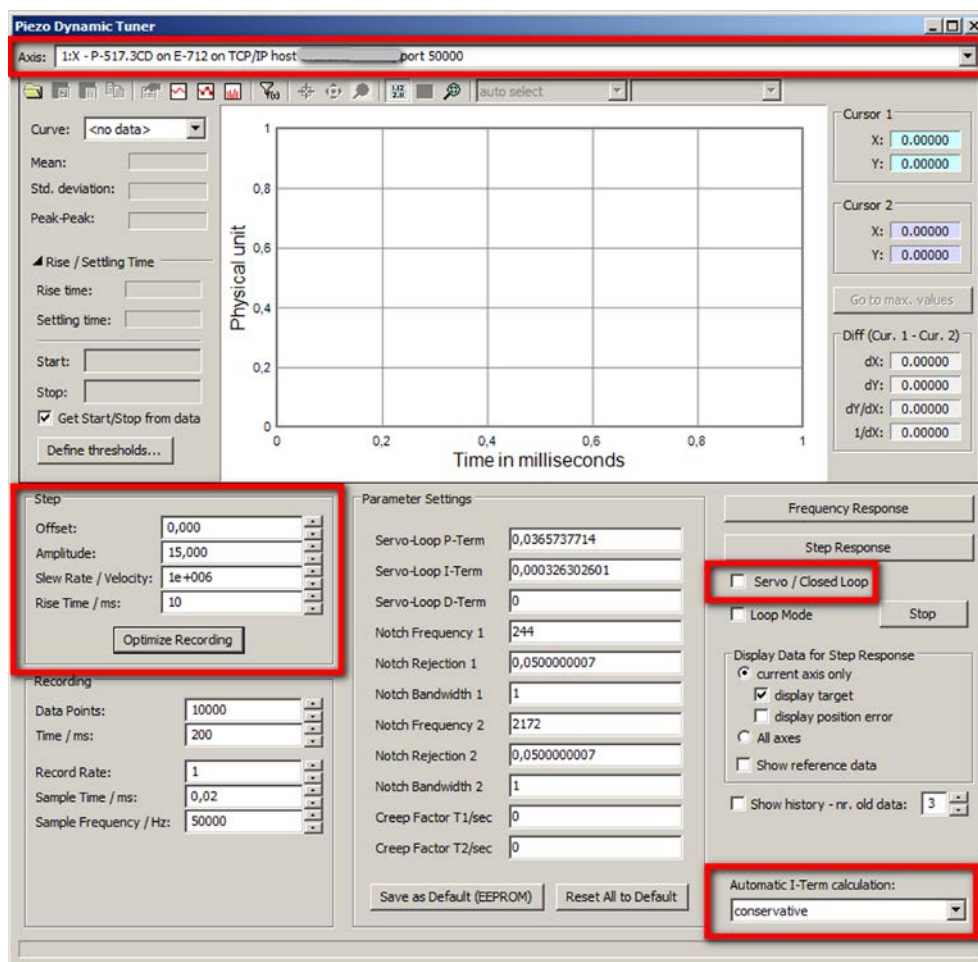
In addition to the measurement described below, you can create a Bode plot: In the PIMikroMove main window, open the *Data Recorder* window via the *E-712... ⇒ Show data recorder ...* menu item. At the bottom of the *Data Recorder* window, enter the *Amplitude* value and click *Estimate* to start the frequency response.

To measure the resonant frequency and adjust the notch filter(s), a frequency response (axis response to an impulse) is recorded in open-loop operation.

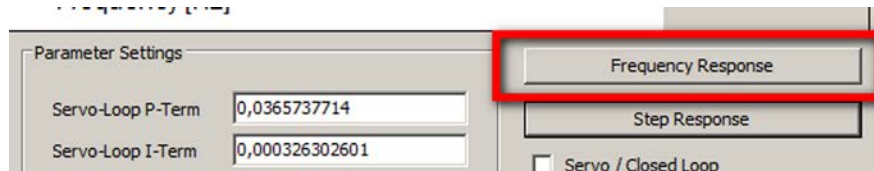
Proceed as follows for each axis:

- 1 Read the “General Notes on Servo-Controller Dynamic Tuning” (p. 219).
- 2 Make sure the stage is mounted in exactly the same way as in the application. The load on the stage is especially important.
- 3 In the main window of PIMikroMove, open the *Piezo Dynamic Tuner* window via the *E-712... ⇒ Dynamic Tuner ...* menu item.
- 4 Configure the frequency response in the *Piezo Dynamic Tuner* window:

- 4.1 Make sure that the correct axis is selected (Axis drop down list).
- 4.2 Enter suitable values for the start value (*Offset:*) and the amplitude (*Amplitude:*) of the impulse in the *Step* panel. The start value should be 0, and the amplitude should be about 10 % of the axis travel range.
- 4.3 Make sure that the axis is in open-loop operation (*Servo / Closed Loop* box is **not** checked).
- 4.4 Select if and how the *Servo-Loop I-Term* is to be adjusted automatically when *Notch Frequency 1* is changed (*Automatic I-Term calculation*).

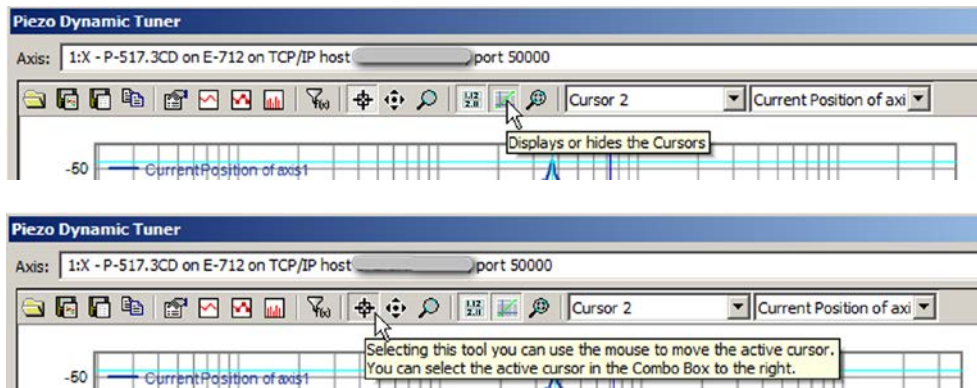


- 5 Perform the frequency response measurement by clicking the *Frequency Response* button in the *Piezo Dynamic Tuner* window.



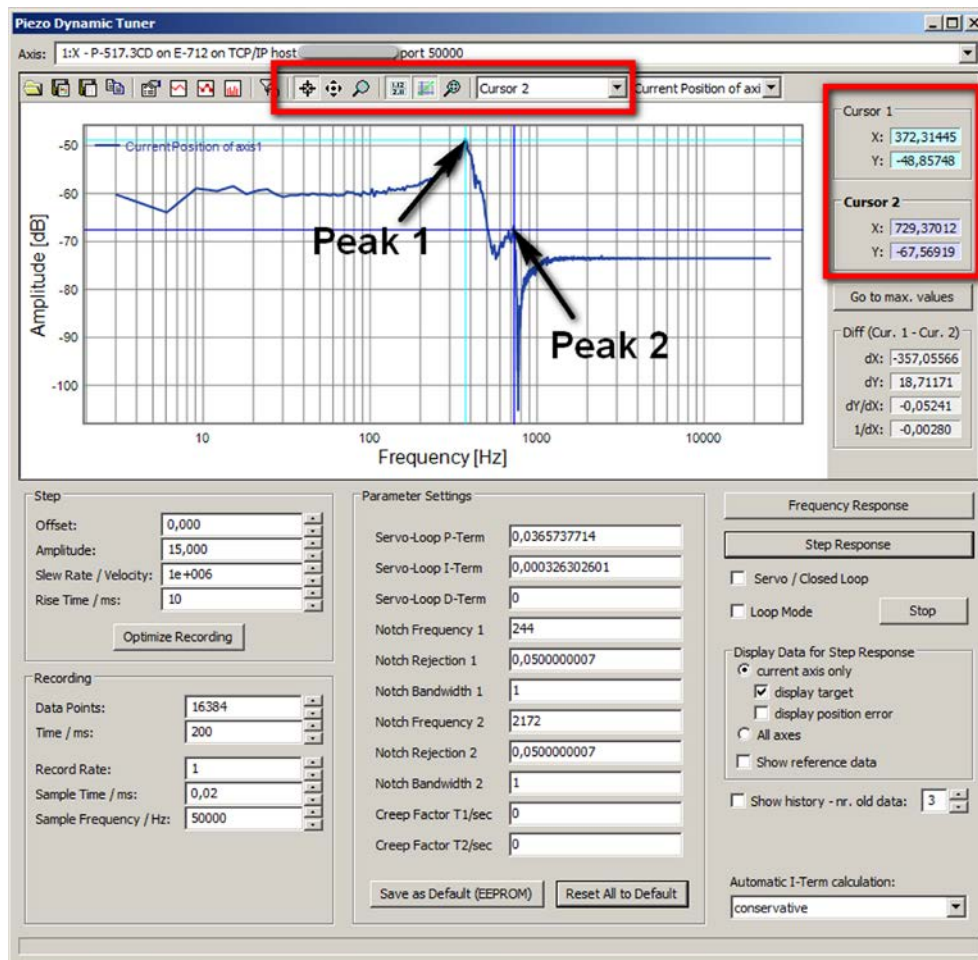
6 Identify the resonant frequency in the *Piezo Dynamic Tuner* window:

6.1 Show cursors and enable cursor motion using the buttons shown in the figures below.



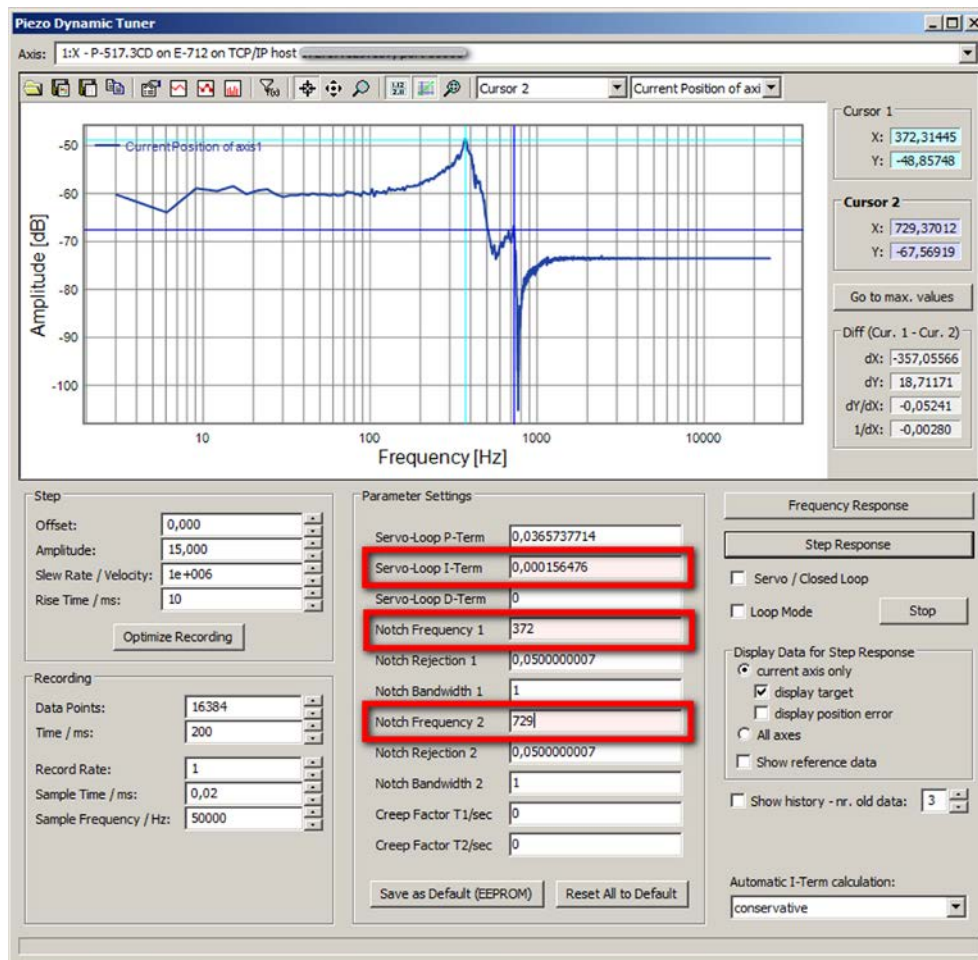
6.2 Identify the resonance peak(s) in the FFT display. To do so, place a cursor on the peak and read out the cursor value which is displayed on the right hand side of the graph. If there is more than one resonance peak, peak 1 is always the one with the lowest frequency.

In the figure below, cursor 1 is at the first resonance peak (372.31445 Hz), and cursor 2 is at the second (next higher) resonance peak (729.37012 Hz).



- 7 If necessary, adjust the notch filter settings in the *Parameter Settings* panel to the measured resonant frequencies (adjustment is necessary if the values significantly differ).
 - 7.1 Enter the frequency value of the first resonance peak in the *Notch Frequency 1* field (in Hz). You can either right-click the field with the mouse and select the value from a menu, or type the value in the field.

 note that depending on the selection for *Automatic I-Term calculation*, the *Servo-Loop I-Term* value is changed too automatically when you change the *Notch Frequency 1* value (for details, see p. 219).
 - 7.2 If you have measured a second resonance peak, enter the frequency value of the second resonance peak in the *Notch Frequency 2* field (in Hz). If the second notch filter is deactivated (rejection value = 1), change the rejection value to 0.05 in the *Notch Rejection 2* field.



8 Save or discard the new settings in the *Parameter Settings* panel:

- If you want to keep the new settings, save them to the non-volatile memory of the E-711/E-712 system by clicking the *Save as Default (EEPROM)* button.
- If you want to discard the new settings and reset the parameter values to their defaults (i.e. to their values from non-volatile memory), click the *Reset All to Default* button.

16.2.4 Checking and Optimizing the Servo-Control Parameters

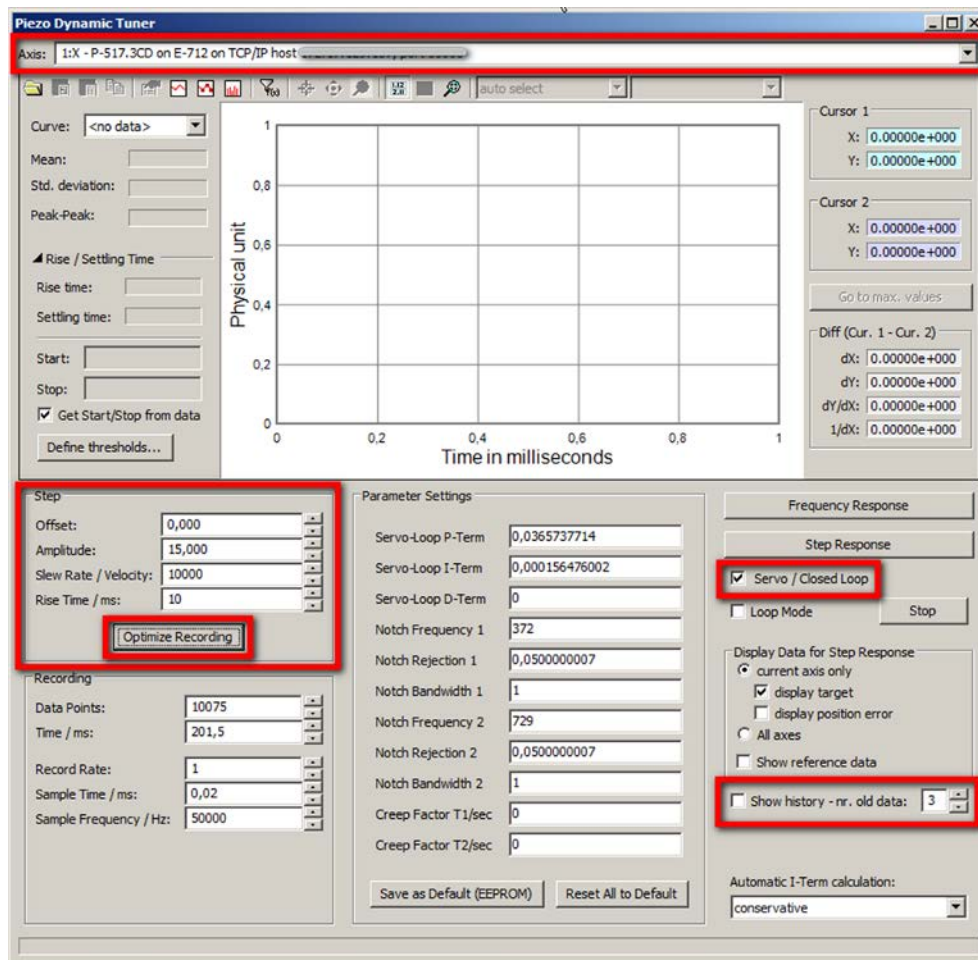
Adjusting the servo-control parameters (P-term, I-term) optimizes the dynamic properties of the system (overshoot and settling time). The optimum settings depends on your application and your requirements.

For further details regarding the PID algorithm, see “Control Algorithms for Closed-Loop Operation”, p. 121.

The optimization of the servo-control parameters is typically done empirically: The response of the axes to a step ("step response") is analyzed under various values in closed-loop operation.

Proceed as follows for each axis:

- 1 Read the "General Notes on Servo-Controller Dynamic Tuning" (p. 219).
- 2 Make sure the stage is mounted in exactly the same way as in the application. The load on the stage is especially important.
- 3 Make sure that the notch filter(s) are properly adjusted. For details, see "Adjusting the Notch Filter(s) in Open-Loop Operation" (p. 221).
- 4 In the main window of PIMikroMove, open the *Piezo Dynamic Tuner* window via the *E-712... ⇒ Dynamic Tuner ...* menu item.
- 5 Configure the step response in the *Piezo Dynamic Tuner* window:
 - 5.1 Make sure that the correct axis is selected (*Axis* drop down list).
 - 5.2 Make sure that the axis is in closed-loop operation (*Servo / Closed Loop* box is checked).
 - 5.3 Enter suitable values for the start value (*Offset:*) and the amplitude (*Amplitude:*) of the step in the *Step* panel. The start value should be 0, and the amplitude should be about 10 % of the axis travel range.
 - 5.4 By clicking the *Optimize Recording* button in the *Step* panel, optimize the number of data recorder points that will be read from the controller when the step response has been performed.
 - 5.5 If you want to compare the results of multiple step response measurements, check the *Show history* box and select the number of old recordings to be displayed.

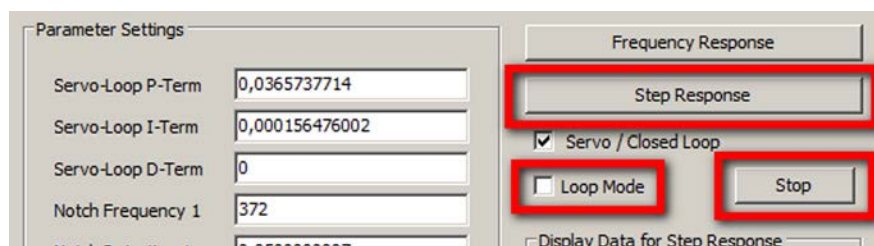


6 Perform and analyze the step response measurement in the *Piezo Dynamic Tuner* window:

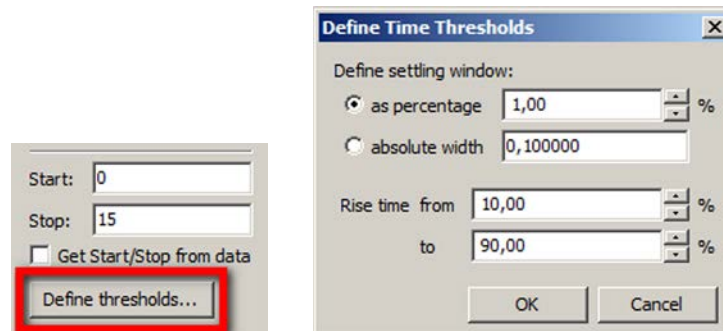
6.1 Optional: Check the *Loop Mode* box to move the axis in a permanent loop.

The loop mode is useful if you want to do the adjustment of the servo-control parameters during the motion. (The loop motion can be stopped at any time by clicking the *Stop* button.)

6.2 Start the step response by clicking the *Step Response* button.

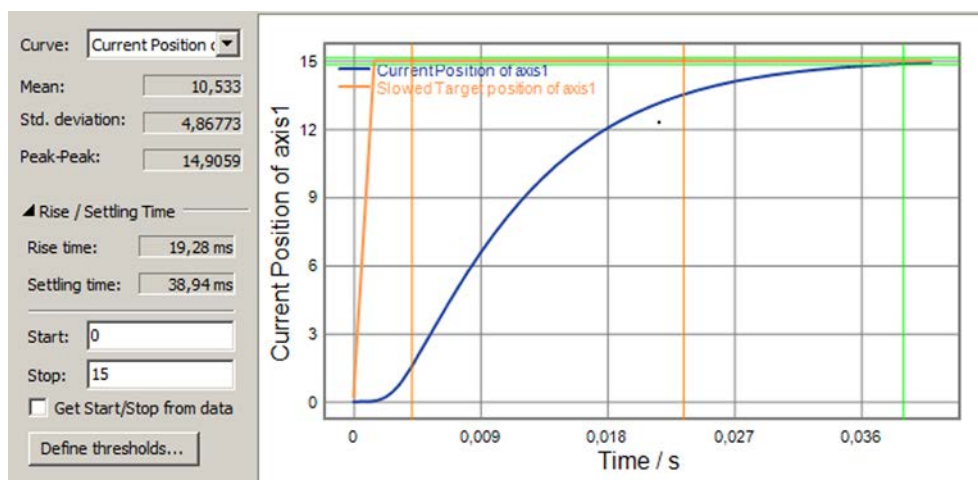


- 6.3 Optional: Click the *Define thresholds...* button to open the Define Time Thresholds window. In the *Define Time Thresholds* window, you can adjust the thresholds which are used by the *Piezo Dynamic Tuner* window to calculate and display the rise time and settling time of the axis, based on the recorded step response measurement.

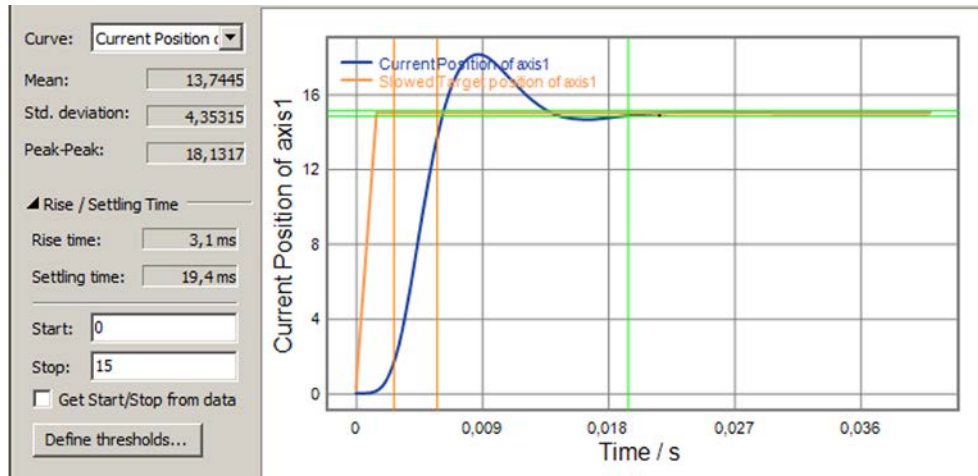


- 6.4 Check the step response result and compare it with the examples shown in the figures below.

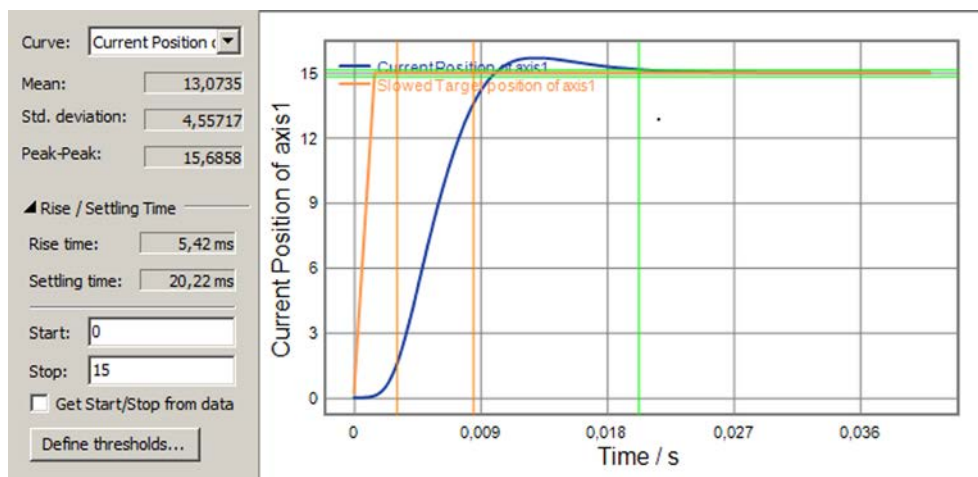
The rise rate of the step response is very low in the figure below. This means that the P term is too low and has to be increased.



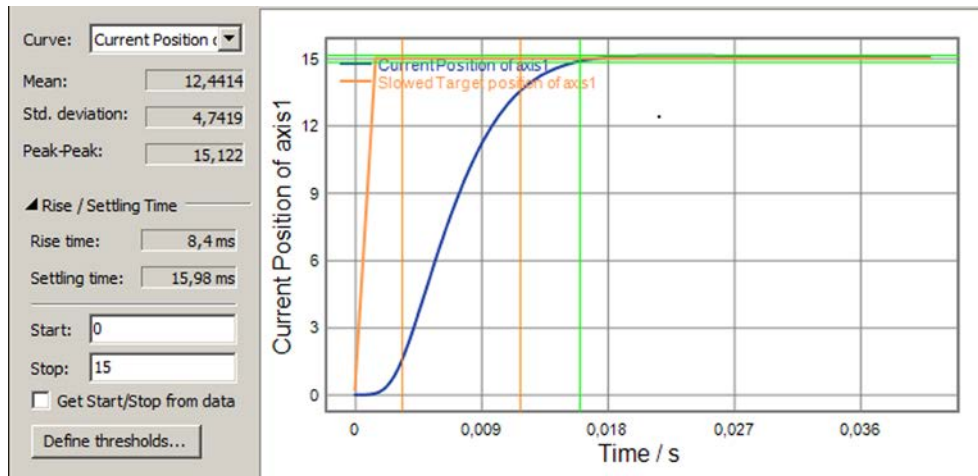
The figure below shows a step response with a high overshoot which means that the P term is too high and has to be decreased.



The figure below shows a step response with a small overshoot which means that the P term is still too high and has to be decreased.



The result of the step response is satisfactory when there is minimum overshoot, and the settling time is not too long, as in the figure below. No changes are required for the servo-control parameters.



7 Save or discard the new settings in the *Parameter Settings* panel:

- If you want to keep the new settings, save them to the non-volatile memory of the E-711/E-712 system by clicking the *Save as Default (EEPROM)* button.
- If you want to discard the new settings and reset the parameter values to their defaults (i.e. to their values from non-volatile memory), click the *Reset All to Default* button.

17 Controller Parameters

17.1 Parameter Handling

To adapt the E-711/E-712 system to your application, you can modify parameter values. The parameters available depend on the controller firmware. With HPA? you can obtain a list of all available parameters with information about each (e.g. short descriptions). The volatile and non-volatile memory parameter values can be read with the SPA? or SEP? commands, respectively.

Note that many parameters are "protected" by higher command levels, as indicated in the "Command Level" column in the "Parameter Overview" table (p. 233). By going to command level 1 using the CCL command, it is possible to change level-1 parameters. Parameters with level 2 or higher are reserved for service personnel.

Using the "general" modification commands SPA, RPA, SEP and WPA, all parameters for which the currently active command level has write permission can be changed in volatile memory (SPA, RPA) or in non-volatile memory (SEP, WPA). It is recommended that any modifications be first made with SPA, and when the controller runs well, saved using WPA. Do not change the current interface settings with SPA—except of the baud rate—because it will not be possible to maintain communication afterwards.

In addition to the "general" modification commands, there are commands which change certain specific parameters. All the commands listed below, except of IFS, change the parameter value only in volatile memory, and WPA must be used to save the value to non-volatile memory. IFS changes and saves the interface parameters directly in non-volatile memory only.

AOS (analog input offset)
ATZ (Autozero Matched Offset as the autozero result)
CTO (trigger output configuration)
DRC (data source and record option)
DPO (internal DDL processing parameters)
IFC (baud rate for RS-232 serial connection)
RTR (record table rate)
VEL (Servo Loop Slew-Rate)
WOS (wave generator output offset)
WTR (wave table rate)

IFS (interface parameters in non-volatile memory)

When a TCP/IP connection is established at which the IP address is obtained from DHCP server, this address will automatically be written to the IP Address parameter (ID 0x11000600) in the E-711/E-712 systems volatile memory. Provided that the current command level is 1 (see CCL command) and you send WPA 100, the new address setting will be saved to volatile

memory as the new default value (together with all other current valid parameter values with level-1 write access).

CAUTION

Incorrect parameter values may lead to improper operation or damage to your hardware. Be careful when changing parameters.

It is strongly recommended to save the parameter values of the E-711/E-712 system to a file on the host PC before you make any changes. This way the original settings can be restored if the new parameter settings will not prove satisfactory. To save the parameter values and to load them back to the E-711/E-712 system, use the *Device Parameter Configuration* window of PIMikroMove. See "Create Backup File for Controller Parameters" (p. 86) for more information.

NOTE

The *Device Parameter Configuration* window of PIMikroMove gives access to parameter values in a more convenient way. Use this window to check/edit the individual parameters. See the PIMikroMove manual for more information.

Each parameter refers to one of the following items (see the "Item Type Concerned" column in the table below):

- Whole system (controller)
- Hardware modules of the controller
- Firmware units
- Logical axes
- PiezoWalk channels
- Input or output signal channels
- Digital output lines
- Data recorder tables

The "Max. No. of Items" column shows the maximum number of items for which the parameter is used. Example: With an E-712.6CD, for parameter 0x02000200 the number of items is "6" which means that this parameter has different values for each of the 6 input signal channels. For parameters which refer to the whole system the maximum number of items is always 1. See "Accessible Items and Their Identifiers" (p. 104) for the item identifiers to use with SPA, SEP or WPA when changing/saving parameter values or when asking for parameter values with the SPA? or SEP? commands.

Values stored in non-volatile memory are power-on defaults, so that the system can be used in the desired way immediately. Note that PI records the data files of every E-711/E-712 system calibrated at the factory for easy restoration of original settings should that ever be necessary.

When the stage is equipped with an ID-chip (located in the stage connector) and connected to the controller for the first time, the values for stage-related parameters will be written from the ID-chip to the volatile and non-volatile memory of the E-711/E-712 system upon controller power-on. You cannot overwrite the parameters in the ID-chip (this can only be done by PI). See "ID-Chip Support / Stage Replacement" (p. 216) for more information. The parameters stored in the ID-chip are marked in the "Notes" column in the table below.

17.2 Parameter Overview

See "Parameter Handling" (p. 231) for the meaning of the individual columns.

The content of the "Max. No. of Items" column is to be interpreted as follows:

- For "Item Type Concerned" = "Input Signal Channel" and "Output Signal Channel", it gives the number of channels for
E-712.3CD / E-712.3CDA
E-712.6CD / E-712.6CDA
E-712.1AM
- For "Item Type Concerned" = "Logical Axis", it gives the number of axes for
E-712.3CD and E-712.3CDA / E-712.6CD and E-712.6CDA
E-712.1AM
- For "Item Type Concerned" = "PiezoWalk Channel", it gives the number of PiezoWalk channels for
E-712.1AM / any other system with PiezoWalk channels, "*" means that the number depends on the configuration of the E-711/E-712 system (no PiezoWalk channels with E-712.3CD, E-712.3CDA, E-712.6CD or E-712.6CDA). A PiezoWalk channel represents a NEXLINE® drive or NEXACT® drive connected to the controller.

- For "Item Type Concerned" = "Hardware Module", it gives the number of slots available in the E-712.R1 chassis. Counting (front view): 1 = leftmost slot , 8 = rightmost slot. The actual number of hardware modules present in the chassis depends on the system configuration, see "Model Survey" (p. 17) for more information.
- For "Item Type Concerned" = "Firmware Unit", it gives the number of different firmware units present in
 E-712.3CD / E-712.3CDA
 E-712.6CD / E-712.6CDA
 E-712.1AM
 An E-712.M1 or .N1 digital computer and interface module has 2 different firmware units installed (CPU and FPGA firmware), and all other modules have one firmware unit installed. All firmware units can be updated separately, see "Firmware Updates" (p. 260) for more information.

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x02000001	1	Input Signal Channel	3 / 7 6 / 10 1	INT	Sensor Type	0 = reserved 1 = absolute (e.g. analog sensor) 2 = incremental sin/cos (analog) 3 = incremental A/B (digital) 4 = absolute BISS/C 5 = absolute ENDAT
0x02000100	1	Input Signal Channel	3 / 7 6 / 10 1	INT	Sensor Range factor	ID-chip The available factors depend on the type of sensor module.
0x02000102	1	Input Signal Channel	3 / 7 6 / 10 1	INT	Sensor Offset factor	ID-chip
0x02000103	1	Input Signal Channel	3 / 7 6 / 10 1	INT	Sensor Cable Compensation	ID-chip

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x02000200	1	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Mech. Correction 1	ID-chip This value is set as new sensor position after a referencing motion started with FRF.
0x02000300	1	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Mech. Correction 2	ID-chip
0x02000400	1	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Mech. Correction 3	ID-chip
0x02000500	1	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Mech. Correction 4	ID-chip
0x02000600	1	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Mech. Correction 5	ID-chip
0x02000a00	1	Input Signal Channel	3 / 7 6 / 10 1	INT	Sensor Reference Mode	Determines how the sensor is to be referenced: 1 = no reference (absolute sensors, no referencing required, RON cannot be set to 0) 2 = reference on negative hardstop 3 = reference on positive hardstop 4 = reference on signed reference mark 5 = reference on impulse
0x02001000	1	Input Signal Channel	3 / 7 6 / 10 1	INT	Sensor Lim, Ref Signal Inversion	Determines if the E-711/E-712 system inverts the reference switch and limit switch signals. For details, see “Signals of Reference and Limit Switches” (p. 126).

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x02001900	3	Input Signal Channel	3 / 7 6 / 10 1	INT	Lim, Ref Signals Detectable	Flag that indicates if signals of limit and reference point switches are evaluated by the firmware (does not indicate if switches are actually present) 0 = no evaluation of switch signals 1 = switch signals are evaluated
0x02001B00	1	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Maximum Position Of Sensor	The parameters are only available if the plausibility check is enabled with the Check Sensor Position Plausibility parameter (ID 0x0e001f00) For further details, see "Plausibility Check of Sensor Position" (p. 215).
0x02001C00	1	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Minimum Position Of Sensor	
0x02001D00	1	Input Signal Channel	3 / 7 6 / 10 1	INT	Validate Sensor Position Range	
0x03000100	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 1 1	
0x03000101	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 2 1	
0x03000102	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 3 1	
0x03000103	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 4 1	
0x03000104	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 5 1	
0x03000105	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 6 1	
0x03000106	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 7 1	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x03000107	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 8 1	
0x03000108	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 9 1	
0x03000200	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 1 2	
0x03000201	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 2 2	
0x03000202	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 3 2	
0x03000203	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 4 2	
0x03000204	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 5 2	
0x03000205	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 6 2	
0x03000206	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 7 2	
0x03000207	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 8 2	
0x03000208	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 9 2	
0x03000300	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 1 3	
0x03000301	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 2 3	
0x03000302	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 3 3	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x03000303	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 4 3	
0x03000304	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 5 3	
0x03000305	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 6 3	
0x03000306	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 7 3	
0x03000307	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 8 3	
0x03000308	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 9 3	
0x03000400	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 1 4	
0x03000401	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 2 4	
0x03000402	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 3 4	
0x03000403	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 4 4	
0x03000404	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 5 4	
0x03000405	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 6 4	
0x03000406	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 7 4	
0x03000407	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 8 4	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x03000408	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 9 4	
0x03000500	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 1 5	
0x03000501	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 2 5	
0x03000502	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 3 5	
0x03000503	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 4 5	
0x03000504	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 5 5	
0x03000505	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 6 5	
0x03000506	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 7 5	
0x03000507	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 8 5	
0x03000508	2	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Sensor Elec. Correction 9 5	
0x03003400	1	Input Signal Channel	3 / 7 6 / 10 1	INT	Input Numerical Format	0 = IEEE 32 bit float format 1 = unsigned integer 16 bit 2 = unsigned integer 24 bit 3 = unsigned integer 32 bit 4 = signed integer 16 bit 5 = signed integer 24 bit 6 = signed integer 32 bit
0x05000000	1	Input Signal Channel	3 / 7 6 / 10 1	INT	Digital Filter Type	ID-chip

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x05000001	1	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	Digital Filter Bandwidth	ID-chip
0x05000002	1	Input Signal Channel	3 / 7 6 / 10 1	INT	Digital Filter Order	ID-chip
0x05000101	1	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	User Filter Param. 1	
0x05000102	1	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	User Filter Param. 2	
0x05000103	1	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	User Filter Param. 3	
0x05000104	1	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	User Filter Param. 4	
0x05000105	1	Input Signal Channel	3 / 7 6 / 10 1	FLOAT	User Filter Param. 5	
0x06000500	1	Logical Axis	3 / 6 1	INT	ADC Channel for Target	
0x06000501	1	Logical Axis	3 / 6 1	FLOAT	Analog Target Offset	
0x06010000	1	Logical Axis	3 / 6 1	FLOAT	Target Generator Maximum Acceleration	For details, see “Profile Generator” (p. 120).
0x06010100	1	Logical Axis	3 / 6 1	FLOAT	Target Generator Maximum Jerk	
0x06010200	1	Logical Axis	3 / 6 1	FLOAT	Time Constant Of Target Generator	
0x06010300	1	Logical Axis	3 / 6 1	INT	Target Generator Enable	
0x06010400	3	Logical Axis	3 / 6 1	FLOAT	Target Generator Maximum Velocity	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes	
0x07000000	1	Logical Axis	3 / 6 1	FLOAT	Range Limit min	ID-chip	
0x07000001	1	Logical Axis	3 / 6 1	FLOAT	Range Limit max	ID-chip	
0x07000200	1	Logical Axis	3 / 6 1	FLOAT	Servo Loop Slew-Rate	ID-chip	
0x07000201	1	Logical Axis	3 / 6 1	FLOAT	Open Loop Slew-Rate		
0x07000204	1	PiezoWalk Channel	1 / *	FLOAT	PiezoWalk Open-Loop Velocity	[µm/s] Used to calculate the piezo voltage slew rate, affects the duration of the relax procedures for PiezoWalk channels	
0x07000300	1	Logical Axis	3 / 6 1	FLOAT	Servo-loop P-Term	ID-chip	For details, see “Control Algorithms for Closed-Loop Operation” (p. 121).
0x07000301	1	Logical Axis	3 / 6 1	FLOAT	Servo-loop I-Term	ID-chip	
0x07000302	1	Logical Axis	3 / 6 1	FLOAT	Servo-loop D-Term	ID-chip	
0x07000307	1	Logical Axis	3 / 6 1	FLOAT	Servo-Loop P-Term for Velocity	For details, see “Control Algorithms for Closed-Loop Operation” (p. 121).	
0x07000308	1	Logical Axis	3 / 6 1	FLOAT	Servo-Loop I-Term for Velocity		
0x07000309	1	Logical Axis	3 / 6 1	FLOAT	Servo-Loop D-Term for Velocity		
0x07000500	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 1	ID-chip	
0x07000501	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 2	ID-chip	
0x07000502	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 3	ID-chip	
0x07000503	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 4	ID-chip	
0x07000504	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 5	ID-chip	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x07000505	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 6	ID-chip
0x07000506	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 7	
0x07000507	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 8	
0x07000508	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 9	
0x07000509	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 10	
0x0700050a	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 11	
0x0700050b	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 12	
0x0700050c	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 13	
0x0700050d	1	Logical Axis	3 / 6 1	FLOAT	Position from Sensor 14	
0x07000600	1	Logical Axis	3 / 6 1	CHAR	Axis Name	ID-chip
0x07000601	2	Logical Axis	3 / 6 1	CHAR	Axis Unit	ID-chip
0x07000800	1	Logical Axis	3 / 6 1	INT	Power Up Servo ON Enable	
0x07000802	1	Logical Axis	3 / 6 1	INT	Power Up AutoZero Enable	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x07000900	1	Logical Axis	3 / 6 1	FLOAT	ON Target Tolerance	ID-chip [μm] Gives the settling window. The settling window is centered around the axis target position. The on-target status of an axis is true when the current axis position is inside the settling window and stays there for at least the settling time (Settling Time parameter, ID 0x07000901). See the separate E712T0010 Technical Note for the special functionality of this parameter with PiezoWalk systems.
0x07000901	1	Logical Axis	3 / 6 1	FLOAT	Settling Time	[s]
0x07000a00	1	Logical Axis	3 / 6 1	FLOAT	AutoZero Low Voltage	ID-chip
0x07000a01	1	Logical Axis	3 / 6 1	FLOAT	AutoZero High Voltage	ID-chip
0x07000c01	1	Logical Axis	3 / 6 1	FLOAT	Default voltage	ID-chip
0x07001005	1	Logical Axis	3 / 6 1	FLOAT	Position Report Scaling	For details, see “Using the Analog Output” (p. 165).
0x07001006	1	Logical Axis	3 / 6 1	FLOAT	Position Report Offset	
0x07001500	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Forward Correction 1	Coefficients of step forward linearization polynomial of piezo-voltage-to-position-calculation. Coefficients are used to compensate for the piezo nonlinearity. If 0x07001500 is set to 1
0x07001501	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Forward Correction 2	
0x07001502	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Forward Correction 3	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x07001503	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Forward Correction 4	and all other coefficients are set to 0, linearization is disabled.
0x07001504	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Forward Correction 5	
0x07001600	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Backward Correction 1	Coefficients of step backward linearization polynomial of piezo-voltage-to-position-calculation. Coefficients are used to compensate for the piezo nonlinearity. If 0x07001600 is set to 1 and all other coefficients are set to 0, linearization is disabled.
0x07001601	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Backward Correction 2	
0x07001602	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Backward Correction 3	
0x07001603	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Backward Correction 4	
0x07001604	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Backward Correction 5	
0x07001900	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Clamp-Free Voltage	Used for nanostepping mode only. See the separate E712T0010 Technical Note for details.
0x07001a00	1	PiezoWalk Channel	1 / *	INT	PiezoWalk Driving Mode	Selects the driving mode: 0 = Full-step mode 1 = Nanostepping mode 2 = Analog mode (if selected, no stepping motion is possible)

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x07011700	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Step Size	[μm] The feed caused by one step cycle, with maximum amplitude of the shearing voltages ($V_{D\text{max}} - V_{D\text{min}}$). Can be determined as follows: Perform n steps using OSM, measure the position difference and divide it by the number of steps.
0x07011800	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Shearing-To-Clamping Ratio	Used for full-step mode only. Must be in the range of 0.01 to 0.5 See the separate E712T0010 Technical Note for details.
0x07011900	1	PiezoWalk Channel	1 / *	FLOAT	NEXLINE Min Step Cycle Time	[s] Minimum time per step cycle. This value will be rounded to a multiple of the sample time. See the separate E712T0010 Technical Note for details.
0x07011b00	1	PiezoWalk Channel	1 / *	FLOAT	PiezoWalk Max Acceleration	[$\mu\text{m}/\text{s}^2$] Maximum acceleration of a NEXLINE® drive.
0x07011d00	1	PiezoWalk Channel	1 / *	INT	PiezoWalk Type	1 = NEXLINE 2 = NEXACT

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x07011e00	1	PiezoWalk Channel	1 / *	FLOAT	NEXACT Drive Weight	Gives the ratio of the shearing voltage amplitude to the clamping voltage amplitude. This ratio influences the feed caused by one step cycle, i.e. the step size. Only used if PiezoWalk Type (ID 0x07011d00) = NEXACT (2).
0x07011f00	2	PiezoWalk Channel	1 / *	INT	PiezoWalk Driven Cycles	Counter for the electrical cycles performed by a PiezoWalk drive.
0x07012400	2	PiezoWalk Channel	1 / *	INT	To PiezoWalk Linked Sensor Chan	Gives the ID of an input signal channel with limit switches that is assigned to the PiezoWalk channel. 0 (default) = no channel assigned
0x07012500	1	PiezoWalk Channel	1 / *	INT	To PiezoWalk Linked Brake Output Chan	Gives the ID of an output signal channel with brake that is assigned to the PiezoWalk channel. The brake of the assigned output signal channel is open as long as the PiezoWalk channel is not in the relaxed state (opened when the servo mode is switched on, closed when the servo mode is switched off). 0 (default) = no channel assigned
0x07012600	1	PiezoWalk Channel	1 / *	FLOAT	Time To Open Brake	Time to open the brake assigned to a PiezoWalk channel, in s To open the brake, for the assigned output signal channel first the lower limit for the piezo voltage output is set, and afterwards the upper limit.

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x07012700	1	PiezoWalk Channel	1 / *	FLOAT	Time To Close Brake	Time to close the brake assigned to a PiezoWalk channel, in s To close the brake, for the assigned output signal channel first the lower limit for the piezo voltage output is set, and afterwards the zero voltage value.
0x07022000	3	Logical Axis	3 / 6 1	INT	Open-Loop Control Mode	0 = position control 1 = velocity control 2 = velocity control for PiezoWalk drives, use OSM and SVA in open-loop operation
0x07030100	1	Logical Axis	3 / 6 1	INT	Closed-Loop Control Mode	Gives the control modes to be used for closed-loop operation. For details, see “Control Algorithms for Closed-Loop Operation” (p. 121).
0x07030200	1	Logical Axis	3 / 6 1	INT	Servo Instability Detection	Configuration of servo instability detection.
0x07030202	3	Logical Axis	3 / 6 1	INT	Servo Instability Indicator	
0x07030300	1	Logical Axis	3 / 6 1	FLOAT	Referencing Velocity	Velocity for reference moves. For further details, see “How to Reference an Axis” (p. 94).
0x07030600	1	Logical Axis	3 / 6 1	FLOAT	Feed Forward Gain	For details, see “Feedforward” (p. 124).
0x07030900	1	Logical Axis	3 / 6 1	INT	Feed Forward Input Channel Index	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes	
0x07030a00	1	Logical Axis	3 / 6 1	FLOAT	On-Target Tolerance To Fix I-Term	[μm] Gives a window in which the I-term of the control algorithm stays fixed. For details, see “Control Algorithms for Closed-Loop Operation” (p. 121).	
0x07030b00	1	Logical Axis	3 / 6 1	FLOAT	HLT Deceleration	Deceleration applied when HLT is sent for an axis, in phys. unit/s ²	
0x07030c00	1	Logical Axis	3 / 6 1	INT	ID-Chip Chan. to Axis Map Ctrl	Mapping of axis IDs when stages with ID chip are connected to the E-711/E-712 system. For details, see “ID-Chip Support / Stage Replacement” (p. 216).	
0x07030d00	1	Logical Axis	3 / 6 1	INT	Zeroing Control Value If I-Term Is Fixed	Determines the control value when the current position is in the window given by parameter 0x07030a00. For details, see “Control Algorithms for Closed-Loop Operation” (p. 121).	
0x08000100	1	Logical Axis	3 / 6 1	FLOAT	Notch frequency 1	ID-chip	For details, see “Notch Filters”, p. 125.
0x08000101	1	Logical Axis	3 / 6 1	FLOAT	Notch frequency 2	ID-chip	
0x08000200	1	Logical Axis	3 / 6 1	FLOAT	Notch Rejection 1	ID-chip	
0x08000201	1	Logical Axis	3 / 6 1	FLOAT	Notch Rejection 2	ID-chip	
0x08000300	1	Logical Axis	3 / 6 1	FLOAT	Notch Bandwidth 1	ID-chip	
0x08000301	1	Logical Axis	3 / 6 1	FLOAT	Notch Bandwidth 2	ID-chip	
0x08000400	1	Logical Axis	3 / 6 1	FLOAT	Creep factor T1/sec	ID-chip	
0x08000401	1	Logical Axis	3 / 6 1	FLOAT	Creep factor T2/sec	ID-chip	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x08000500	1	Logical Axis	3 / 6 1	INT	Enable Notch in Open Loop	For details, see “Notch Filters”, p. 125.
0x08000600	1	Logical Axis	3 / 6 1	INT	Notch Filter Calculation Method	
0x09000000	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 1	ID-chip
0x09000001	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 2	ID-chip
0x09000002	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 3	ID-chip
0x09000003	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 4	ID-chip
0x09000004	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 5	ID-chip
0x09000005	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 6	ID-chip
0x09000006	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 7	ID-chip
0x09000007	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 8	ID-chip
0x09000008	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 9	
0x09000009	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 10	
0x0900000a	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 11	
0x0900000b	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 12	
0x0900000c	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 13	
0x0900000d	1	Logical Axis	3 / 6 1	FLOAT	Driving Factor of Piezo 14	
0x09000100	1	PiezoWalk Channel	1 / *	INT	PiezoWalk Shear Channel 1	Gives the piezo amplifier channel to be used for the first pair of shearing segments of the NEXLINE® or NEXACT® drive

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x09000101	1	PiezoWalk Channel	1 / *	INT	PiezoWalk Shear Channel 2	Gives the piezo amplifier channel to be used for the second pair of shearing segments of the NEXLINE® or NEXACT® drive
0x09000200	1	PiezoWalk Channel	1 / *	INT	PiezoWalk Clamp Channel 1	Gives the piezo amplifier channel to be used for the first pair of clamping segments of the NEXLINE® or NEXACT® drive
0x09000201	1	PiezoWalk Channel	1 / *	INT	PiezoWalk Clamp Channel 2	Gives the piezo amplifier channel to be used for the second pair of clamping segments of the NEXLINE® or NEXACT® drive
0x0a000003	1	Output Signal Channel	4 / 8 8 / 12 4	INT	Select Output type	For details, see “Using the Analog Output” (p. 165)
0x0a000004	1	Output Signal Channel	4 / 8 8 / 12 4	INT	Select Output index	
0x0b000007	2	Output Signal Channel	4 / 8 8 / 12 4	FLOAT	Min Output Voltage of Amplifier	
0x0b000008	2	Output Signal Channel	4 / 8 8 / 12 4	FLOAT	Max Output Voltage of Amplifier	
0x0b000009	2	Output Signal Channel	4 / 8 8 / 12 4	FLOAT	Voltage of Amplifier with zero to Dac	
0x0b00000a	1	Output Signal Channel	4 / 8 8 / 12 4	FLOAT	Voltage Offset of Amplifier	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x0b000500	1	Output Signal Channel	4 / 8 8 / 12 4	INT	Output Numerical Format	0 = IEEE 32 bit float format 1 = unsigned integer 16 bit 2 = unsigned integer 24 bit 3 = unsigned integer 32 bit 4 = signed integer 16 bit (a zero is zero) 5 = signed integer 24 bit (a zero is zero) 6 = signed integer 32 bit (a zero is zero) 7 = raw format (used for driver/DAC issues)
0x0b000800	1	Output Signal Channel	4 / 8 8 / 12 4	INT	Input Index To Feed Back	Can be used to test SPI channels and data transfer: The parameter value gives the ID of an input signal channel. This input signal channel is connected to the output signal channel to provide feedback on the output.
0x0B000900	1 / 3	Output Signal Channel	4 / 8 8 / 12 4	FLOAT	Max Output Current (A)	For systems that support DC motors (E-711.C82 motor driver module is present), The command level for write access is 1 for channels that belong to the E-711.C82 motor driver module(s), and 3 for all other channels.
0x0c000000	1	Output Signal Channel	4 / 8 8 / 12 4	FLOAT	Soft Voltage Low Limit	ID-chip Gives the lower limit for the piezo voltage output
0x0c000001	1	Output Signal Channel	4 / 8 8 / 12 4	FLOAT	Soft Voltage High Limit	ID-chip Gives the upper limit for the piezo voltage output.
0x0d000000	2	System	1	CHAR	Device S/N	
0x0d000100	2	Hardware Module	8	CHAR	Hardware S/N	
0x0d000200	2	Hardware Module	8	CHAR	Hardware Name	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x0e000100	2	System	1	FLOAT	Sensor Sampling Time	
0x0e000200	3	System	1	FLOAT	Servo Update Time	
0x0e000301	1	System	1	INT	Disable Error 10	This parameter can be used to avoid that error code 10 is set when axes are stopped with the STP, #24 or HLT commands. 0 = OFF (Error code 10 is set.) 1 = ON (Error code 10 is not set.)
0x0e000400	1	System	1	INT	DDL license	
0x0e000401	3	System	1	INT	DDL license valid	
0x0e000b00	3	System	1	INT	Number of input channels	
0x0e000b01	3	System	1	INT	Number of output channels	
0x0e000b02	2	System	1	INT	Number of system axes	
0x0e000b03	3	System	1	INT	Number of sensor channels	
0x0e000b04	3	System	1	INT	Number of piezo channels	
0x0e000b05	3	System	1	INT	Number of trigger outputs	
0x0e000b06	3	System	1	INT	Number Of PiezoWalk Channels	Maximum possible number of PiezoWalk channels.
0x0e000b07	3	System	1	INT	Number of conf. PiezoWalk chan.	Number of PiezoWalk channels which are actually configured for use.

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x0e000e00	1	Logical Axis	3 / 6 1	INT	Adv. Piezo Control License	Advanced Piezo Control (APC) license number, must be valid before APC can be used. Advanced Piezo Control is an alternative control algorithm for closed-loop operation of piezo actuator systems. The APC license must be expressly ordered (order# E-712.U1). The Closed-Loop Control Mode parameter (ID 0x07030100) must be set to 2. See Technical Note E712T0007 for more information.
0x0e000f00	3	Logical Axis	3 / 6 1	INT	Adv. Piezo Control License-Valid Flag	Check if APC license activation was successful
0x0e001100	1	System	1	INT	Digital IO Synchronisation	Enables/disables synchronization of digital I/Os: 1 = enabled 0 = disabled
0x0e001200	1	System	1	INT	PCI Busmaster DMA Enable	Enables/disables PCI bus mastering DMA for reading sensors and writing amplifiers 1 = enabled 0 = disabled
0x0e001300	1	System	1	INT	Number Of SPI Inputs	Number of additional input signal channels and output signal channels to be provided by the E-711/E-712 system for the SPI slave interface. For details, see "Accessible Items and Their Identifiers" (p. 104).
0x0e001400	1	System	1	INT	Number Of SPI Outputs	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x0e001500	1	System	1	INT	Reboot on DIO Input	The value of the parameter enables/disables the Reset input on pin 19 of the MDR20 socket (details see p. 285).
0x0e001d00	1	System	1	INT	Trigger Input Filter Enable	The value of the parameter enables/disables a filter for the digital trigger input as follows: 0 = OFF: Filter is disabled. Processing of the trigger input is done immediately. 1 = ON: Filter is enabled to reduce the influence of noise at the trigger input line. Default setting.
0x0e001e00	1	System	1	INT	Disconnect Analog Target Input When Stopping	Determines the behaviour when the analog input is used as a control source. See p. 158 for details.
0x0e001f00	1	System	1	INT	Check Sensor Position Plausibility	Enables oder disables plausibility check of sensor position. For further details, see "Plausibility Check of Sensor Position" (p. 215).
0x0e002000	1	System	1	INT	Move To Last Commanded Position	Determines the target positions to be set when switching on the servo mode: 0 = The target positions are set to the current positions of the axes (default). 1= The target positions are set to the last commanded target positions for the axes. This can cause an abrupt motion.

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x0e002200	1	System	1	INT	Use Autofocus Functions	Enables or disables the autofocus functionality. 0 = Autofocus functionality disabled 1 = Autofocus functionality enabled. Additional commands and parameters are available, see E712T0017 Technical Note for details.
0x0f000000	1	Input Signal Channel	3 / 7 6 / 10 1	INT	Power Up Read ID-Chip	For details, see “ID-Chip Support / Stage Replacement” (p. 216).
0x0f000100	1	Input Signal Channel	3 / 7 6 / 10 1	CHAR	Stage Type	ID-chip
0x0f000200	1	Input Signal Channel	3 / 7 6 / 10 1	CHAR	Stage Serial Number	ID-chip
0x0f000300	1	Input Signal Channel	3 / 7 6 / 10 1	CHAR	Stage assembly Date	ID-chip
0x11000400	1	System	1	INT	Uart Baudrate	For details, see “Communication” (p. 128).
0x11000600	1	System	1	CHAR	IP-Address	
0x11000700	1	System	1	CHAR	IP-Mask	
0x11000800	1	System	1	INT	IP-Configuration	
0x11000b00	2	System	1	CHAR	MAC-Address	
0x11000c00	1	System	1	INT	Max. TCP/IP Connections	
0x13000004	3	System	1	INT	Max Wave Points	For details, see “Wave Generator” (p. 172).
0x13000109	1	System	1	INT	Wave Generator Table Rate	
0x1300010a	3	System	1	INT	Number of Waves	
0x1300010b	1	Logical Axis	3 / 6 1	FLOAT	Wave Offset	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x13000202	1	System	1	INT	Wave Multi Start By Trigger	
0x14000001	1	Logical Axis	3 / 6 1	INT	DDL repeat number	For details, see “Dynamic Digital Linearization (DDL)” (p. 189).
0x14000006	1	Logical Axis	3 / 6 1	FLOAT	Time Delay Max	
0x14000007	1	Logical Axis	3 / 6 1	FLOAT	Time Delay Min	
0x14000008	1	Logical Axis	3 / 6 1	INT	Time Delay Change Rule	
0x1400000a	1	Logical Axis	3 / 6 1	INT	DDL Zero Gain Number	
0x1400000b	3	System	1	INT	Max DDL Points	
0x16000000	1	System	1	INT	Data Recorder Table Rate	For details, see “Data Recording” (p. 138).
0x16000100	3	System	1	INT	Max Number of Data Recorder Channels	
0x16000200	3	System	1	INT	Data Recorder Max Points	
0x16000300	1	System	1	INT	Data Recorder Chan Number	
0x16000700	1	Data recorder table	12	INT	DRC Data Source	
0x16000701	1	Data recorder table	12	INT	DRC Record Option	
0x18000201	1	Digital Output Line	7	FLOAT	CTO Trigger Step	For details, see “Configuring Trigger Output” (p. 142).
0x18000202	1	Digital Output Line	7	INT	CTO Axis	
0x18000203	1	Digital Output Line	7	INT	CTO Trigger Mode	
0x18000205	1	Digital Output Line	7	FLOAT	CTO Min. Threshold	
0x18000206	1	Digital Output Line	7	FLOAT	CTO Max. Threshold	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0x18000207	1	Digital Output Line	7	INT	CTO Polarity	
0x18000208	1	Digital Output Line	7	FLOAT	CTO Start Threshold	
0x18000209	1	Digital Output Line	7	FLOAT	CTO Stop Threshold	
0x1a00000n	1	System	1	FLOAT	Max. Position Error to Stop <i>n</i>	Safety shutdown Configuration and activation/deactivation of the safety shutdown functionality. For details, see “Safety Shutdown” (p. 211).
0x1a000100	1	System	1	INT	Number of Trigger Input to Stop	
0x1a00020n	1	System	1	FLOAT	Absolute Input Value to Stop <i>n</i>	
0x1a00030n	1	System	1	INT	Usage of Limit Switch <i>n</i>	
0x1a00040n	1	System	1	INT	Usage of Position Error to Stop <i>n</i>	
0x1a00050n	1	System	1	INT	Observe Sensor Signal <i>n</i>	
0x1a000600	1	System	1	INT	Safety Shutdown Activation Register	
0xffff0001	2	Firmware Unit	4 / 5 6 / 7 3	INT	firmware mark	
0xffff0002	2	Firmware Unit	4 / 5 6 / 7 3	INT	firmware CRC	
0xffff0003	2	Firmware Unit	4 / 5 6 / 7 3	INT	firmware desc. CRC	
0xffff0004	2	Firmware Unit	4 / 5 6 / 7 3	INT	firmware desc. version	
0xffff0006	2	Firmware Unit	4 / 5 6 / 7 3	CHAR	firmware matchcode	

Parameter ID (hexa-decimal)	Command Level for write access	Item Type Concerned	Max. No. of Items	Data Type	Parameter Description	Notes
0xffff0007	2	Firmware Unit	4 / 5 6 / 7 3	CHAR	hardware matchcode	
0xffff0008	2	Firmware Unit	4 / 5 6 / 7 3	INT	firmware version	
0xffff000b	2	Firmware Unit	4 / 5 6 / 7 3	INT	firmware max. size	
0xffff000c	2	Firmware Unit	4 / 5 6 / 7 3	CHAR	firmware device	
0xffff000d	2	Firmware Unit	4 / 5 6 / 7 3	CHAR	short description of firmware	
0xffff000e	2	Firmware Unit	4 / 5 6 / 7 3	CHAR	date of firmware	
0xffff000f	2	Firmware Unit	4 / 5 6 / 7 3	CHAR	firmware developer	
0xffff0010	2	Firmware Unit	4 / 5 6 / 7 3	INT	firmware length	
0xffff0011	2	Firmware Unit	4 / 5 6 / 7 3	INT	firmware compatibility	
0xffff0012	2	Firmware Unit	4 / 5 6 / 7 3	INT	firmware rel. address	
0xffff0013	2	Firmware Unit	4 / 5 6 / 7 3	CHAR	firmware device type	
0xffff0014	2	Firmware Unit	4 / 5 6 / 7 3	INT	hardware revision	
0xffff0015	2	Firmware Unit	4 / 5 6 / 7 3	INT	firmware dest. addr.	
0xffff0016	2	Firmware Unit	4 / 5 6 / 7 3	INT	firmware configuration	

18 Maintenance

18.1 Cleaning

- 1 Switch off the E-711/E-712 system and remove the line cord.
- 2 Wait a minute to be sure that any residual voltage has dissipated.
- 3 Clean the housing surfaces of the E-711/E-712 system using mild detergents or disinfectant solutions.

18.2 AC Power and Line Fuses



CAUTION

Both fuses are active and have to be checked if there is a fault.

The power connection and line fuses are located on the rear panel of the chassis. To access the line power fuses, proceed as follows:

- 1 Switch off the E-711/E-712 system and remove the line cord.
- 2 Wait a minute to be sure that any residual voltage has dissipated.
- 3 Pry open the door that covers the fuse carrier and pry out the fuse carrier (see figures below).
- 4 Be sure to replace both fuses with fuses of the suitable type (IEC standard):
E-712.R5 chassis: 2 x T3.15AH, 250 V*
E-712.R1, .R2, .R3, .R4 chassis: 2 x T1.6AH, 250 V*
* Unless otherwise noted on the type label on the rear panel of the chassis.

Note: IEC-standard fuses are designed to carry the nominal current indefinitely. Other fuse rating standards differ.

- 5 Reinstall the carrier and close the door.



18.3 Firmware Updates

The overall firmware revision of your E-711/E-712 system can be identified in the answer of the IDN? command or the *IDN? command.

Each hardware module in the E-711/E-712 system has its own firmware which can be updated separately.

Detailed information regarding the individual firmware can be obtained from the parameters 0xffff0001 to 0xffff0016, see "Parameter Overview" (p. 233) for a detailed parameter list. Query with the SPA? or SEP? command.

All firmware updates can be made by running the Firmware Update Wizard on the host computer, using the same communication interfaces as for normal operation. The Firmware Update Wizard is available on the E-712 CD and is included in the "complete" software installation (p. 68).

The Firmware Update Wizard guides you through the update of the firmware for the hardware modules of your E-711/E-712 system.



Figure 57: The Firmware Update Wizard Start Screen

19 Troubleshooting

Communication with controller does not work

Communication cable is wrong or defective

⇒ Check cable. Does it work properly with another device?

For RS-232, a null-modem cable must be used.

For TCP/IP connections: Connect the controller to an Ethernet connector in the PC using the included cross-over network cable. When connecting to a network hub or router, it might be necessary to use a straight-through network cable.

The power-on/reboot sequence had not finished

⇒ Wait at least 40 seconds after power-on or reboot before you try to establish communications or to send commands. Note that even if the Power LED glows permanently, it might take another period of about 30 seconds before communication between controller and host PC can be established, if you have a network without DHCP server or a point-to-point connection via TCP/IP.

The interface is not configured correctly

⇒ With the RS-232 interface, check port and baud rate (depending on your controller, the baud rate may be set via DIP switches on the front panel or via a controller parameter). It is recommended that the host PC have a "genuine" RS-232 interface on board. If the host PC uses a USB-to-serial adapter instead, data loss could occur during communication, especially when transferring large amounts of data.

⇒ With the TCP/IP connection, connect the controller to a network access point before you power it on. Check IP address and IP mask (the settings of the devices in the network must be compatible, e.g. the IP address of each device must be unique). Make sure that your network administrator has not set the network to forbid unknown devices like the E-711/E-712 system to log on. Note that the maximum number of TCP/IP connections can be configured via the Max. TCP/IP Connections parameter (ID 0x11000c00; default: 1).

Controller was power-cycled or rebooted

⇒ With TCP/IP and USB connections, communication cannot be maintained after the E-711/E-712 system is power-cycled or rebooted. The connection must then be closed and reopened.

Another program is using the interface

⇒ Close the other program.

Specific software has problems

⇒ See if the system works with some other software, e.g. a terminal or development environment. You can, for example, test the communication by simply starting a terminal program, e.g. PI Terminal, and entering commands like *IDN? or HLP?. Note that multi-character commands are transferred as terminated by a LF (line feed) character and are executed only after the LF is received.

Stage does not move

Cable not connected properly

⇒ Check the connecting cable(s)

Stage or stage cable is defective

⇒ Exchange stage with a working stage to test a new combination of controller and stage.

Wrong command or wrong syntax

⇒ Check the error code with the ERR? command. "Error Codes" in the GCS commands manual PZ233E gives the complete error reference.

Wrong axis commanded

⇒ Check if the correct axis identifier is used and if the commanded axis is that of the desired stage (axis identifier also required with single-axis systems!)

Move commands or wave generator output provoke errors and are ignored

⇒ The axis motion can result from multiple control sources (see "Control Value Generation" (p. 114) for details). The sources have different write priorities:

Motion commands like MOV, MVR, SVA, SVR, IMP and STE are not allowed (will cause an error) when analog control input or wave generator output are active.

It is possible to configure an axis for control by an analog input line while the wave generator output is active for that axis. In this case, the wave generator will continue running, but its output will no longer be used for control value generation. As long as the corresponding axis is set up to be commanded by analog control input, you can stop the wave generator output, but not restart it.

The analog control input is ignored

⇒ When the analog input is used as control source and the axis motion is stopped with STP or #24, the analog input channel may be disconnected from the axis. To recommence commanding the axis via the analog input, the corresponding input signal channel must be reconnected to the axis. See "How to work with the Analog Input" (p. 100) for more information.

Offset between axis position range and voltage range is too large so that the commanded control value cannot be realized

⇒ For systems with linear piezo actuators and capacitive or strain gauge sensors, both the range of sensor position values and the range of the output drive voltages are limited. If mechanical drift of the piezo actuator causes too great a shift in the relation between these ranges, then the usable closed-loop travel range will be reduced. Such an offset can be compensated by the AutoZero function. If AutoZero is compatible with your application, use the ATZ command or the AutoZero functionality of PIMikroMove.

Incorrect configuration

⇒ Check the parameter settings on the controller with the SPA? and SEP? commands.

The high voltage output of the controller is deactivated, while the communication with the controller is still possible

⇒ With amplifier modules, the high voltage output is deactivated automatically when the internal temperature sensor detects 75 °C. The

output will be reactivated automatically when the internal temperature has fallen to 60 °C.

Wait a few minutes to let the E-711/E-712 system cool down. Note that the wave generator output will continue even if the high voltage output is deactivated, i.e. if a certain number of output cycles was set, they may have already finished when the high voltage output is reactivated.

How to avoid overheating:

Keep the ambient temperature at a noncritical value: Note that the difference between ambient temperature and internal temperature of the E-711/E-712 system normally is about 20 Centigrade (36 Fahrenheit) degrees.

Place the system in a location with adequate ventilation. Allow at least 15 cm clearance from the rear of the unit and 1 cm from the bottom (ensured by the feet of the chassis). Never cover any ventilation openings as this will impede ventilation.

When using the wave generator, it is recommended to reduce the frequency and/or the amplitude and/or the output duration to avoid overheating. See "Wave Generator" (p. 172) for more information.

Unsatisfactory system performance

The sensor values are not reliable, and the whole system is instable.

⇒ Only thermally stable systems can have the best performance. For a thermally stable system, power on the E-711/E-712 system at least one hour before you start working with it.

The waveform output is incomplete, the last points are missing

⇒ All waveform output is synchronized because there is a common pulse generator used by all wave generators. For that reasons, wave tables which are supposed to run at the same time (each with one wave generator) should have the same length. If the wave tables have different lengths, an output cycle will comprise only the number of points given by the shortest table. This means that all waveform output is cut to the length of the shortest waveform currently running.

A phase shift between the trigger output and the actual position is observed when the wave generator is running with the "Use DDL" or "Use and reinitialize DDL" options.

⇒ If the actual position (measured with external device) is ahead of the trigger signal, check the cut-off-frequency f_g of the IIR low-pass filter used for

digital filtering and increase it to shorten the duration of the digital filtering.

The duration of the signal processing for a capacitive sensor results from two portions:

- Duration of the analog sensor processing
- Duration of the digital filtering. If the Digital Filter Type parameter, ID 0x05000000, is set to "IIR low-pass filter, 2nd order", the required time depends on the cut-off-frequency f_g of the IIR low-pass filter which is given by the Digital Filter Bandwidth parameter, ID 0x05000001. For signal frequencies $f < f_g/2$, the duration of the filtering can be estimated as follows:
 $t \approx 0.216 / f_g$
If required, increase f_g . Examples:
 $f_g = 300 \text{ Hz: } t = 0.72 \text{ ms}$
 $f_g = 1000 \text{ Hz: } t = 0.216 \text{ ms}$

Custom software accessing PI drivers does not run.

Wrong combination of driver routines/VIs

⇒ Check if system runs with Terminal program. If yes read the software manual and compare sample code from the E-712 CD to check the necessary driver routines.

20 Customer Service

For inquiries and orders, contact your PI sales engineer or send us an e-mail (info@pi.ws).

If you have questions concerning your system, have the following information ready:

- Product codes and serial numbers of all products in the system
- Firmware version of the controller (if present)
- Version of drivers and / or PC software (if present)
- Operating system on the PC (if present)

If possible: Take photographs or make videos of your system that can be sent to our customer service department if requested.

The latest versions of the user manuals are available for download on our website (www.pi.ws).

Only PI service personnel must repair the E-711/E-712 system.

21 Old Equipment Disposal

In accordance with the applicable EU law, electrical and electronic equipment may not be disposed of with unsorted municipal wastes in the member states of the EU.

When disposing of your old equipment, observe the international, national and local rules and regulations.

To meet the manufacturer's product responsibility with regard to this product, Physik Instrumente (PI) GmbH & Co. KG will ensure environmentally correct disposal of old PI equipment that was first put into circulation after 13 August 2005, free of charge.

If you have such old equipment from PI, you can send it to the following address postage-free:

Physik Instrumente (PI) GmbH & Co. KG
Auf der Roemerstr. 1
76228 Karlsruhe, Germany



22 Pin Assignments

For the pinout of the digital sensor signal transmission, see the corresponding Technical Note: E711T0002 for E-711.OCT/.OCT0 / E711T0007 for E-711.OET/.OET0.

For pinout of the E-711.IP PIO interface module see the separate E711T0001 Technical Note.

22.1 E-711.SC3H: Socket for Piezo Actuators and Capacitive Sensors

The socket for the piezo actuators and the capacitive sensors is located on the E-711.SC3H sensor module.

Connector type: Sub-D mix 25W3 socket for 3 coax lines and 22 single pins

Pin	Signal	Function
Coax inner lines:		
A1	input	Sensor Probe Ch 2
A2	input	Sensor Probe Ch 3
A3	input	Sensor Probe Ch 1
Standard pins:		
1	output	Sensor Target Ch 2
2	output	Sensor Target Ch 3
3	GND	ID-Chip GND Axis 1
4	bidirectional	ID-Chip Axis 1
5	GND	ID-Chip GND Axis 2
6	bidirectional	ID-Chip Axis 2
7	output	Piezo Ch 4 +
8	output	Piezo Ch 3 +
9	output	Piezo Ch 2 +
10	output	Piezo Ch 1 +
11	output	Sensor Target Ch 1
12	GND	Sensor Target Ch 2 shield
13	GND	Sensor Target Ch 3 shield
14	-	nc
15	-	nc
16	GND	ID-Chip GND Axis 3
17	bidirectional	ID-Chip Axis 3
18	output	Piezo Ch 4 -
19	output	Piezo Ch 3 -
20	output	Piezo Ch 2 -
21	output	Piezo Ch 1 -
22	GND	Sensor Target Ch 1 shield



Note:

Sensor Probe and Sensor Target are the connections of the capacitive sensor in the mechanics.

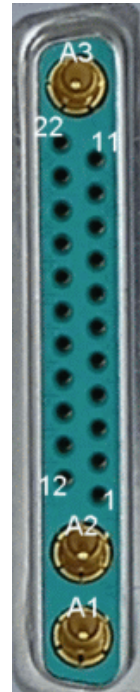
The Piezo lines carry the piezo voltage for the actuators in the mechanics, in the range of -30 to 135 V.

22.2 E-711.AL41: Socket for Piezo Actuators

The socket for the piezo actuators is located on the E-711.AL41 amplifier module.

Connector type: Sub-D mix 25W3 socket for 3 coax lines and 22 single pins

Pin	Signal	Function
Coax inner lines:		
A1	-	nc
A2	-	nc
A3	-	nc
Standard pins:		
1	-	nc
2	-	nc
3	-	nc
4	-	nc
5	-	nc
6	-	nc
7	output	Piezo Ch 4 +
8	output	Piezo Ch 3 +
9	output	Piezo Ch 2 +
10	output	Piezo Ch 1 +
11	-	nc
12	-	nc
13	-	nc
14	-	nc
15	-	nc
16	-	nc
17	-	nc
18	output	Piezo Ch 4 -
19	output	Piezo Ch 3 -
20	output	Piezo Ch 2 -
21	output	Piezo Ch 1 -
22	-	nc



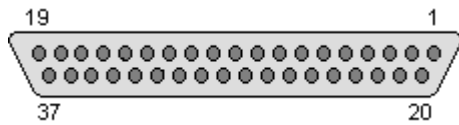
Note:

The Piezo lines carry the piezo voltage for the actuators in the mechanics, in the range of -30 to 135 V.

22.3 E-711.SS3: Socket for Piezo Actuators and Strain Gauge Sensors

The socket for the piezo actuators and the strain gauge sensors (SGS) is located on the E-711.SS3 sensor module.

Connector type: Sub-D37 (f) socket



Pin	Signal	Function
1	Input / Output	PT1000 +
2	GND	GND
3	Bidirectional	ID Chip 2 Data
4	Bidirectional	ID Chip 3 Data
5	GND	GND
6	Input	SGS 4 -
7	GND	SGS 4 GND
8	Input	SGS 3 -
9	GND	SGS 3 GND
10	Input	SGS 2 -
11	GND	SGS 2 GND
12	Input	SGS 1 -
13	GND	SGS 1 GND
14	-	Not connected
15	-	Not connected
16	Output	Piezo 1 +
17	Output	Piezo 2 +
18	Output	Piezo 3 +
19	Output	Piezo 4 +
20	Input	PT1000 -
21	Bidirectional	ID Chip 1 Data
22	GND	GND
23	Bidirectional	ID Chip 4 Data
24	Input	SGS 4 +
25	Output	SGS 4 Reference
26	Input	SGS 3 +
27	Output	SGS 3 Reference
28	Input	SGS 2 +
29	Output	SGS 2 Reference
30	Input	SGS 1 +
31	Output	SGS 1 Reference
32	-	Not connected
33	-	Not connected
34	Output	Piezo 1 -
35	Output	Piezo 2 -
36	Output	Piezo 3 -
37	Output	Piezo 4 -

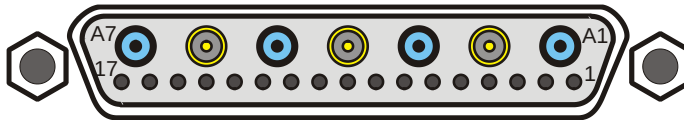
Note:

The Piezo lines carry the piezo voltage for the actuators in the mechanics, in the range of -30 to 135 V.

22.4 E-711.AM4: Socket for NEXLINE® Drive or Piezo Actuators

The socket is located on the E-711.AM4 amplifier module.

Connector type: Sub-D mix socket 24W7 for 7 coax lines and 17 single pins



Pin	Signal	Function
Coax inner lines:		
A1	output	Piezo Ch 1 +
A2	nc	nc
A3	output	Piezo Ch 2 +
A4	nc	nc
A5	output	Piezo Ch 3 +
A6	nc	nc
A7	output	Piezo Ch 4 +
Standard pins:		
1	output	Piezo Ch 1 -
2	-	reserved
3	-	reserved
4	nc	nc
5	nc	nc
6	output	Piezo Ch 2 -
7	-	reserved
8	-	reserved
9	nc	nc
10	nc	nc
11	output	Piezo Ch 3 -
12	-	reserved
13	-	reserved
14	nc	nc
15	nc	nc
16	nc	nc
17	output	Piezo Ch 4 -

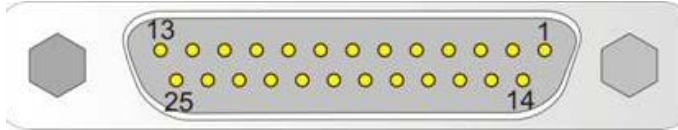
Note:

The Piezo lines carry piezo voltages in the range of -250 to 250 V. They can be used for the shearing and clamping segments of the NEXLINE® stack actuator pairs in a NEXLINE® drive, or for up to four conventional piezo actuators.

22.5 E-711.AM5: Socket for NEXLINE® Drive or Piezo Actuators

The socket is located on the E-711.AM5 amplifier module.

Connector type: Sub-D 25-pin (f)



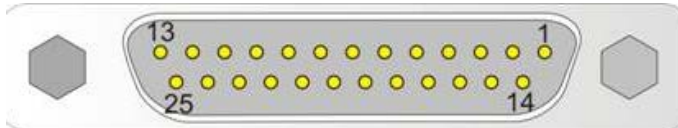
Pin	Signal	Function
1	output	Piezo Ch 1 +
2	-	internal use
3	-	internal use
4	output	Piezo Ch 2 +
5	-	internal use
6	-	internal use
7	output	Piezo Ch 3 +
8	GND	GND
9	GND	GND
10	output	Piezo Ch 4 +
11	nc	nc
12	-	internal use
13	-	internal use
14	-	internal use
15	output	Piezo Ch 1 -
16	nc	internal use
17	nc	internal use
18	output	Piezo Ch 2 -
19	-	internal use
20	-	internal use
21	output	Piezo Ch 3 -
22	nc	nc
23	-	internal use
24	output	Piezo Ch 4 -
25	-	internal use

Note:

The Piezo lines carry piezo voltages in the range of -250 to 250 V. They can be used for the shearing and clamping segments of the NEXLINE® stack actuator pairs in a NEXLINE® drive, or for up to four conventional piezo actuators.

22.6 E-711.AM5A and .AM5B: Socket for NEXLINE® Drive with Encoder

The socket for a NEXLINE® drive with encoder is located on the E-711.AM5A amplifier module (incremental encoder) or on the E-711.AM5B amplifier module (absolute measuring encoder via BiSS interface).
Connector type: Sub-D 25-pin (f)



Pin	Signal	Function
1	output	Piezo Ch 1 +
2	output	5 V Out
3	output	5 V Out
4	output	Piezo Ch 2 +
5	-	nc
6	-	nc
7	output	Piezo Ch 3 +
8	GND	GND
9	GND	GND
10	output	Piezo Ch 4 +
11	-	nc
12	input	Ref-
13	input	Ref+
14	input	Enc Positive Limit
15	output	Piezo Ch 1 -
16	input	E-711.AM5A: Sin+ E-711.AM5B: MA+
17	input	E-711.AM5A: Sin- E-711.AM5B: MA-
18	output	Piezo Ch 2 -
19	input	E-711.AM5A: Cos+ E-711.AM5B: SLI+
20	input	E-711.AM5A: Cos- E-711.AM5B: SLI-
21	output	Piezo Ch 3 -
22	-	nc
23	input	Enc Negative Limit
24	output	Piezo Ch 4 -
25	bidirectional	ID-Chip

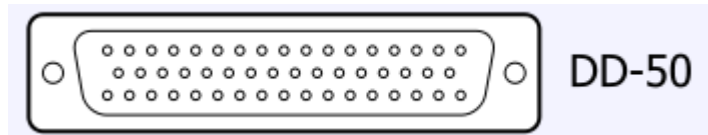
Note:

The Piezo lines carry the piezo voltages for the shearing and clamping segments of the NEXLINE® stack actuator pairs in the NEXLINE® drive, in the range of -250 to 250 V.

22.7 E-712.R5: Socket for NEXLINE® Drives or Piezo Actuators

The socket is located on the front panel of the E-712.R5 chassis near the ON/OFF switch. It carries the output voltage of up to 6 E-711.AM6 amplifier modules that are present in the E-712.R5 chassis.

Connector type: Sub-D 50-pin (f) High Density



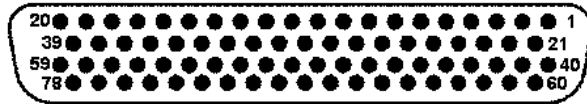
Pin	Signal		Pin	Signal
1	Amp 1 Piezo Ch 1 -		26	Amp 4 Piezo Ch 1 -
2	Amp 1 Piezo Ch 1 +		27	Amp 4 Piezo Ch 1 +
3	Amp 1 Piezo Ch 2 -		28	Amp 4 Piezo Ch 2 -
4	Amp 1 Piezo Ch 2 +		29	Amp 4 Piezo Ch 2 +
5	Amp 1 Piezo Ch 3 -		30	Amp 4 Piezo Ch 3 -
6	Amp 1 Piezo Ch 3 +		31	Amp 4 Piezo Ch 3 +
7	Amp 1 Piezo Ch 4 -		32	Amp 4 Piezo Ch 4 -
8	Amp 1 Piezo Ch 4 +		33	Amp 4 Piezo Ch 4 +
9	Amp 2 Piezo Ch 1 -		34	Amp 5 Piezo Ch 1 -
10	Amp 2 Piezo Ch 1 +		35	Amp 5 Piezo Ch 1 +
11	Amp 2 Piezo Ch 2 -		36	Amp 5 Piezo Ch 2 -
12	Amp 2 Piezo Ch 2 +		37	Amp 5 Piezo Ch 2 +
13	Amp 2 Piezo Ch 3 -		38	Amp 5 Piezo Ch 3 -
14	Amp 2 Piezo Ch 3 +		39	Amp 5 Piezo Ch 3 +
15	Amp 2 Piezo Ch 4 -		40	Amp 5 Piezo Ch 4 -
16	Amp 2 Piezo Ch 4 +		41	Amp 5 Piezo Ch 4 +
17	internal use		42	Amp 6 Piezo Ch 1 -
18	Amp 3 Piezo Ch 1 -		43	Amp 6 Piezo Ch 1 +
19	Amp 3 Piezo Ch 1 +		44	Amp 6 Piezo Ch 2 -
20	Amp 3 Piezo Ch 2 -		45	Amp 6 Piezo Ch 2 +
21	Amp 3 Piezo Ch 2 +		46	Amp 6 Piezo Ch 3 -
22	Amp 3 Piezo Ch 3 -		47	Amp 6 Piezo Ch 3 +
23	Amp 3 Piezo Ch 3 +		48	Amp 6 Piezo Ch 4 -
24	Amp 3 Piezo Ch 4 -		49	Amp 6 Piezo Ch 4 +
25	Amp 3 Piezo Ch 4 +		50	internal use

Note: The Piezo lines carry piezo voltages in the range of -250 to 250 V. They can be used for the shearing and clamping segments of the NEXLINE® stack actuator pairs in up to 6 NEXLINE® drives, or for up to 24 conventional piezo actuators.

22.8 E-712.R5: Socket for NEXACT® Drives

The socket is located on the front panel of the E-712.R5 chassis. It carries the output voltage for up to 6 NEXACT® drives (the appropriate number of E-711.AN40 amplifier modules must be present in the E-712.R5 chassis).

Connector type: Sub-D 78-pin (f) High Density



Pin	Pin	Signal	Pin	Pin	Signal
1		Amp 1 CH1+	40		Amp 4 CH1+
	21	Amp 1 CH1-		60	Amp 4 CH1-
2		Amp 1 CH2+	41		Amp 4 CH2+
	22	Amp 1 CH2-		61	Amp 4 CH2-
3		Amp 1 CH3+	42		Amp 4 CH3+
	23	Amp 1 CH3-		62	Amp 4 CH3-
4		Amp 1 CH4+	43		Amp 4 CH4+
	24	Amp 1 CH4-		63	Amp 4 CH4-
5		Amp 1 enable*	44		Amp 4 enable*
	25	n.c.		64	n.c.
6		n.c.	45		n.c.
	26	n.c.		65	n.c.
7		Amp 2 CH1+	46		Amp 5 CH1+
	27	Amp 2 CH1-		66	Amp 5 CH1-
8		Amp 2 CH2+	47		Amp 5 CH2+
	28	Amp 2 CH2-		67	Amp 5 CH2-
9		Amp 2 CH3+	48		Amp 5 CH3+
	29	Amp 2 CH3-		68	Amp 5 CH3-
10		Amp 2 CH4+	49		Amp 5 CH4+
	30	Amp 2 CH4-		69	Amp 5 CH4-
11		Amp 2 enable*	50		Amp 5 enable*
	31	n.c.		70	n.c.
12		n.c.	51		n.c.
	32	n.c.		71	n.c.
13		Amp 3 CH1+	52		Amp 6 CH1+
	33	Amp 3 CH1-		72	Amp 6 CH1-
14		Amp 3 CH2+	53		Amp 6 CH2+
	34	Amp 3 CH2-		73	Amp 6 CH2-
15		Amp 3 CH3+	54		Amp 6 CH3+
	35	Amp 3 CH3-		74	Amp 6 CH3-
16		Amp 3 CH4+	55		Amp 6 CH4+
	36	Amp 3 CH4-		75	Amp 6 CH4-
17		Amp 3 enable*	56		Amp 6 enable*
	37	n.c.		76	n.c.
18		n.c.	57		n.c.

Pin	Pin	Signal	Pin	Pin	Signal
	38	n.c.		77	n.c.
19		n.c.	58		n.c.
	39	n.c.		78	n.c.
20		n.c.	59		n.c.

*This pin is connected to GND in the connector shell of the NEXACT® drive to enable the amplifiers. If this connection is removed, all piezo voltages will be zero.

The CH1 to CH4 lines carry the piezo voltages for the actuator segments of the NEXACT® drives, in the range of -10 to +45 V.

22.9 E-711.AN4: Socket for NEXACT® Drive

The socket for the NEXACT® drive is located on the E-711.AN4 amplifier module.

Connector type: Sub-D 15-pin (f) High Density

Pin	Signal direction	Function
1	output	Piezo 1
2	output	Piezo 3
3	nc	nc
4	nc	nc
5	nc	nc
6	output	Piezo 0
7	output	Piezo 2
8	input	AMP (amplifier enable)*
9	nc	nc
10	nc	nc
11	AGND	Piezo GND
12	AGND	Piezo GND
13	AGND	Piezo GND
14	AGND	Piezo GND
15	GND	Digital GND



*This pin is connected to GND in the connector shell of the NEXACT® drive to enable the amplifiers. If this connection is removed, all piezo voltages will be zero.

22.10 E-711.C82: Socket for DC Motors

Sockets for DC motors are located on the E-711.C82 DC motor driver module.

Connector type: Sub-D 15-pin (f)

Pin	Signal direction	Function
1	output	Programmable motor brake (0 or + 5 V)
2	output	Motor + (differential; power PWM); for stages without PWM amplifier
3	output	PWM magnitude (TTL); for stages with PWM amplifier
4	output	+5 V
5	input	Positive limit switch
6	-	reserved
7	input	A- / Sin- / BISS MA1
8	input	B- / Cos- / BISS SL1
9	output	Motor – (differential; power PWM); for stages without PWM amplifier
10	GND	Power GND
11	output	PWM sign (TTL); for stages with PWM amplifier
12	input	Negative limit switch
13	input	Reference point switch
14	input	A+ / Sin+ / BISS MA1
15	input	B+ / Cos+ / BISS SL1



The E-711.C82 supports the following encoder types:

- Incremental sensor: AB (quadrature) single-ended or differential TTL signal acc. to RS-422; 60 MHz
- Incremental sensor: Analog sin/cos; 1 Vpp
- Absolute measuring sensor via 32-bit BiSS interface

The encoder type is selectable via the Sensor Type parameter (ID 0x02000001).

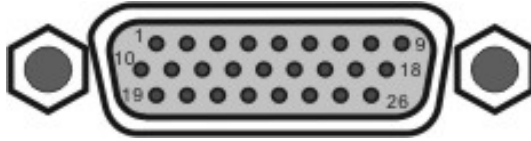
22.11 E-711.SA3 and E-711.SA6: Socket for Incremental Sensors

The INCR. SENSOR socket is located on the E-711.SA3 and E-711.SA6 sensor modules. Sensors cannot be connected directly to that socket but must be connected via an E-711.SAx sensor signal splitter box of the suitable type (x = P, H or N depending on the sensor type) and an E-711.SAX connection cable.

Connector type:

E-711.SA3: 1 x Sub-D 26-pin (m) High Density

E-711.SA6: 2 x Sub-D 26-pin (m) High Density

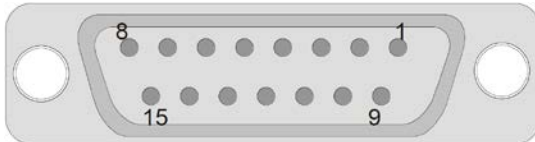


Pin Number	Type	Function
1	IN	Encoder A1(+)
2	IN	Encoder B1(+)
3	IN	REF CH1 (+)
4	IN	Encoder A2(+)
5	IN	Encoder B2(+)
6	IN	REF CH2 (+)
7	IN	Encoder A3(+)
8	IN	Encoder B3(+)
9	IN	REF CH3(+)
10	IN	Encoder A1(-)
11	IN	Encoder B1(-)
12	IN	REF CH1 (-)
13	IN	Encoder A2(-)
14	IN	Encoder B2(-)
15	IN	REF CH2 (-)
16	IN	Encoder A3(-)
17	IN	Encoder B3(-)
18	IN	REF CH3(-)
19	IN	Positive Limit CH1
20	IN	Negative Limit CH1
21	IN	Positive Limit CH2
22	IN	Negative Limit CH2
23	IN	Positive Limit CH3
24	IN	Negative Limit CH3
25	OUT	5V
26	GND	GND

22.12 E-711.SAH: Sockets for Incremental Sensors

The “Sensor 1” to “Sensor 3” sockets are located on the E-711.SAH sensor signal splitter box. One Heidenhain incremental sensor can be connected per socket.

Connector type: Sub-D 15-pin (f)



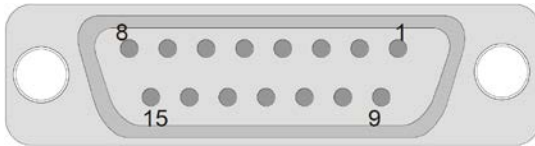
Pin Number	Type	Function
1	IN	Encoder A(+)
2	GND	AGND
3	IN	Encoder B(+)
4	OUT	5V
5	NC	NC
6	IN	Negative Limit
7	IN	REF(-)
8	IN	Positive Limit
9	IN	Encoder A(-)
10	GND	AGND
11	IN	Encoder B(-)
12	OUT	5V
13	NC	NC
14	IN	REF(+)
15	NC	NC

The “Sensor In” socket (Sub-D 26-pin (f) High Density) on the E-711.SAH sensor signal splitter box has the same pinout as the INCR. SENSOR socket(s) on the E-711.SA3 and E-711.SA6 modules (see p. 277).

22.13 E-711.SAN: Sockets for Incremental Sensors

The “Sensor 1” to “Sensor 3” sockets are located on the E-711.SAN sensor signal splitter box. One Numeric Jena incremental sensor can be connected per socket.

Connector type: Sub-D 15-pin (f)



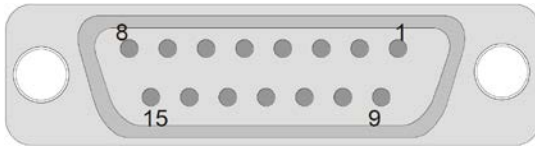
Pin	Type	Usage
1	NC	NC
2	IN	Negative Limit
3	NC	NC
4	IN	REF(-)
5	IN	Encoder B(-)
6	IN	Encoder A(-)
7	NC	NC
8	OUT	5 V
9	GND	AGND
10	IN	Positive Limit
11	NC	NC
12	IN	REF(+)
13	IN	Encoder B(+)
14	IN	Encoder A(+)
15	GND	shield

The “Sensor In” socket (Sub-D 26-pin (f) High Density) on the E-711.SAN sensor signal splitter box has the same pinout as the INCR. SENSOR socket(s) on the E-711.SA3 and E-711.SA6 modules (see p. 277).

22.14 E-711.SAP: Sockets for Incremental Sensors

The “Sensor 1” to “Sensor 3” sockets are located on the E-711.SAP sensor signal splitter box. One PIONe incremental sensor can be connected per socket.

Connector type: Sub-D 15-pin (f)



Pin Number	Type	Function
1	OUT	5 V
2	GND	AGND
3	IN	Encoder A(+)
4	IN	Encoder A(-)
5	NC	NC
6	IN	Encoder B(+)
7	IN	Encoder B(-)
8	IN	Negative Limit
9	NC	NC
10	IN	REF (+)
11	NC	NC
12	IN	REF (-)
13	NC	NC
14	NC	NC
15	IN	Positive Limit

The “Sensor In” socket (Sub-D 26-pin (f) High Density) on the E-711.SAP sensor signal splitter box has the same pinout as the INCR. SENSOR socket(s) on the E-711.SA3 and E-711.SA6 modules (see p. 277).

22.15 E-711.0ATS: Sockets for Encoders with BiSS Interface

The “Sensor 1” to “Sensor 8” sockets are located on the E-711.0ATS sensor box. One absolute measuring or incremental encoder with 32-bit BiSS interface can be connected per socket.

Connector type: Sub-D 9-pin (f)



Pin Number	Type	Function
1	GND	GND
2	IN	MA+
3	IN	MA-
4	OUT	5V Sensor
5	OUT	5V Sensor
6	IN	SLI+
7	IN	SLI-
8	IN	GND
9	IN	GND

The “SPI” socket on the E-711.0ATS sensor box has the same pinout as the SPI sockets on the E-711.iS3 and E-711.iS6 SPI interface modules (see p. 284).

22.16 E-711.SE3: Sockets for Capacitive Sensor Probe

Up to three PISeca single-electrode capacitive sensors can be connected to the E-711.SE3 module. The sensor probes are connected to the Ch1 to Ch3 sockets, while the target surfaces of all sensors must be connected to the GND grounding stud.

Connector type for Ch1 to Ch3: Triaxial socket, LEMO ECP.00.650.NLL.543

Pin	Function
Center Pin (Core)	Capacitive Sense Line
Inner shield	Active Shielding
Outer shield	Measurement Ground

22.17 Analog In Sockets

The analog input lines In 1 to In 4 are located on the E-711.IA4 analog interface module. The identifiers representing these analog input lines in the firmware of the E-711/E-712 system depend on the number of capacitive sensor channels present in the E-711/E-712 system. See "Accessible Items and Their Identifiers" (p. 104) for more information.

Analog input lines which are not used should be deactivated to avoid interferences. See "Deactivation of Unused Analog Input Lines" (p. 100) for more information.

Connector type: LEMO EPG.00.302.NLN

Pin	Signal	Function
1	input	A_{IN+} Analog differential input
2	input	A_{IN-} Analog differential input



The socket case is connected to GND.

$|A_{IN+} - A_{IN-}| \leq 10 \text{ V}$ in the range of -10 to +10 V

Resolution: 18 bit

Bandwidth: 25 kHz

Max. impedance of the signal source: 250 ohm

A larger impedance of the signal source will increase the noise and hence reduce the resolution.

22.18 Analog Out Sockets

The analog output lines Out 1 to Out 4 are located on E-711.IA4 analog interface module.

Connector type: LEMO EPG.00.302.NLN

Pin	Signal	Function
1	output	A_{OUT} Analog output, -12.775 to 12.775 V
2	GND	GND



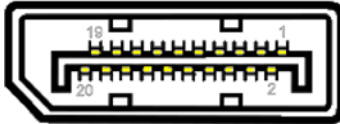
Resolution: 16 bit

Bandwidth: 12 kHz

22.19 E-711.iS3 and E-711.iS6: SPI Socket

The sockets for connection of SPI slaves (e.g., sensor boxes from PI) are located on the E-711.iS3 and E-711.iS6 SPI modules.

Connector type: DisplayPort socket



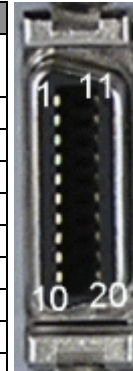
Pin	Direction	Signal Name
20		nc
19	Input	LDAT-
18	Input	LDAT+
17	Input	CS-
16		GND
15	Input	CS+
14		nc
13		GND
12	Input	SCLK+
11		nc
10	Input	SCLK-
9	Input	MOSI+
8		nc
7	Input	MOSI-
6	Output	MISO+
5		GND
4	Output	MISO-
3	Output	DCLK+
2		GND
1	Output	DCLK-

22.20 Digital I/O Socket

The socket for the digital I/O lines is located on the E-712.M1/.N1 digital processor module.

Connector type: MDR20

Pin	Signal	Function
1	output	ServoTrigger (TTL, digital output of the servo cycle; not accessible by command)
2	output	OUT1 (TTL, active high; identifier is 1)
3	output	OUT2 (TTL, active high; identifier is 2)
4	output	OUT3 (TTL, active high; identifier is 3)
5	GND	GND
6	output	OUT4 (TTL, active high; identifier is 4)
7	output	OUT5 (TTL, active high; identifier is 5)
8	output	OUT6 (TTL, active high; identifier is 6)
9	output	OUT7 (TTL, active high; identifier is 7)
10	-	reserved
11	input	IN1 (TTL, active high; identifier is 1)
12	input	IN2 (TTL, active high; identifier is 2)
13	input	IN3 (TTL, active high; identifier is 3)
14	input	IN4 (TTL, active high; identifier is 4)
15	GND	GND
16	input	IN5 (TTL, active high; identifier is 5)
17	input	IN6 (TTL, active high; identifier is 6)
18	input	IN7 (TTL, active high; identifier is 7)
19	input	IN8 (TTL, active high; identifier is 8)
20	-	reserved



Digital inputs:

- TTL (low: 0 to 0.8 V, high: 2 to 5 V, max.: 5 V)
- When nothing is connected to a digital input, the signal level is high due to an internal pull-up with 10 kohm resistor.
- The digital input lines can be used for the following purposes:
 - o Safety shutdown: configure the lines IN1 to IN8 via parameters, see p. 211 for details
 - o Start data recording: configure the lines IN1 to IN8 with DRT
 - o Trigger the wave generator output (lines IN1 and IN2) and stop it (line IN2), see p. 184 and the WGO command for details
- Line IN8 (pin 19) can be configured as reset input using the Reboot On DIO Input parameter (ID 0x0e001500). Changes of the parameter value become effective immediately. The value of the parameter enables/disables the Reset input as follows:
 - o 0 = OFF: Reset input is disabled (default setting)
 - o 1 = ON: Reset input is enabled. If the signal level on the Reset input becomes low, the E-711/E-712 signal is rebooted (same behaviour as with the RBT command).

Digital outputs:

- High level:
 - o at -2 mA output current => min. 2.2 V
 - o at -0.1 mA output current => min. 3.0 V

- Low level:
 - o at +2 mA output current => max. 0.6 V
 - o at +0.1 mA output current => max. 0.21 V
- The digital output lines OUT1 to OUT7 can be configured with the CTO and TWS commands for triggering tasks.
- The servo cycle output on pin 1 is not accessible for commands.

22.21 RS-232 Socket

The RS-232 socket is located on the E-712.M1 digital computer and interface module.

Connector type: Sub-D 9 pin (m)

Pin	Function
1	nc
2	RXD receive data
3	TXD send data
4	nc
5	DGND ground
6	nc
7	RTS Hardware handshake, output
8	CTS Hardware handshake, input
9	nc

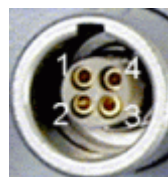


22.22 Sync In Socket

The SYNC IN socket is located on the E-712.M1 digital computer and interface module.

Connector type: LEMO EPG.00.304.NLN

Pin	Function
1	SYNCBASE_I_P (LVDS, 4.8 MHz)
2	SYNCBASE_I_N (LVDS, 4.8 MHz)
3	SYNC100_I_P (LVDS, 100 kHz)
4	SYNC100_I_N (LVDS, 100 kHz)



NOTE

If you apply external synchronization signals to the Sync In socket, make sure that these signals match the following specifications:

The tolerance of the 100 kHz input signal is ± 2 kHz.

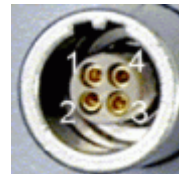
The 4.8 MHz signal must be synchronized with the 100 kHz signal, and both signals must be in a fixed ratio to each other.

22.23 Sync Out Socket

The SYNC OUT socket is located on the E-712.M1 digital computer and interface module.

Connector type: LEMO EPA.00.304.NLN

Pin	Function
1	SYNCBASE_O_P (LVDS, 4.8 MHz)
2	SYNCBASE_O_N (LVDS, 4.8 MHz)
3	SYNC100_O_P (LVDS, 100 kHz)
4	SYNC100_O_N (LVDS, 100 kHz)



23 EC Declaration of Conformity

For the E-711/E-712 Modular Digital Multi-Channel Controller System, an EC Declaration of Conformity has been issued in accordance with the following European directives:
Low Voltage Directive
EMC Directive
RoHS Directive

The applied standards certifying the conformity are listed below.
Safety (Low Voltage Directive): EN 61010-1
EMC: EN 61326-1
RoHS: EN 50581

